

P0820-TERNIUM SIDERAR

Sistema de embalaje automático de pallets

Esquemas Eléctricos

5	5	PROJECT	SIDERCROM EMBALAJE	SHEET SHEETS
4	4	DRAWING NO.	001	001 / 131
3	3	DESIGNED BY	TERNIUM SIDERAR	
2	2	REVIEWED BY		
1	1	REV. CLIENTE	REV. INT. P&F	
		 P&F S.A.	Av. Belgrano 2272 CHISIDVID Don Torcuato Buenos Aires Argentina	
		JOB	29/06/2015	
		DEVISION	PLANT SIDERAR	
		LOCATION/ACTIVITY		

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3	INDICE PRINCIPAL
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30	INPUTS, OUTPUTS
53	HARDWIRED CONTROL CIRCUIT
80	INTERCONNECT DIAGRAM
90	CONTROL PANEL AND BACKPLATE LAYOUTS
130	BILL OF MATERIAL

PROJECT SIDERCROM EMBALAJE		SHEET SHEETS 002 131	
DRAWING NO. 002		TITLE INDICE GENERAL	
DESIGNED BY L.A.	DESIGNED BY E.G.	DIVISION PLANTA SIDERCROM	
REV. CLIENTE	REV. INT. P&F	LOCATION/ACTIVITY	
Av. Belgrano 2272 (BELLEROS) San Torcuato Buenos Aires Argentina		DATE 29/10/2015	SCALE
JOB		DIVISION	

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015	ALIMENTACION MOTORES	380V	
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023	ALIMENTACION 24VDC	SWITCH ETHERNET	
024			
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027	RELE DE SEGURIDAD	PUERTAS	ESTACION #3
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030	INPUTS AND OUTPUTS		
031	MODULO INPUTS	RACK 1 – SLOT 4	1756–IB32
032	MODULO INPUTS	RACK 1 – SLOT 5	1756–IB32
033	MODULO INPUTS	RACK 1 – SLOT 6	1756–IB32
034	MODULO INPUTS	RACK 1 – SLOT 7	1756–IB32
035	MODULO INPUTS	RACK 1 – SLOT 8	1756–IB32
036			
037	MODULO OUTPUTS	RACK 1 – SLOT 9	1756–OW16I
038	MODULO OUTPUTS	RACK 1 – SLOT 10	1756–OW16I
039	MODULO OUTPUTS	RACK 1 – SLOT 11	1756–OW16I
040	MODULO OUTPUTS	RACK 1 – SLOT 12	1756–OW16I
041			
042	MODULO INPUTS	RACK 2 – SLOT 1	1756–IB32
043	MODULO INPUTS	RACK 2 – SLOT 2	1756–IB32
044			
045			
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048			
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051			
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053	CARATULA	HARDWIRED CONTROL CIRCUIT	
054			
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056			
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061	VARIADORES DE FRECUENCIA	3 Y 4	
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072	VARIADORES DE FRECUENCIA	25	
073			
074			

PLANO	DESCRIPCION 1	DESCRIPCION 2	DESCRIPCION 3
075	CONTROL EXTRACURSO	SUNCHADORA	
076			
077			
078			
079			
080	INTERCONNECT DIAGRAM		
081	RED DE ETHERNET		
082	DISTRIBUCION MULTIPAR	SEÑALES	
083	DISTRIBUCION CABLES	POTENCIA	
084	DISTRIBUCION	ROBOT Y CONTROLADOR	
085	MANDO A DISTANCIA		
086	LECTOR CODIGO DE BARRAS	EST #3	
087	LECTOR CODIGO DE BARRAS	EST #5	
088	EST#2	BALANZA	CELDA DE CARGA
089	EST#5	BALANZA	CELDA DE CARGA
090	CARATULA	CONTROL PANEL AND	BACKPLATE LAYOUTS
091	TOPOGRAFICO	TABLERO COMANDO	
092	TOPOGRAFICO	TABLERO POTENCIA	
093			
094	TOPOGRAFICO	TABLERO	JB ACCESO 1
095	TOPOGRAFICO	TABLERO	JB ACCESO 2
096	TOPOGRAFICO	TABLERO	JB ACCESO 3
097	TOPOGRAFICO	TABLERO	JB ACCESO 4
098	TOPOGRAFICO	BOTONERA TIJERA	CORTE DE FILM
099	TOPOGRAFICO	TABLERO DISTRIBUCION	JB #1
100	TOPOGRAFICO	TABLERO DISTRIBUCION	JB #2
101	TOPOGRAFICO	TABLERO DISTRIBUCION	JB #3
102	TOPOGRAFICO	TABLERO DISTRIBUCION	JB #4
103	TOPOGRAFICO	TABLERO DISTRIBUCION	JB #5
104	TOPOGRAFICO	TABLERO NEUMATICA	JB #6
105	TOPOGRAFICO	TABLERO NEUMATICA	JB #7
106	TOPOGRAFICO	TABLERO DISTRIBUCION	JB #8
107	CABEZAL SUNCHADOR	TITAN K120–19M	
108	TOPOGRAFICO	TABLERO PANEL VIEW	PV #1
109	TOPOGRAFICO	TABLERO PANEL VIEW	PV #2
110	TOPOGRAFICO	TABLERO PANEL VIEW	PV #3
111	TABLERO GRIPPER ROBOT 1	JB ROBOT #1	
112	TABLERO GRIPPER ROBOT 2	JB ROBOT #2	
113			
114	ELECTRICAL INSTALLATION	EST #1 Y #2	
115	ELECTRICAL INSTALLATION	EST #3 Y #4	
116	ELECTRICAL INSTALLATION	RETENEDOR	ESQUINERO EST#3
117	ELECTRICAL INSTALLATION	TG SUNCHADORA	
118	ELECTRICAL INSTALLATION	EST #5 Y #6	
119			
120	DISTRIBUCION	TABLEROS	
121			
122	NEUMATICA	ESTACION #3	
123	NEUMATICA	ESTACION #5	
124	NEUMATICA	GRIPPER ROBOT#1	
125	NEUMATICA	GRIPPER ROBOT#2	
126			
127			
128			
129			
130	CARATULA	BILL OF MATERIAL	
131	BILL OF MATERIAL		

5	4	3	2	1	PROJECT SIDERCROM EMBALAJE		DRAWING NO. 003		SHEET / SHEETS 003 / 131		
REV. CLIENTE		REV. INT. P&F		DESIGNED BY: E.A.		CHECKED BY: D.F.		DIVISION PLANTA SIDEROCROM		LOCATION/ACTIVITY	
		Av. Belgrano 2272 CNEU/IVVD San Torcuato Buenos Aires Argentina		DATE 29/14/215		SCALE					

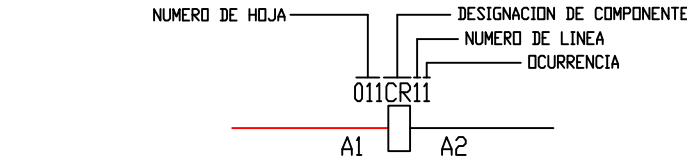
DRAWING STANDARDS

1. Etiquetas identificación de componente:

PREFIJO: Se refiere a la identificación del equipo, normalmente es usado para separar. TAGS dispositivo puede o no contener PREFIJO.

Identificación de componente: Se compone del número de página, número de línea, tipo de componente y ocurrencia en línea (por varios elementos en una línea).

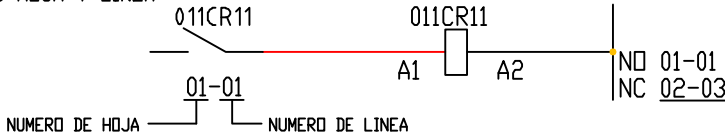
Ejemplo: _011CR11, Es un relé de control el cual se encuentra en la línea 1 de la Hoja 11 con ocurrencia 1.



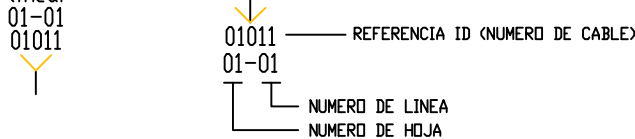
2. Referencia cruzada:

IDENTIFICA donde un componente SE ENCUENTRA EN EL JUEGO DE DIBUJO. Se utiliza para indicar que los cables ir fuera de la página, donde varias instancias del mismo componente se encuentran, y donde los contactos del relé están en relación a su bobina asociada..

REFERENCIA bobina y contacto CRUZ - números adyacentes a la bobina Y EN CONTACTO indican los números HOJA Y LINEA

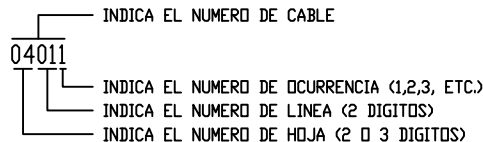


Fuera de página de referencias cruzadas - números adyacentes A DESTINO Y SÍMBOLOS INDICAN HOJA y números de línea.



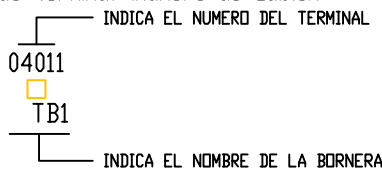
3. CABLE (generación de números):

Identifica el número de CABLE, número se compone de Hoja Número, número de línea y número de ocurrencia. Ejemplo: 04011, es un hilo que comienza en la página 04 línea 01 y es la primera aparición. MÁS DE 10 cables en una línea que no está permitido.



4. TERMINALES:

Identifica el número de bloque de bornes y punto de conexión. CONSTA DE DESIGNACIÓN terminal, y un número de terminal (número de cable).



5. Designación de COLORES en Cable

CABLE COLOR	DESIGNACIÓN	CABLE COLOR	DESIGNACIÓN
BLACK (NEGRO)	BLK	GREEN (VERDE)	GRN
BROWN (MARRÓN)	BRN	BLUE (AZUL)	BLU
RED (ROJO)	RED	VIOLET (VIOLETA)	VID
ORANGE (NARANJA)	ORG	GRAY (GRIS)	GRY
YELLOW (AMARILLO)	YEL	WHITE (BLANCO)	WHT

6. Código de colores en CABLE

Codificación de colores del cableado deberá ajustarse a NFPA79 salvo indicación contraria.

NEGRO - Cableado de las tensiones de línea (CIRCUITOS DE 380 VAC).

ROJO - Cableado de control en menor voltaje de la línea (CIRCUITOS DE 220 / 110 VAC).

AZUL - Cableado de control (CIRCUITOS DE 24VDC).

AMARILLO - Cableado que pueden estar bajo tensión de fuentes externas (CIRCUITOS DE SEGURIDAD).

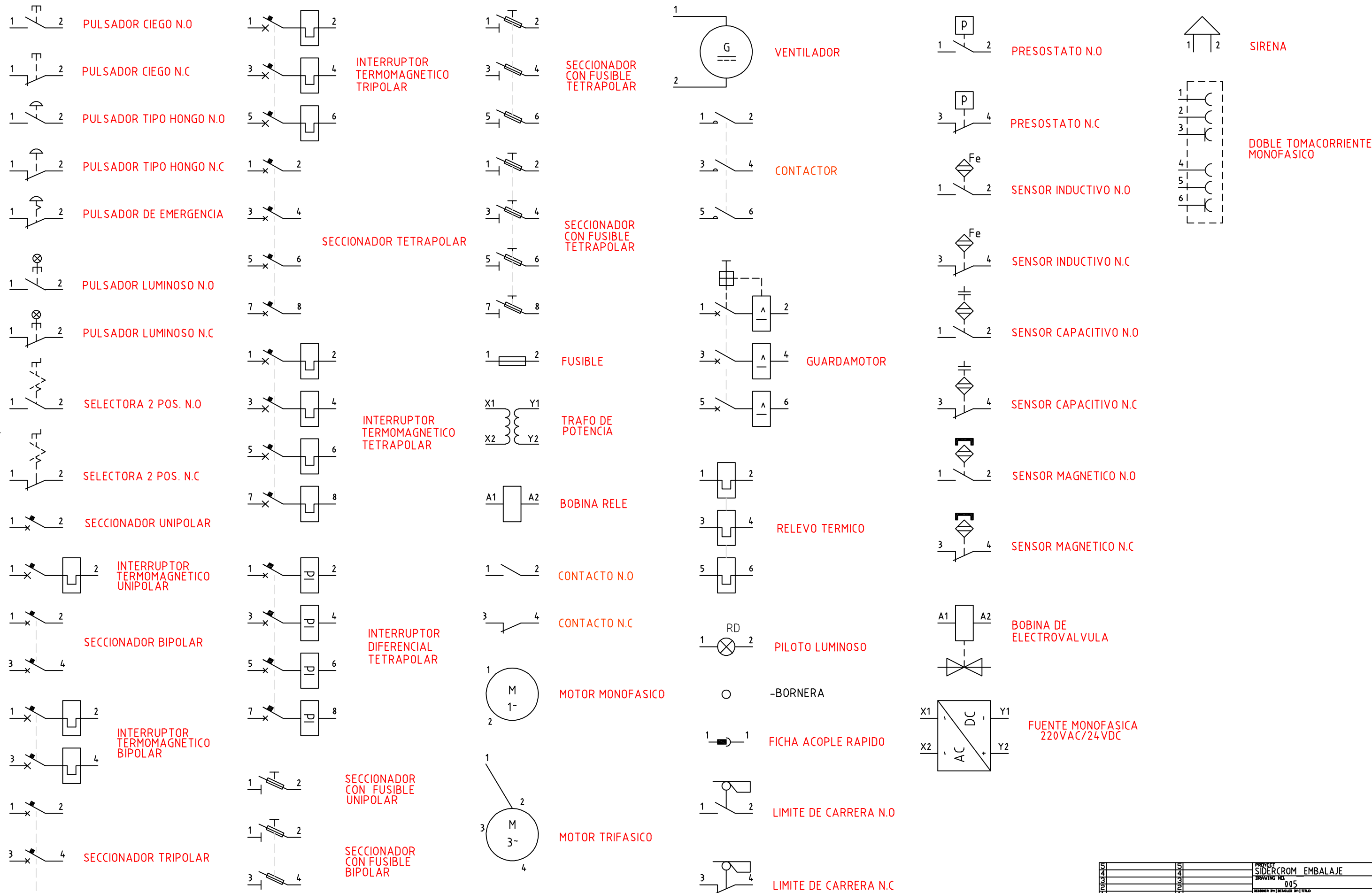
VERDE/AMARILLO - Cableado que lleva al potencial de tierra.

DESIGNACION DE COMPONENTES

DESCRIPTION	DESIGNATION	DESCRIPTION	DESIGNATION
ADJUSTABLE RESISTOR	POT	POWER SWITCH GEAR	CB,S
AMPLIFIER	AR	PROTECTIVE DEVICE	FU
ARRESTOR	E	REACTOR	ACR,X,SX
BATTERY	BATT	RECTIFIER (BRIDGE)	REC
BELL	ABE	RECTIFIER (SCR)	SCR
BRAKE	MP, BRK	RESISTOR	RES
BUSBAR	W	RESOLVER	B
BUZZER	ABU	RECEPTOR	RE
CABLE	CAB	ROTATING FREQUENCY GENERATOR	G,MG
CABLE BALANCING NETWORK	Z	ROTATING GENERATOR	G
CAPACITOR	CAP	SEMICONDUCTOR	D,CR,Q
CIRCUIT BREAKER	CI	SIGNAL GENERATOR	A
CLUTCH	MP	SOLDERING TERMINAL STRIP	TB
CONTROL (RELAY)	CR	SOLENOID	SL
CONTROL (TIME DELAY RELAY)	TR	STATIC FREQUENCY CHANGER	A
CONTROL (CONTACTOR)	MC	SUPPLY	HR
CONTROLLER (PROGRAMABLE LOGIC)	PLC	SUPPLY DEVICE	A,PS
CONVERTER	A,U,MG	SUPRESSOR (AC DIODE)	E
DIODE	D	SUPRESSOR (DC DIODE)	E
DIODE (ZENER)	DZ	SUPRESSOR (RC TYPE)	E
DYNAMOTOR	MG	SWITCH - AUXILIARY	S
FAN	F	SWITCH - CABLE OPERATED	CDS
FUSE	FU	SWITCH - CONTROL	S
GENERATOR	G	SWITCH - DISCONNECT	DS
GUARDAMOTOR	CB	SWITCH - FLOAT	FS
HORN	AH	SWITCH - FOOT OPERATED	FTS
INDUCTOR	L	SWITCH - LIMIT	LS
INVERTER	A,U,PS,MG	SWITCH - PRESSURE	PS
ISOLATOR	AT	SWITCH - PROXIMITY	PRS
JUMPER WIRE	W	SWITCH - PUSHBUTTON	PB
LIGHT (FLUORESCENT)	H	SWITCH - PUSHBUTTON ILLUMINATED	IPB
LIGHT (NEON)	NLT	SWITCH - ROTARY	RSS
LIGHT (PILOT)	LT	SWITCH - SELECTOR	SS
LOUD SPEAKER	SP	SWITCH - TEMPERATURE	TS
METER	IND	SYNCHRO	B
METER (AMMETER)	AM	TERMINAL	TB
METER (SHUNT)	MSH	TERMINAL BOARD	TB
METER (VOLTAGE)	VM	TEST JACK	TB,PL
MOTOR	M	THERMISTOR	RT
OSCILLATOR	Y,G	THERMOCOUPLE	TC
OVERLOAD	OL	TRANSDUCER	A,BT
OVERVOLTAGE DISCHARGE TYPE	E	TRANSFORMER (POTENCIA)	T
PHOTOELECTRIC CELL	PSC	TRANSFORMER (CURRENT)	T,CT
PLUG	PL	TRANSFORMER (VOLTAGE)	T,PT
PLUG (DISCONNECTING)	PL	TRANSISTOR	Q
PLUG (SOCKET)	PL	VARIATOR (MOV)	MOV
PNEUMATIC VALVE	MP,PV		
POTENTIOMETER	POT		

5	5	PROJECT	SIDERCROM EMBALAJE
4	4	DRAWING NO.	004
3	3	REVISION BY	004
2	2	REVISION BY	004
1	1	REVISION BY	004
REV. CLIENTE	REV. INT. P&F	DATE	20/04/2015
Av. Belgrano 2272 (BOLIVAR)	Av. Belgrano 2272 (BOLIVAR)	LOCATION/ACTIVITY	
Argentina	Argentina		

SIMBOLOGIA



POWER CIRCUITS

PROJECT SIDERCROM EMBALAJE		DRAWING NO. 010		SHEET SHEETS 010 1131	
DESIGNED BY LH		CHECKED BY E.A.		TITLE CARATULA POWER CIRCUITS	
REV. CLIENTE		REV. INT. P&F		DIVISION	
P&F		Av. Belgrano 2272 C1811JVD Buenos Aires Argentina		PLANT/SIDERCROM LOCATION/ACTIVITY	
JOB		DATE 29/14/2015		SCALE	

013 01

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013 03

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013 20

L1 014.02
L2 014.03
L3 014.03

TB POTENCIA
-GM013051

TB POTENCIA
-GM013052

TB POTENCIA
-GM013053

TB POTENCIA
-GM013054

TB POTENCIA
-GM013055

TB POTENCIA
-GM013056

TB POTENCIA
-K03706

TB POTENCIA
-K03707

TB POTENCIA
-K03708

TB POTENCIA
-K03709

TB POTENCIA
-K03710

TB POTENCIA
-K03711

037.06
CONTACTOR

037.07
CONTACTOR

037.08
CONTACTOR

037.09
CONTACTOR

037.10
CONTACTOR

037.11
CONTACTOR

POWERFLEX 525
VARIADOR PF525_A020007

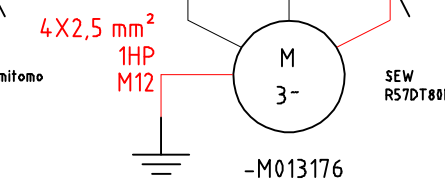
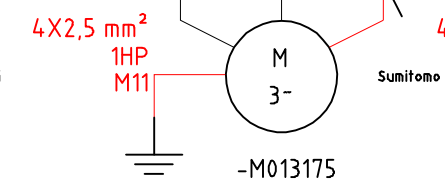
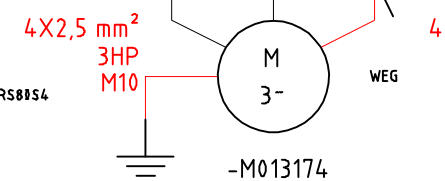
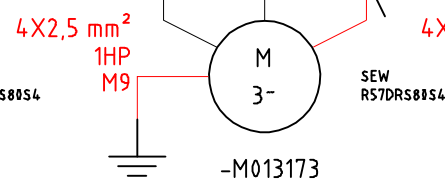
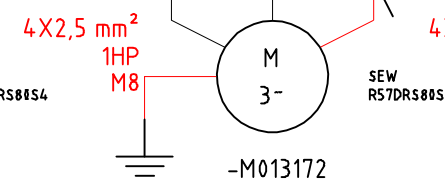
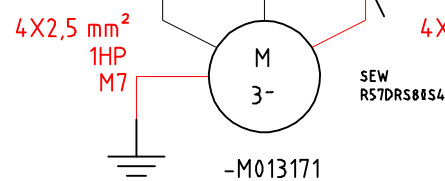
POWERFLEX 525
VARIADOR PF525_A020008

POWERFLEX 525
VARIADOR PF525_A020009

POWERFLEX 525
VARIADOR PF525_A020010

POWERFLEX 525
VARIADOR PF525_A020011

POWERFLEX 525
VARIADOR PF525_A020012



TRANSPORTADOR DE DESCARTE #2

TRANSPORTADOR BUFFER INTERMEDIO #1

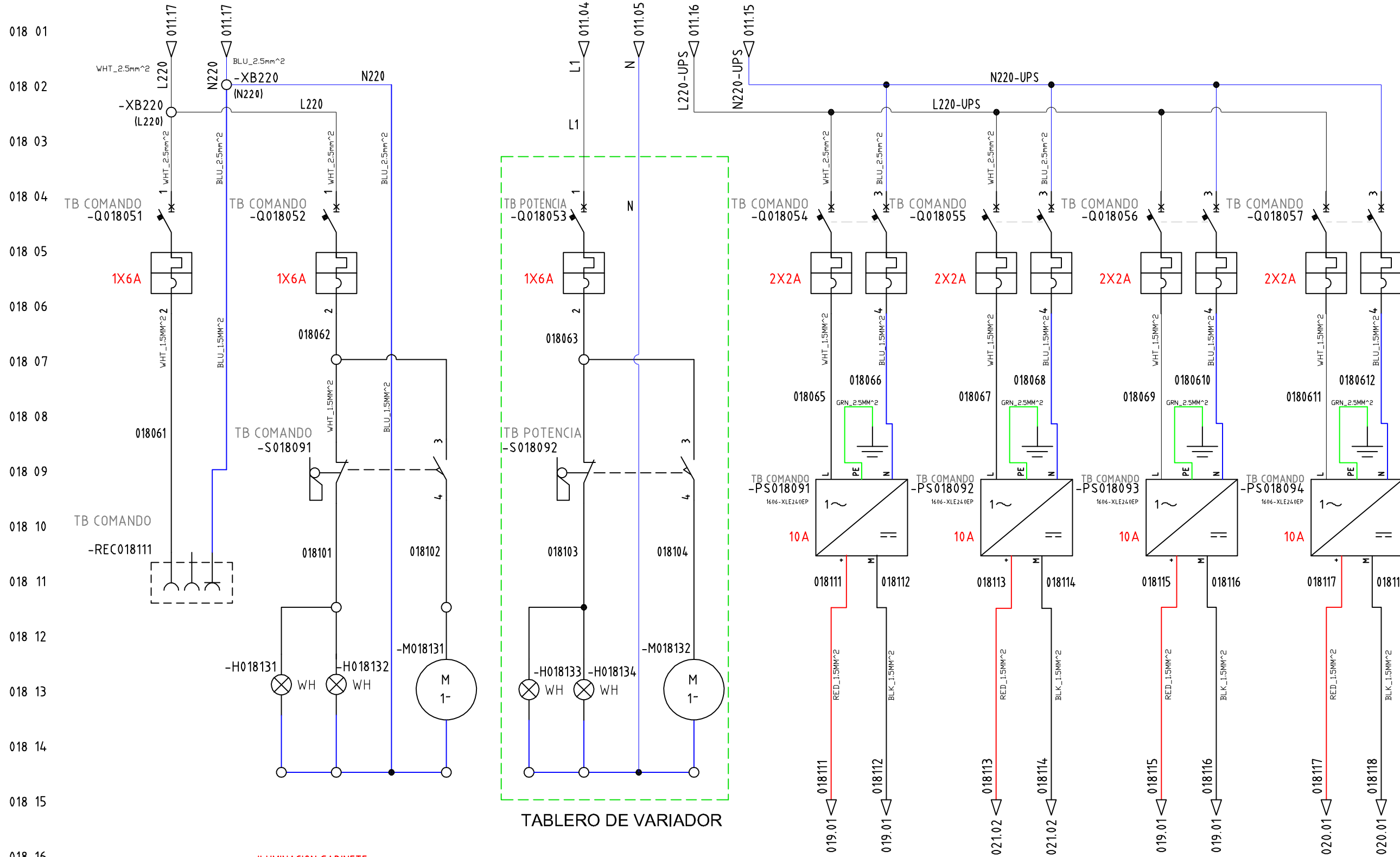
TRANSPORTADOR GIRATORIO DE SUNCHADO

TRANSPORTADOR DE ESTACIÓN 3

TRANSPORTADOR BUFFER INTERMEDIO #2

TRANSPORTADOR CON BALANZA

PROYECTO	SIDERCROM EMBALAJE	REVISOR	EA	FECHA	013	SHEET SHEETS	013 1131
DISEÑO		REVISOR	EA	FECHA			
REV. CLIENTE	REV. INT. P&F	CHIEF	EA	FECHA			
ORDERED BY	Av. Belgrano 2272 CHIEF/DVD	SAFETY	EA	FECHA			
DATE	29/14/2015	SAFETY	EA	FECHA			
JOB	PLANTA SIDERCROM	SAFETY	EA	FECHA			
	LOCATION/ACTIVITY	SAFETY	EA	FECHA			



TOMACORRIENTE
220VCA GABINETE

ILUMINACION GABINETE

REFRIGERACION
GABINETE
PLC

ILUMINACION GABINETE

REFRIGERACION
GABINETE
VARIADORES

FUENTE ALIMENTACION 24VDC
INPUTS RACK 1

FUENTE ALIMENTACION 24VDC
SEGURIDAD

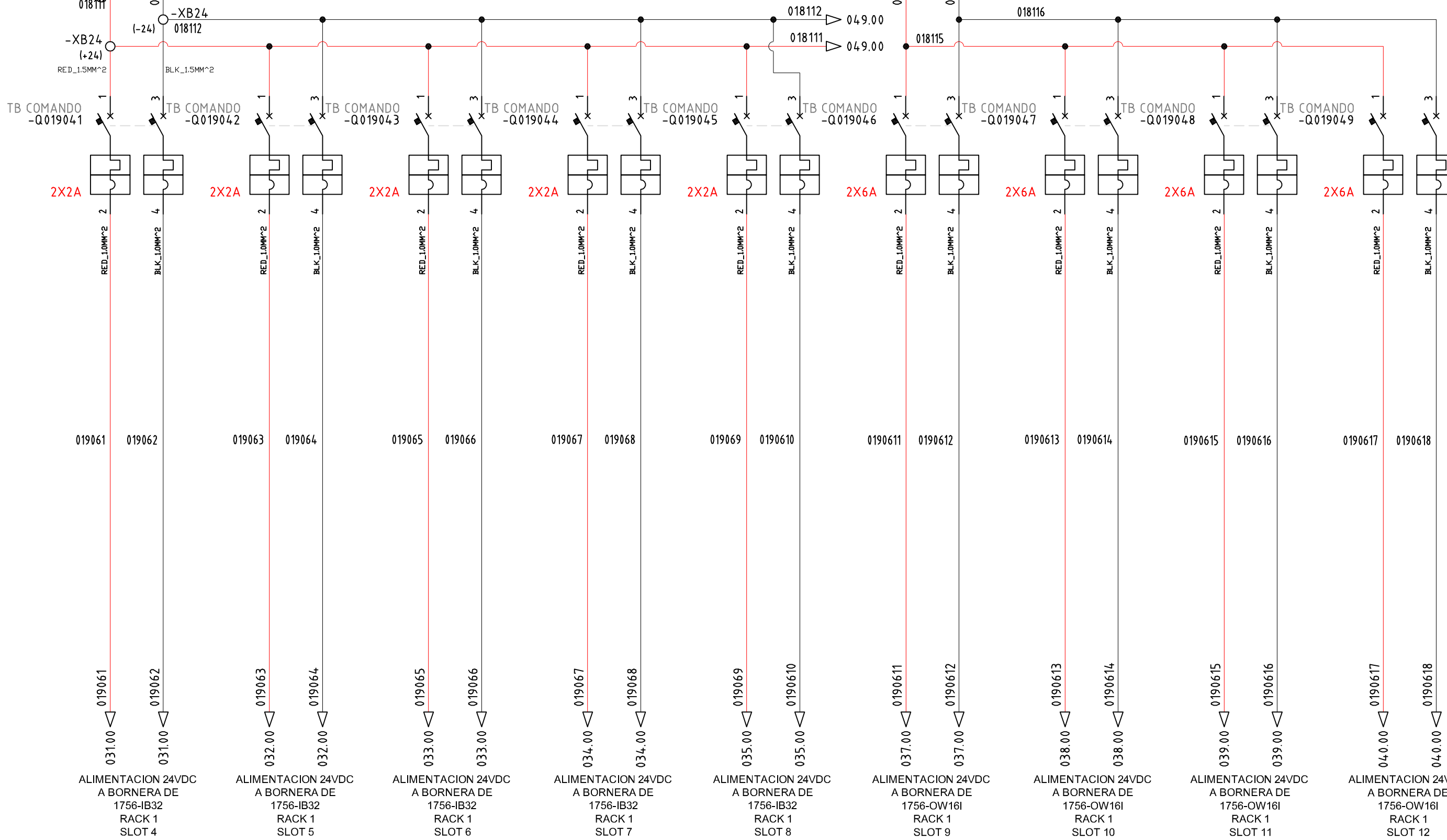
FUENTE ALIMENTACION 24VDC
OUTPUT RACK 1

FUENTE ALIMENTACION 24VDC
INPUTS/OUTPUT RACK 2

5	5	PROYECTO	SIDERCROM EMBALAJE
4	4	DISEÑO	018
3	3	REVISOR	018 1131
2	2	REVISOR	018 1131
1	1	REVISOR	018 1131
REV. CLIENTE	REV. INT. P&F	ALIMENTACION PRINCIPAL	220VAC
Av. Belgrano 2272 (CHILIVAD)	Don Torcuato Buenos Aires	PLANTA SIDERCROM	LOCATION/ACTIVITY
29/14/2015	29/14/2015	DIVISION	

019 01
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PROJECT: SIDEROCROM EMBALAJE		DRAWING NO: 019		SHEET / SHEETS: 019 / 131	
REVISION: REV. CLIENTE		REVISION: REV. INT. P&F		REVISION: REV. INT. P&F	
CHECKED BY: D.F.		CHECKED BY: D.F.		CHECKED BY: D.F.	
DATE: 29/10/2015		DATE: 29/10/2015		DATE: 29/10/2015	
DIVISION: PLANTA SIDEROCROM		DIVISION: PLANTA SIDEROCROM		DIVISION: PLANTA SIDEROCROM	
LOCATION/ACTIVITY:		LOCATION/ACTIVITY:		LOCATION/ACTIVITY:	

020 01
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TB COMANDO
-Q020041

TB COMANDO
-Q020042

TB COMANDO
-Q020043

TB COMANDO
-Q020044

2X2A

2X2A

2X6A

2X6A

ALIMENTACION 24VDC
A BORNERA DE
1756-IB32
RACK 2
SLOT 1

ALIMENTACION 24VDC
A BORNERA DE
1756-IB32
RACK 2
SLOT 2

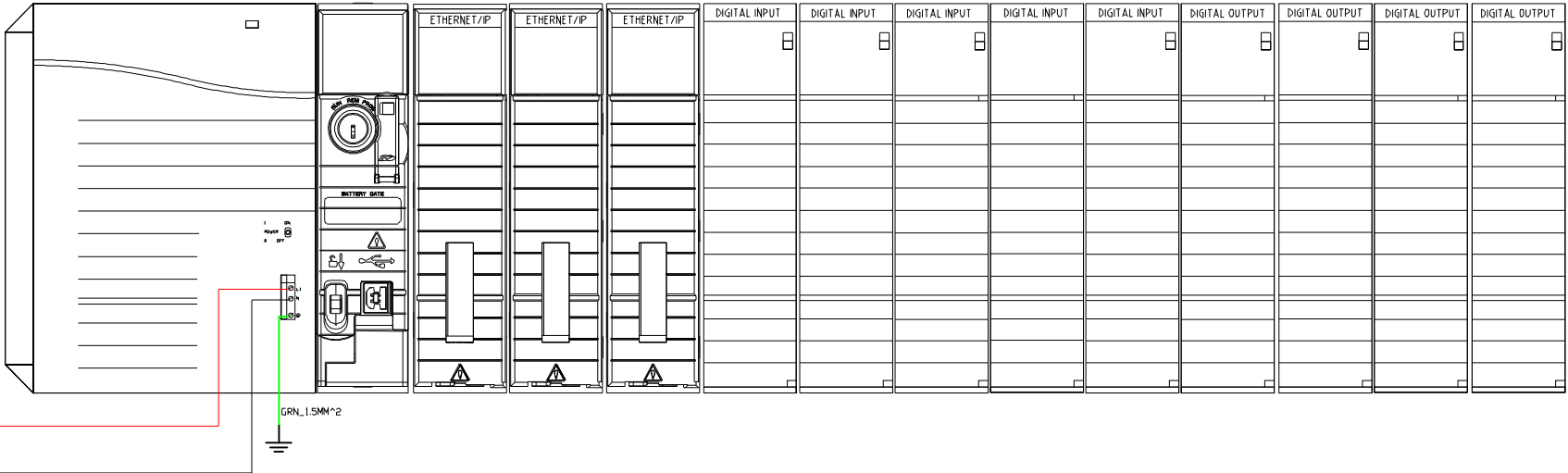
ALIMENTACION 24VDC
A BORNERA DE
1756-OW16I
RACK 2
SLOT 3

ALIMENTACION 24VDC
A BORNERA DE
1756-OW16I
RACK 2
SLOT 4

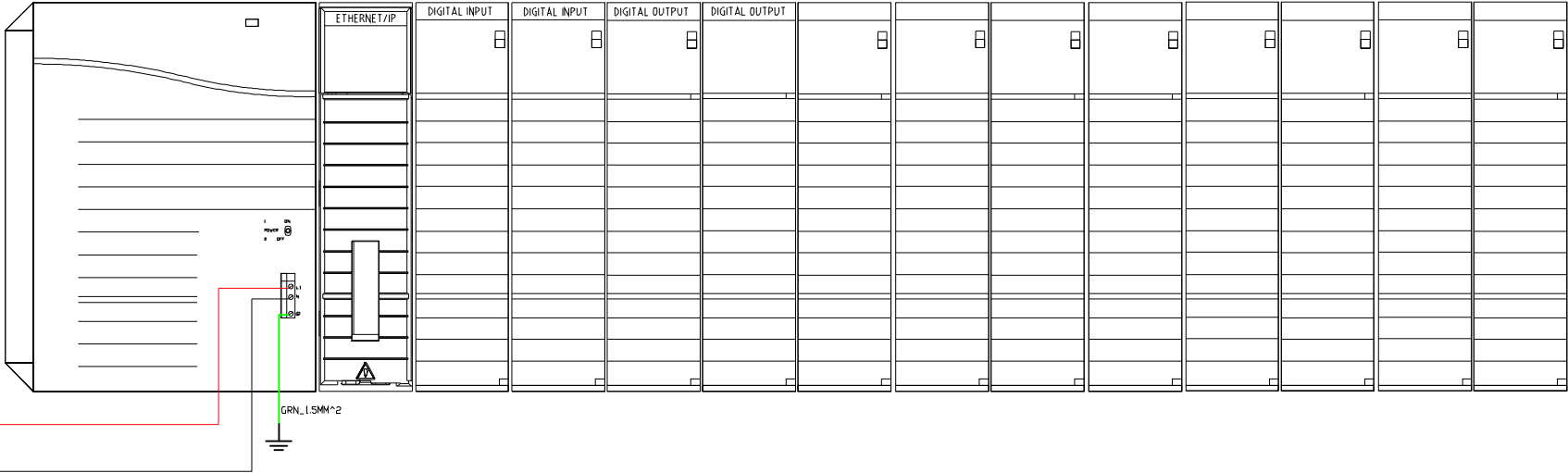
PROJECT SIDERCROM EMBALAJE		DRAWING NO. 020		SHEET SHEETS 020 131	
DESIGNED BY J.L.		CHECKED BY D.F.		DIVISION PLANTA SIDERCROM	
REV. CLIENTE		REV. INT. P&F		ALIMENTACION PRINCIPAL 24VDC	
P&F		28/14/215		LOCATION/ACTIVITY	

ALIMENTACION 24VDC PLC

CHASSIS 1

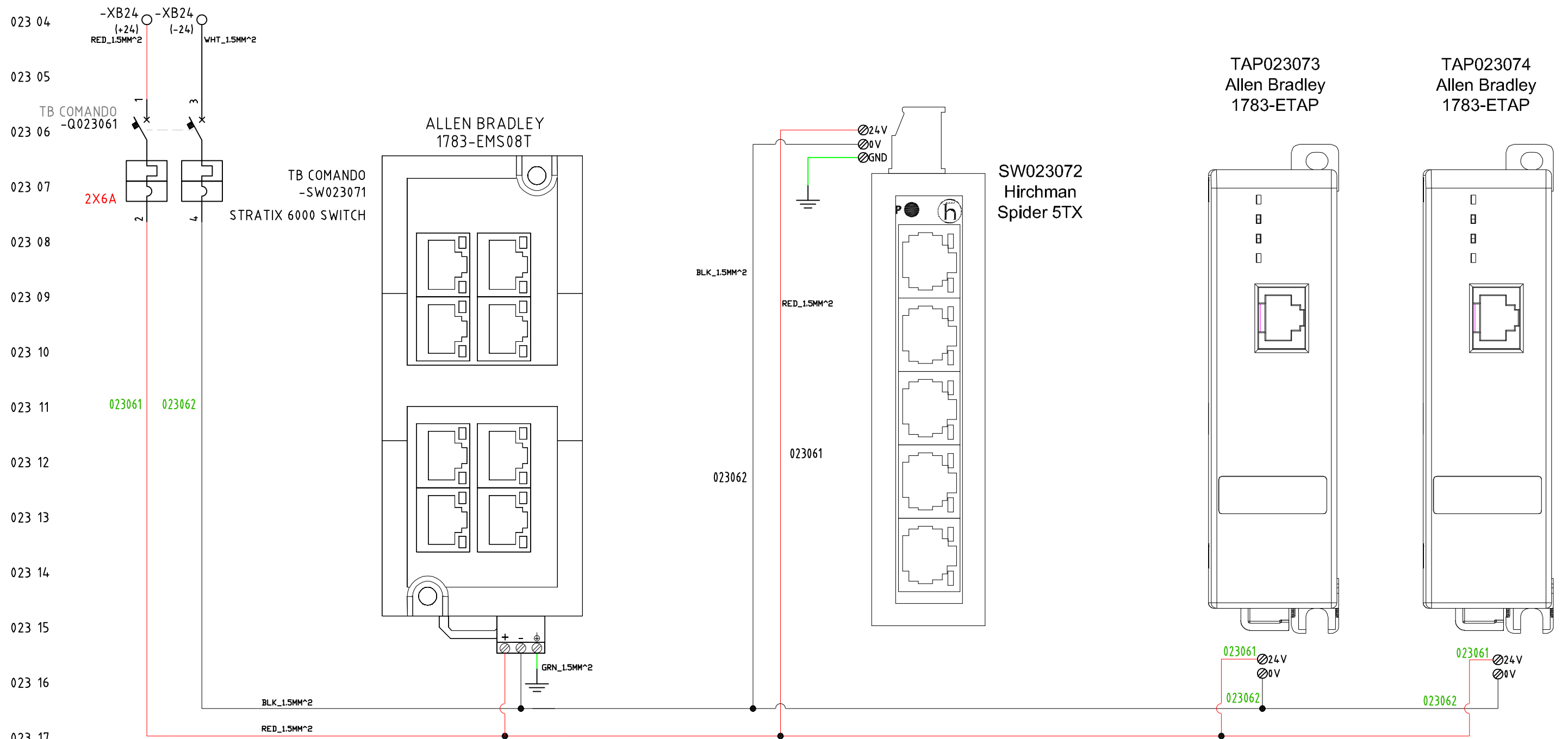


CHASSIS 2



ALIMENTACION 24VDC

SWITCH Y ETAP



C O D E		5	PRIORITY	SIDEROCROM	EMBALAJE	
		4	DRAWING	023		SHEET/SHEET
		3				023 131
		2	DESIGNED BY	DETAILS BY	TITLE	
		1	J.N.	E.A.	ALIMENTACION 24VDC	
REV. CLIENTE		REV. INT. P&F		SWITCH ETHERNET		
		Av. Delgado 2272 (B618)VD Buen Torcuato Buenos Aires Argentina				
COT.		DATE	19/M/2015	WORLD	DIVISIONES	
					PLANT, SIDEROCROM	
					TURKEY, CROATIA, ITALY,	

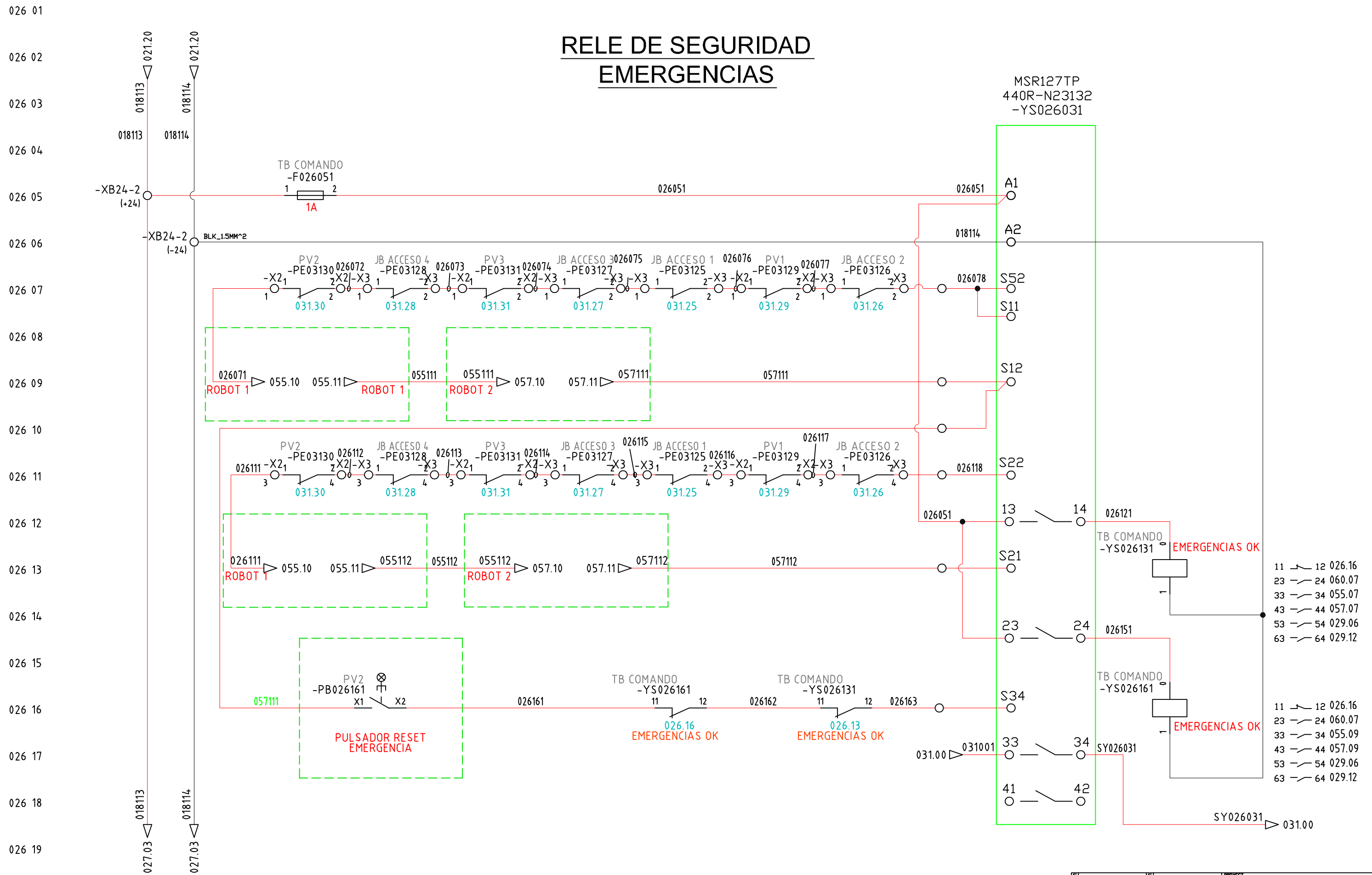
HARDWIRED SAFETY
CIRCUITS

PROJECT SIDERCROM EMBALAJE		SHEET SHEETS 025 131	
DRAWING NO. 025		TITLE CARATULA HARDWIRED SAFETY CIRCUIT	
REV. CLIENTE	REV. INT. P&F	DESIGNED BY D.F.	CHECKED BY E.G.
Av. Belgrano 2272 (BOLIVAR) Don Torcuato Buenos Aires Argentina		DATE 29/14/2015	DIVISION PLANTA SIDERCROM LOCATION/ACTIVITY

RELE DE SEGURIDAD

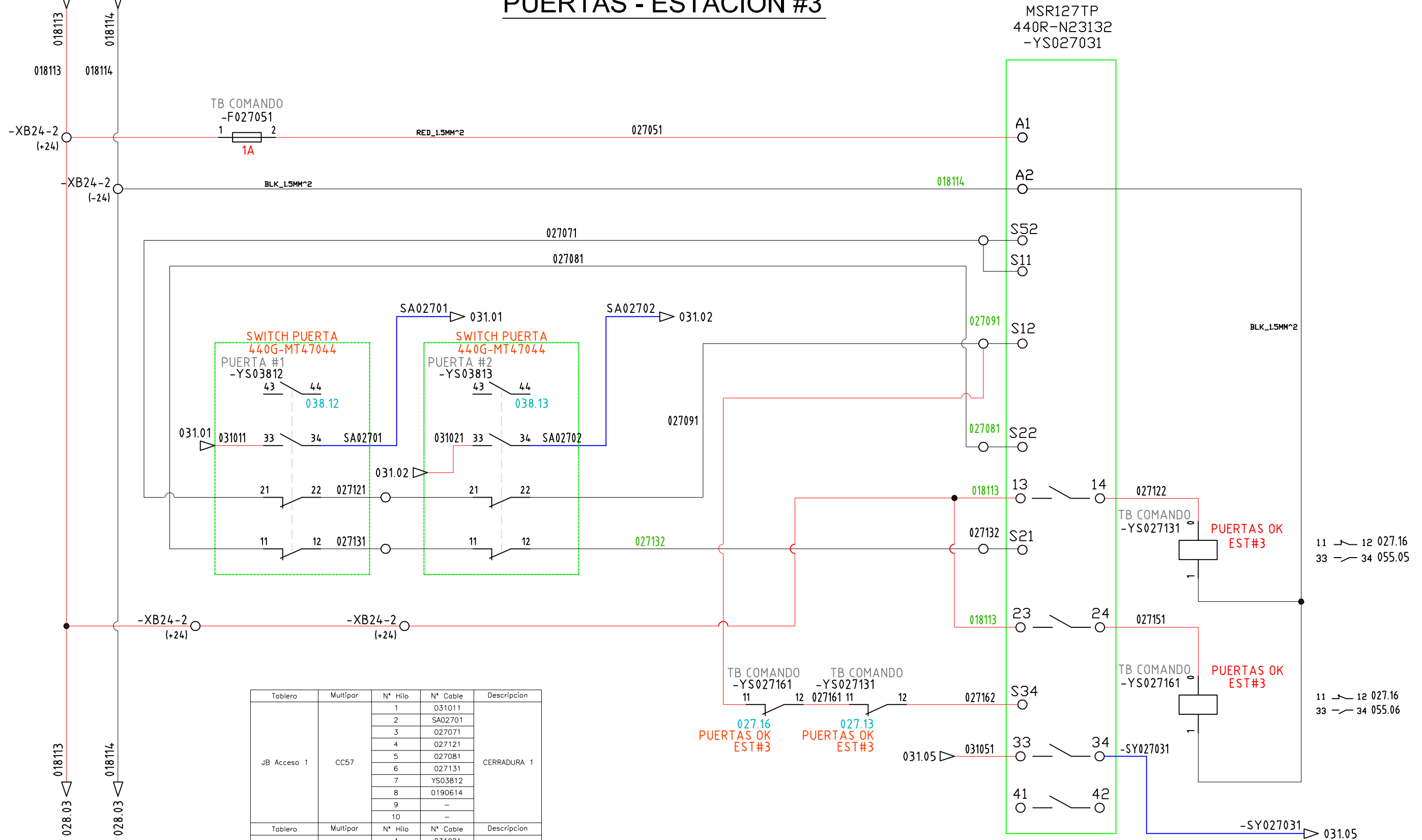
EMERGENCIAS

MSR127TP
440R-N23132
-YS026031



5	5	PROYECTO	SIDERCROM EMBALAJE	SHEET	SERIES
4	4	DESCRIPCION	026	026	131
3	3	ESTADO	DETALLADO	FIELD	
2	2	REV. CLIENTE	REV. INT. P&F	FELE DE SEGURIDAD EMERGENCIAS	
1	1	 	Av. Belgrano 2072 C1611HVD Don Torcuato Buenos Aires Argentina.	D.F. DATED: 24/01/2015 SHEETED BY: SOLID DIVISION: PLANT/SIDERCROM PROJECT/PROCESS/STORY:	

RELE DE SEGURIDAD PUERTAS - ESTACION #3



Tablero	Multipar	N° Hilo	N° Cable	Descripcion
JB Acceso 1	CC57	1	031011	CERRADURA 1
		2	SA02701	
		3	027071	
		4	027121	
		5	027081	
		6	027131	
		7	YS03812	
		8	0190614	
		9	—	
		10	—	
Tablero	Multipar	N° Hilo	N° Cable	Descripcion
JB Acceso 2	CC58	1	031021	CERRADURA 2
		2	SA02702	
		3	027121	
		4	027091	
		5	027131	
		6	027132	
		7	YS03813	
		8	0190614	
		9	—	
		10	—	

RELE DE SEGURIDAD PUERTAS - ESTACION #5

MSR127TP
440R-N23132
-YS028031

TB COMANDO
-F028051
1A

BLK_1.5MM^2

028071

028081

SA02803 031.03

SA02804 031.04

SWITCH PUERTA
440G-MT47044
PUERTA #3
-YS03814

SWITCH PUERTA
440G-MT47044
PUERTA #4
-YS03815

43 44
038.14

43 44
038.15

031.03 031031 33 34 SA02803

031.04 031041 33 34 SA02804

028091

018114

028071 S52

S11

028091 S12

028081 S22

028132 S21

028162 S34

031.06 031061

028122

TB COMANDO
-YS028131

PUERTAS OK
EST#5

028151

TB COMANDO
-YS028161

PUERTAS OK
EST#5

-SY028031

-SY028031 031.06

11 12 028.16
43 44 057.05

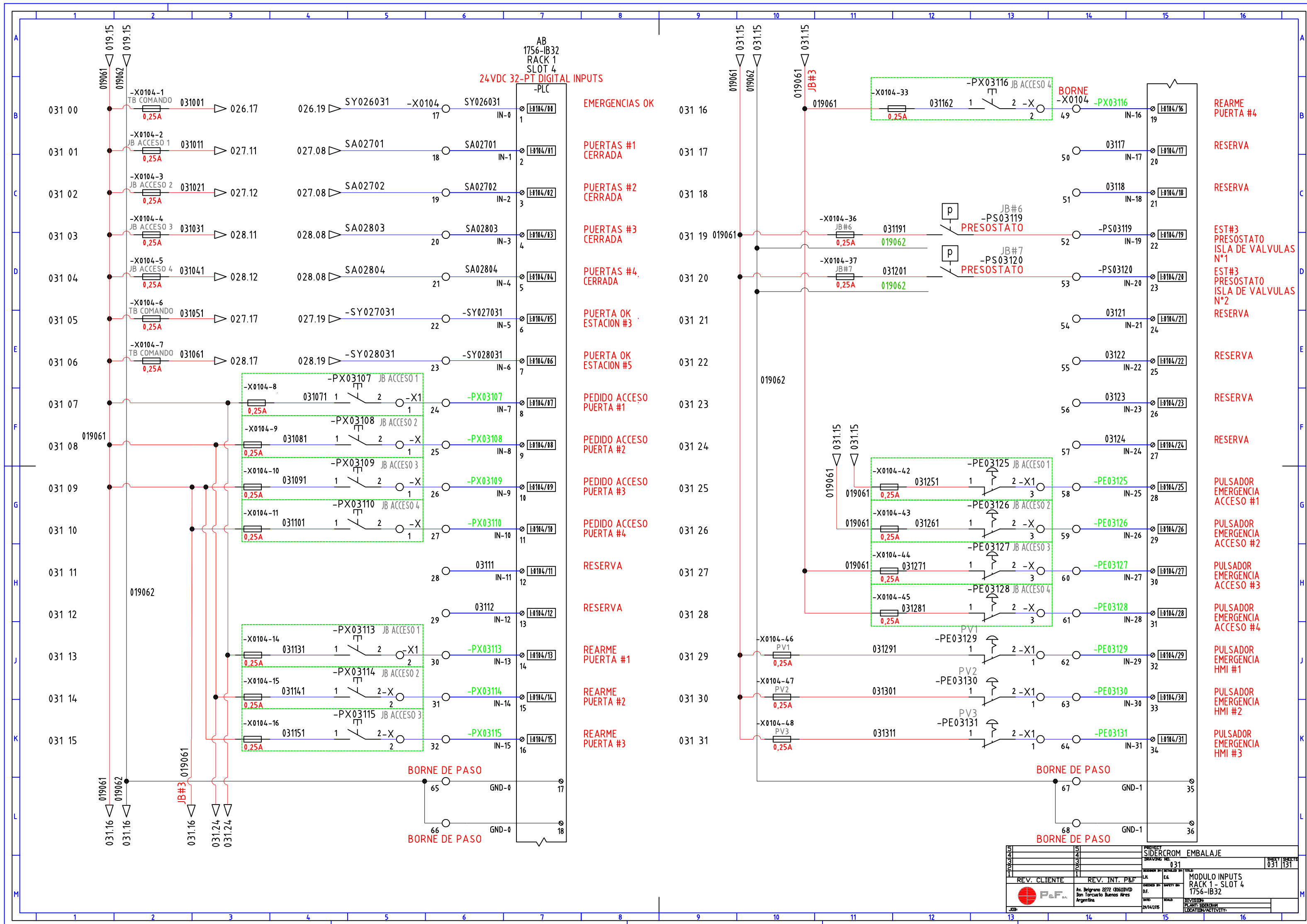
11 12 028.16
43 44 057.06

Tablero	Multipar	N° Hilo	N° Cable	Descripcion
JB Acceso 3	CC59	1	031031	CERRADURA 3
		2	SA02803	
		3	028071	
		4	028121	
		5	028081	
		6	028131	
		7	YS03814	
		8	0190614	
		9	-	
		10	-	
Tablero	Multipar	N° Hilo	N° Cable	Descripcion
JB Acceso 4	CC60	1	031041	CERRADURA 4
		2	SA02804	
		3	028121	
		4	028091	
		5	028131	
		6	028132	
		7	YS03815	
		8	0190614	
		9	-	
		10	-	

5	5	PROJECT	SIDERCROM EMBALAJE
4	4	DRAWING NO.	028
3	3	REVISION BY	REVISOR BY
2	2	DATE	DATE
1	1	REV. CLIENTE	REV. INT. P&F
		Av. Belgrano 2272 (BOLIVAR) Don Torcuato Buenos Aires Argentina	SAFETY BR. DIVISION PLANTA SIDERCROM LOCATION/ACTIVITY
		20/10/2015	
		14	15
		16	17

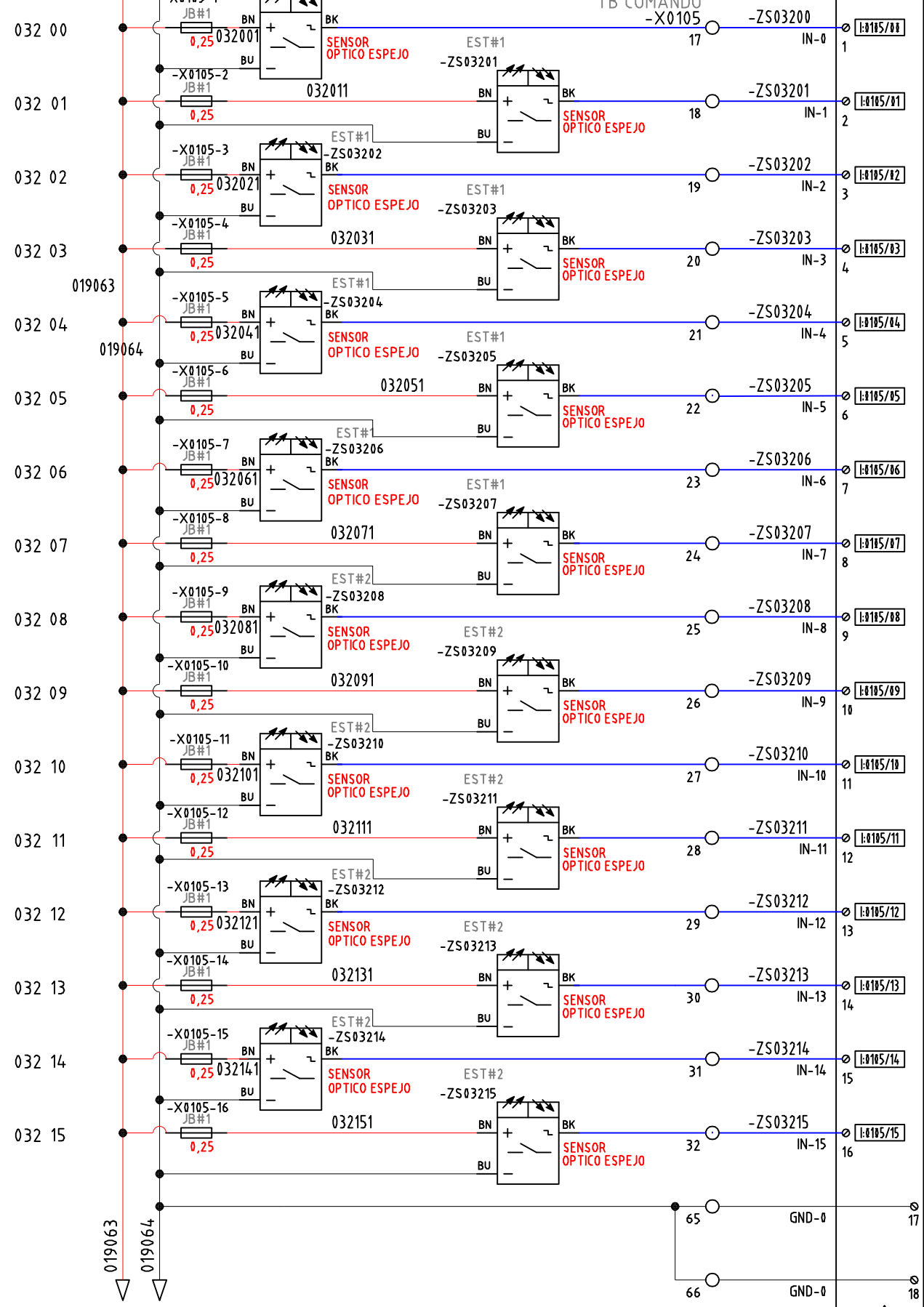
INPUTS AND OUTPUTS

PROJECT SIDERCROM EMBALAJE		SHEET SHEETS 030 131	
DRAWING NO. 030		TITLE INPUTS AND OUTPUTS	
DESIGNED BY L.A.	DETAILS BY E.A.	CHECKED BY D.F.	
REV. CLIENTE		REV. INT. P&F	
P&F		Av. Belgrano 2272 (BOLIVAR) Don Torcuato Buenos Aires Argentina	
JOB	DATE 29/14/2015	DIVISION PLANTA SIDERCROM LOCATION/ACTIVITY	

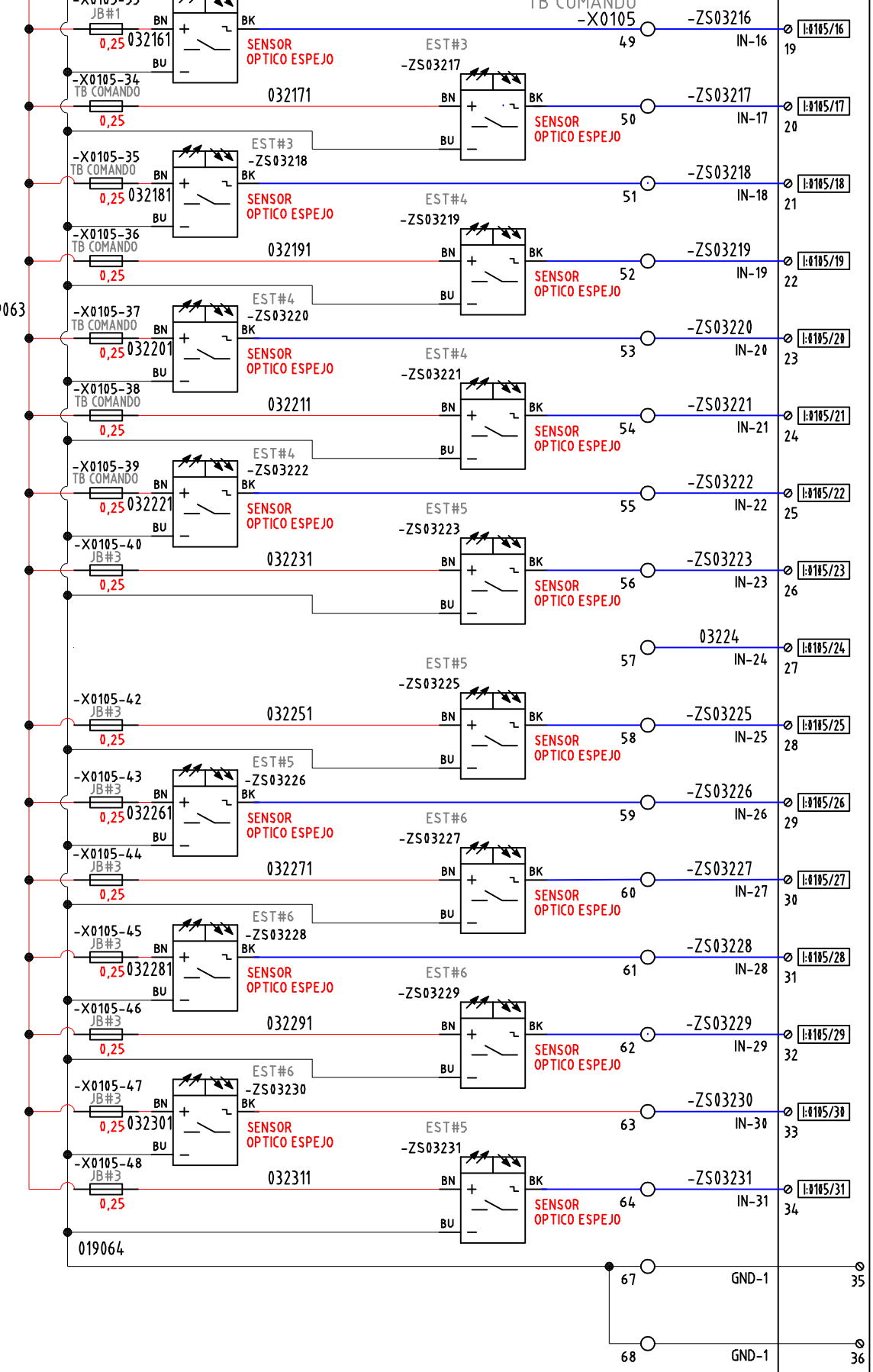


24VDC 32-PT DIGITAL INPUTS

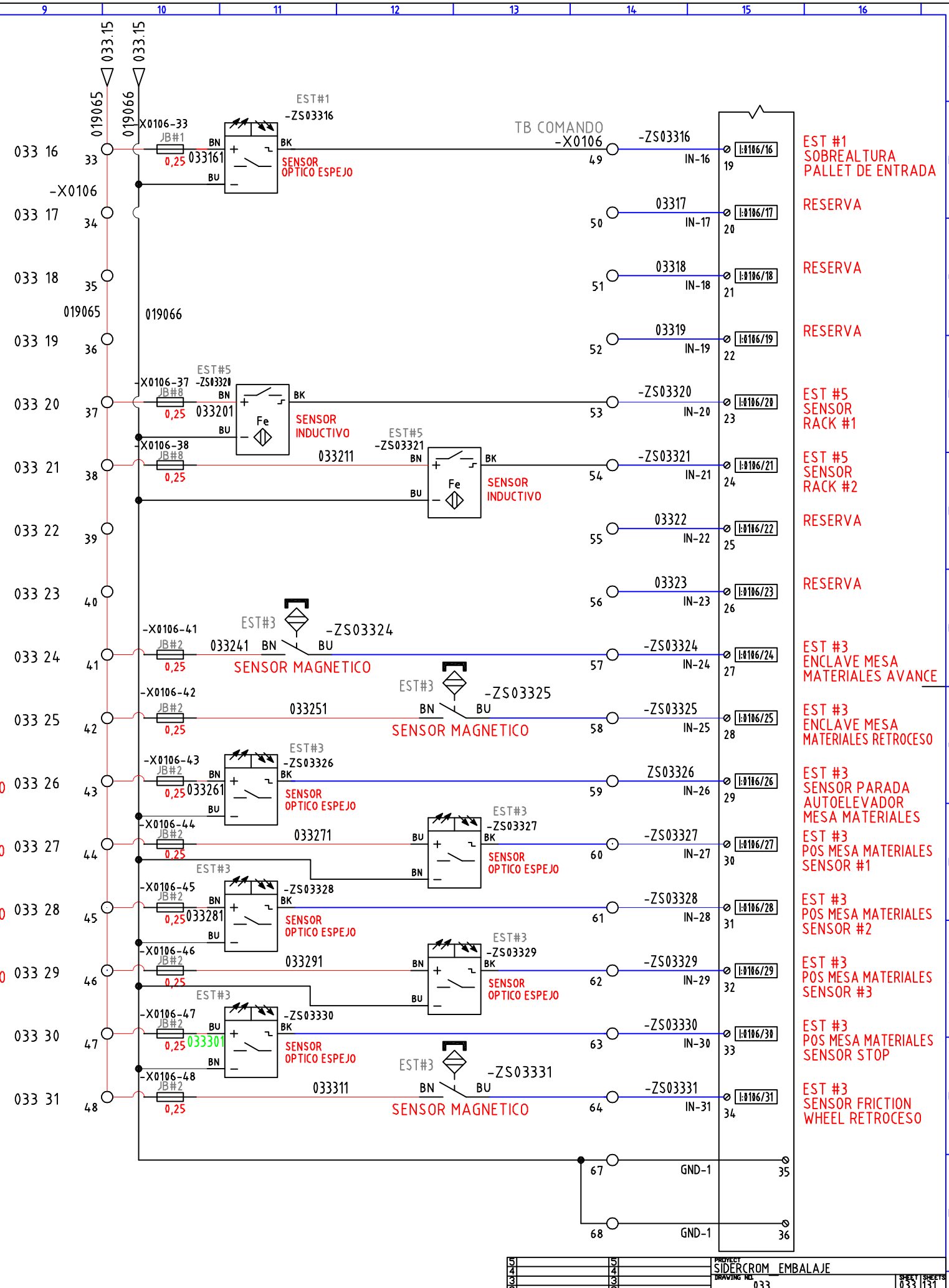
AB
1756-IB32
RACK 1
SLOT 5



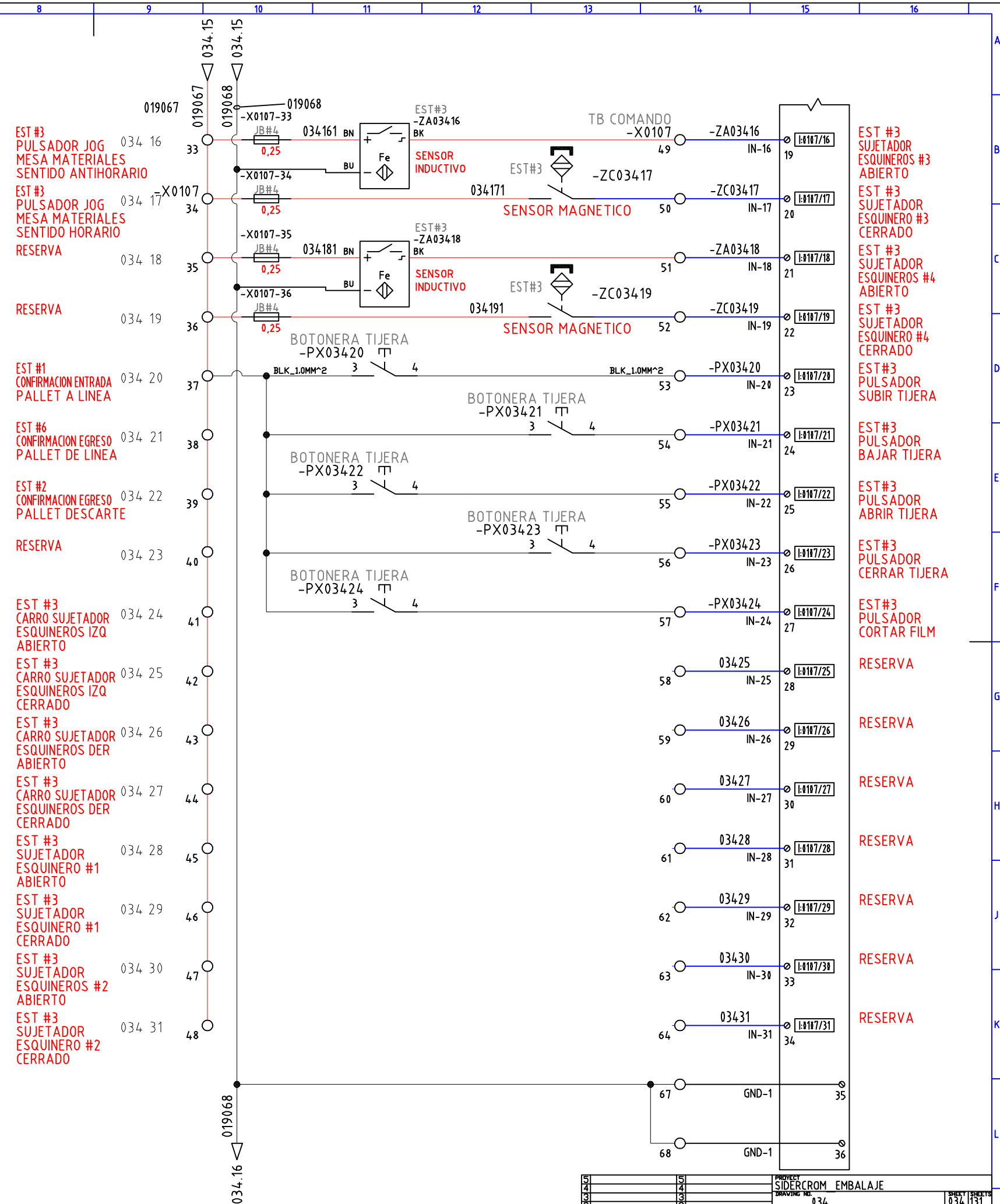
- EST#1 PRESENCIA PALLET POS#1
- EST #1 STOP PALLET POS #1
- EST #1 PRESENCIA PALLET POS #2
- EST #1 STOP PALLET POS #2
- EST #1 PRESENCIA PALLET POS#3
- EST#1 STOP PALLET POS#3
- EST#1 PRESENCIA PALLET POS#4
- EST#1 STOP PALLET POS #4
- EST #2 PRESENCIA PALLET POS#5 (TG)
- EST #2 POSICION STOP TRASERO POS #5 (TG)
- EST #2 POSICION STOP DELANTERO POS #5 (TG)
- EST #2 PRESENCIA PALLET POS #6
- EST #2 STOP TRASERO PALLET POS #6
- EST #2 STOP DELANTERO PALLET POS #6
- EST #2 PRESENCIA PALLET POS #7
- EST #2 STOP TRASERO PALLET POS #7



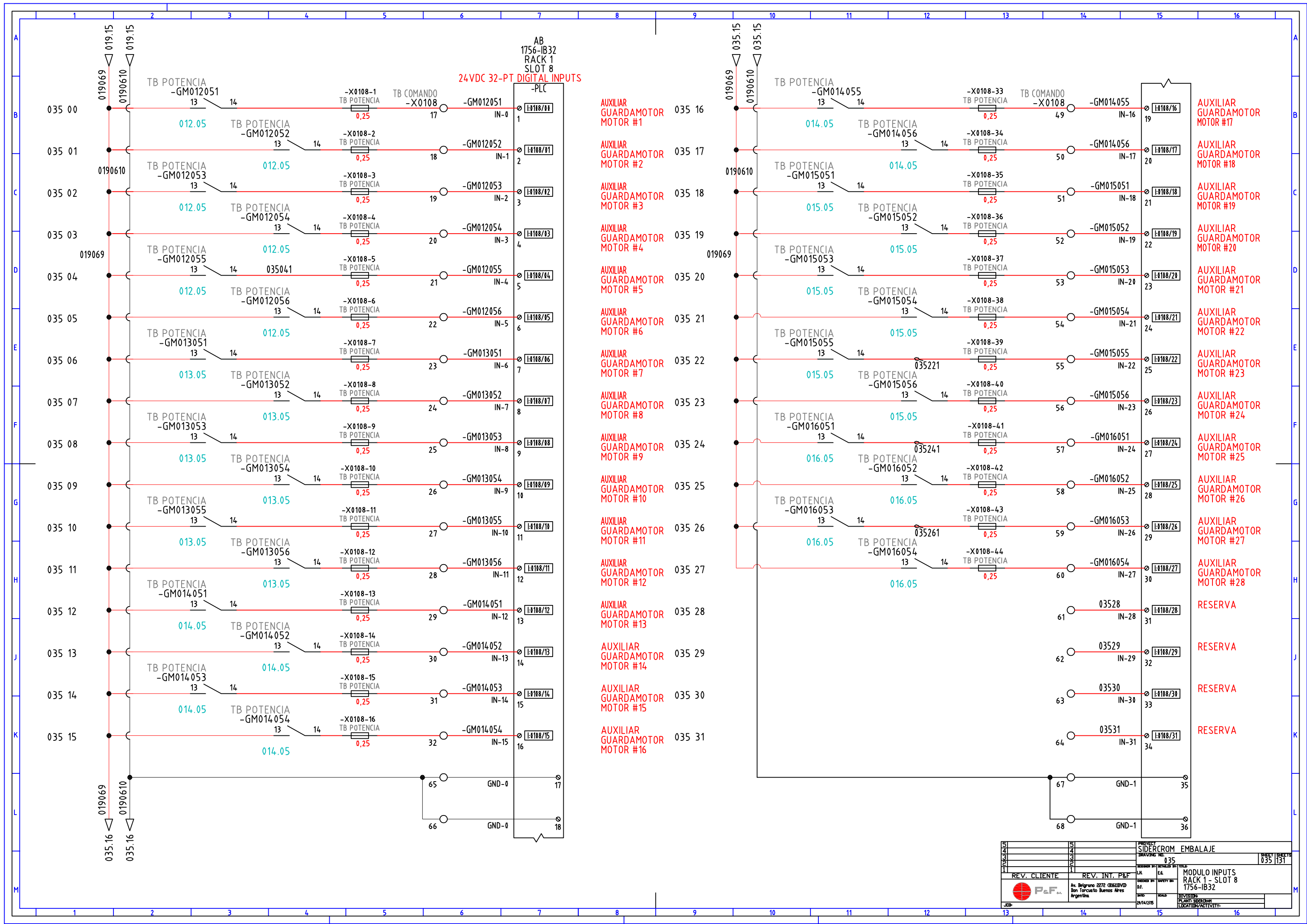
- EST #2 STOP DELANTERO PALLET POS #7
- EST #3 PRESENCIA PALLET POS #8
- EST#3 POSICION STOP POS #8
- EST #4 PRESENCIA PALLET POS #9
- EST #4 STOP PALLET POS #9
- EST #4 PRESENCIA PALLET POS #10
- EST #4 STOP PALLET POS #10
- EST #5 PRESENCIA PALLET POS #11 (TG)
- RESERVA
- EST #5 PRESENCIA PALLET POS #12
- EST #5 STOP PALLET POS #12
- EST #6 PRESENCIA PALLET POS #13
- EST #6 STOP PALLET POS #13
- EST #6 PRESENCIA PALLET POS #14
- EST #6 STOP PALLET POS #14
- EST #5 STOP PALLET CORTO POS #11(TG)

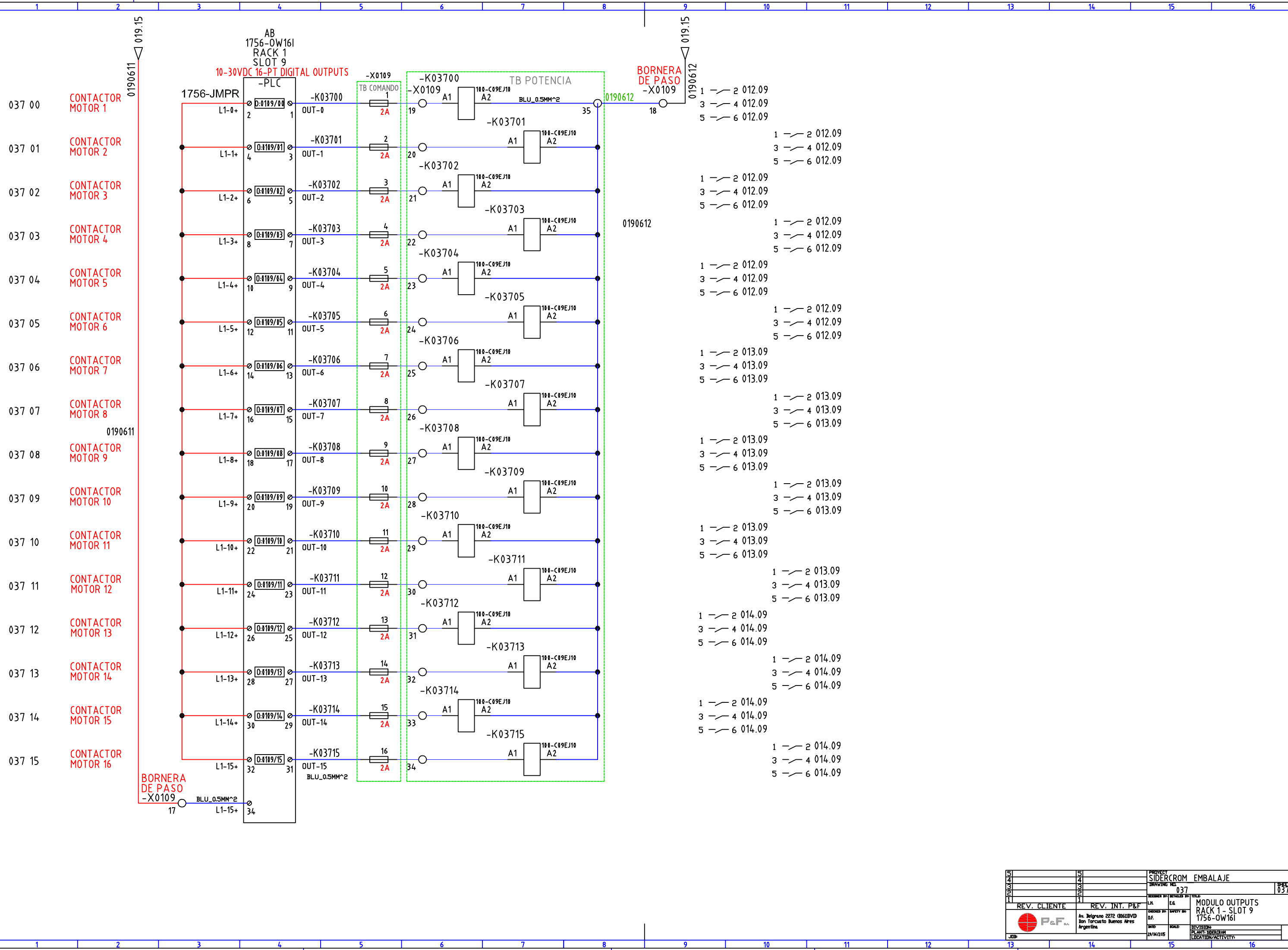


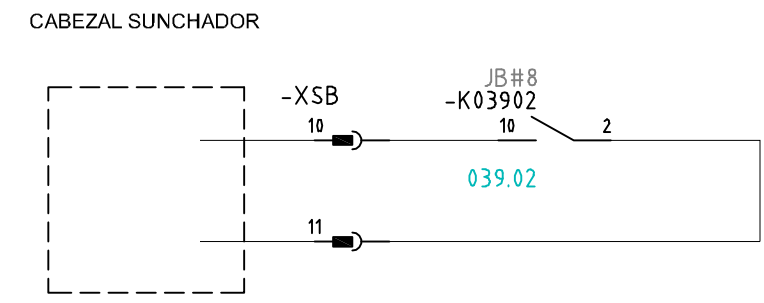
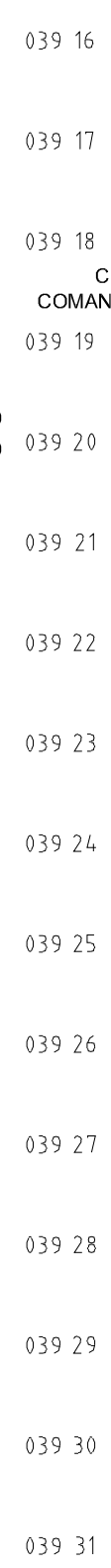
5		5		PROPERTY		
4		4		SIDERCROM	EMBALAJE	
3		3		DRAWING NO.	033	SHEET 1
2		2				131
1		1		DESIGNED BY	REVIEWED BY	DATE
	REV. CLIENTE		REV. INT. P&F	DATE	SCALE	
	 Av. Belgrano 2272 (B8611)VDI Don Torcuato de Buenos Aires Argentina			DATE	SCALE	
				DATE	SCALE	
				DATE	SCALE	
				DATE	SCALE	



5	5	PROJECT		SIDEROCROM EMBALAJE		SHEET SHEETS	
4	4	DRAWING NO.		034		034 131	
3	3	DESIGN BY		DETAILS BY		TITLE	
2	2	L.H.		E.G.		MODULO INPUTS	
1	1	CHECKED BY		SAFETY BY		RACK 1 - SLOT 7	
		D.F.				1756-IB32	
		DATE		SCALE		DIVISION	
		29/04/215				PLANT: SIDEROCROM	
						LOCATION/ACTIVITY:	
JOB:							





[illegible]

040 00 EST #3 MESA MATRIALES FRICTION WHEEL AVANZA

040 01 EST #3 MESA MATRIALES FRICTION WHEEL RETROCEDE

040 02 EST #3 ENCLAVE MESA MATERIALES AVANCE

040 03 EST #3 ENCLAVE MESA MATERIALES RETROCESO

040 04 RESERVA

040 05 EST #1 BALIZA VERDE

040 06 EST #1 BALIZA ROJA

040 07 EST #3 BALIZA ROJA ACCESO 1

040 08 EST #3 BALIZA ROJA ACCESO 2

040 09 EST #4 BALIZA VERDE TABLERO PRINCIPAL

040 10 EST #4 BALIZA AMBAR TABLERO PRINCIPAL

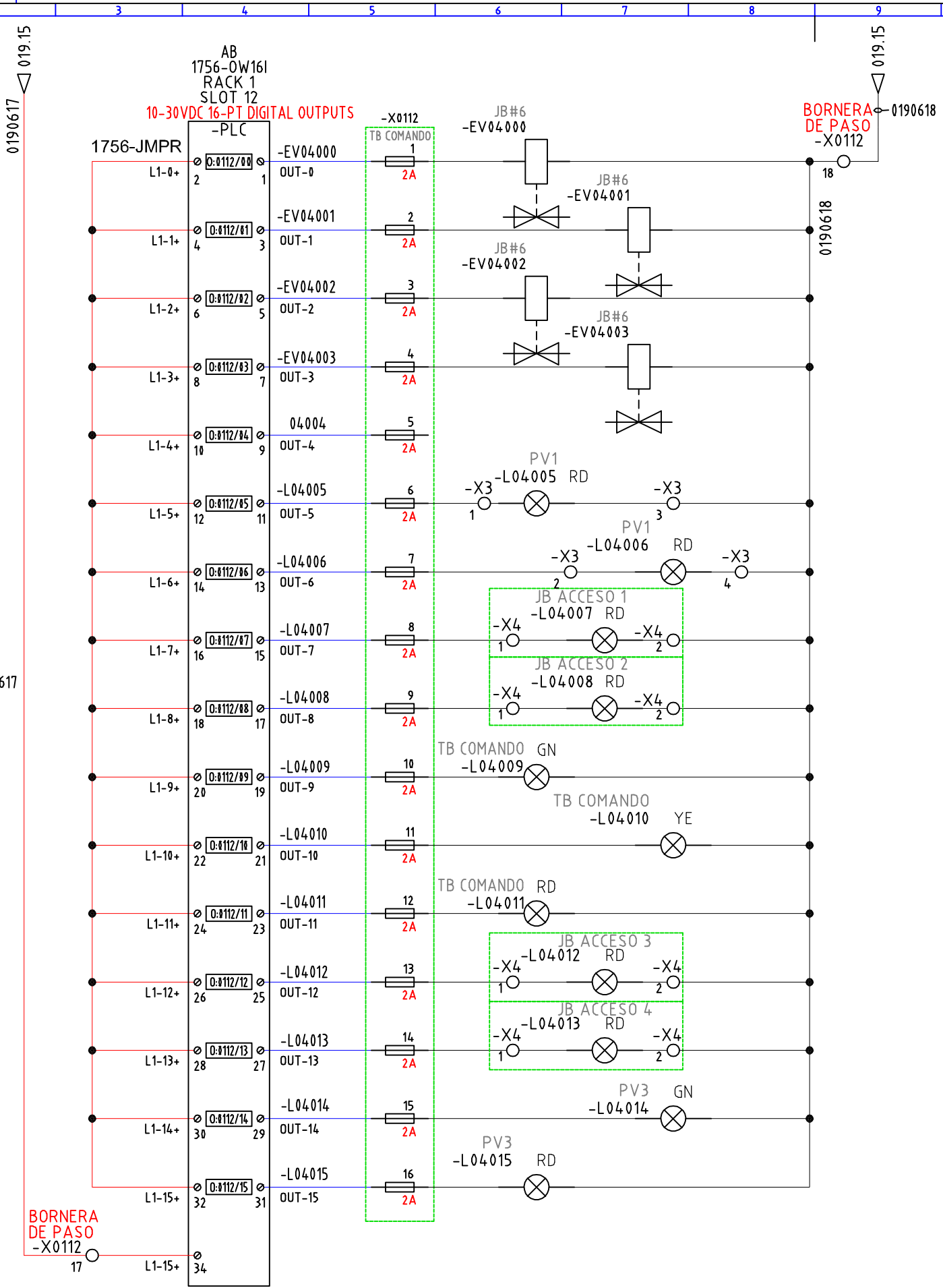
040 11 EST #4 BALIZA ROJA TABLERO PRINCIPAL

040 12 EST #3 BALIZA ROJA ACCESO 3

040 13 EST #3 BALIZA ROJA ACCESO 4

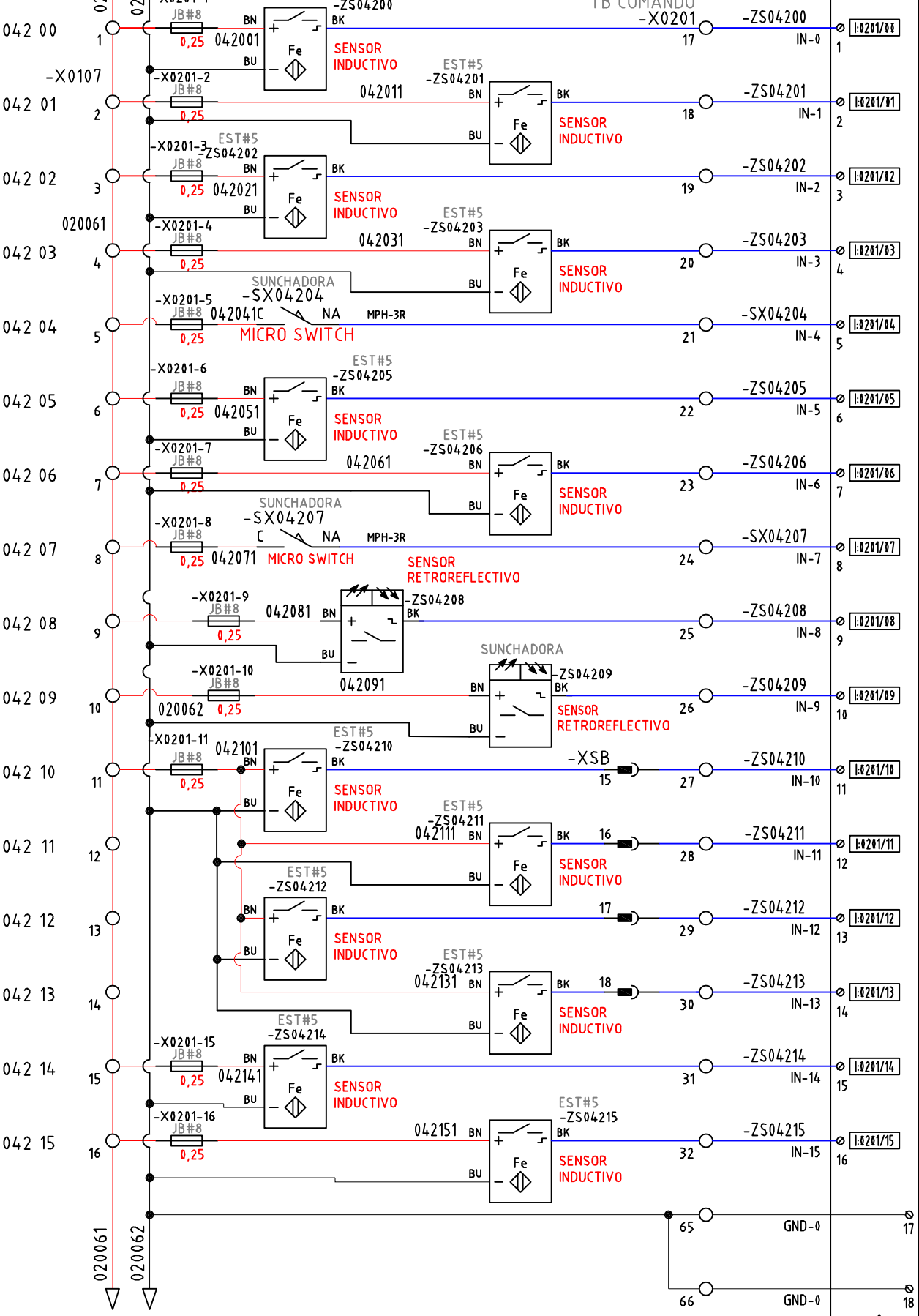
040 14 EST #6 BALIZA VERDE

040 15 EST #6 BALIZA ROJA

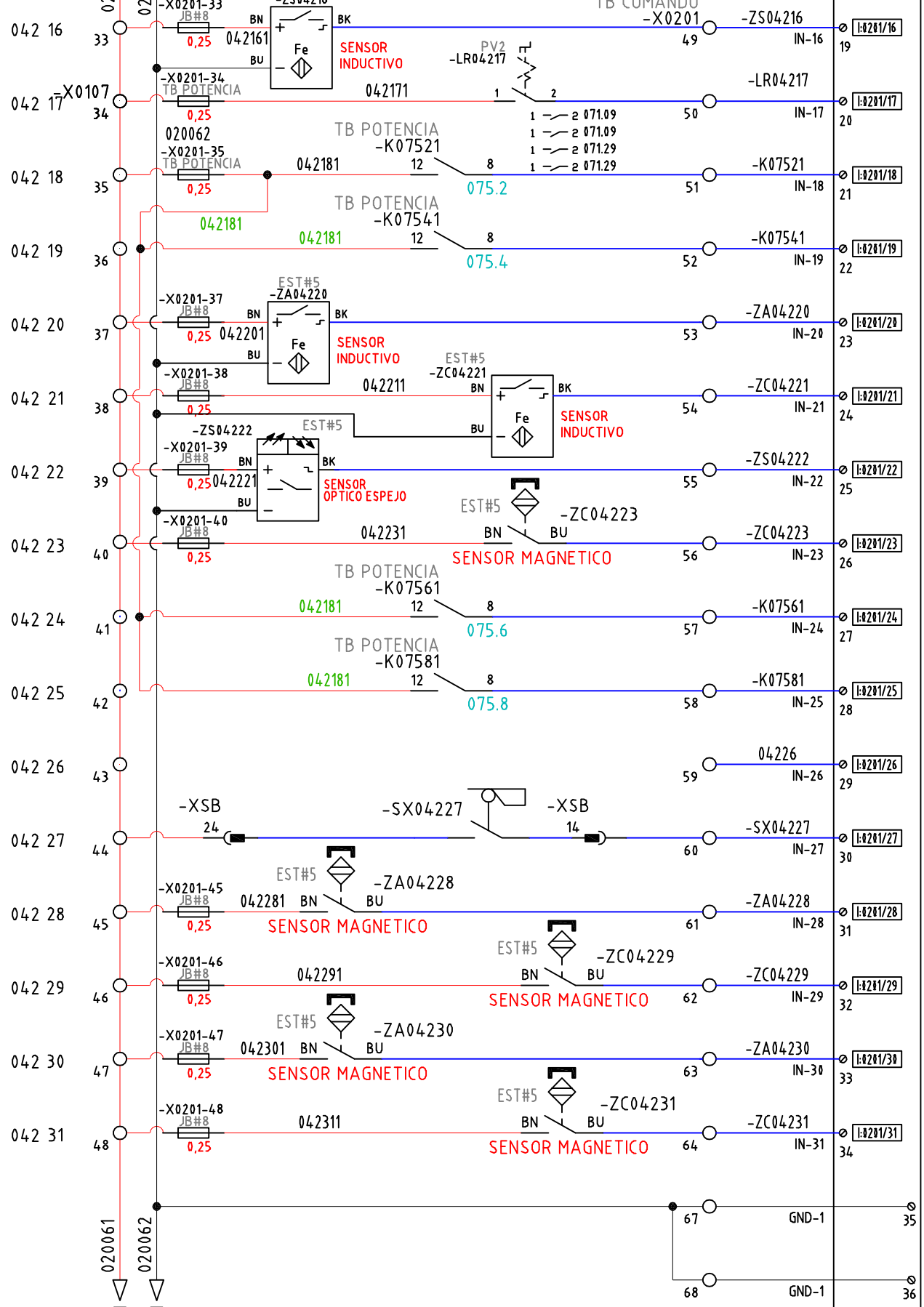


5	5	PROJECT	SIDERCROM EMBALAJE
4	4	DRAWING NO.	040
3	3	REVISION BY	040
2	2	REVISION BY	040
1	1	REVISION BY	040
REV. CLIENTE	REV. INT. P&F	DATE	20/10/2015
Av. Belgrano 2272 (BELLEROS)	Av. Belgrano 2272 (BELLEROS)	Av. Belgrano 2272 (BELLEROS)	Av. Belgrano 2272 (BELLEROS)
Argentina	Argentina	Argentina	Argentina
PLANTA SIDERCROM	PLANTA SIDERCROM	PLANTA SIDERCROM	PLANTA SIDERCROM
LOCATION/ACTIVITY	LOCATION/ACTIVITY	LOCATION/ACTIVITY	LOCATION/ACTIVITY

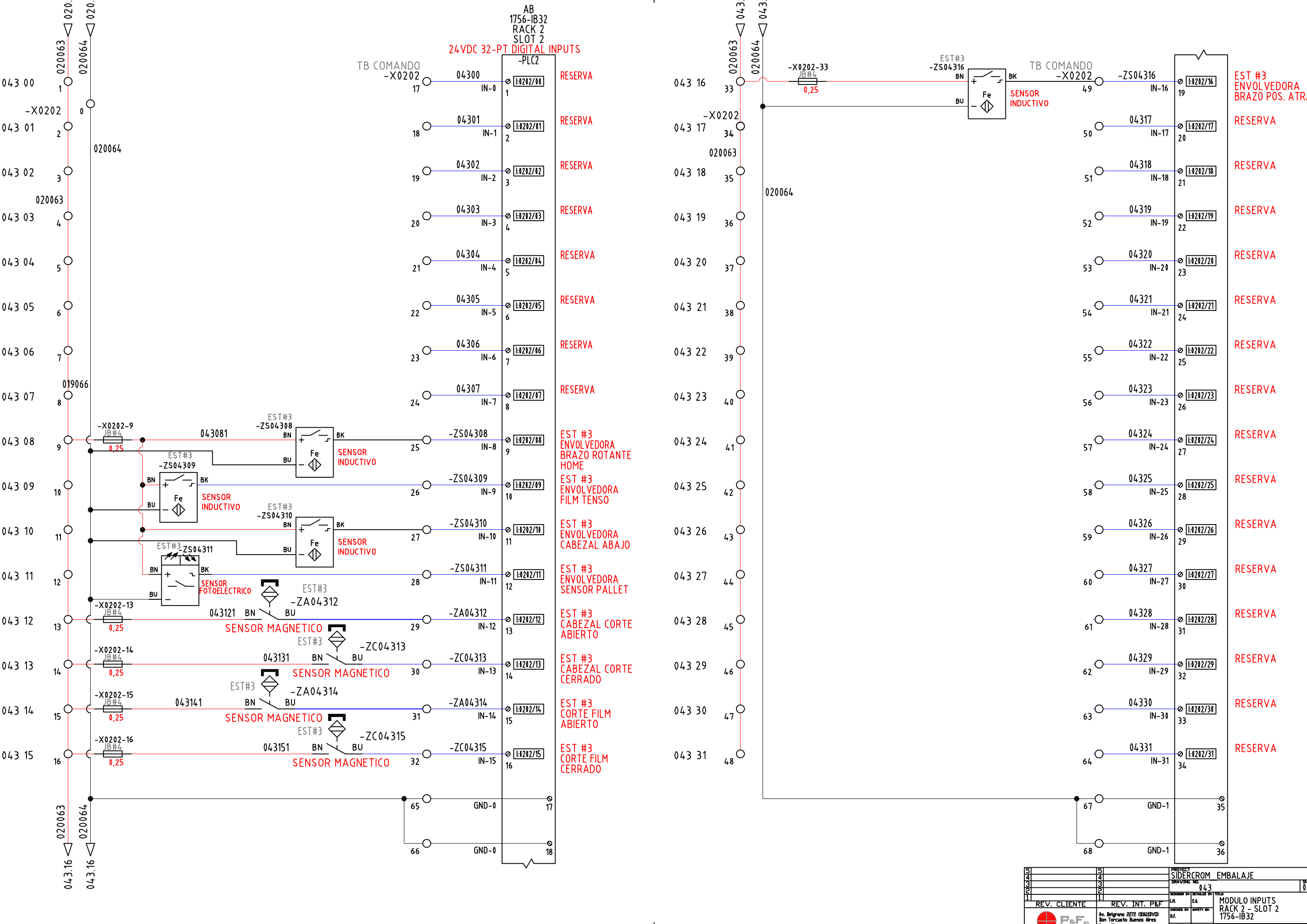
24VDC 32-PT DIGITAL INPUTS



- EST #5 SUNCHADORA LANZA CANAL CERRADO
- EST #5 SUNCHADORA LANZA CANAL ABIERTO
- EST #5 SUNCHADORA LANZA APROX. CANAL ABIERTO
- EST #5 SUNCHADORA LANZA APROX. CANAL CERRADO
- EST #5 SUNCHADORA PRESENCIA DE PALLET
- EST #5 SUNCHADORA TOPE AVANCE TRANSVERSAL
- EST #5 SUNCHADORA TOPE RETROCESO TRANSVERSAL
- EST #5 SUNCHADORA PARADA PRECISA
- EST #5 SUNCHADORA SENSOR RANURA SOBRE LANZA #1
- EST #5 SUNCHADORA SENSOR RANURA SOBRE LANZA #2
- EST #5 CABEZAL SUNCHADORA SENSOR B81
- EST #5 CABEZAL SUNCHADORA SENSOR B82
- EST #5 CABEZAL SUNCHADORA SENSOR B83
- EST #5 CABEZAL SUNCHADORA SENSOR B84
- EST #5 SENSOR DE BAJO INSUMO FLEJE
- EST #5 FALTA INSUMO FLEJE



- EST5 SENSOR DE BUFFER SUNCHO VACIO
- SUNCHADORA LLAVE SELECTORA 2 POS. BY-PASS EXTRACURSO
- SUNCHADORA EXTRACURSO MOV. TRANSVERSAL ADELANTE
- SUNCHADORA EXTRACURSO MOV. TRANSVERSAL ATRAS
- EST #5 SUNCHADORA FIN DE CARRERA ARRIBA
- EST #5 SUNCHADORA FIN DE CARRERA ABAJO
- EST #5 SUNCHADORA APROXIMACION
- EST #5 FRICTION WHEEL SUNCHADORA RETROCESO
- SUNCHADORA EXTRACURSO PUENTE ELEVADOR ARRIBA
- SUNCHADORA EXTRACURSO PUENTE ELEVADOR ABAJO
- RESERVA
- SUNCHADORA SERIE TERMISTORES CABEZAL
- EST #5 SUJETADOR ESQUINERO #1 ABIERTO
- EST #5 SUJETADOR ESQUINERO #1 CERRADO
- EST #5 SUJETADOR ESQUINERO #2 ABIERTO
- EST #5 SUJETADOR ESQUINERO #2 CERRADO



046 00 EST #5
CABEZAL SUNCHADORA
MOTOR M26
MARCHA

046 01 EST #5
CABEZAL SUNCHADORA
MOTOR M26
CONTRAMARCHA

046 02 EST #5
CABEZAL SUNCHADORA
MOTOR 27
MARCHA

046 03 EST #5
CABEZAL SUNCHADORA
MOTOR M27
CONTRAMARCHA

046 04 EST #5
CABEZAL SUNCHADORA
MOTOR 28
MARCHA

046 05 EST #5
FRENO PUENTE
SUNCHADORA

046 06 EST #5
FRENO PORTICO
SUNCHADORA

046 07 EST #5
FRENO
TG2

046 08 EST #5
PRESIONADOR
ESQUINERO #1
AVANZA

046 09 EST #5
PRESIONADOR
ESQUINERO #1
RETROCEDE

046 10 EST #5
PRESIONADOR
ESQUINERO #2
AVANZA

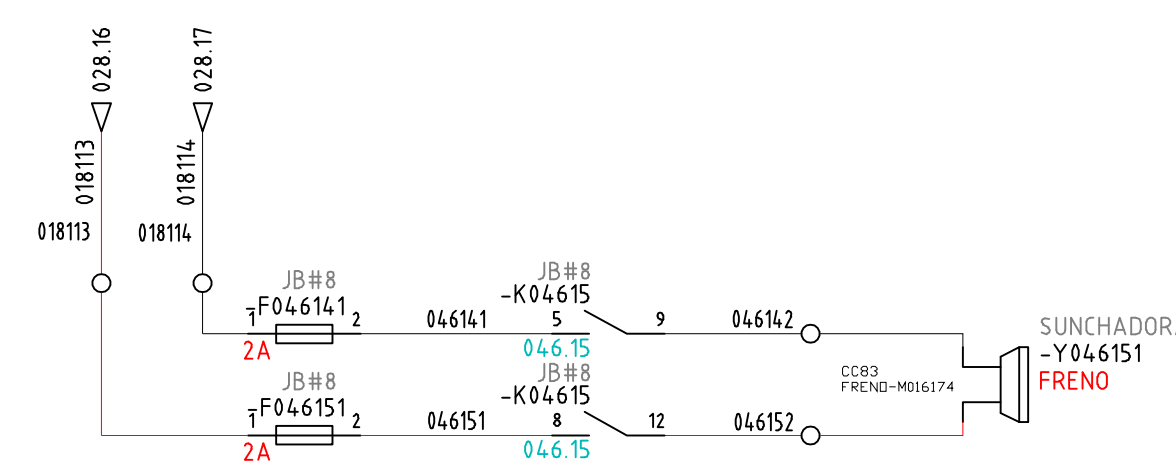
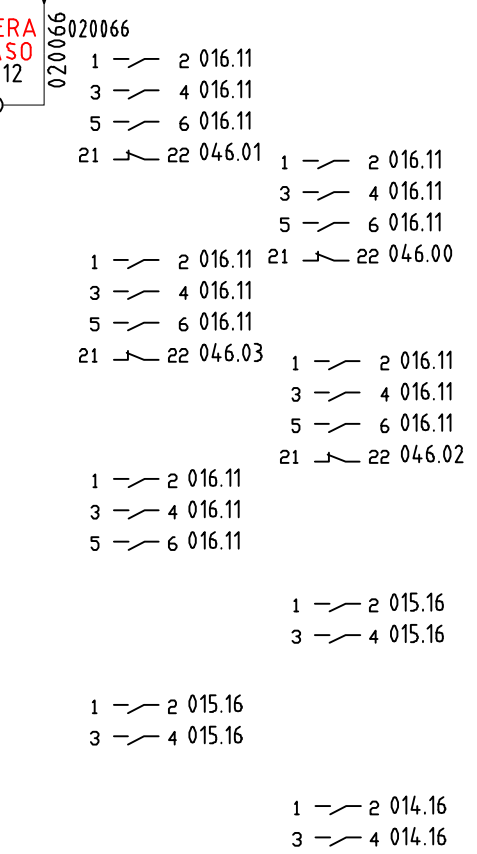
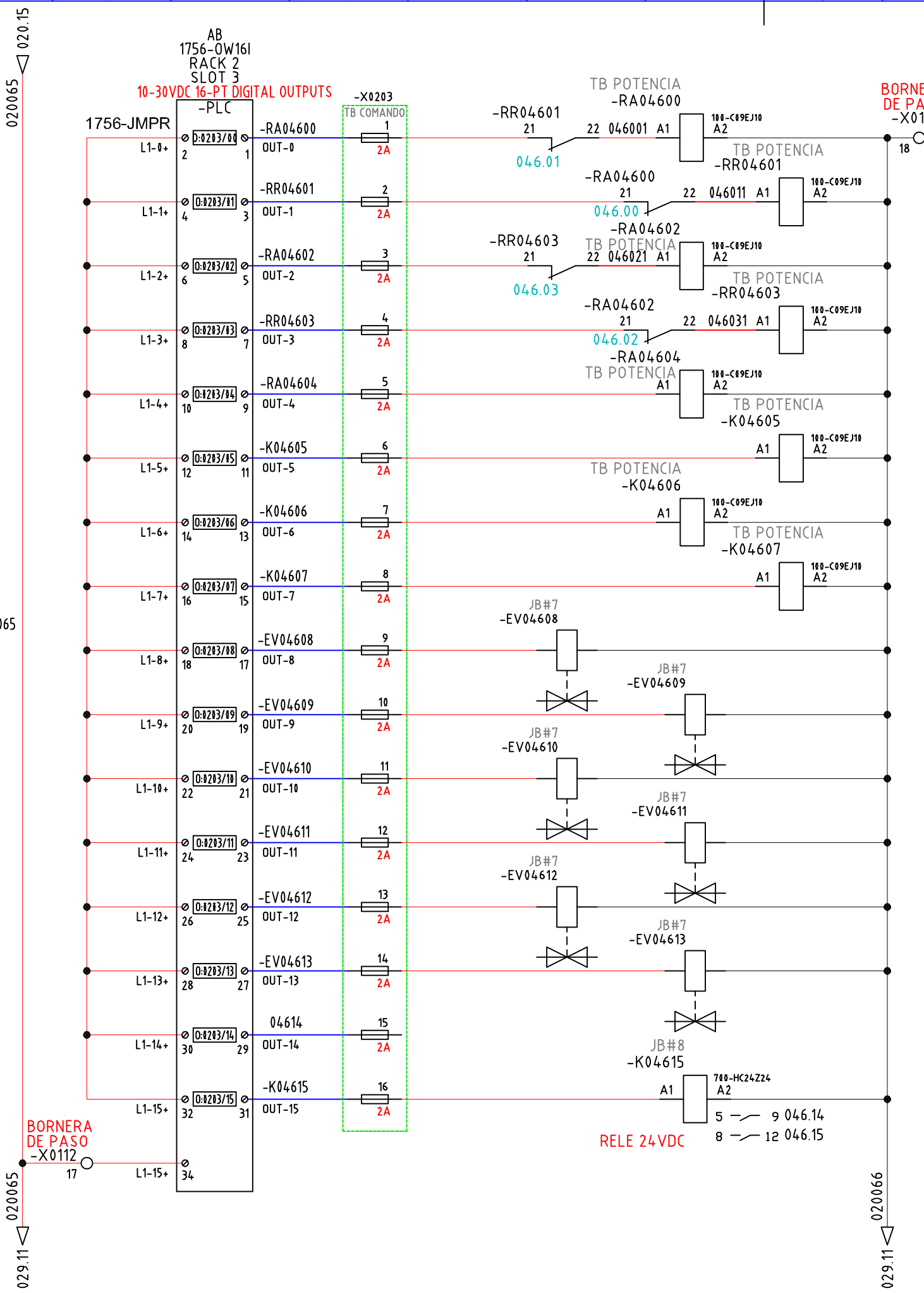
046 11 EST #5
PRESIONADOR
ESQUINERO #2
RETROCEDE

046 12 EST #5
FRICTION WHEEL
SUNCHADORA
AVANCE

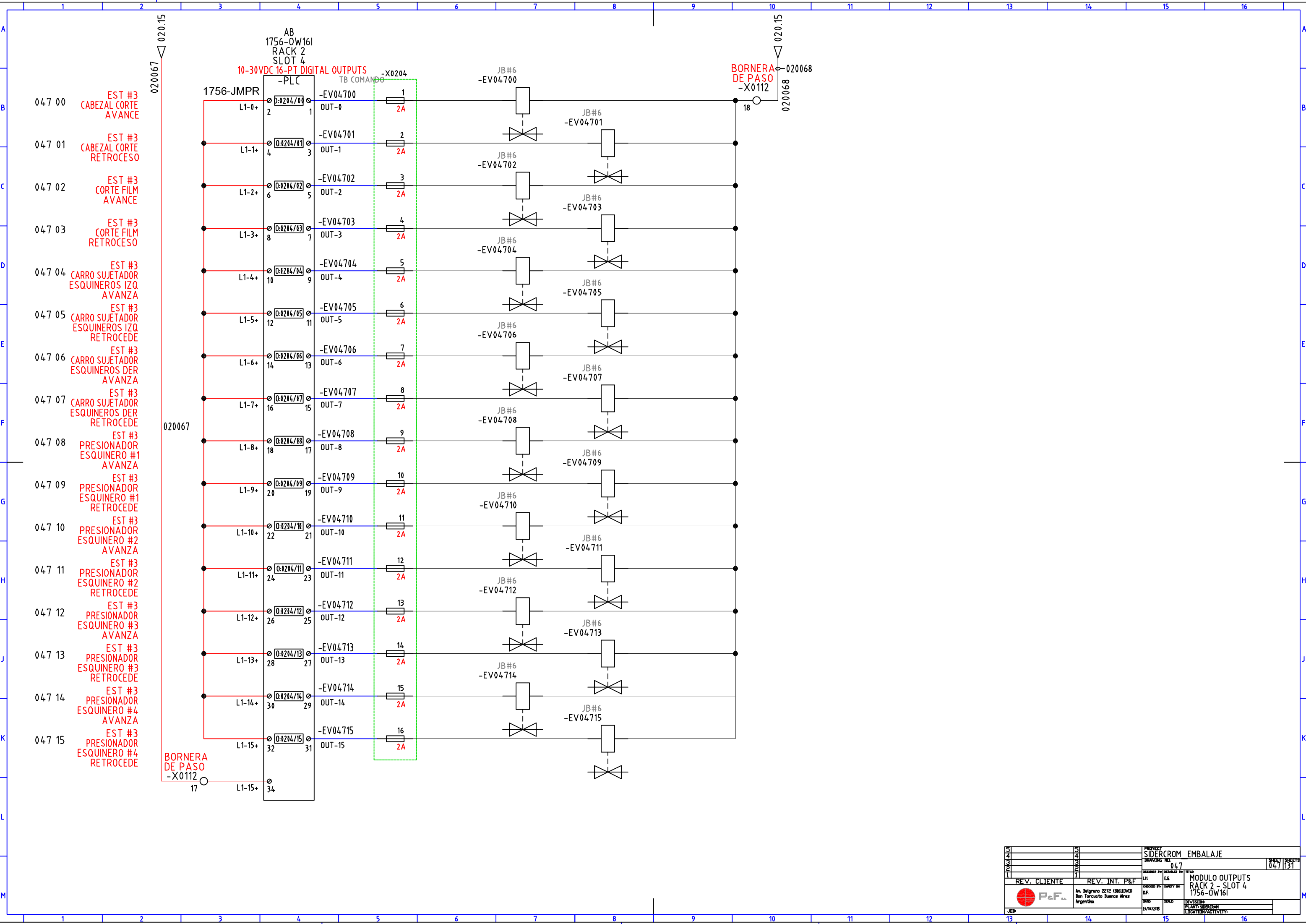
046 13 EST #5
FRICTION WHEEL
SUNCHADORA
RETROCESO

046 14 RESERVA

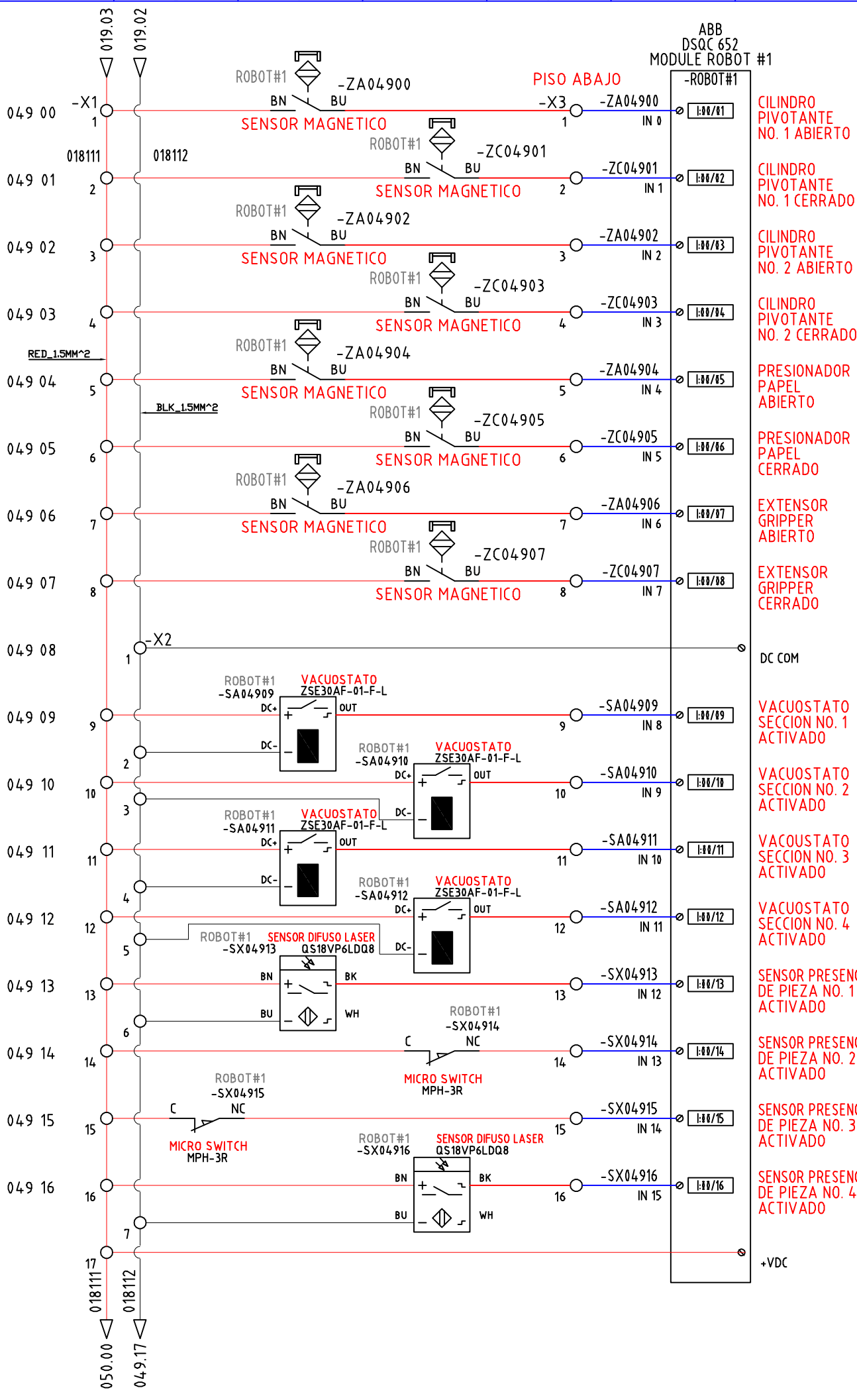
046 15 EST #5
FRENO CARRETE
SUNCHADORA



5	5	PROJECT	SIDERCROM EMBALAJE
4	4	DRAWING NO.	046
3	3	REVISION BY	REVISION BY
2	2	DATE	DATE
1	1	REV. CLIENTE	REV. INT. P&F
		Av. Belgrano 2272 (CHUQUID)	Don Torcuato Buenos Aires
		DATE	DATE
		PLANT	PLANT
		LOCATION/ACTIVITY	LOCATION/ACTIVITY



5	5	PROJECT	SIDERCROM EMBALAJE
4	4	DRAWING NO.	047
3	3	REVISION BY	047
2	2	REVISION NO.	047
1	1	REVISION DATE	047
REV. CLIENTE	REV. INT. P&F	DATE	047
Av. Belgrano 2272 (BELLEROS)	Don Torcuato Buenos Aires	DATE	047
PLANTA SIDERCROM	LOCATION/ACTIVITY	DATE	047



CILINDRO PIVOTANTE NO. 1 ABIERTO

CILINDRO PIVOTANTE NO. 1 CERRADO

CILINDRO PIVOTANTE NO. 2 ABIERTO

CILINDRO PIVOTANTE NO. 2 CERRADO

PRESIONADOR PAPEL ABIERTO

PRESIONADOR PAPEL CERRADO

EXTENSOR GRIPPER ABIERTO

EXTENSOR GRIPPER CERRADO

VACUOSTATO SECCION NO. 1 ACTIVADO

VACUOSTATO SECCION NO. 2 ACTIVADO

VACUOSTATO SECCION NO. 3 ACTIVADO

VACUOSTATO SECCION NO. 4 ACTIVADO

SENSOR PRESENCIA DE PIEZA NO. 1 ACTIVADO

SENSOR PRESENCIA DE PIEZA NO. 2 ACTIVADO

SENSOR PRESENCIA DE PIEZA NO. 3 ACTIVADO

SENSOR PRESENCIA DE PIEZA NO. 4 ACTIVADO

049 17

049 18

049 19

049 20

049 21

049 22

049 23

049 24

049 25

049 26

049 27

049 28

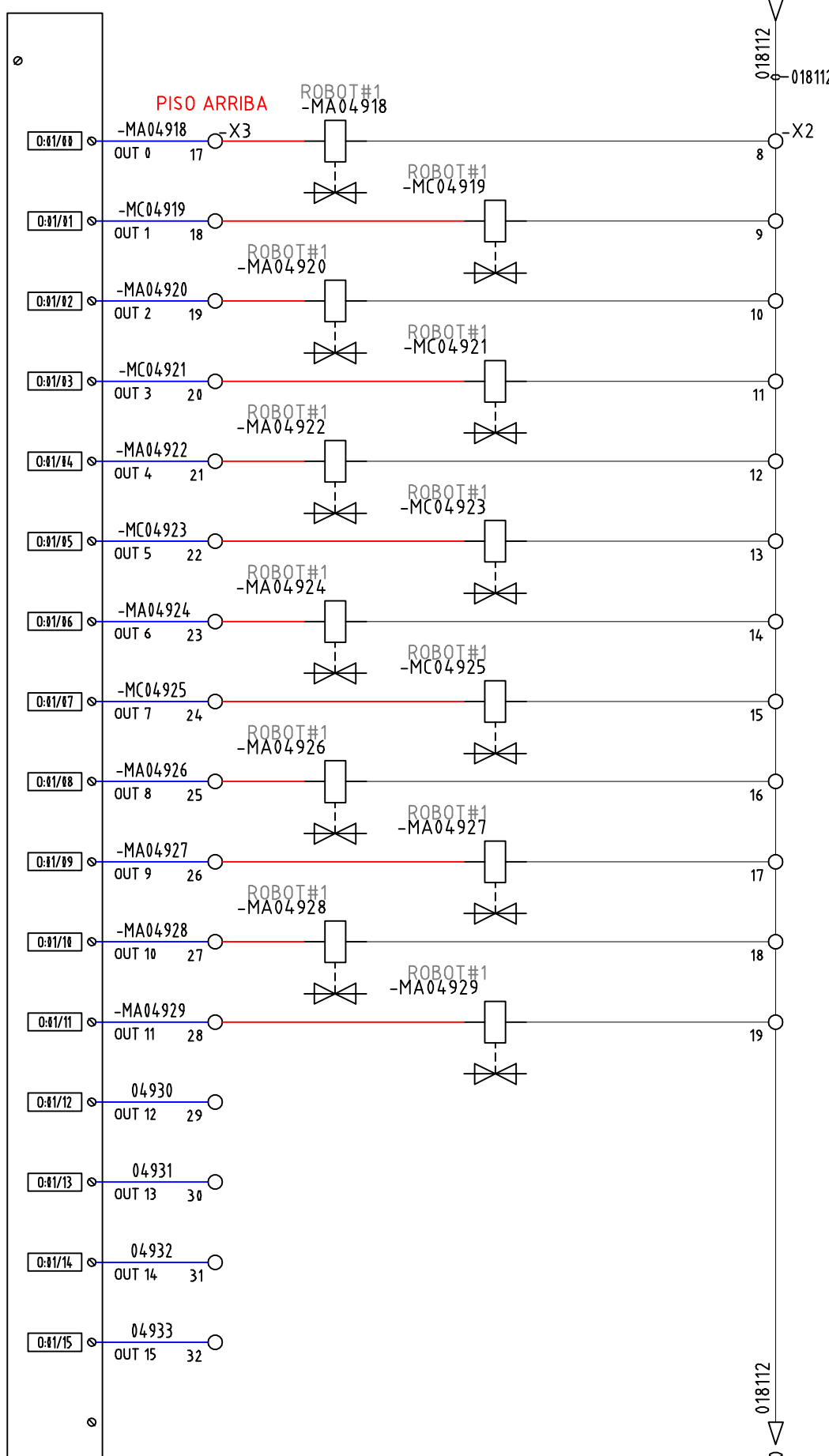
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049 30

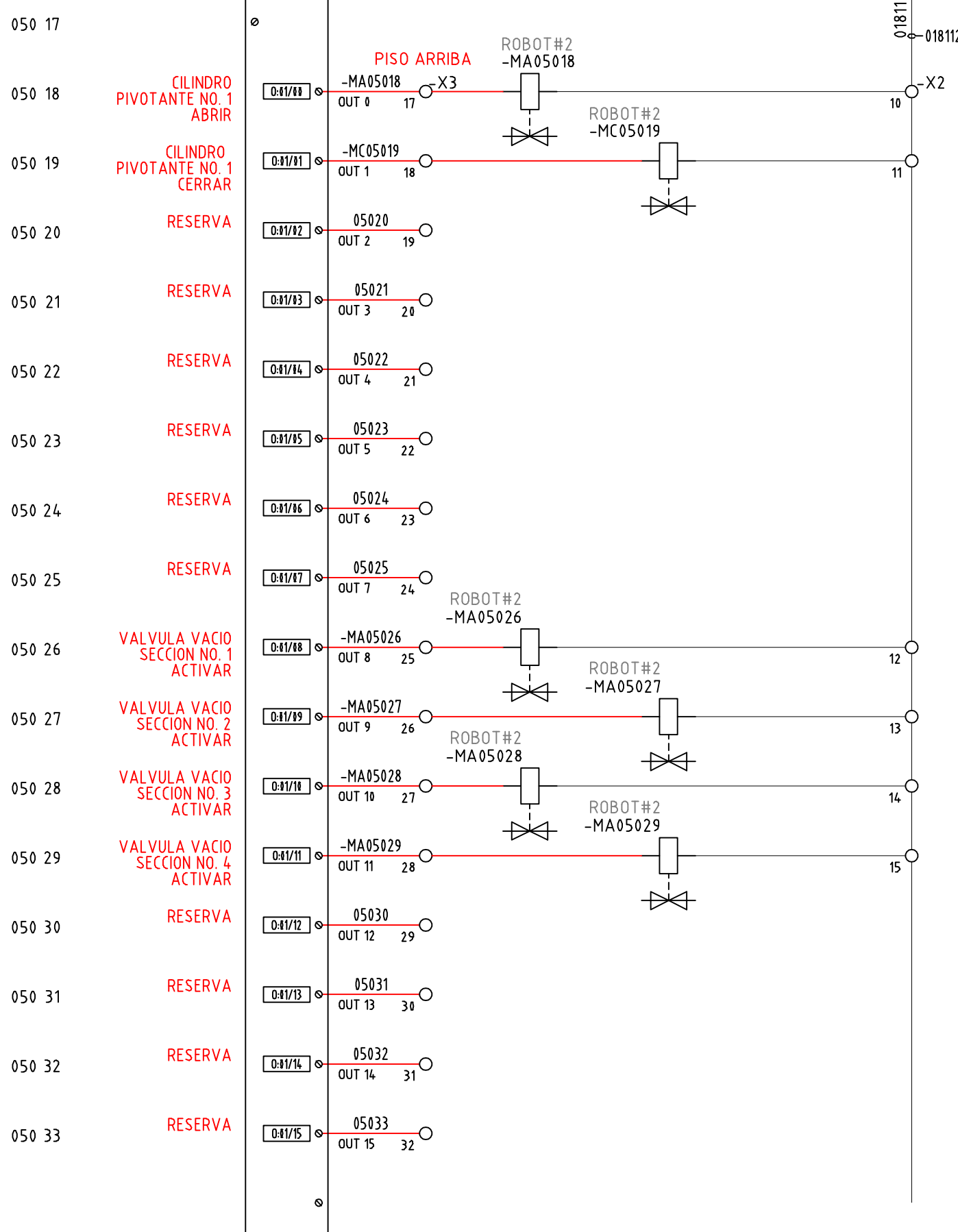
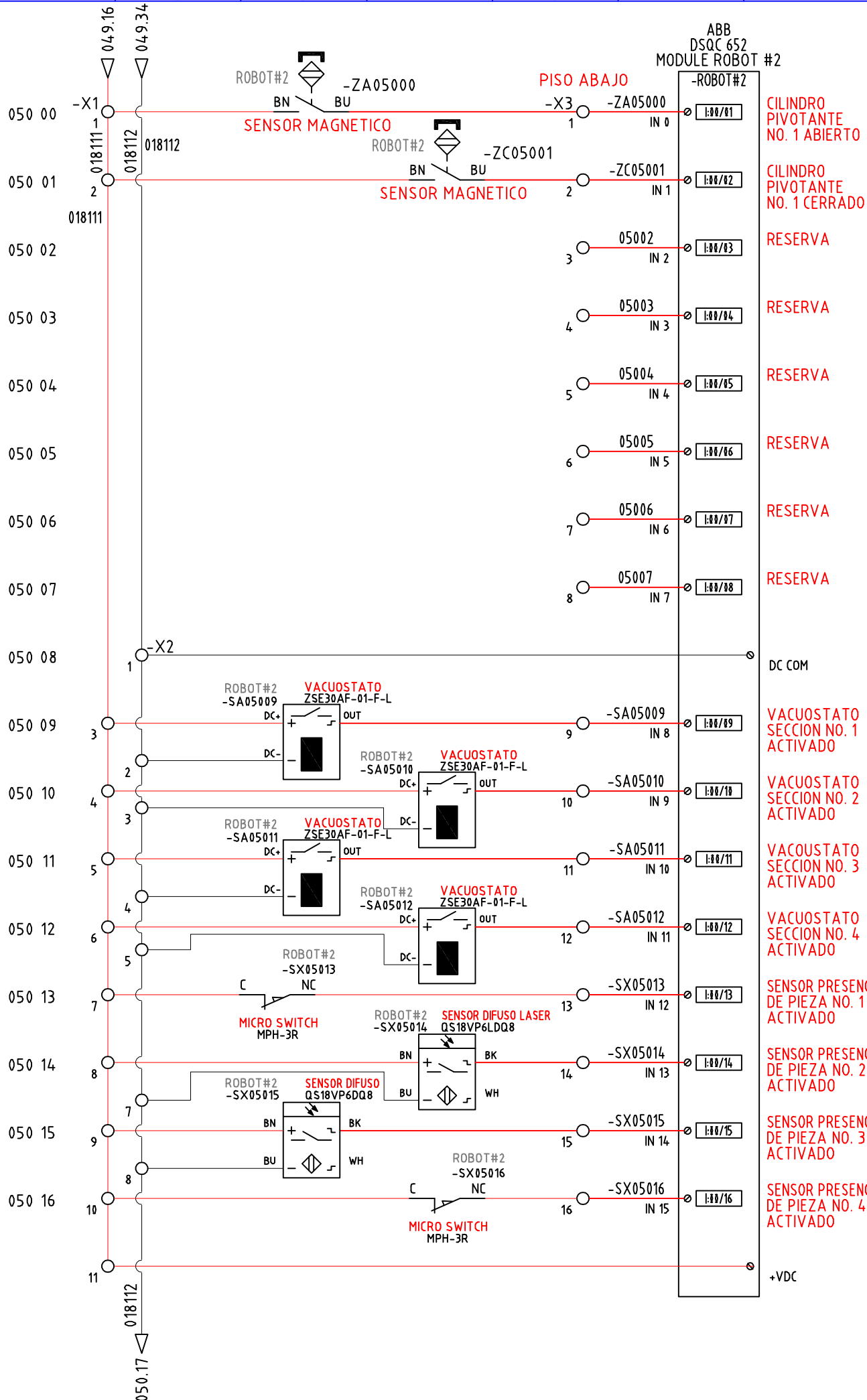
049 31

049 32

049 33



5	5	PROYECTO	SIDERCROM EMBALAJE
4	4	DRAWING NO.	049
3	3	REVISOR	049
2	2	REVISOR	049
1	1	REVISOR	049
REV. CLIENTE	REV. INT. P&F	LA	EA
CHIEF BY D.F.	SAFETY BY	DAVID	DAVID
29/04/2015	29/04/2015	DIVISION	PLANTA SIDERCROM
JOB	JOB	LOCATION/ACTIVITY	LOCATION/ACTIVITY

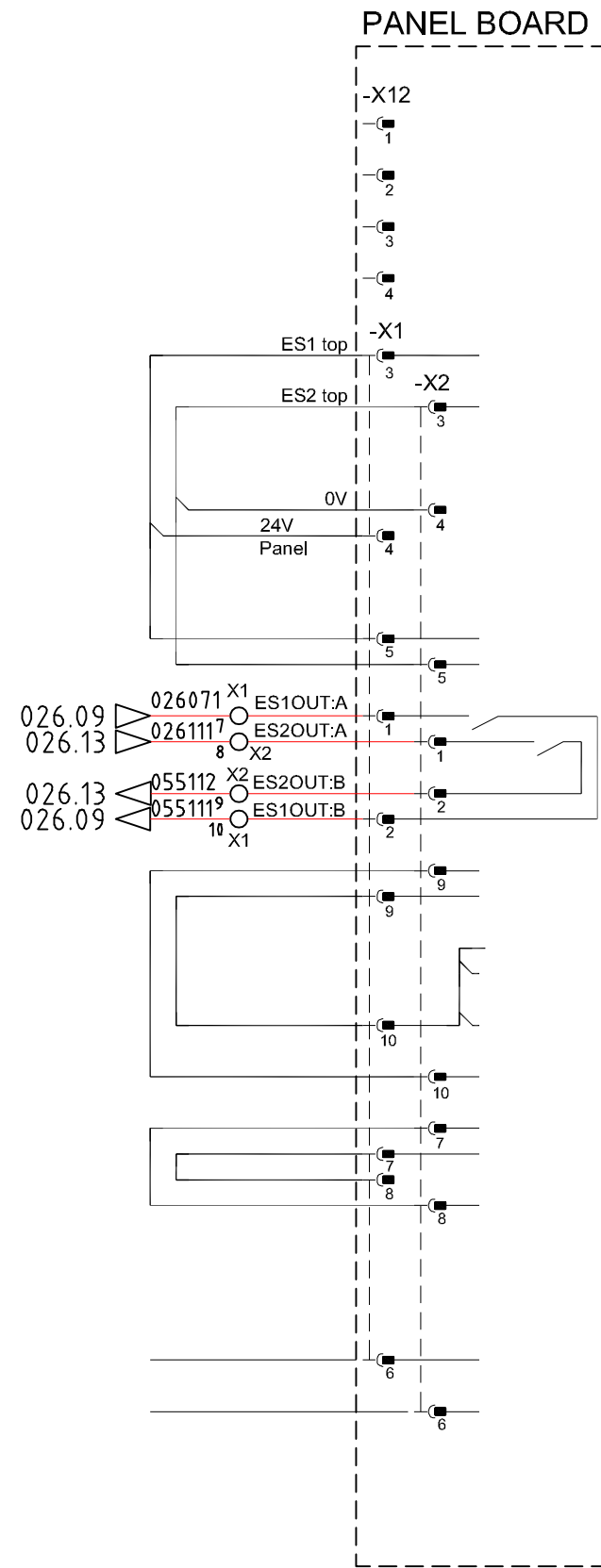


5	5	PROYECTO	SIDERCROM EMBALAJE
4	4	DRAWING NO.	050
3	3	REVISOR	050
2	2	REVISOR	050
1	1	REVISOR	050
REV. CLIENTE	REV. INT. P&F	LA	EA
Av. Belgrano 2272 (BOLIVAR)	Don Torcuato Buenos Aires	SAFETY NO.	MODULO I/O IRC5
DATE	20/04/2015	DATE	ROBOT #2
JOB		DIVISION	PLANTA SIDERCROM
		LOCATION/ACTIVITY	

HARDWIRED CONTROL CIRCUITS

PROJECT SIDERCROM EMBALAJE		DRAWING NO. 053		SHEET SHEETS 053 / 131	
DESIGNED BY L.A.		CHECKED BY D.F.		DATE 29/11/2015	
REV. CLIENTE		REV. INT. P&F		DIVISION PLANTA SIDERCROM	
P&F		Av. Belgrano 2272 (BOLIVAR) Don Torcuato Buenos Aires Argentina		LOCATION/ACTIVITY	

PANEL BOARD
ROBOT 1

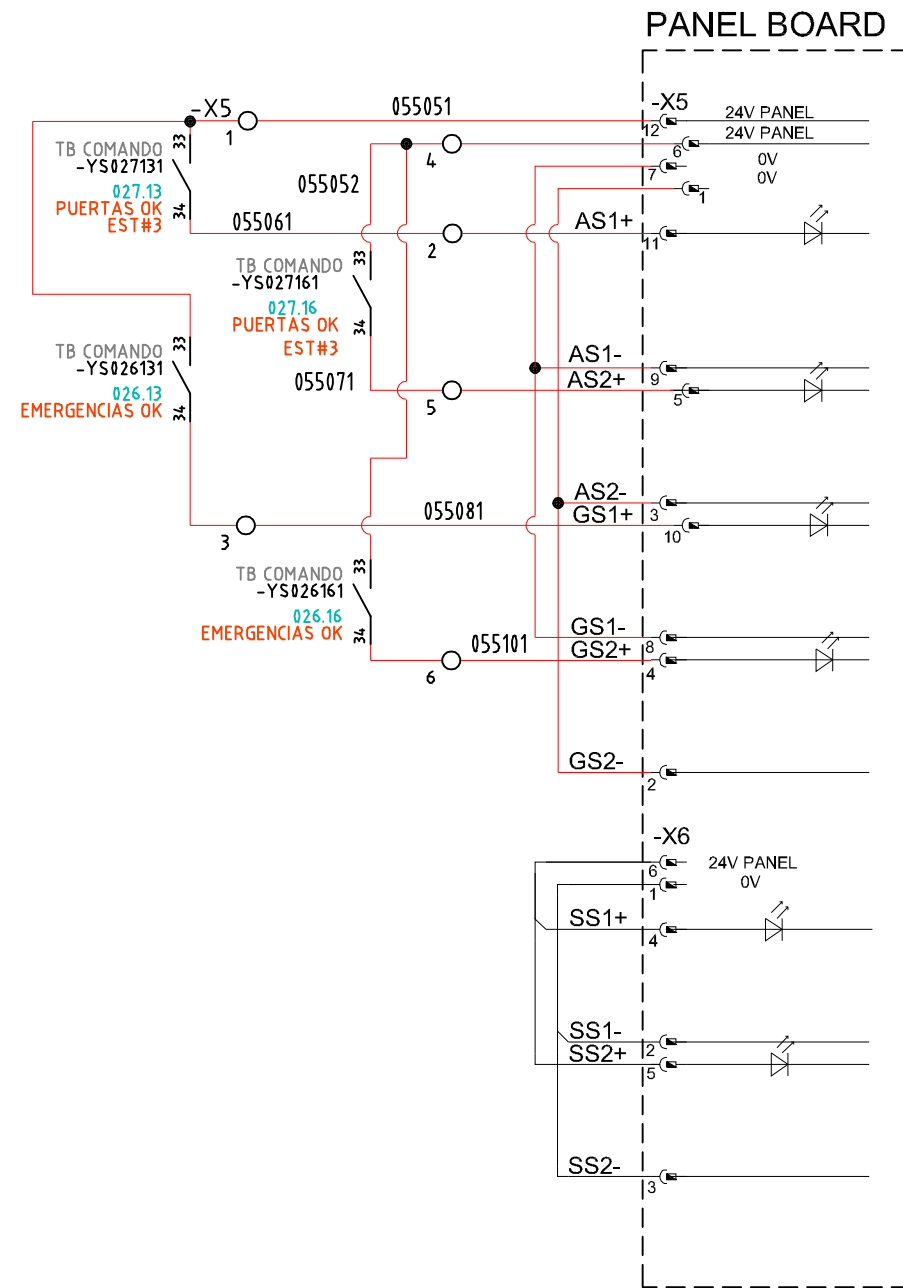
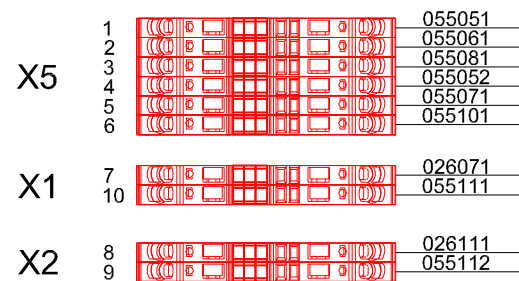


AUTO STOP

GENERAL STOP

SUPERIOR STOP

BLOQUE DE BORNERAS

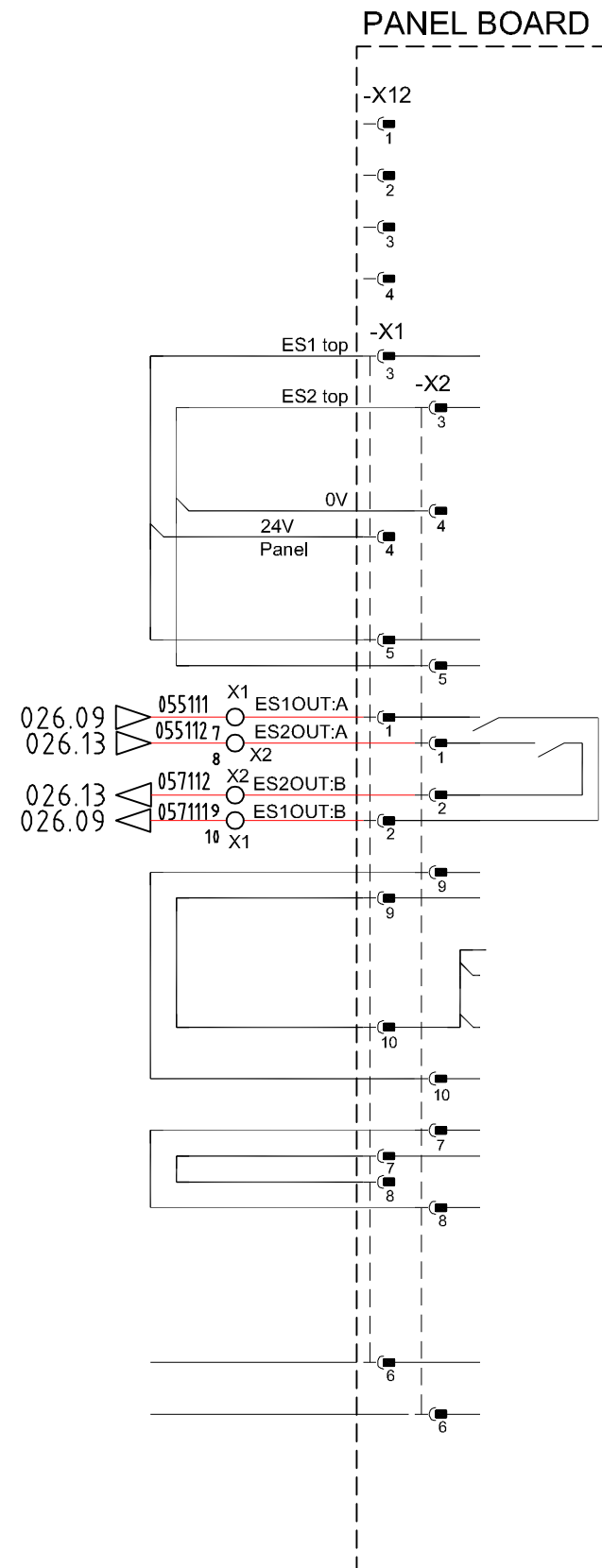


Tablero	Multipar	N° Hilo	N° Cable	Componente	Tags Bornera
TBP	CC37	2	055051	YS027131	X5 - 1
		3	055061	YS027131	X5 - 2
		4	055081	YS026131	X5 - 3
		5	055052	YS027161	X5 - 4
		6	055071	YS027161	X5 - 5
		7	055101	YS026161	X5 - 6
		8	026071	-	X1 - 7
		9	026111	-	X1 - 10
		10	055112	-	X2 - 8
		11	055111	-	X2 - 9
		12	-	-	-

5	5	PROYECTO	SIDERCROM EMBALAJE
4	4	DRAWING NO.	055
3	3	DESIGN BY / DESIGNED BY	055 1131
2	2	REV. CLIENTE	REV. INT. P&F
1	1	CHIEF BY	SAFETY BY
P&F		PLANTA SIDERCROM	
Av. Belgrano 2272 (CHILIVUD)		Buenos Aires	
Argentina		DIVISION	
29/14/215		PLANTA SIDERCROM	
		LOCATION/ACTIVITY	

PANEL BOARD

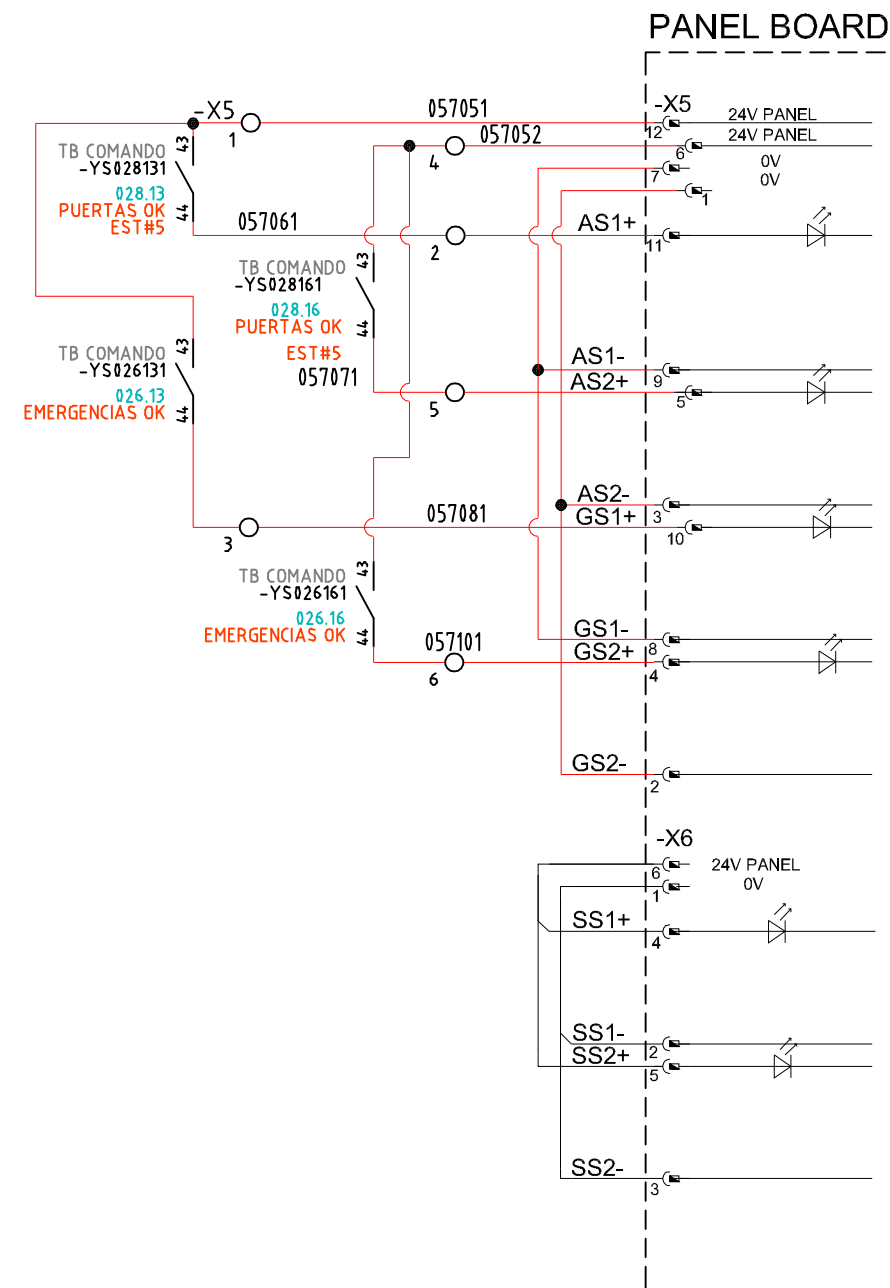
ROBOT 2



AUTO STOP

GENERAL STOP

SUPERIOR STOP

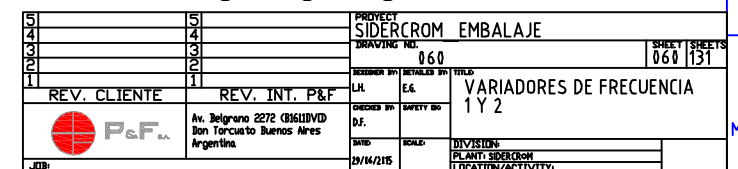
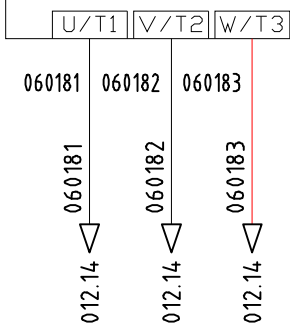


BLOQUE DE BORNERAS

	1		057051
X5	2		057061
	3		057081
	4		057052
	5		057071
	6		057101
	7		055111
X1	10		057111
	8		055112
X2	9		057112

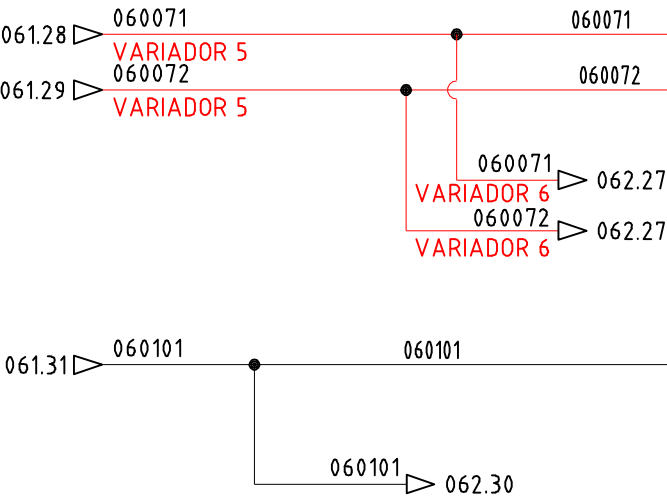
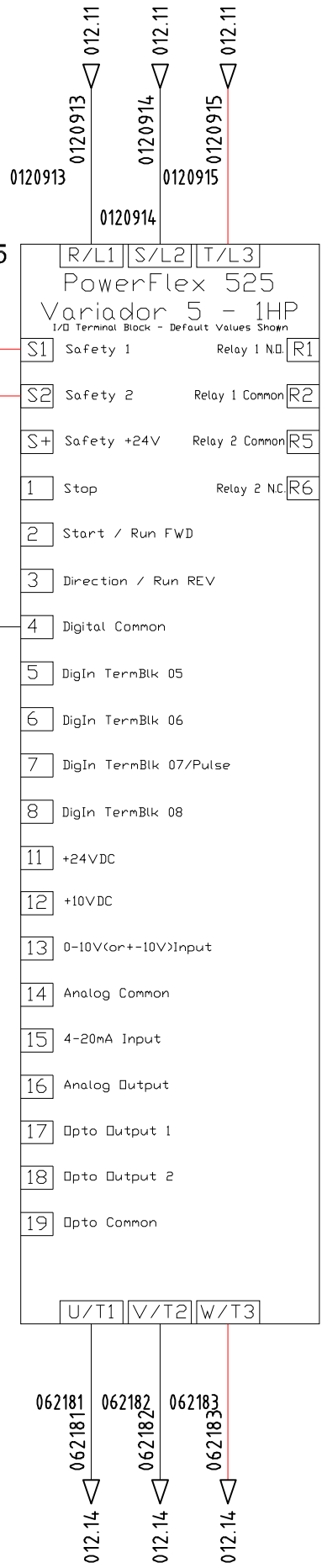
Tablero	Multipar	N° Hilo	N° Cable	Componente	Tags Bornero
TBP	CC38	1	057051	YS028131	X5 – 1
		2	057061	YS028131	X5 – 2
		3	057081	YS026131	X5 – 3
		4	057052	YS028161	X5 – 4
		5	057071	YS028161	X5 – 5
		6	057101	YS026161	X5 – 6
		7	055111	–	X1 – 7
		8	055112	–	X1 – 10
		9	057111	–	X2 – 8
		10	057112	–	X2 – 9
		11	–	–	–

5	5	PROJECT	SIDEROCROM EMBALAJE		SHEET	SHEETS		
4	4	DRAWING NO.	057				057	131
3	3	DESIGNED BY	DETAILS BY	TITLE			PANEL BOARD ROBOT 2	
2	2	J.H.	E.C.					
1	1	CHECKED BY	SAFETY NO.					
		D.F.						
REV. CLIENTE		REV. INT. P&F						
 P&F S.A.		Av. Belgrano 2272 D6111MD Don Torcuato Buenos Aires Argentina						
JOB		DATE		SCALE				
		21/14/215		DIVISION PLANTA SIDEROCROM LOCALIDAD/ACTIVIDAD				



062 01
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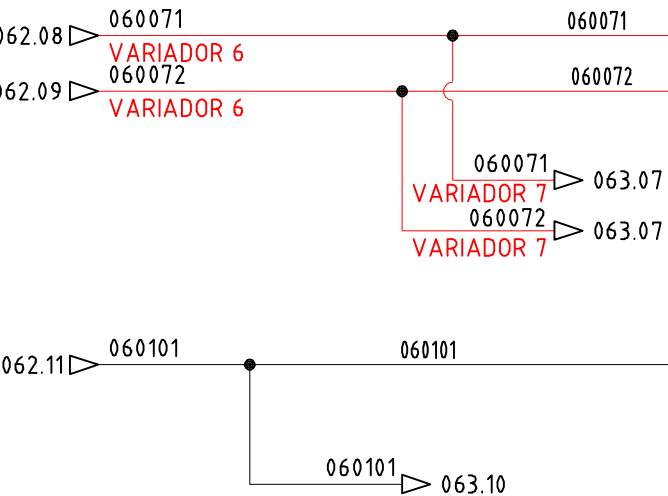
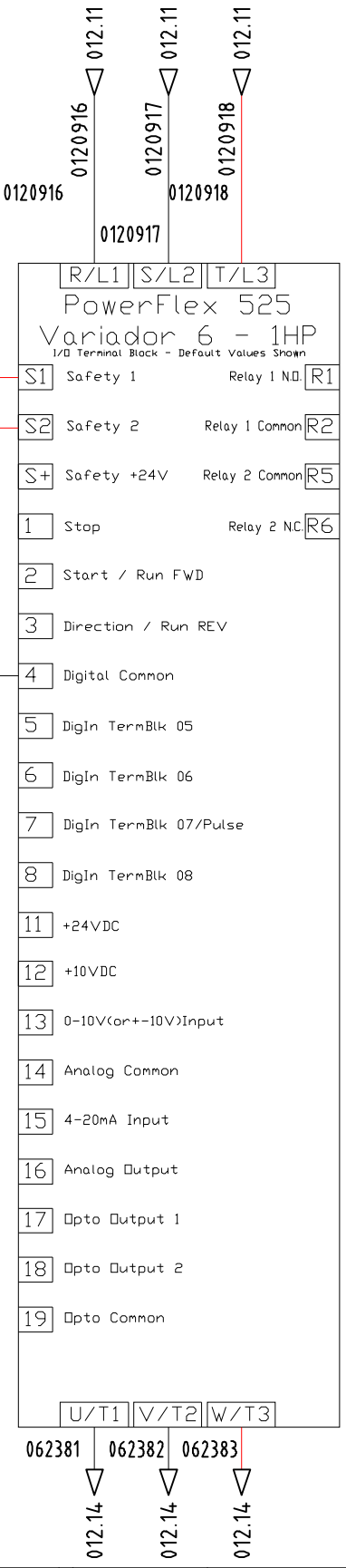
TRANSPORTADOR GIRATORIO DE DESCARTE
ESTACION #2 - TG01
PF525 A020005



DIRECCION IP: 192.168.1.14/16
N° Cat.:25B-D2P3N114

062 21
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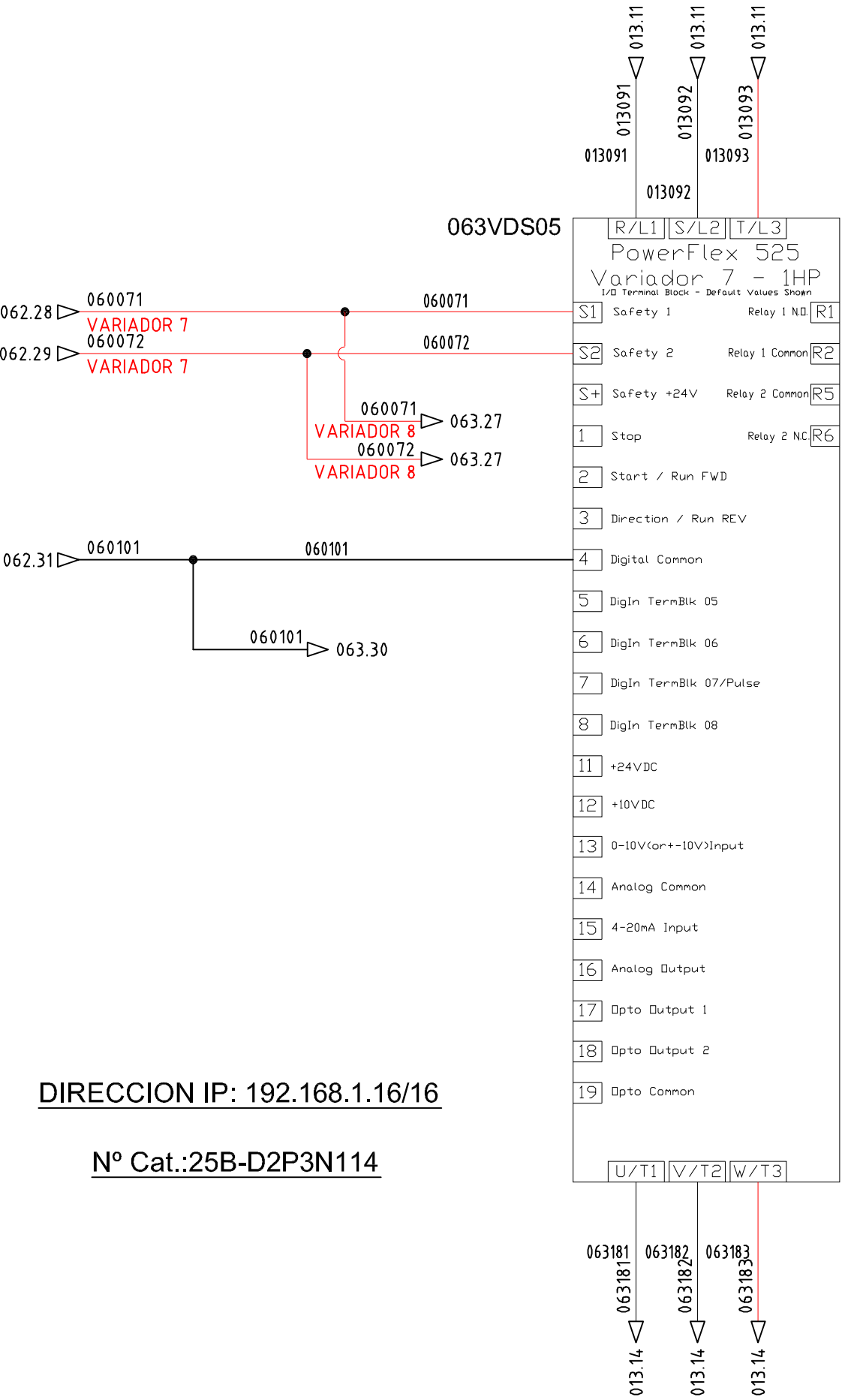
TRANSPORTADOR DE DESCARTE #1
ESTACION #2 - MESA #3
PF525 A020006



DIRECCION IP: 192.168.1.15/16
N° Cat.:25B-D2P3N114

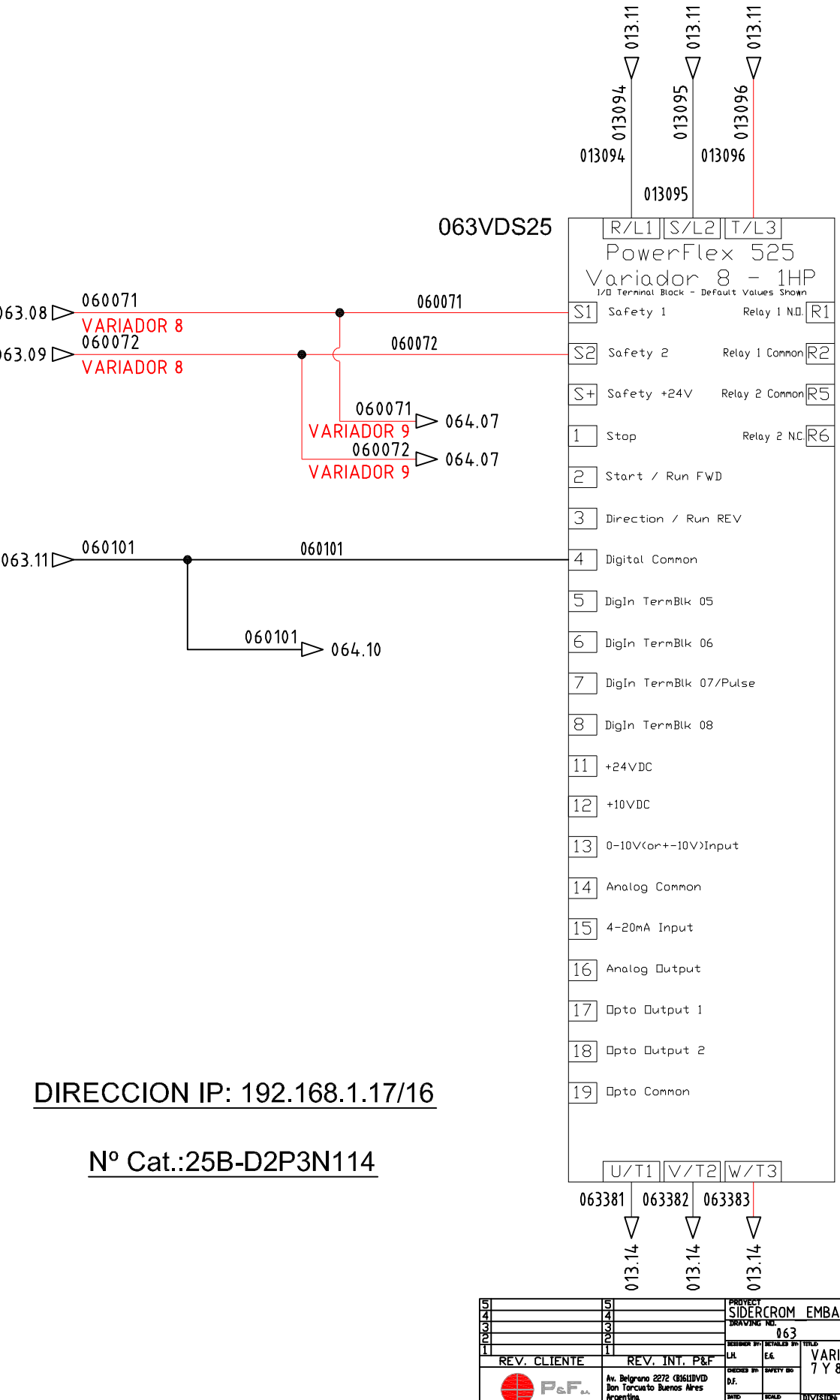
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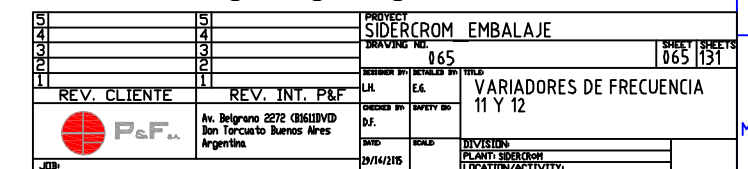
TRANSPORTADOR DE DESCARTE #2
ESTACION #2 - MESA #4
PF525 A020007

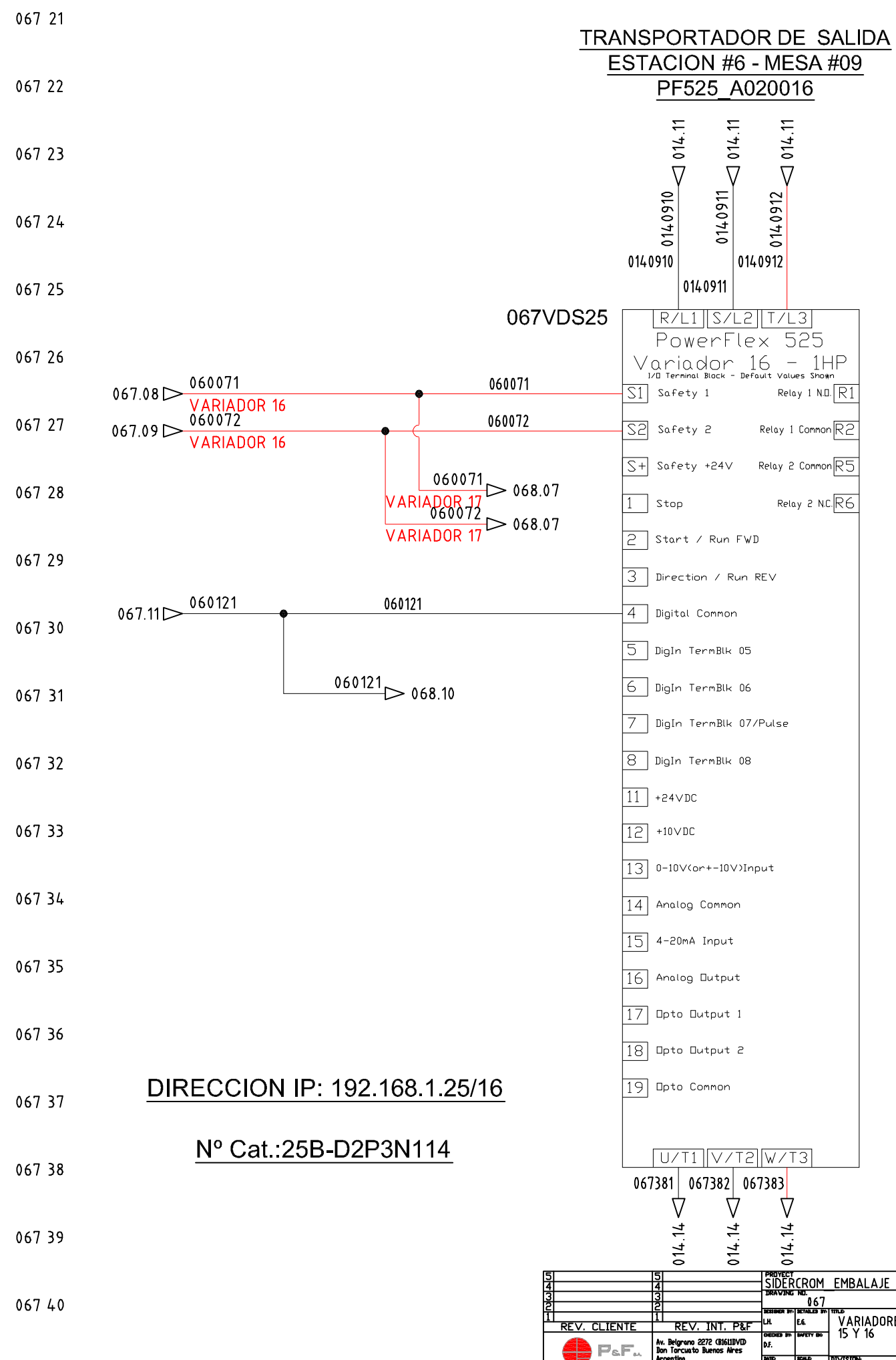
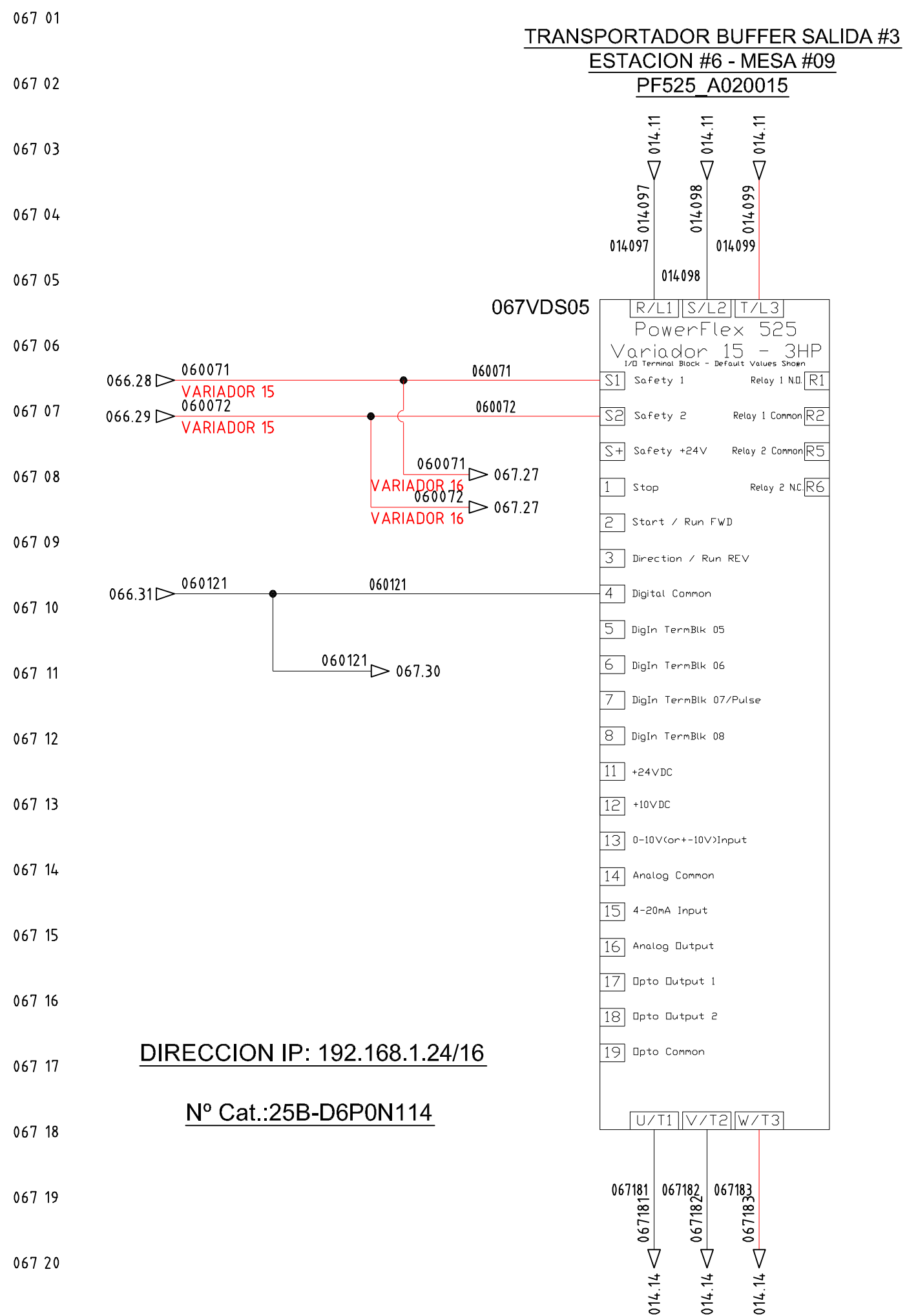


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TRANSPORTADOR
ESTACION #3 - MESA RODILLOS #5
PF525 A020008



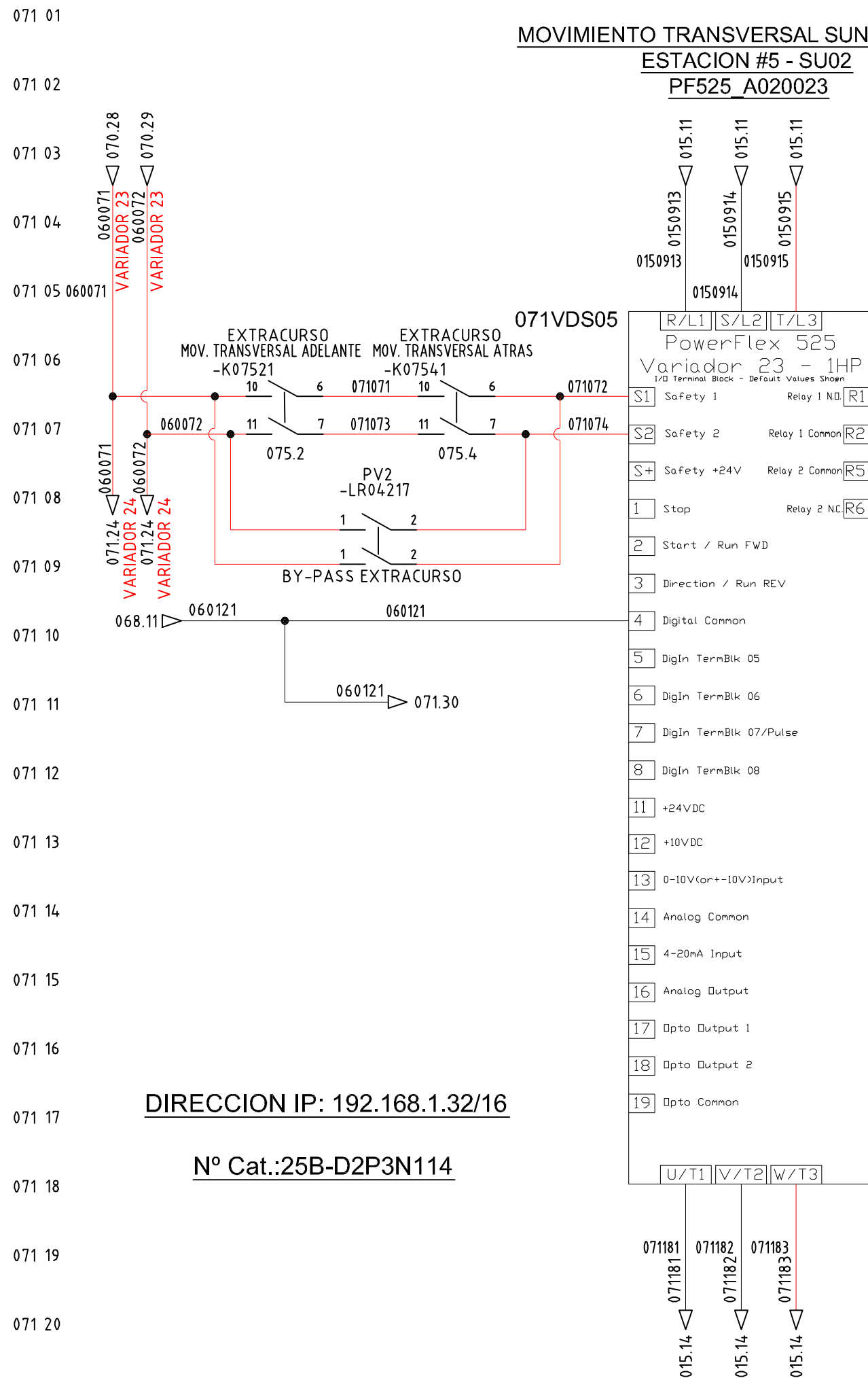




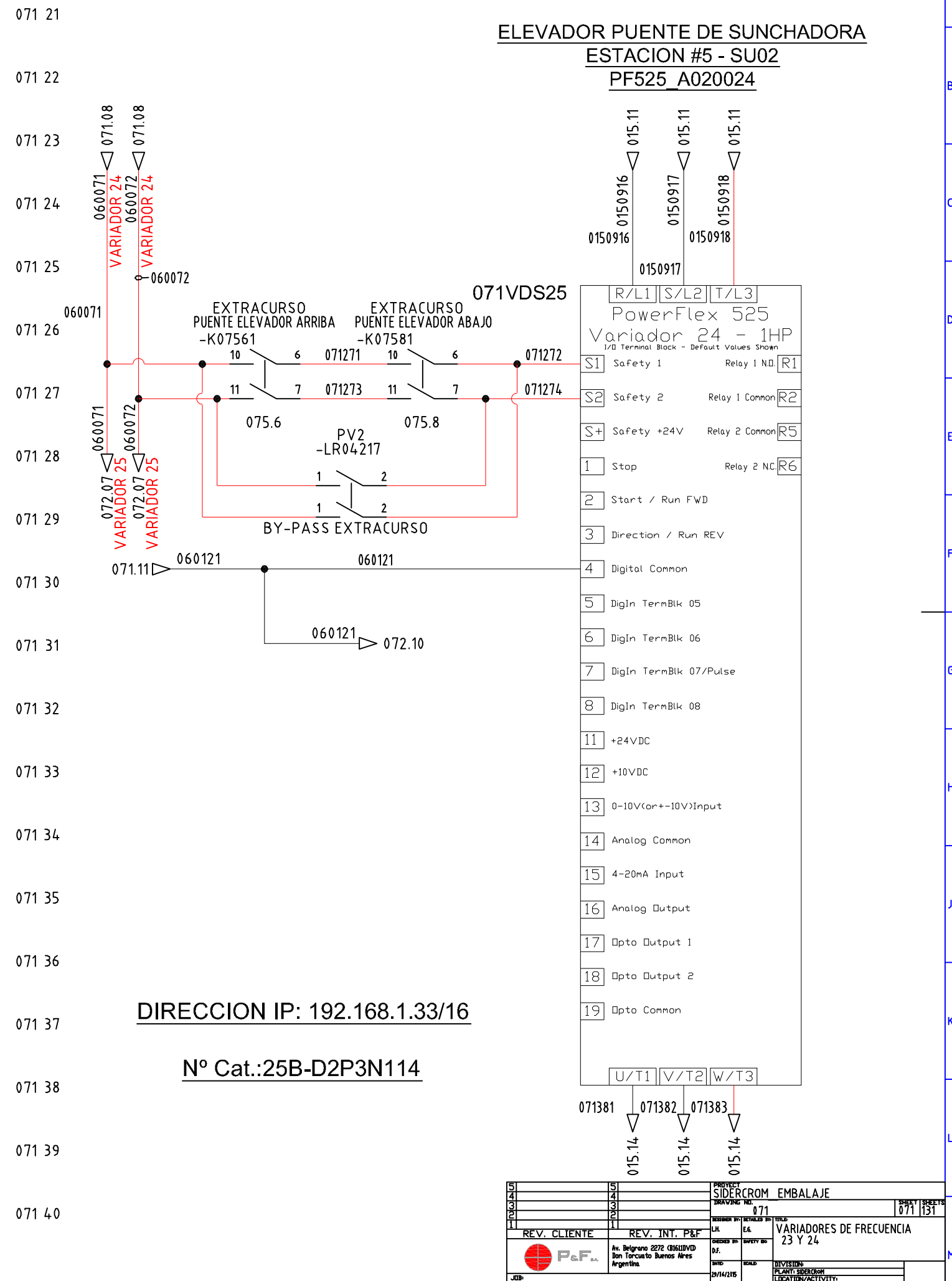


5	5	PROJECT		SIDEROCROM EMBALAJE					
4	4	DRAWING NO.		070		SHEET		131	
3	3	SECTION BY		070		SHEET		131	
2	2	LIT.		E.G.		VARIADORES DE FRECUENCIA			
1	1	REV. CLIENTE		REV. INT. P.F.		21 Y 22			
 P&F		Av. Belgrano 2272 (B1611)VD Don Torcuato Buenos Aires Argentina		DRESSES BY D.F.		SAFETY BY D.F.		DIVISION PLANT: SIDEROCROM LOCAL: PLANTA 1000	
DATE		20/14/215		GOLD					
JTB:									

MOVIMIENTO TRANSVERSAL SUNCHADORA
ESTACION #5 - SU02
PF525 A020023

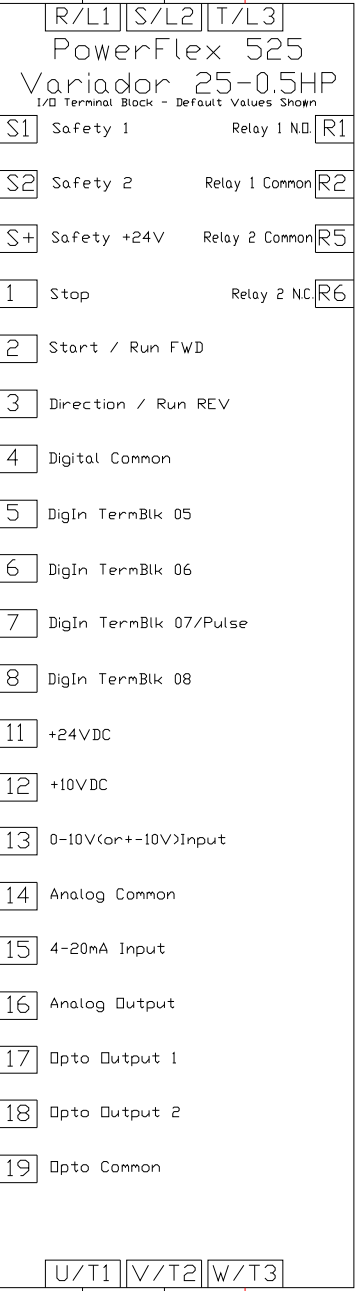


ELEVADOR PUENTE DE SUNCHADORA
ESTACION #5 - SU02
PF525 A020024



LANZA DE SUNCHADORA
ESTACION #5 - SU01
PF525 A020025

072VDS05



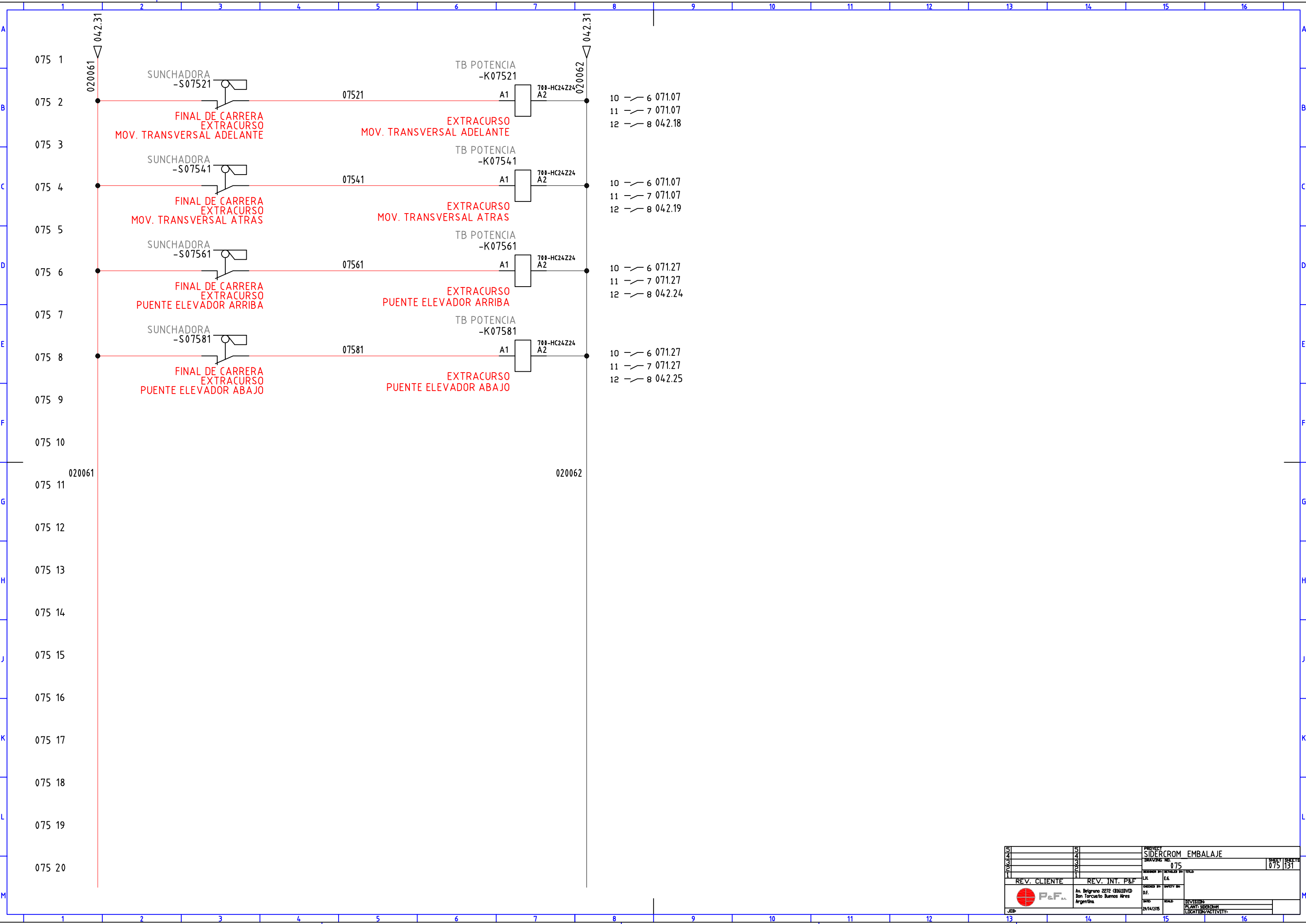
071.28 ▷ 060071 060071
VARIADOR 25
071.28 ▷ 060072 060072
VARIADOR 25

071.31 ▷ 060121 060121

DIRECCION IP: 192.168.1.34/16

Nº Cat.:25B-D1P4N114

5	5	PROJECT	SIDERCROM EMBALAJE
4	4	DRAWING NO.	072
3	3	DESIGNER BY	DESIGNED BY
2	2	1	1
1	1	REV. CLIENTE	REV. INT. P&F
		ORDERED BY	SAFETY NO.
		DATE	20/14/2015
		BY	20/14/2015
		LOCATION	PLANTA SIDERCROM
		ACTIVITY	

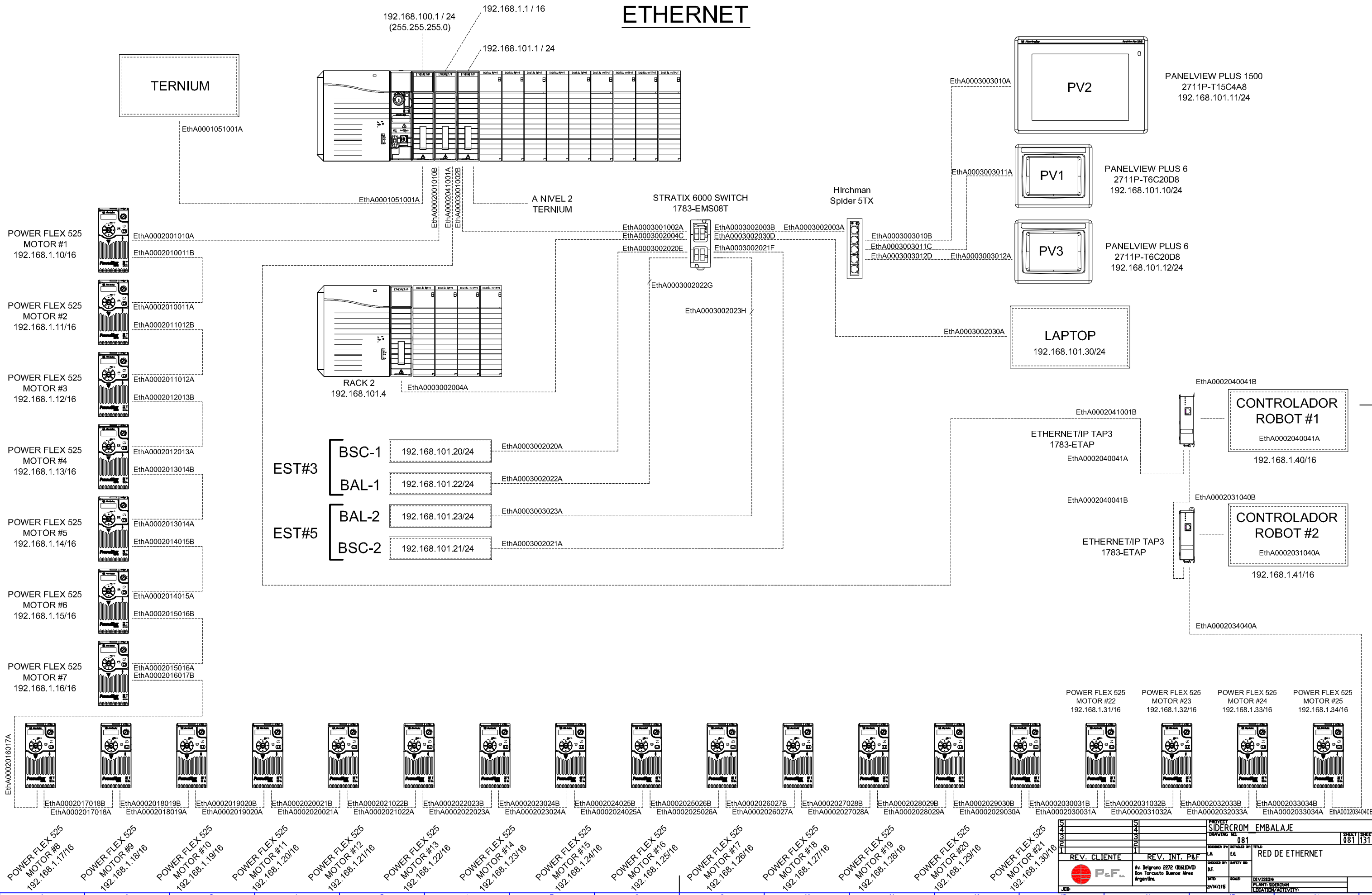


PROJECT: SIDERCROM EMBALAJE		DRAWING NO: 075		SHEET SHEETS: 075 131	
REV. CLIENTE		REV. INT. P&F		DESIGNER BY: DESIGNED BY: TITLO:	
P&F		Av. Belgrano 2272 C1040WVD Buenos Aires Argentina		CHECKED BY: D.F. SAFETY NO: DIVISION: PLANTA SIDERCROM LOCATION/ACTIVITY:	
JOB:		DATE: 29/14/2015		SCALE:	

INTERCONNECT DIAGRAM

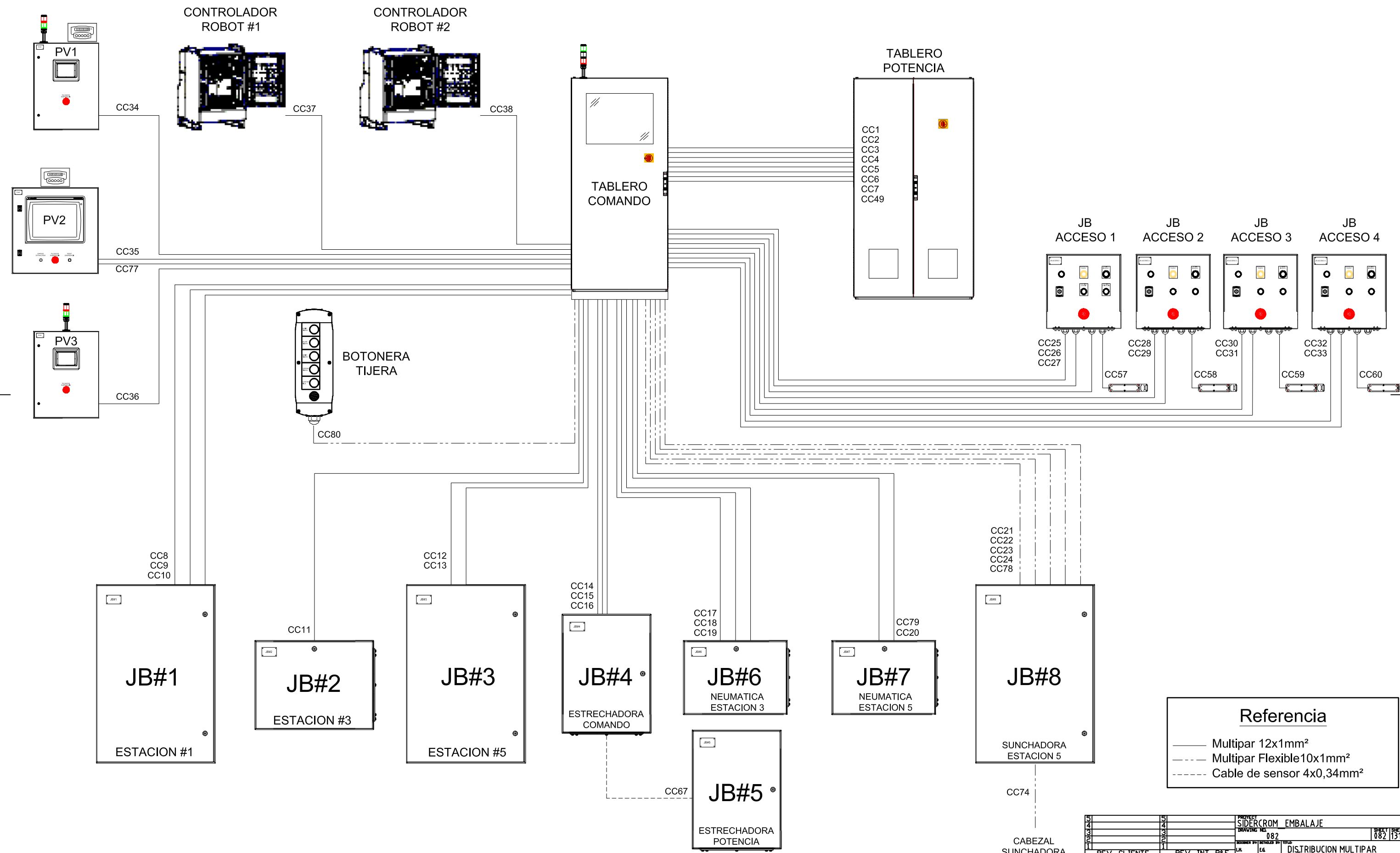
PROJECT SIDERCROM EMBALAJE		SHEET SHEETS 080 131	
DRAWING NO. 080		TITLE INTERCONNECT DIAGRAM	
DESIGNED BY L.A.	DESIGNED BY E.A.	DIVISION PLANTA SIDERCROM	
CHECKED BY D.F.	CHECKED BY D.F.	LOCATION/ACTIVITY	
DATE 29/11/2015	DATE 29/11/2015	DATE 29/11/2015	
JOB P&F		JOB P&F	

ARQUITECTURA RED ETHERNET



PROYECTO: SIDEROCROM EMBALAJE		SHEET 081	
DISEÑADO POR: []		SHEET 081	
REVISADO POR: []		SHEET 081	
APROBADO POR: []		SHEET 081	
TÍTULO: RED DE ETHERNET		SHEET 081	
AUTOR: []		SHEET 081	
FECHA: 21/11/15		SHEET 081	
LUGAR: []		SHEET 081	
DIVISION: PLANTA SIDEROCROM		SHEET 081	
LOCALIDAD: []		SHEET 081	

DISTRIBUCION DE SEÑALES

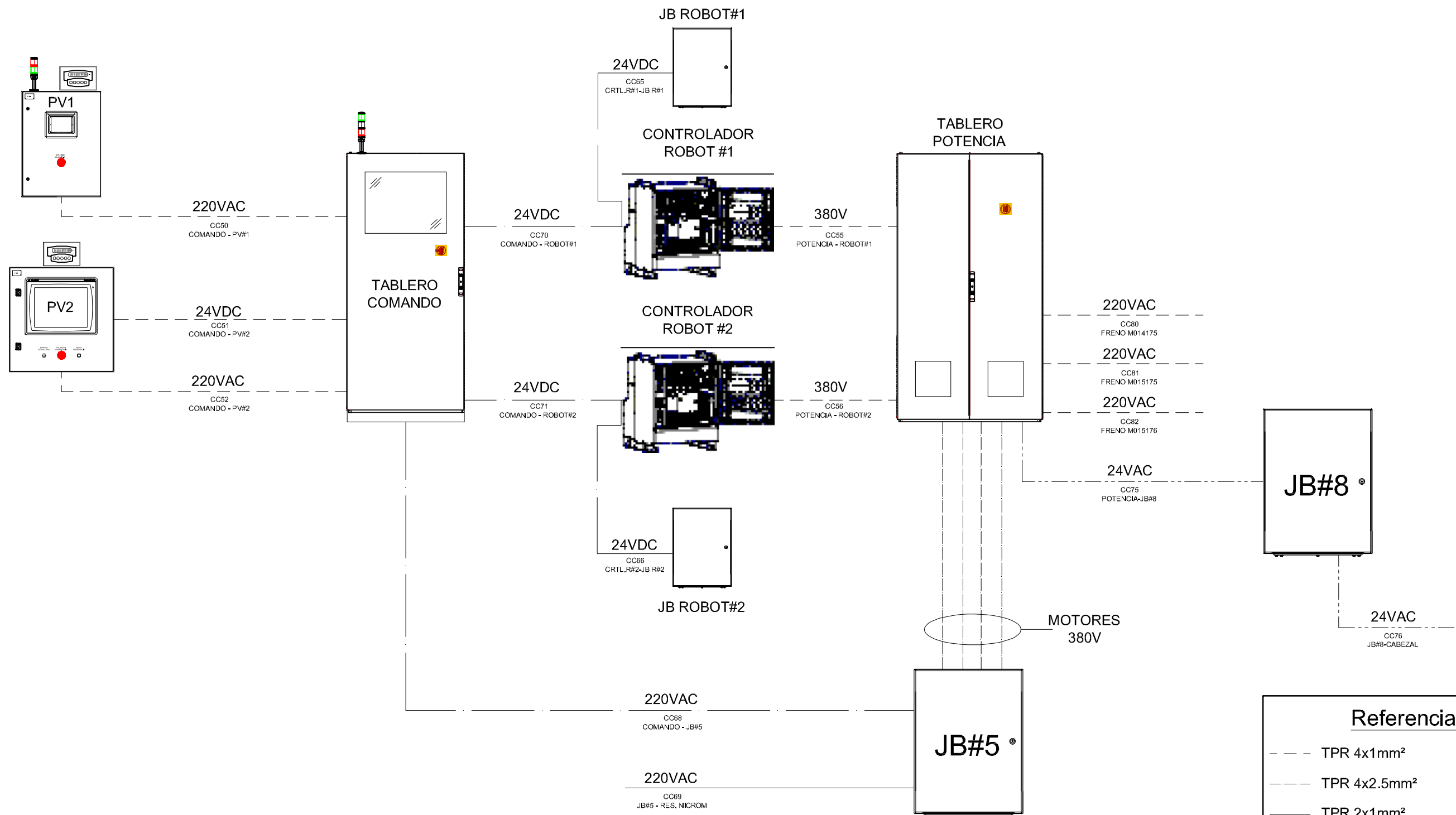


Referencia

- Multipar 12x1mm²
- Multipar Flexible 10x1mm²
- Cable de sensor 4x0,34mm²

PROYECTO	SIDERCROM EMBALAJE	REVISOR	082	SHEET SHEETS	082 131
DRAWING NO.	082	REVISOR	082	REVISOR	082
REV. CLIENTE	REV. INT. P&F	CHK.	EA	SAFETY	SAFETY
Av. Belgrano 2272 (BOLIVAR)	Don Torcuato Buenos Aires	ARGENTINA	ARGENTINA	ARGENTINA	ARGENTINA
DATE	20/10/2015	DATE	20/10/2015	DATE	20/10/2015
PLANTILLA	SIDERCROM	PLANTILLA	SIDERCROM	PLANTILLA	SIDERCROM
LOCATION/ACTIVITY		LOCATION/ACTIVITY		LOCATION/ACTIVITY	

DISTRIBUCION DE POTENCIA



Referencia

- TPR 4x1mm²
- TPR 4x2.5mm²
- TPR 2x1mm²
- TPR 3x1.5mm²

5	5	PROYECTO	SIDERCROM EMBALAJE
4	4	DRAWING NO.	083
3	3	REVISOR	083
2	2	REVISOR	083
1	1	REVISOR	083
REV. CLIENTE	REV. INT. P&F	LA	EA
Av. Belgrano 2272 (BOLIVAR)	Don Torcuato Buenos Aires	Argentina	Argentina
DATE	20/14/2015	SOLD	DIVISION
PLANT	SIDERCROM	LOCATION/ACTIVITY	

CONEXION
CONTROLADOR Y ROBOT

Robot #1 - 192.168.1.40/16 - EthA0002034040A
Robot #2 - 192.168.1.41/16 - EthA0002040041B

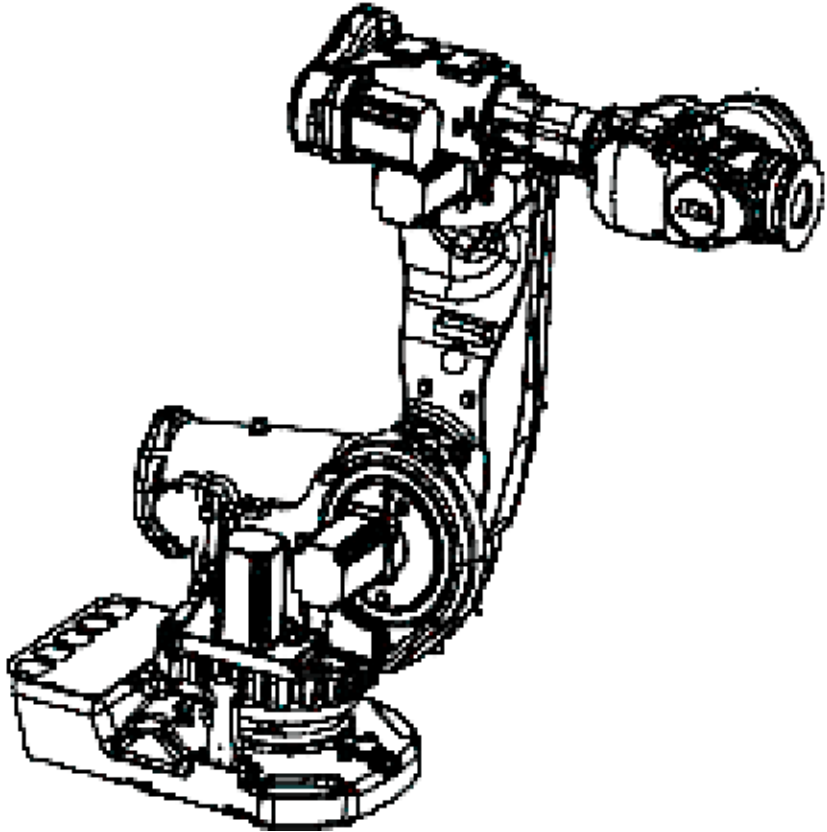
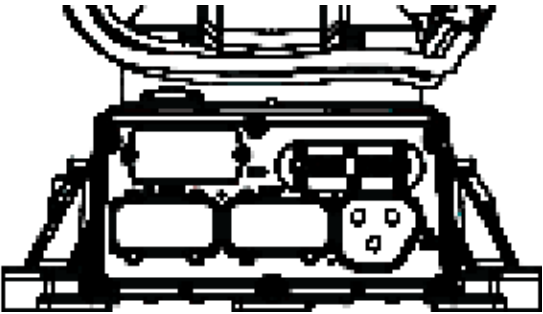
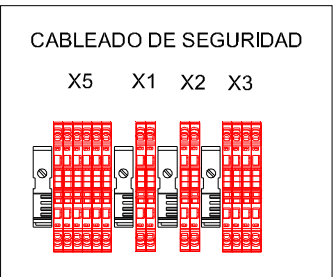
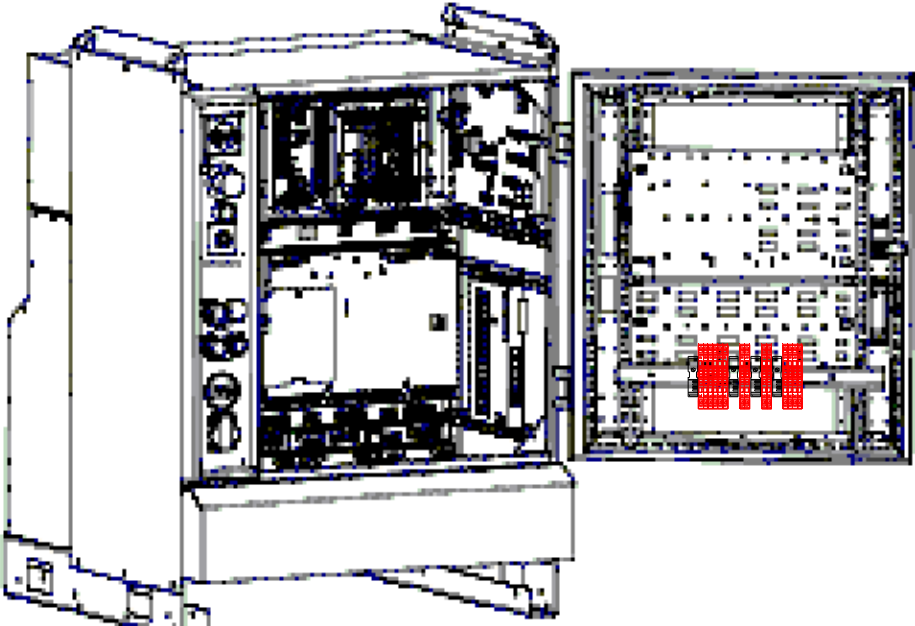
24VDC
ROBOT#1 - CC70
ROBOT#2 - CC71

CABLE TPR 4X1MM - 24VDC

CABLE ETHERNET FTP CAT. 6

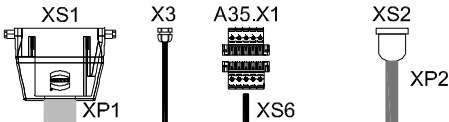
X3

X4
LAN 2 - 10/100



Robot #1 - 192.168.1.40/16 - DnetA0002031040A
Robot #2 - 192.168.1.41/16 - DnetA0002040041A

24VDC
ROBOT#1 - CC65
ROBOT#2 - CC66



SIGNAL CABLE

CABLE DEVICENET

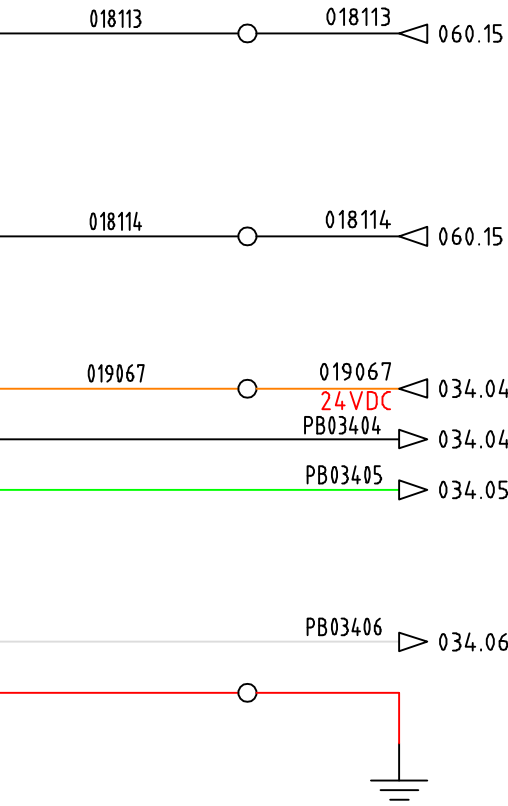
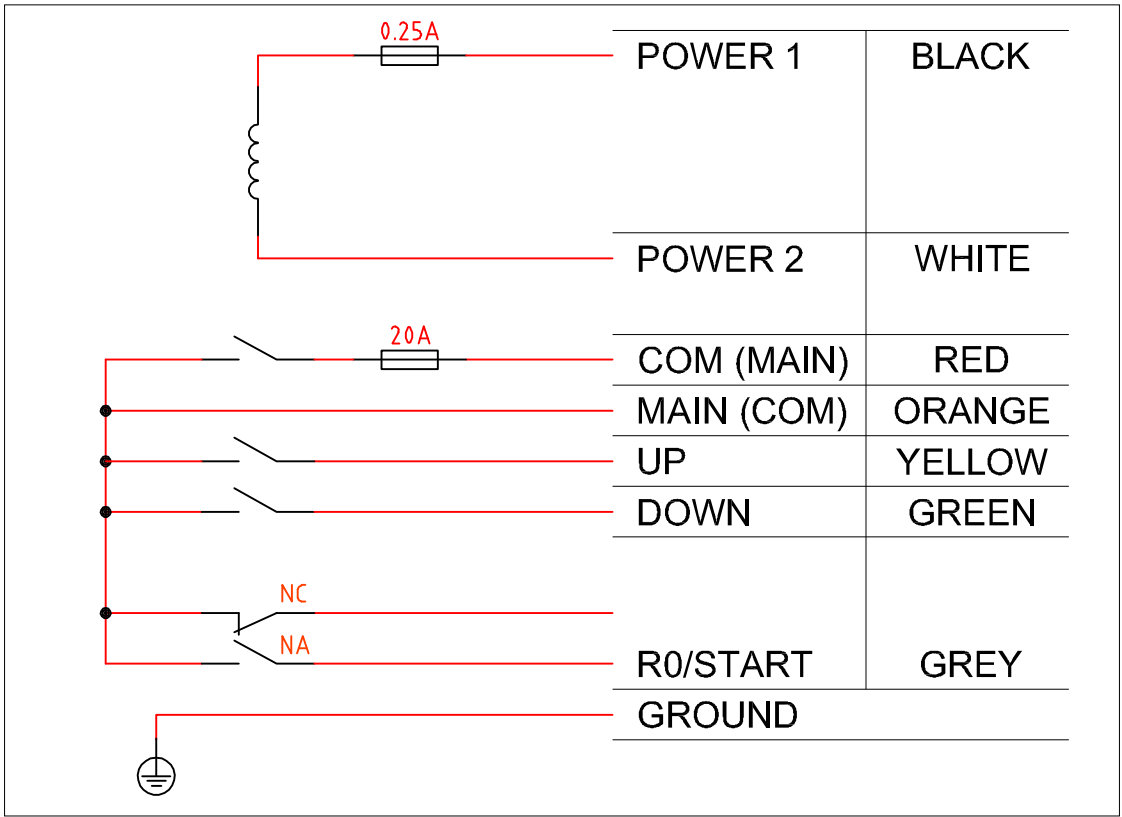
CABLE TPR 4X1MM - 24VDC

POWER CABLE

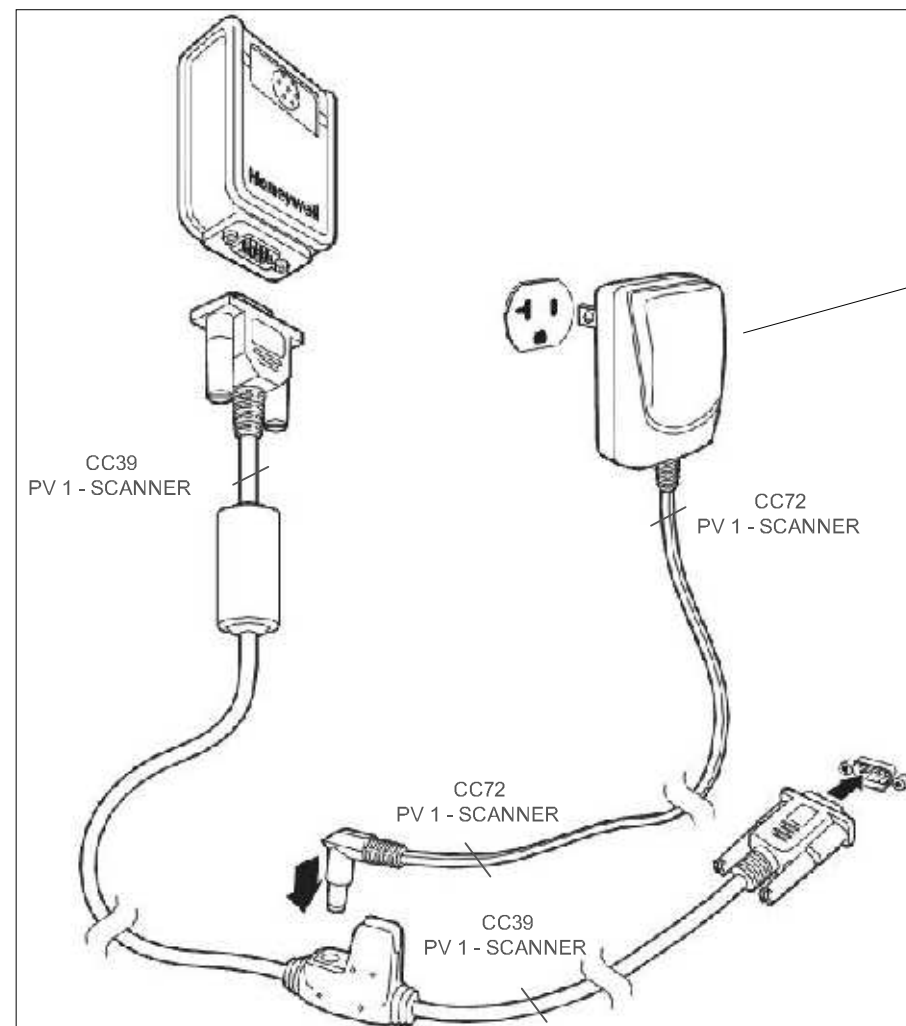
5	5	PROYECTO	SIDERCROM EMBALAJE
4	4	DRAWING NO.	084
3	3	DESIGN BY	084
2	2	DETAIL BY	084
1	1	REV. CLIENTE	REV. INT. P&F
		ORDER BY	SAFETY NO.
		DATE	29/14/2015
		AV. Belgrano 2272 CHESLUPVD	DIVISION
		Bon Torcuato Buenos Aires	PLANTA SIDERCROM
		Argentina	LOCATION/ACTIVITY

MANDO A DISTANCIA

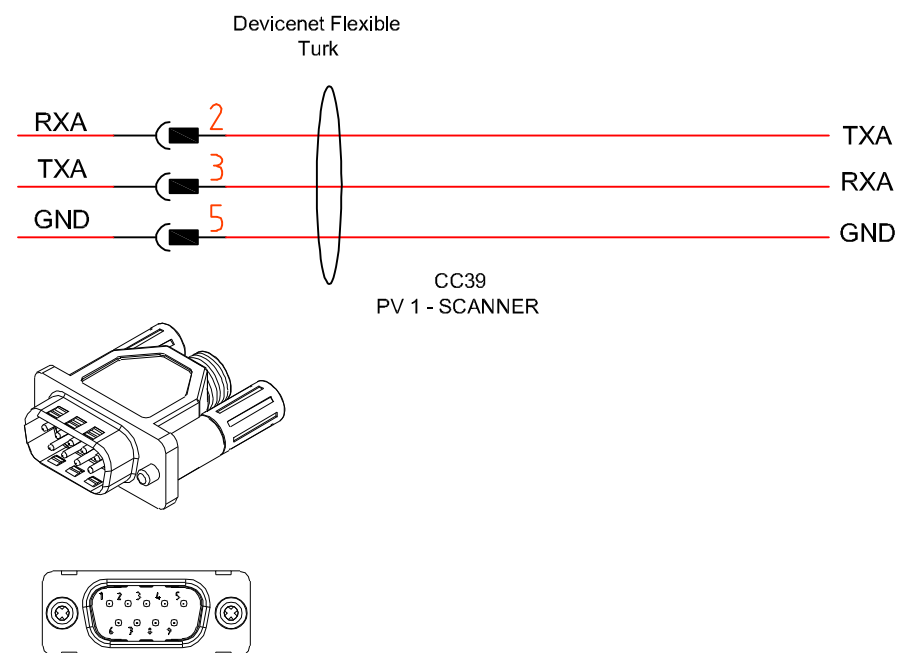
MODEL NO.: SAGA1-L4



LECTOR CODIGO DE BARRAS
ESTACION #3



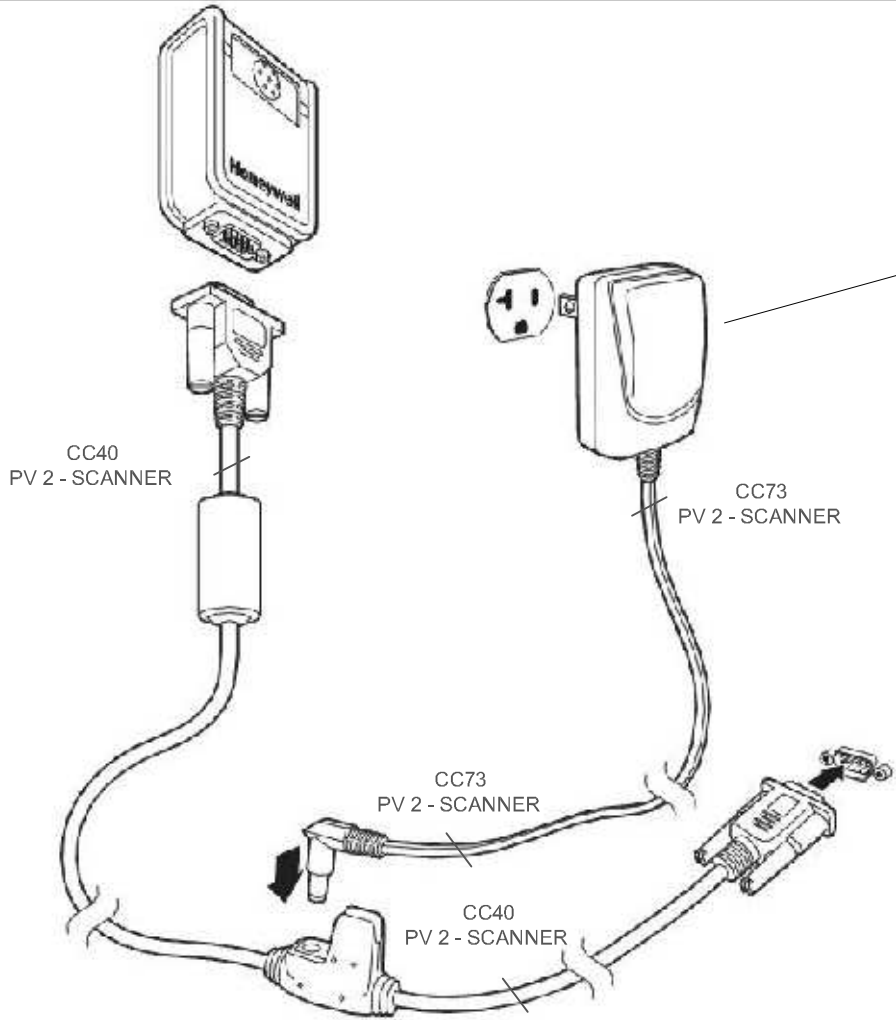
Alimentacion desde PanelView #1

[illegible]

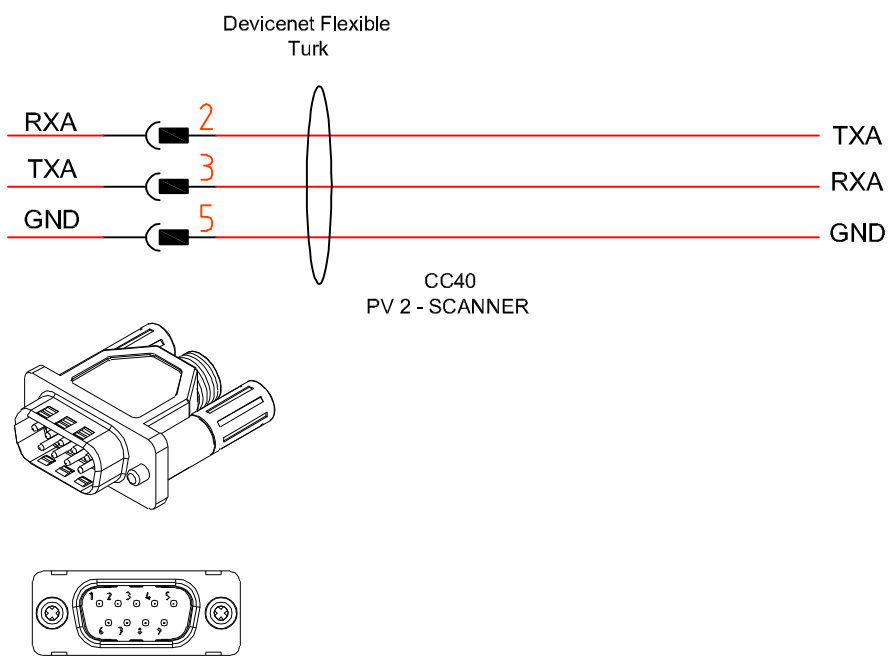
V _{in}	V _{in}	–	GND
R _x A	T _x A	–	–
EXEMYS			
TRA–	TRA+	RA+	RA–
I/O 0	I/O 1	I/O 2	I/O 3

5	5	PROJECT		SIDERCROM EMBALAJE			
4	4	DRAWING NO.		086		SHEET 131	
3	3	REVISION BY		DETAILS BY		TITLO	
2	2	L.H.		E.G.		LECTOR CODIGO DE BARRAS	
1	1	CHECKED BY		SAFETY NO.		EST #3	
		D.F.		BATCH		DIVISION	
		DATE		SOLD		PLANTS SIDERCROM	
		19/12/15				PLANTAS SIDERCROM S.A. S.R.L.	
		JTB					

LECTOR CODIGO DE BARRAS
ESTACION #4



Alimentacion desde PanelView #2

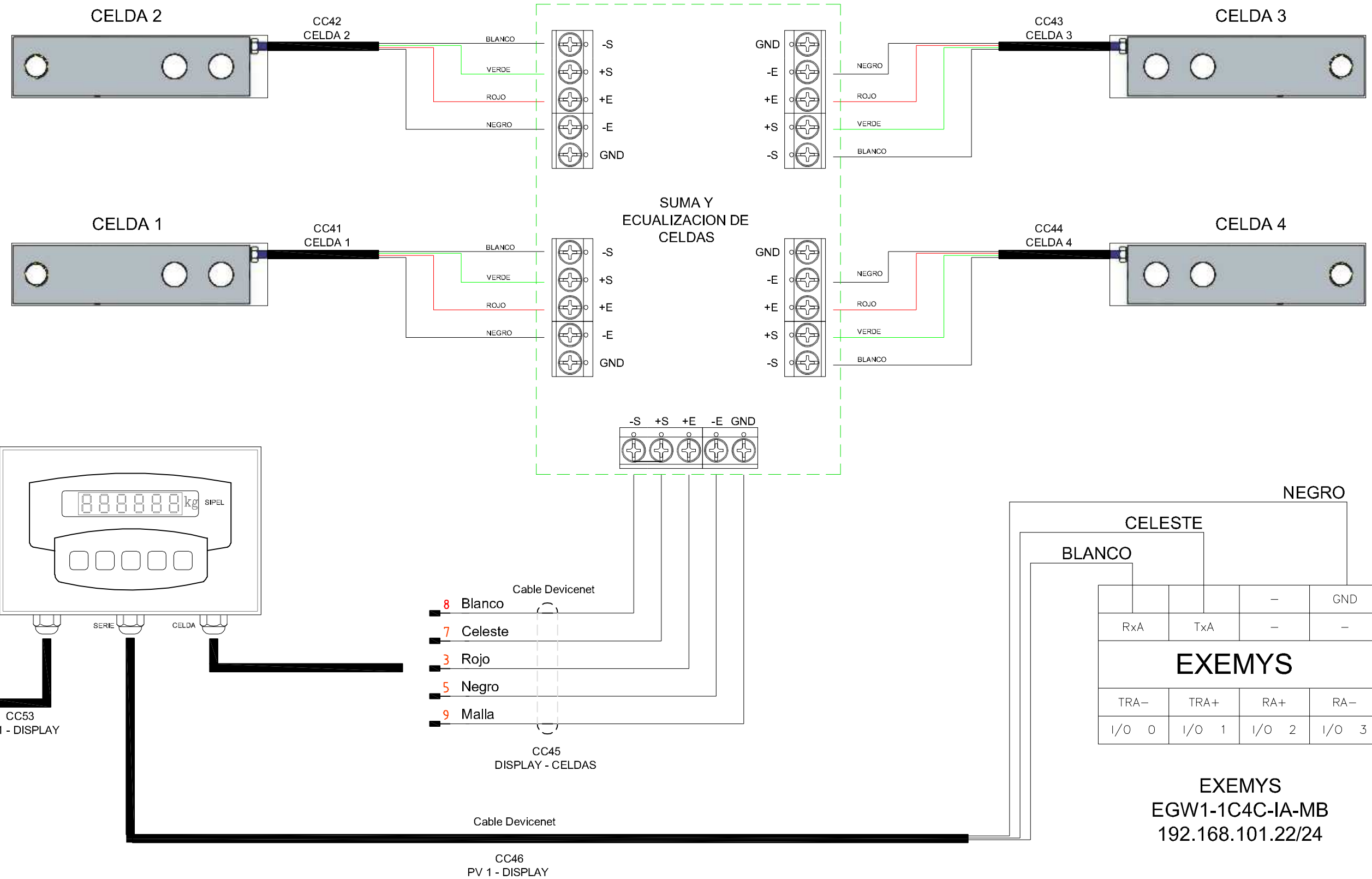


○ ○ ○ ○
○ ○ ○ ○
SSE232
○ ○ ○ ○
○ ○ ○ ○

Vin	Vin	—	GND
RxA	TxA	—	—
EXEMYS			
TRA—	TRA+	RA+	RA—
I/O 0	I/O 1	I/O 2	I/O 3

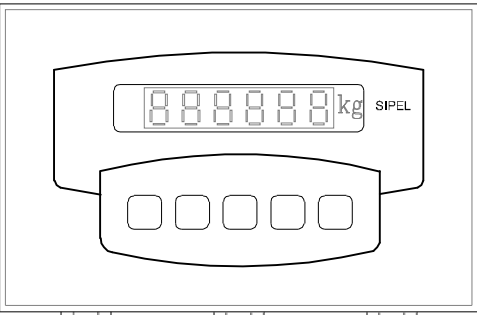
BALANZA Y CELDAS DE CARGA

ESTACION #2



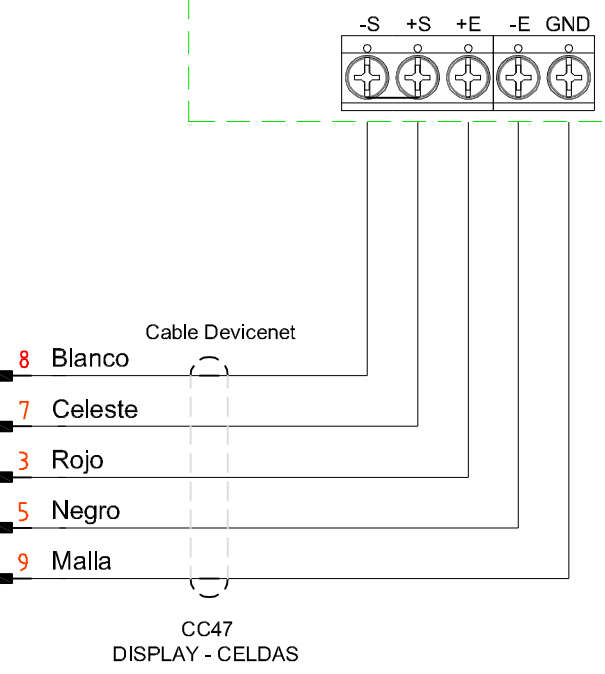
BALANZA Y CELDAS DE CARGA
ESTACION #5

MESA CON CAJA DE
CONEXION DE CELDAS
DE CARGA

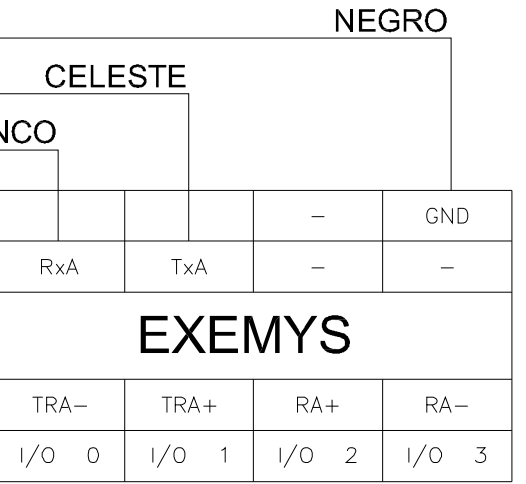


CC54
PV 2 - DISPLAY

ALIMENTACION 220VAC
DESDE PV#1



Cable Devicenet
CC48
PV 2 - DISPLAY

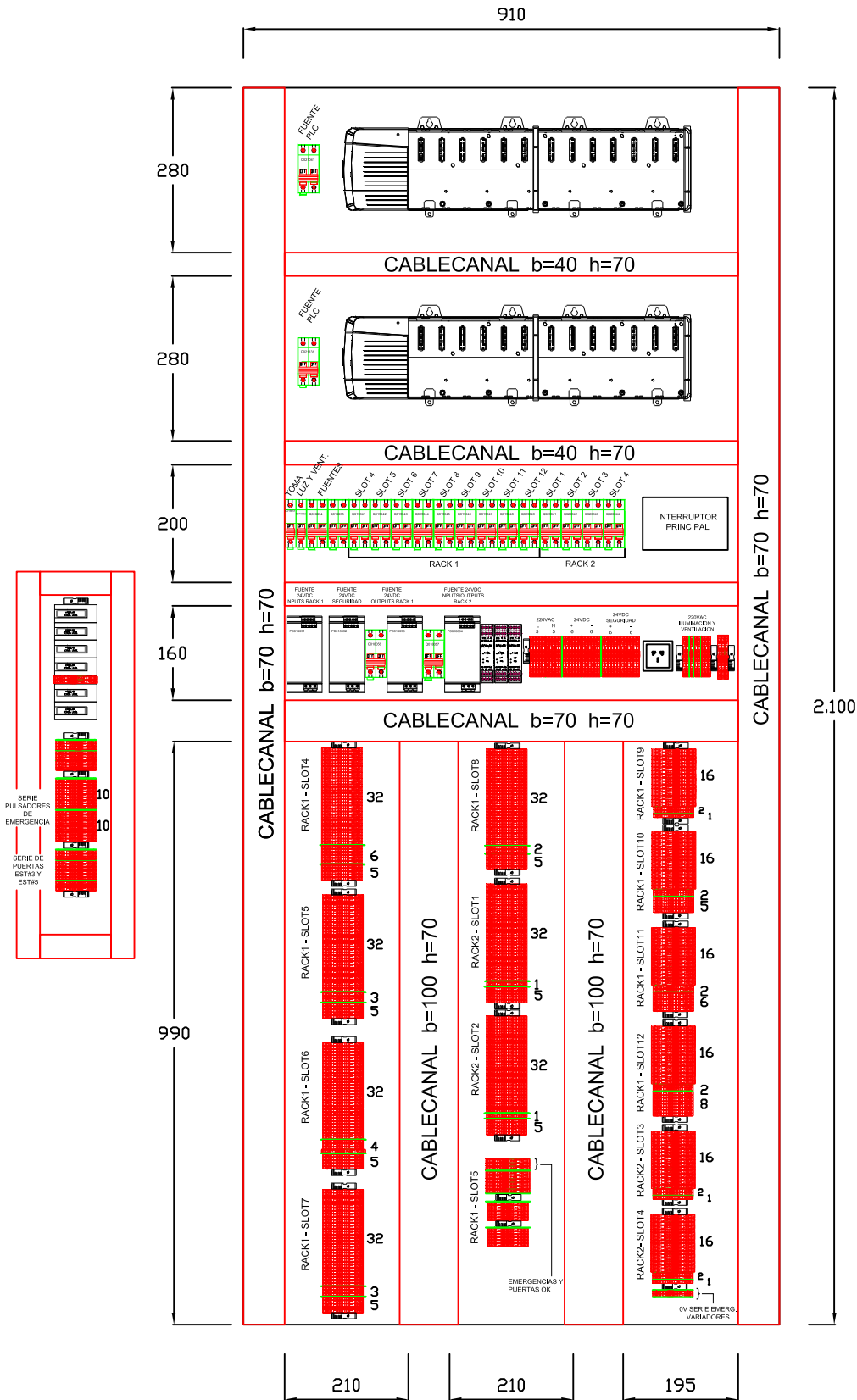
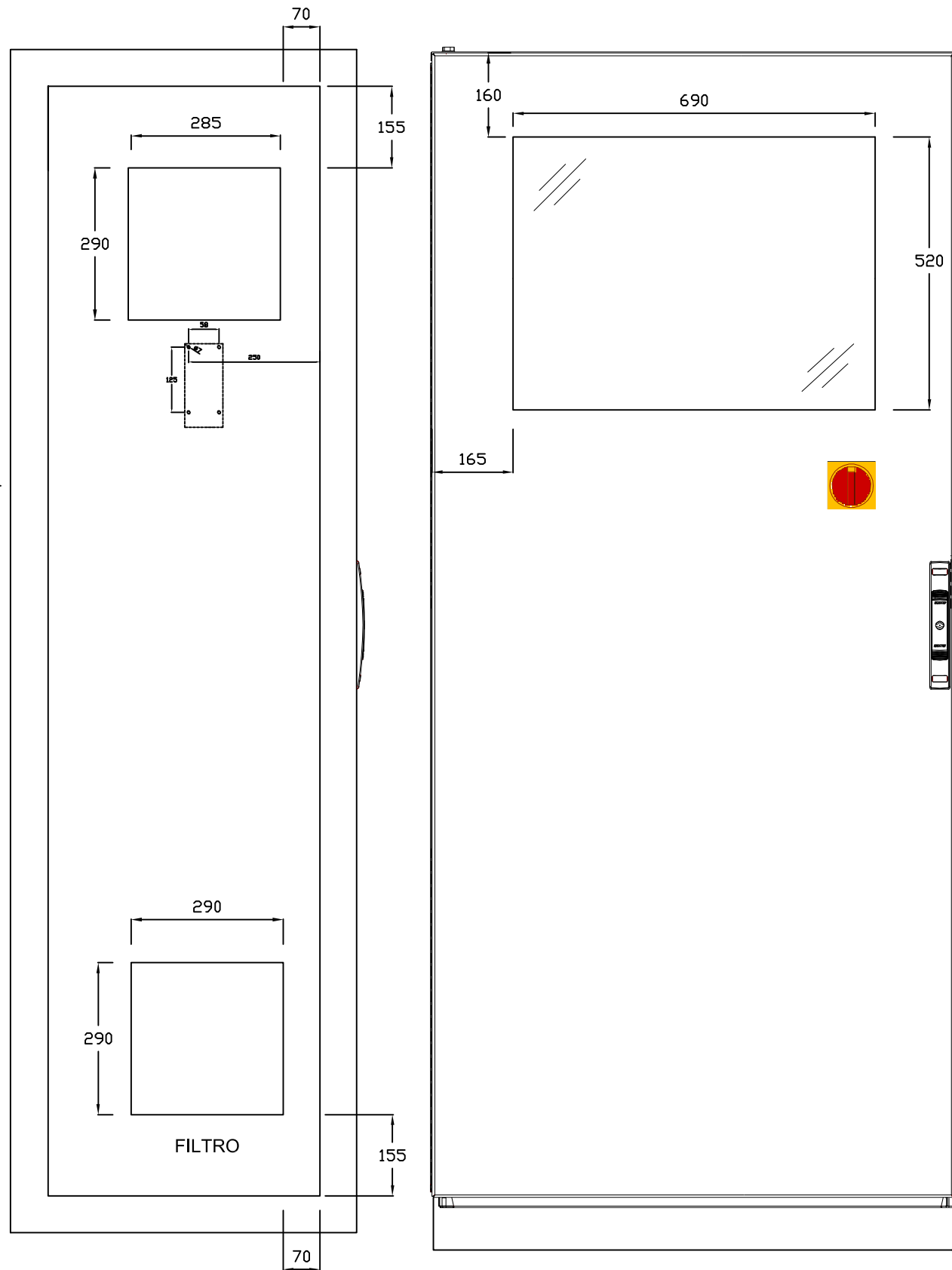


5	5	PROJECT	SIDERCROM EMBALAJE	
4	4	DRAWING NO.	089	SHEET SHEETS
3	3			089 131
2	2	DESIGN BY: DESIGNED BY: FIELD:		
1	1	REV. CLIENTE	REV. INT. P&F	
		Av. Belgrano 2272 CNEU/DVD	Don Torcuato Buenos Aires	
		Argentina		
		DATE	29/14/2015	
		SCALE		
		DIVISION	PLANTA SIDERCROM	
		LOCATION/ACTIVITY		

CONTROL PANEL AND
BACKPLATE LAYOUTS

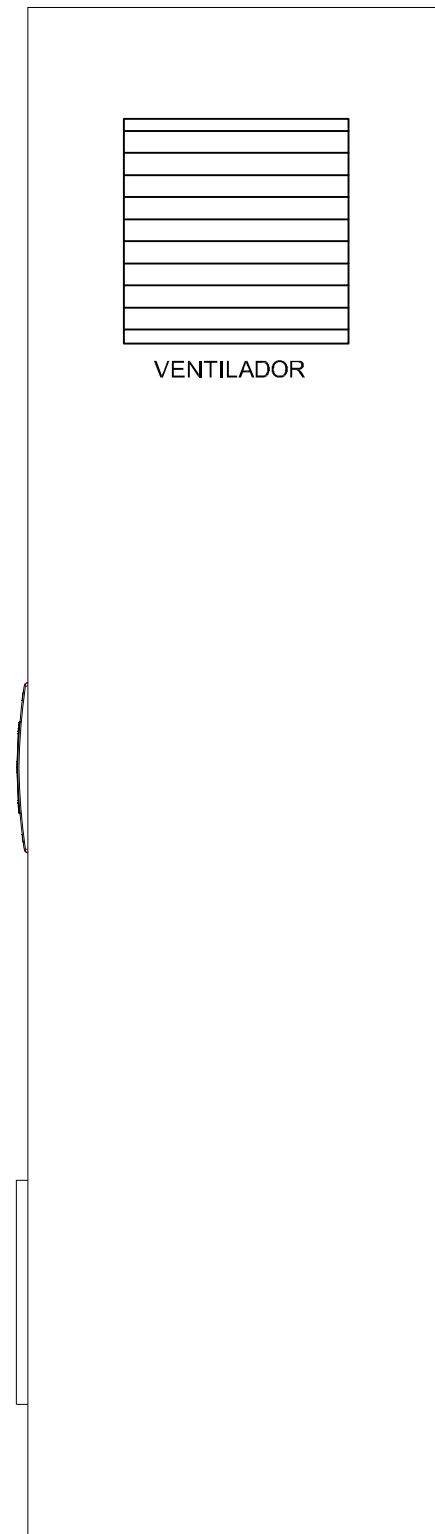
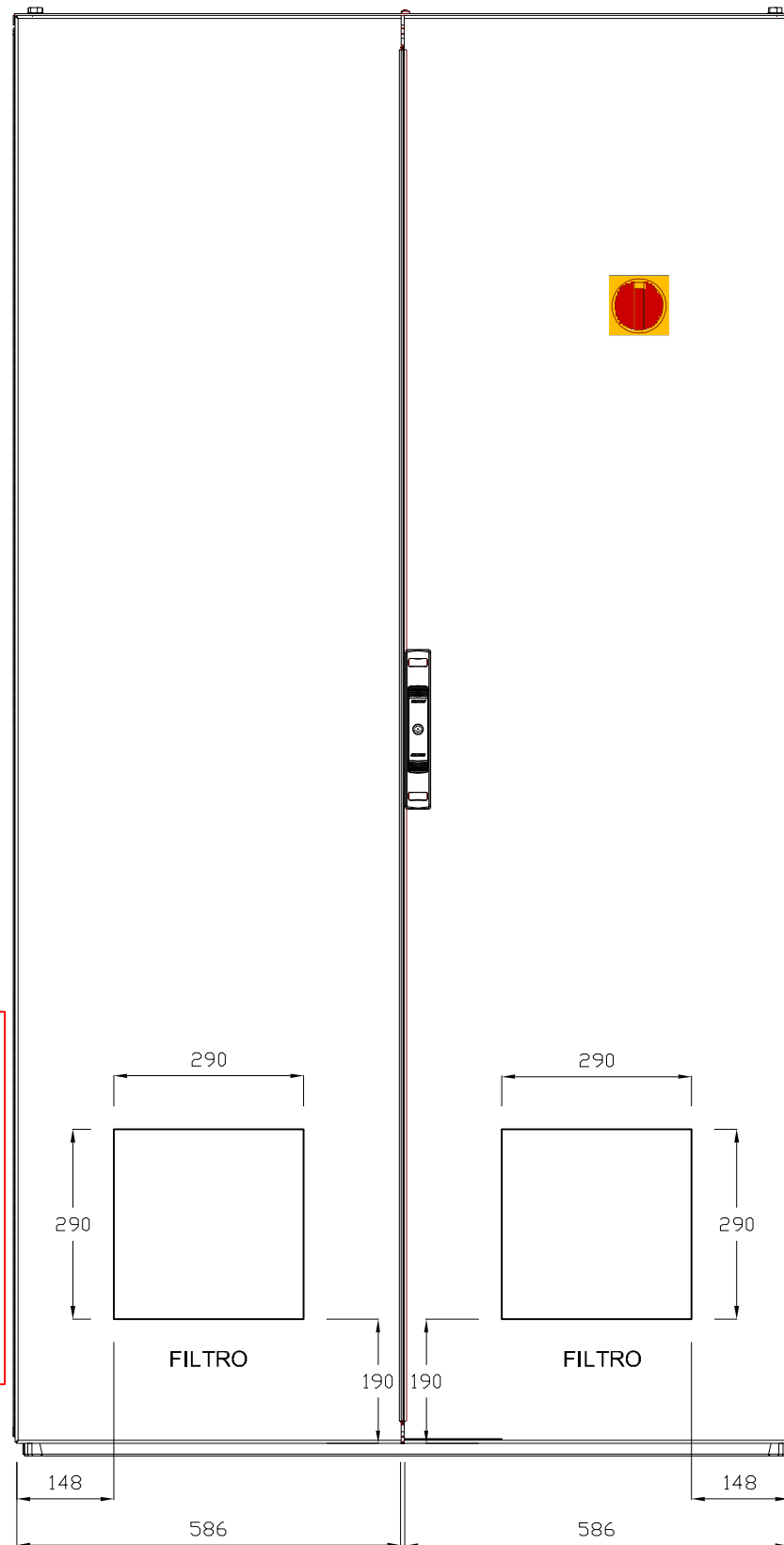
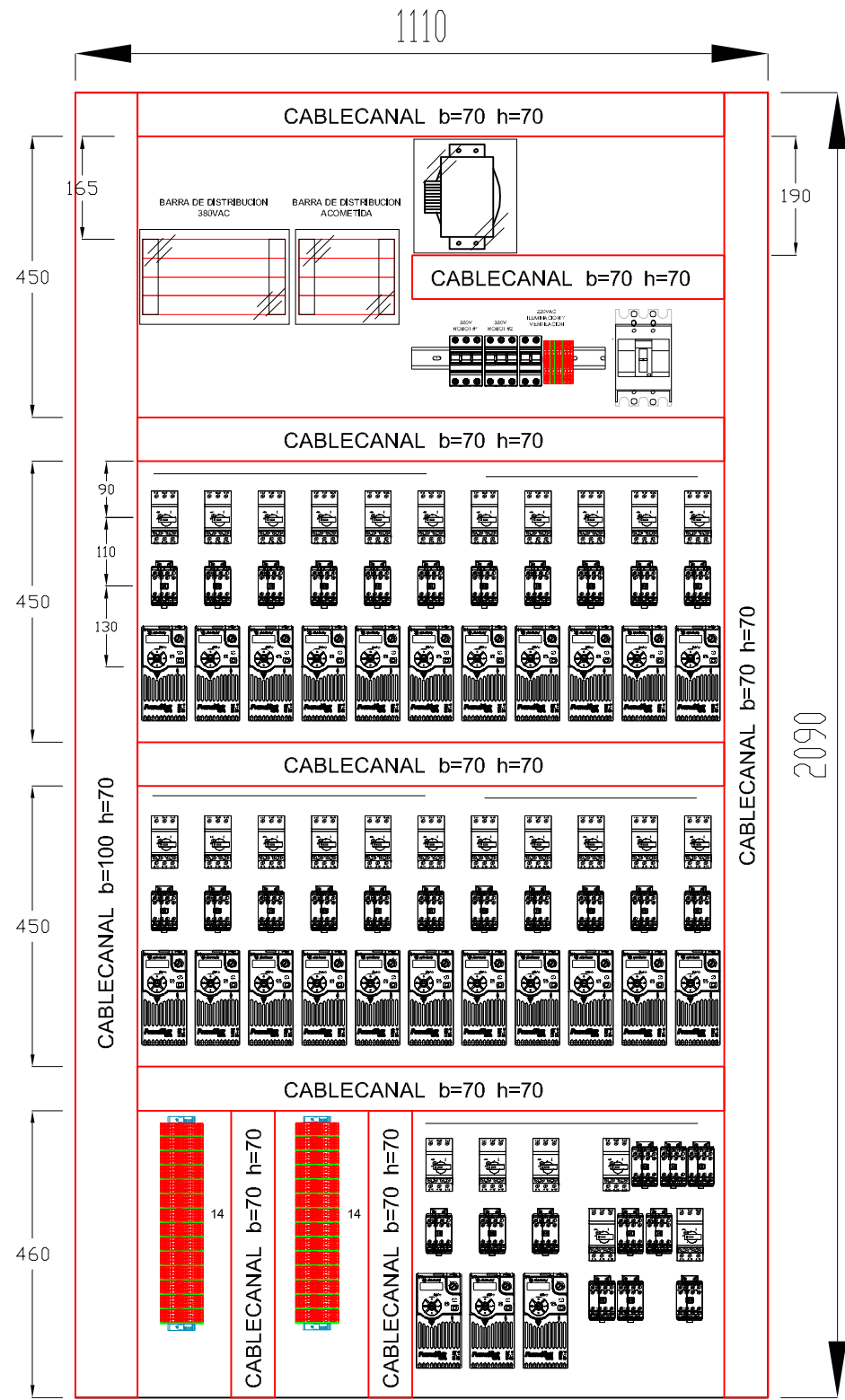
PROJECT SIDERCROM EMBALAJE		DRAWING NO. 090		SHEET / SHEETS 090 / 131	
DESIGNED BY L.H.		DETAILS BY E.G.		TITLE CARATULA CONTROL PANEL AND BACKPLATE LAYOUTS	
REV. CLIENTE		REV. INT. P&F		CHECKED BY D.F.	
P&F		Av. Belgrano 2272 (BOLIVAR) Don Torcuato Buenos Aires Argentina		DIVISION PLANTA SIDERCROM LOCATION/ACTIVITY	
JOB		DATE 29/14/2015		SOLD	

Tablero 2200X1000X660
Placa de Montaje 2100X910



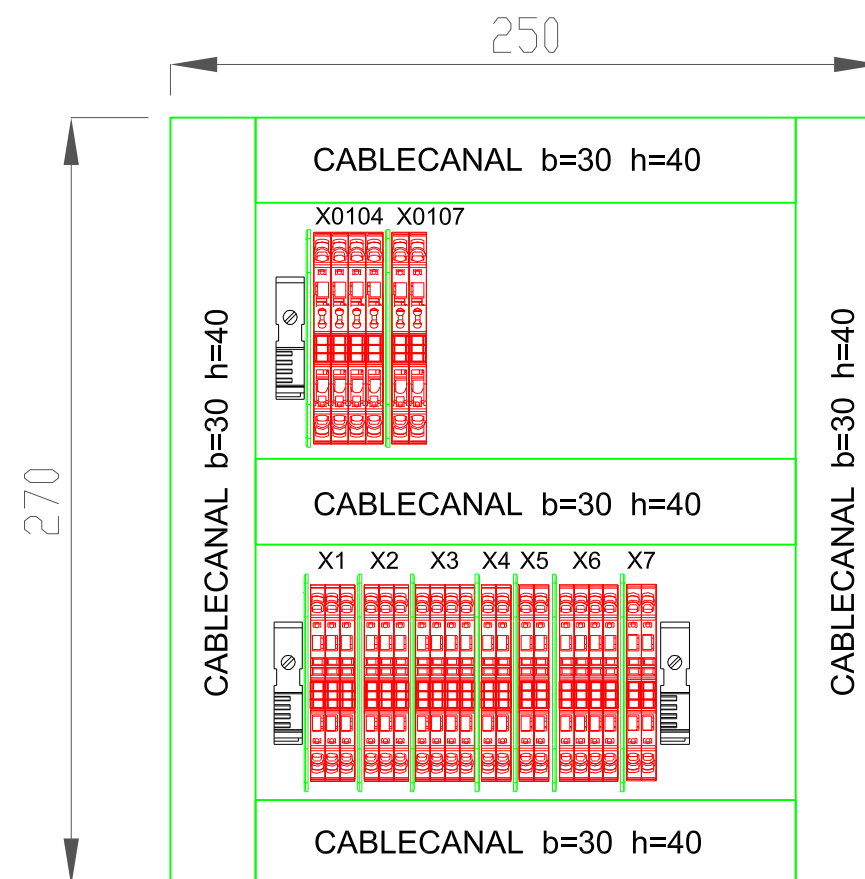
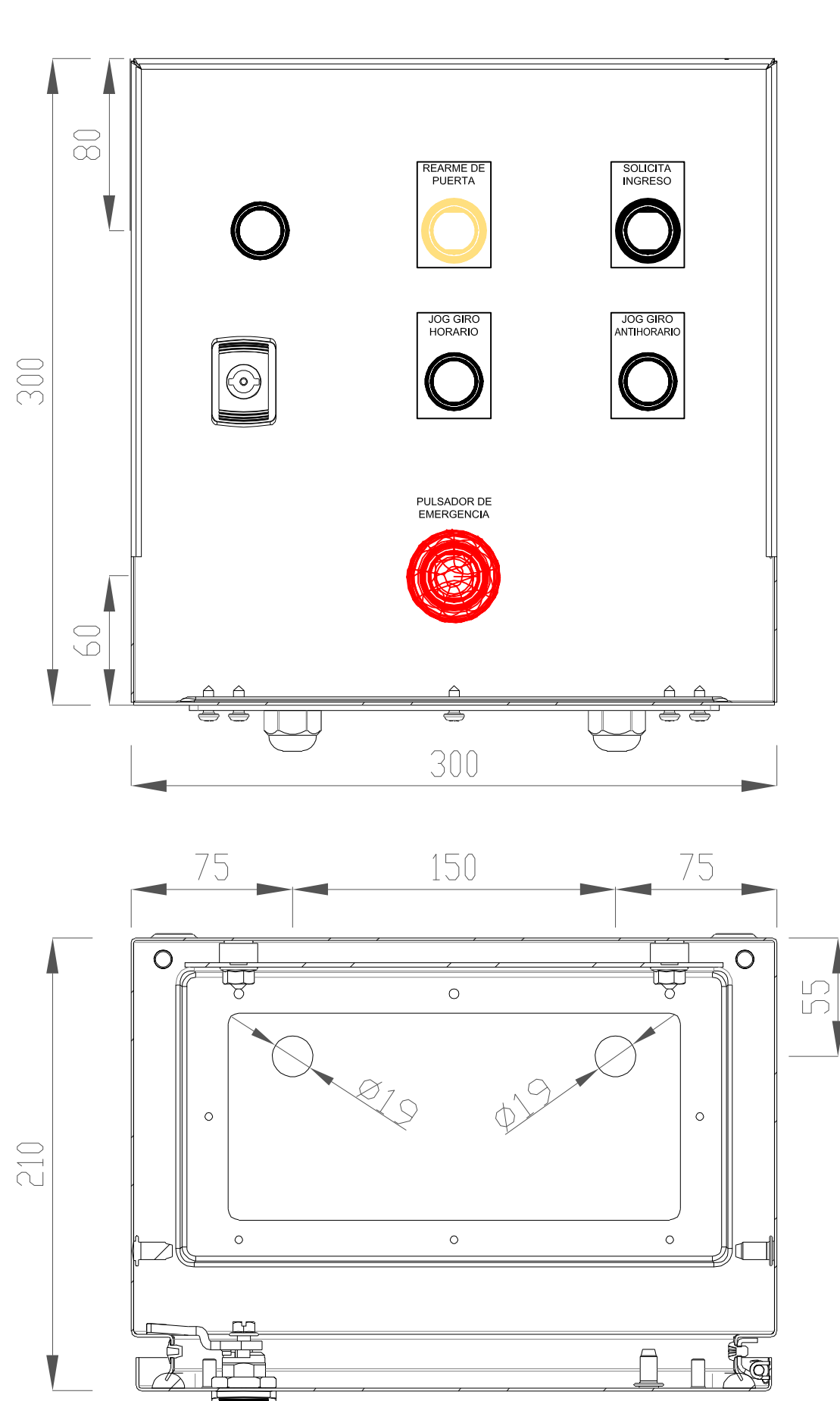
Cable 12x1mm ²	ID	Multipar	N° Hilo	N° Cable	Componente
1	CC1	TB Comando – TB Potencia	2	060071	Rele Seg. Reserva 24VDC
			3	060072	
			4	Reserva	
			5	019069	
			6	GM012051	
			7	GM012052	
			8	GM012053	
			9	GM012054	
			10	GM012055	
			11	GM012056	
			12	GM013051	
			2	CC2	
3	GM013053				
4	GM013054				
5	GM013055				
6	GM013056				
7	GM014051				
8	GM014052				
9	GM014053				
10	GM014054				
11	GM014055				
12	GM014056				
3	CC3	TB Comando – TB Potencia			2
			3	GM015052	
			4	GM015053	
			5	GM015054	
			6	GM015055	
			7	GM015056	
			8	GM016051	
			9	GM016052	
			10	GM016053	
			11	GM016054	
			12	0190611	
			4	CC4	TB Comando – TB Potencia
3	K037011				
4	K037021				
5	K037031				
6	K037041				
7	K037051				
8	K037061				
9	K037071				
10	K037081				
11	K037091				
12	K037101				
5	CC5	TB Comando – TB Potencia			
			3	K037121	
			4	K037131	
			5	K037141	
			6	K037151	
			7	190614	
			8	K038001	
			9	K038011	
			10	K038021	
			11	K038031	
			12	K038041	
			6	CC6	TB Comando – TB Potencia
3	K038061				
4	K038071				
5	K038081				
6	020066				
7	RA046001				
8	RA046011				
9	RA046021				
10	RA046031				
11	RA046041				
12	RA046051				
7	CC7	TB Comando – TB Potencia			
			3	RA046071	
			4	Reserva	
			5	060101	
			6	060121	
			7	Reserva	
			8	Reserva	
			9	Reserva	
			10	Reserva	
			11	Reserva	
			12	Reserva	
			8	CC9	TB Comando – TB Potencia
3	020062				
4	07521				
5	07541				
6	07561				
7	07581				
8	K07521				
9	K07541				
10	K07561				
11	K07581				
12	-				

Tablero 2200X1200X600
Placa de Montaje 2100X1110



5	5	PROJECT: SIDEROCROM EMBALAJE			
4	4	DRAWING: ND 092		SHEET 092	SHEETS 131
3	3	DESIGNED BY: 092			
2	2	TITLE: TOPOGRAFICO TABLERO POTENCIA			
1	1	L.H. E.G.			
		CHECKED BY: SAFETY: 092			
		D.F.			
REV. CLIENTE		REV. INT. P&F			
		Av. Belgrano 2272 (B1610)VD Don Torcuato Buenos Aires Argentina			
JOB:		DATE: 20/11/15		DIVISION: PLANT: SIDEROCROM LOCATION/ACTIVITY:	

TABLERO - JB ACCESO 1

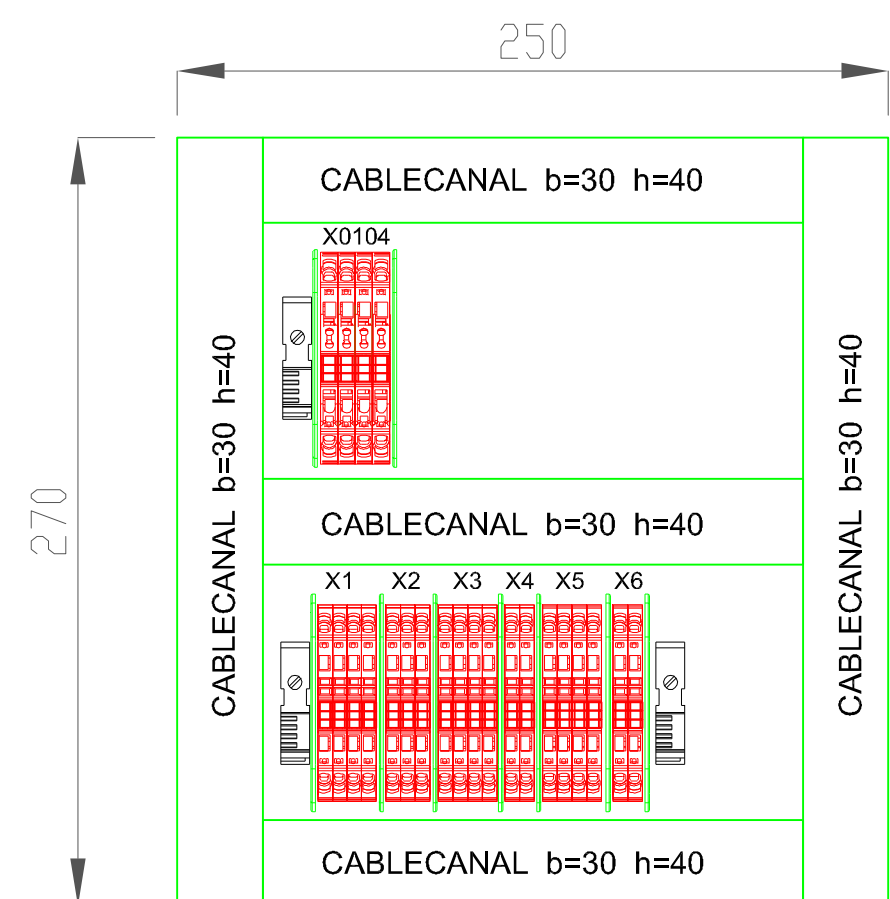
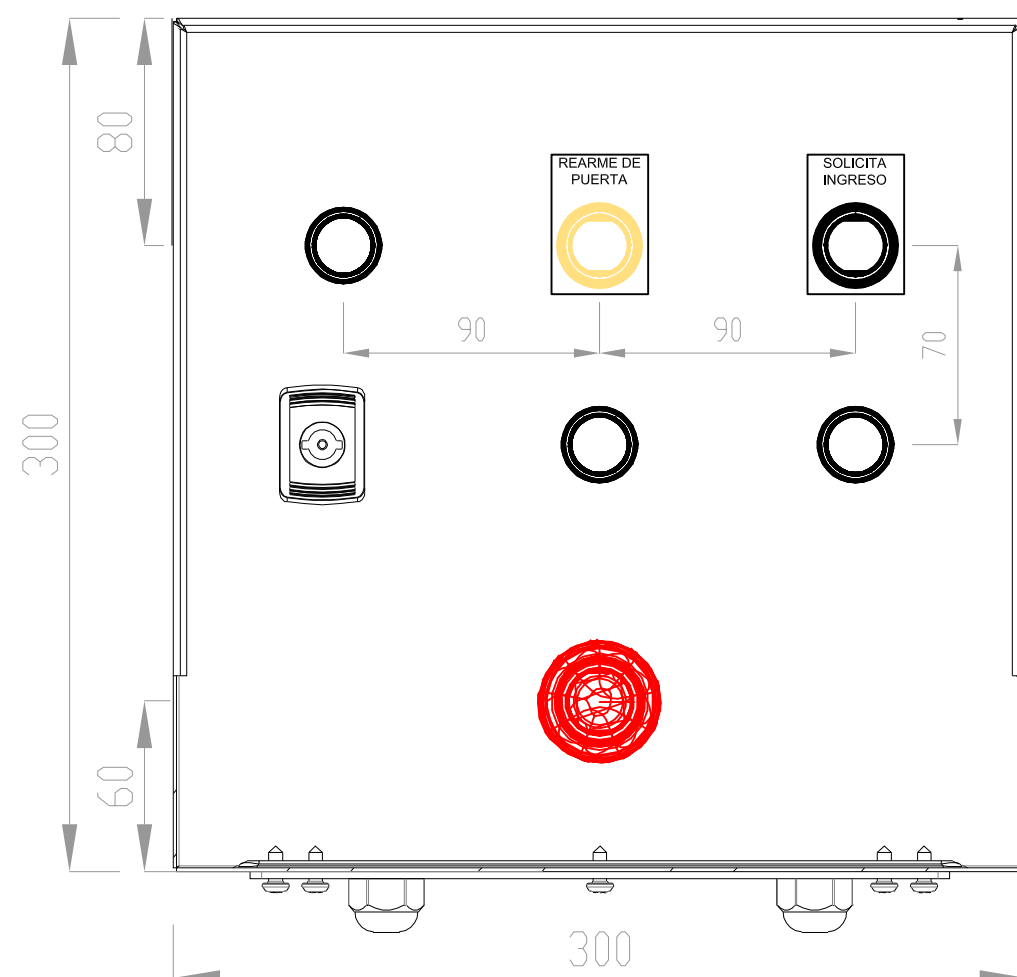


Tablero 300x300x210
Eldon - MAS0303021R5

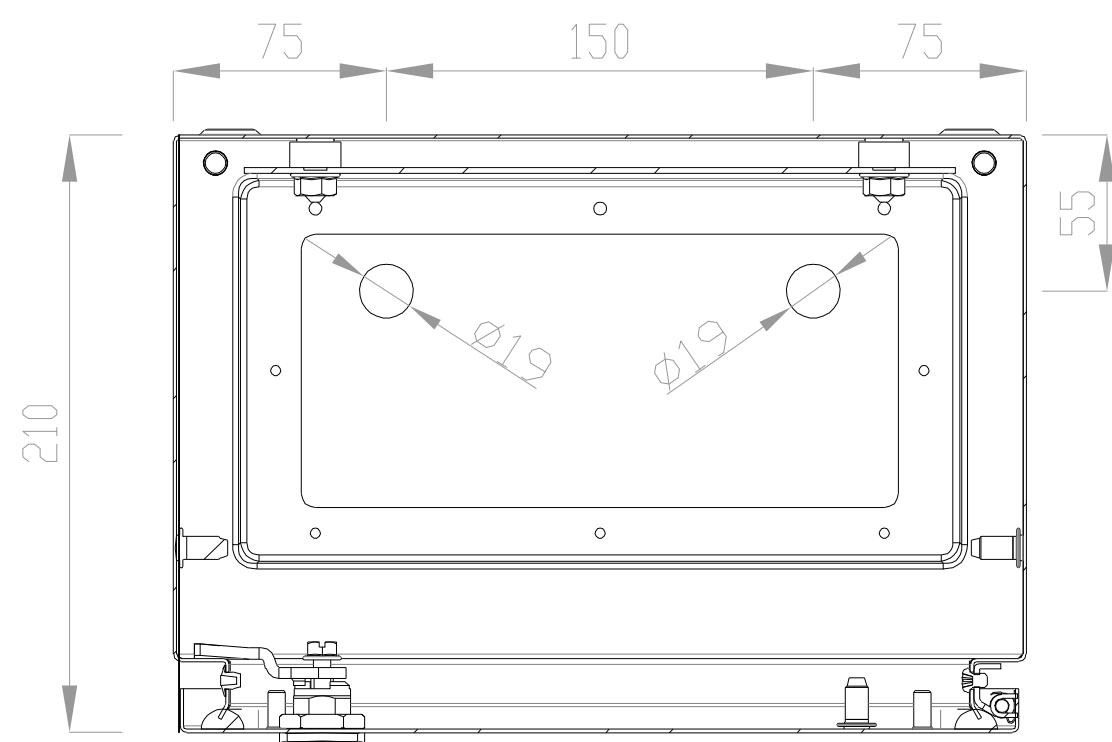
Tablero	Multipor	N° Hilo	N° Cable	Bornero	Componente	Descripción	
JB Acceso 1	CC25	2	019061	X0104	PE03125	24VDC – Rack 1 / Slot 4	
		3	026076	X3		IN – Pulsador de Emerg. Acceso 1	
		4	026116	X3		IN – Pulsador de Emerg. Acceso 1	
		5	026075	X3		OUT – Pulsador de Emerg. Acceso 1	
		6	026115	X3		OUT – Pulsador de Emerg. Acceso 1	
		7	PE03125	X1		Pulsador de Emergencia Acceso 1	
		8	PX03107	X1	PX03107	Pedido de Acceso – Puerta 1	
		9	PX03113	X1	PX03113	Rearme de Puerta – Puerta 1	
		10	027071	X6	YS03812	IN Canal Seg.1 Puerta 1	
		11	027081	X6		IN Canal Seg.1 Puerta 1	
		12	027121	X6		OUT Canal Seg.2 Puerta 1	
		2	027131	X6		OUT Canal Seg.2 Puerta 1	
	CC26	3	YS03812	X1		Puerta 1 Cerrada	
		4	–		RESERVA		
		5	019067	X0107		24V – Rack 1 / Slot 7	
		6	PB03400	X5	PB03400	Pulsador Jog giro antihorario	
		7	PB03401	X5	PB03401	Pulsador Jog giro horario	
		8	–		RESERVA		
		9	–		RESERVA		
		10	0190614	X7		0V – Rack 1 / Slot 10	
		11	YS03812	X7	YS03812	Cerradura de Puerta 1	
		12	–		RESERVA		
		CC27	2	0190616	X2		0V – Rack 1 / Slot 11
			3	L03908	X2	L03908	Led Verde TB Acceso 1
	4		L03909	X2	L03909	Led Rojo TB Acceso 1	
	5		–		RESERVA		
	6		0190618	X4		0V – Rack 1 / Slot 12	
	7		L04007	X4	L04007	Baliza roja Acceso 1	
	8		–		RESERVA		
	9		–		RESERVA		
	10		–		RESERVA		
	11		–		RESERVA		
	12		–		RESERVA		

5 4 3 2 1			PROJECT SIDEROM EMBALAJE		
			DRAWING NO 094		SHEET SHEETS 094 131
			DESIGNER BY LH	DETAILS BY EG	TITLE TOPOGRAFICO TABLERO JB ACCESO 1
			CHECKED BY D.F.	SAFETY BY	
	REV. CLIENTE	REV. INT. P&F	 Av. Belgrano 2272 (B1610VD) Don Torcuato Buenos Aires Argentina		
JOB	DATE: 3/11/12 SCALE: DIVISION PLANT: SIDEROM LOCATION: ACTIVITI				

TABLERO - JB ACCESO 2



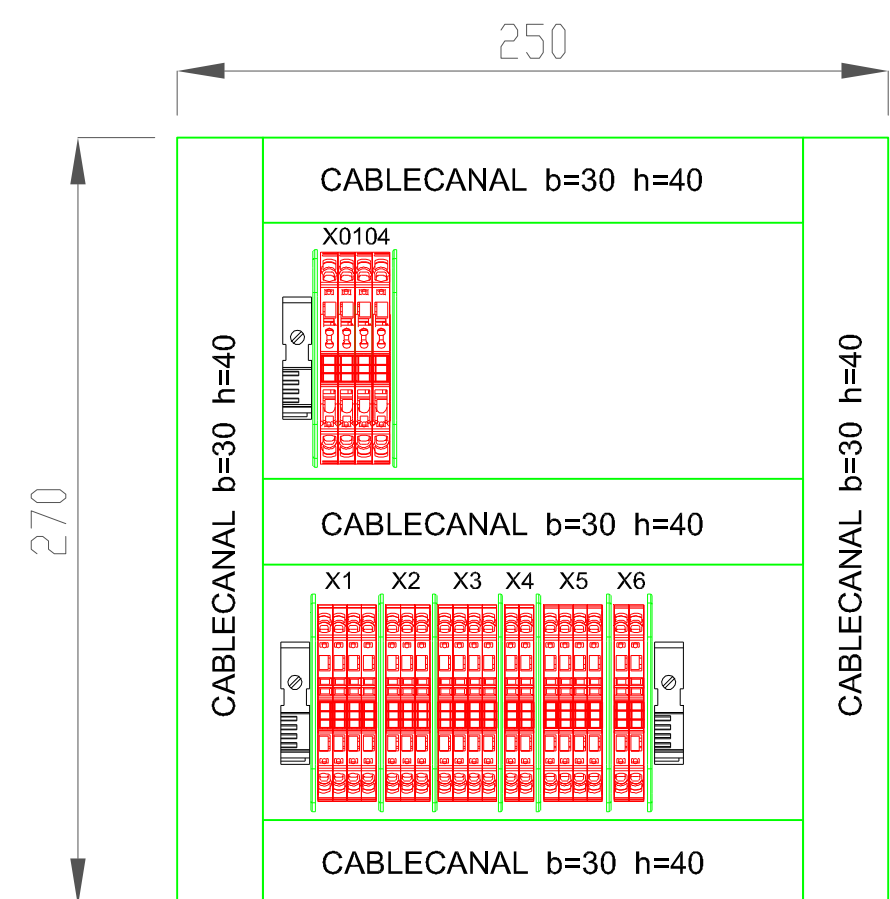
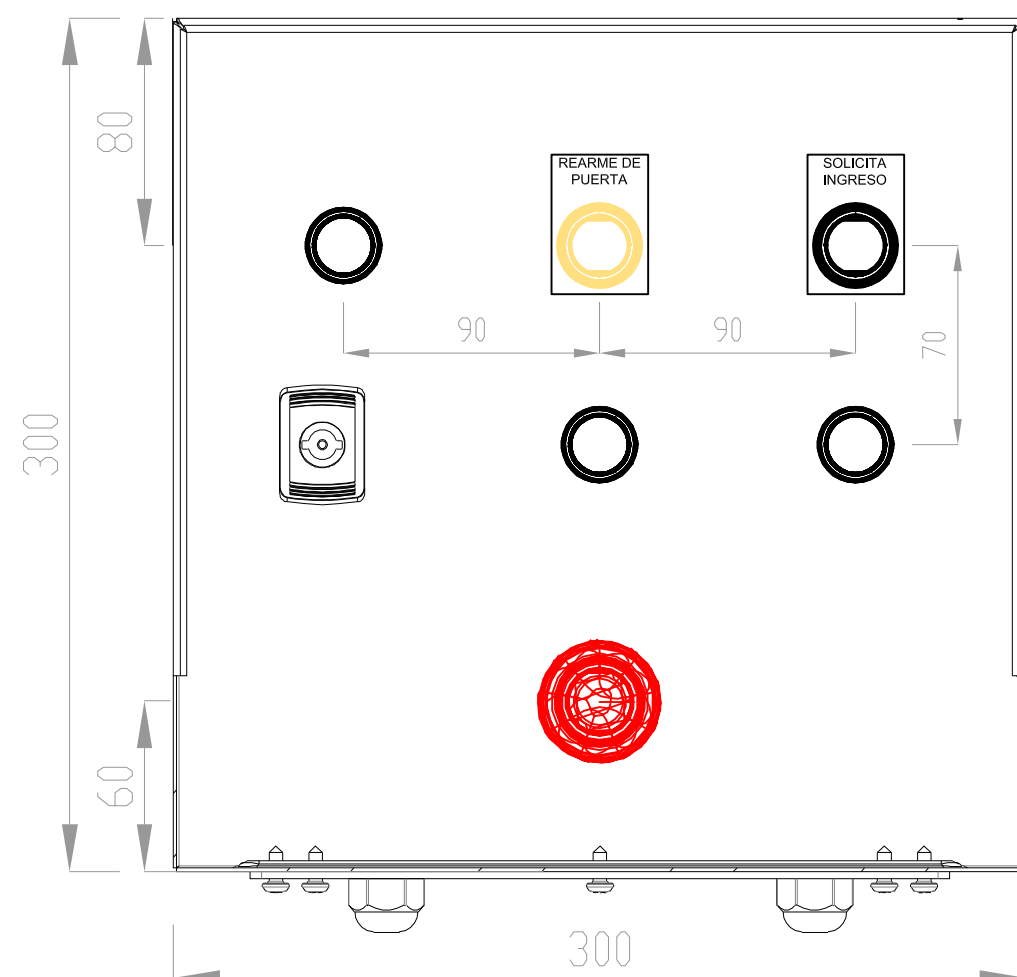
Tablero 300x300x210
Eldon - MAS0303021R5



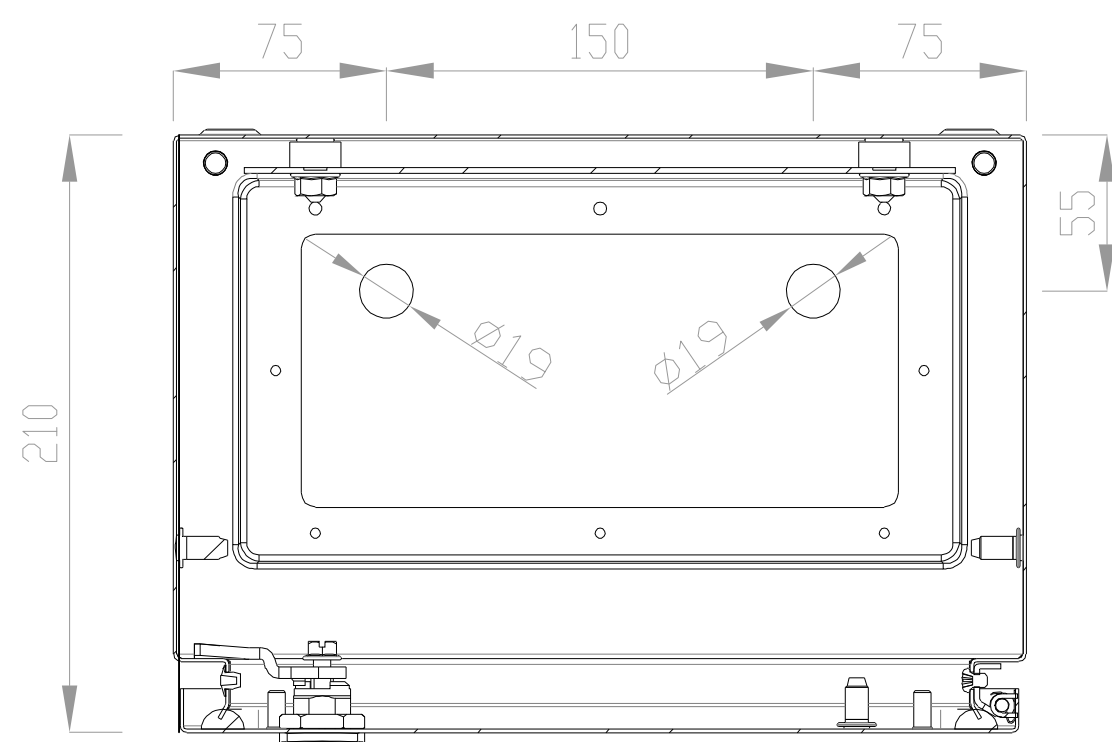
Tablero	Tablero	N° Hilo	N° Cable	Tags Bornera	Componente	Descripcion
JB Acceso 2	CC28	2	019061	X0104	PE03126	24VDC - Rack 1 / Slot 4
		3	026078	X3		IN - Pulsador de Emerg. Acceso 2
		4	026118	X3		IN - Pulsador de Emerg. Acceso 2
		5	026077	X3		OUT - Pulsador de Emerg. Acceso 2
		6	026117	X3		OUT - Pulsador de Emerg. Acceso 2
		7	PE03126	X1		Pulsador de Emergencia Acceso 2
		8	PX03108	X1	PX03108	Pedido de Acceso - Puerta 2
		9	PX03114	X1	PX03115	Rearme de Puerta - Puerta 2
		10	027121	X5	YS03813	IN Canal Seg.1 Puerta 2
		11	027131	X5		IN Canal Seg.1 Puerta 2
		12	027091	X5		OUT Canal Seg.2 Puerta 2
	CC29	2	027132	X5	YS03813	OUT Canal Seg.2 Puerta 2
		3	SA02702	X1		Puerta 2 Cerrada
		4	0190614	X6		0V - Rack 1 / Slot 10
		5	YS03813	X6	YS03813	Cerradura de Puerta 2
		6	0190616	X2		0V - Rack 1 / Slot 11
		7	L03910	X2	L03910	Led Blanco TB Acceso 2
		8	L03911	X2	L03911	Led Amarillo TB Acceso 2
		9	0190618	X4		0V - Rack 1 / Slot 12
		10	L04008	X4	L04008	Baliza roja Acceso 2
		11	-	-	RESERVA	-
		12	-	-	RESERVA	-

5	5	PROYECTO	SIDERCROM EMBALAJE
4	4	DRAWING NO.	095
3	3	DESIGNER BY	TOPOGRAFICO
2	2	DETAIL BY	TABLERO
1	1	REV. CLIENTE	JB ACCESO 2
P&F		REV. INT. P&F	
Av. Belgrano 2272 (B1611DND)		CHECKED BY	
San Torcuato Buenos Aires		DATE	23/10/2015
JCB		SCALE	
		DIVISION	
		PLANT: SIDERCROM	
		LOCATION/ACTIVITY:	

TABLERO - JB ACCESO 3

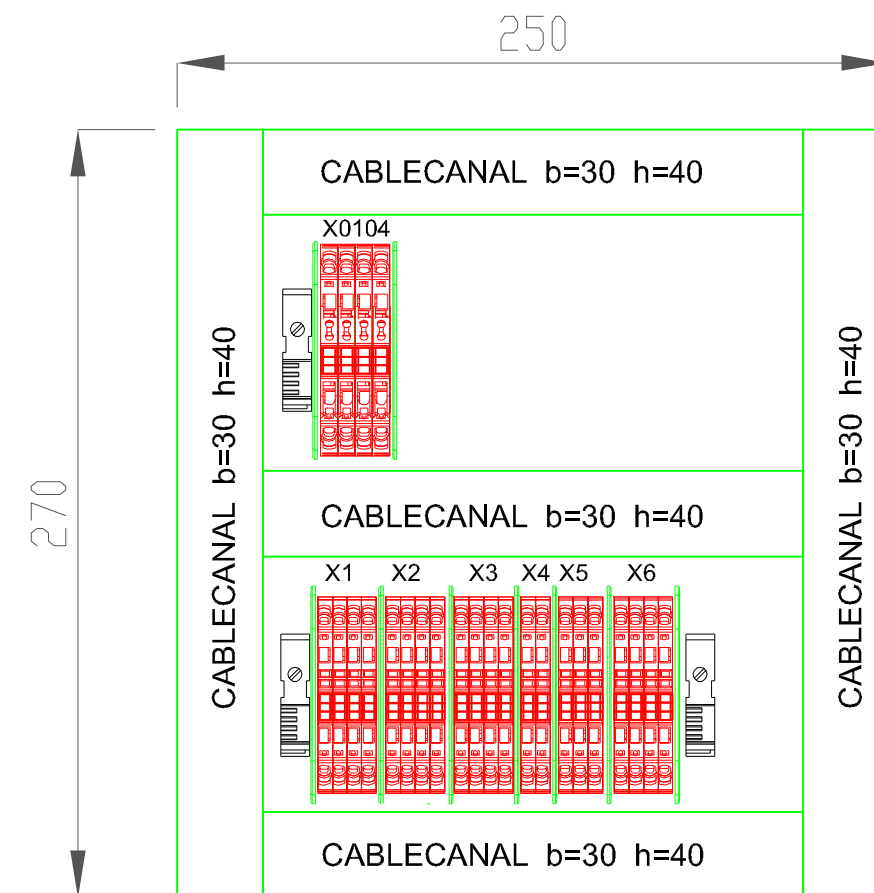
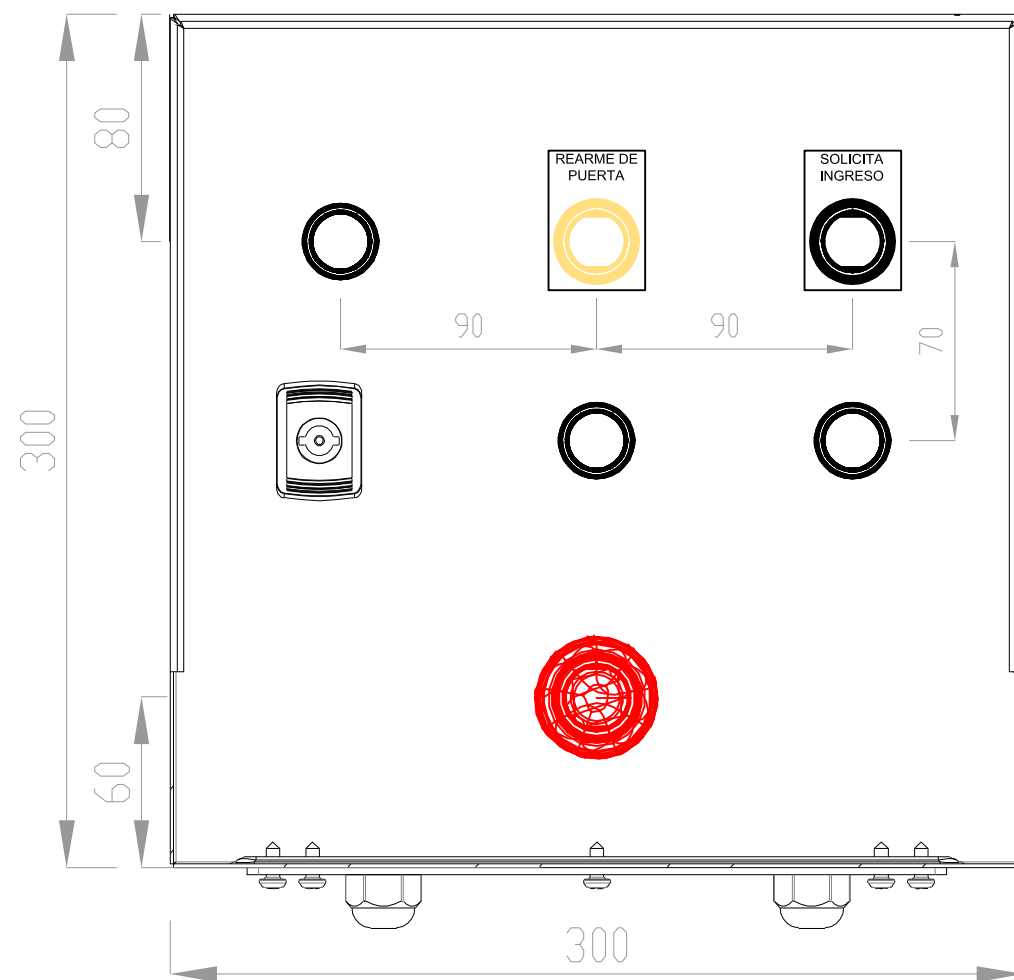


Tablero 300x300x210
Eldon - MAS0303021R5

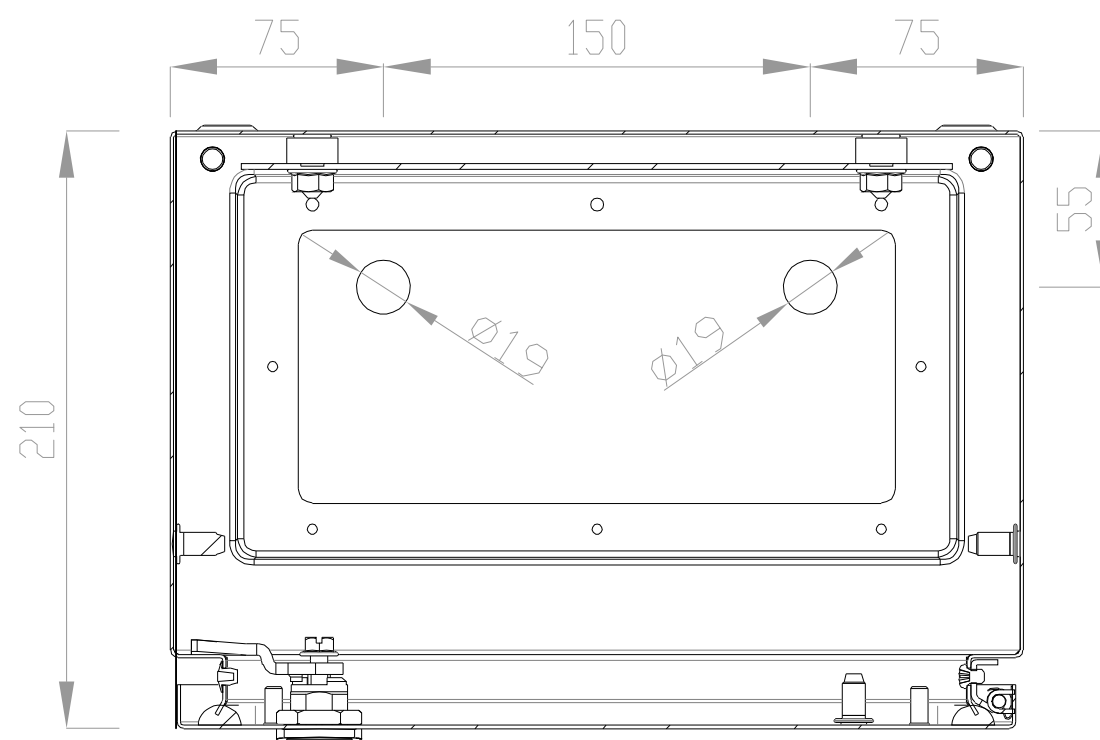


Tablero	Multipar	N° Hilo	N° Cable	Tags Bornera	Componente	Descripcion
JB Acceso 3	CC30	2	019061	X0104	PE03127	24VDC - Rack 1 / Slot 4
		3	026075	X3		IN - Pulsador de Emerg. Acceso 3
		4	026115	X3		IN - Pulsador de Emerg. Acceso 3
		5	026074	X3		OUT - Pulsador de Emerg. Acceso 3
		6	026114	X3		OUT - Pulsador de Emerg. Acceso 3
		7	PE03127	X1		Pulsador de Emergencia Acceso 3
		8	PX03109	X1	PX03109	Pedido de Acceso - Puerta 3
		9	PX03115	X1	PX03115	Rearme de Puerta - Puerta 3
		10	028071	X5	YS03814	IN Canal Seg.1 Puerta 3
		11	028081	X5		IN Canal Seg.1 Puerta 3
		12	028121	X5		OUT Canal Seg.2 Puerta 3
		2	028131	X5		OUT Canal Seg.2 Puerta 3
	CC31	3	SA02803	X1		Puerta 3 Cerrada
		4	0190614	X6		0V - Rack 1 / Slot 10
		5	YS03814	X6	YS03814	Cerradura de Puerta 3
		6	0190616	X2		0V - Rack 1 / Slot 11
		7	L03912	X2	L03912	Led Blanco TB Acceso 3
		8	L03913	X6	L03913	Led Amarillo TB Acceso 3
		9	0190618	X4		0V - Rack 1 / Slot 12
		10	L04012	X4	L04012	Baliza roja Acceso 3
		11	-	-	RESERVA	-
		12	-	-	RESERVA	-

TABLERO - JB ACCESO 4



Tablero 300x300x210
Eldon - MAS0303021R5

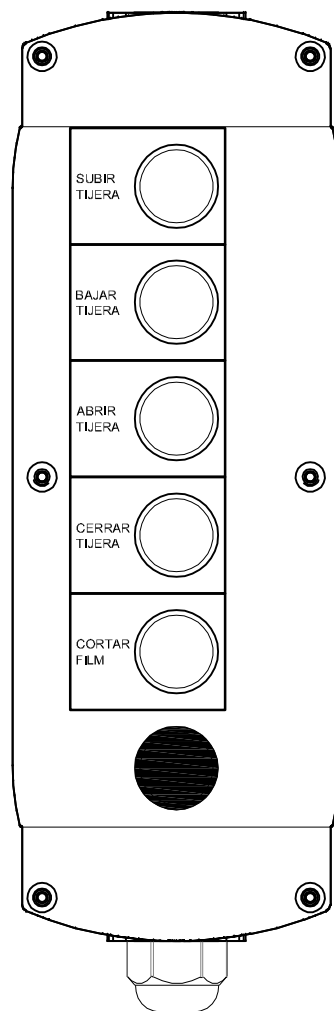


Tablero	Multipar	N° Hilo	N° Cable	Tags Borneo	Componente	Descripcion
JB Acceso 4	CC32	2	019061			24VDC – Rack 1 / Slot 4
		3	026073	X1	PE03128	IN – Pulsador de Emerg. Acceso 4
		4	026113	X1		IN – Pulsador de Emerg. Acceso 4
		5	026072	X1		OUT – Pulsador de Emerg. Acceso 4
		6	026112	X1		OUT – Pulsador de Emerg. Acceso 4
		7	PE03128	X2		Pulsador de Emergencia Acceso 4
		8	PX03110	X2	PX03110	Pedido de Acceso – Puerta 4
		9	PX03116	X2	PX03116	Rearme de Puerta – Puerta 4
		10	028121	X3	YS03815	IN Canal Seg.1 Puerta 4
		11	028131	X3		IN Canal Seg.1 Puerta 4
	12	028091	X3		OUT Canal Seg.2 Puerta 4	
	CC33	2	028132	X3		OUT Canal Seg.2 Puerta 4
		3	SA02804	X2		Puerta 4 Cerrada
		4	0190614	X4		0V – Rack 1 / Slot 10
		5	YS03815	X4	YS03815	Cerradura de Puerta 4
		6	0190616	X5		0V – Rack 1 / Slot 11
		7	L03914	X5	L03914	Led Blanco TB Acceso 4
		8	L03915	X5	L03915	Led Amarillo TB Acceso 4
		9	0190618	X6		0V – Rack 1 / Slot 12
		10	L04013	X6	L04013	Baliza roja Acceso 4
11		–	-	RESERVA	–	
12	–	-	RESERVA	–		

5	4	3	PROPERTY	SIDERCROM EMBALAJE		
4	3	2	DRAWING NO.	097		SHEET 131
3	2	1	DESIGNER BY	DETAILS BY	TITLE	
2	1		LH	EG	TOPOGRAFICO	
1			CHECKED BY	SAFETY OF	TABLERO	
			D.F.		J4 ACCESO 4	
			DATE:	SCALE:	DIVISION:	
			3/14/1915		PLANT - SIDERCROM	
					LOCATION/ACTIVITY:	

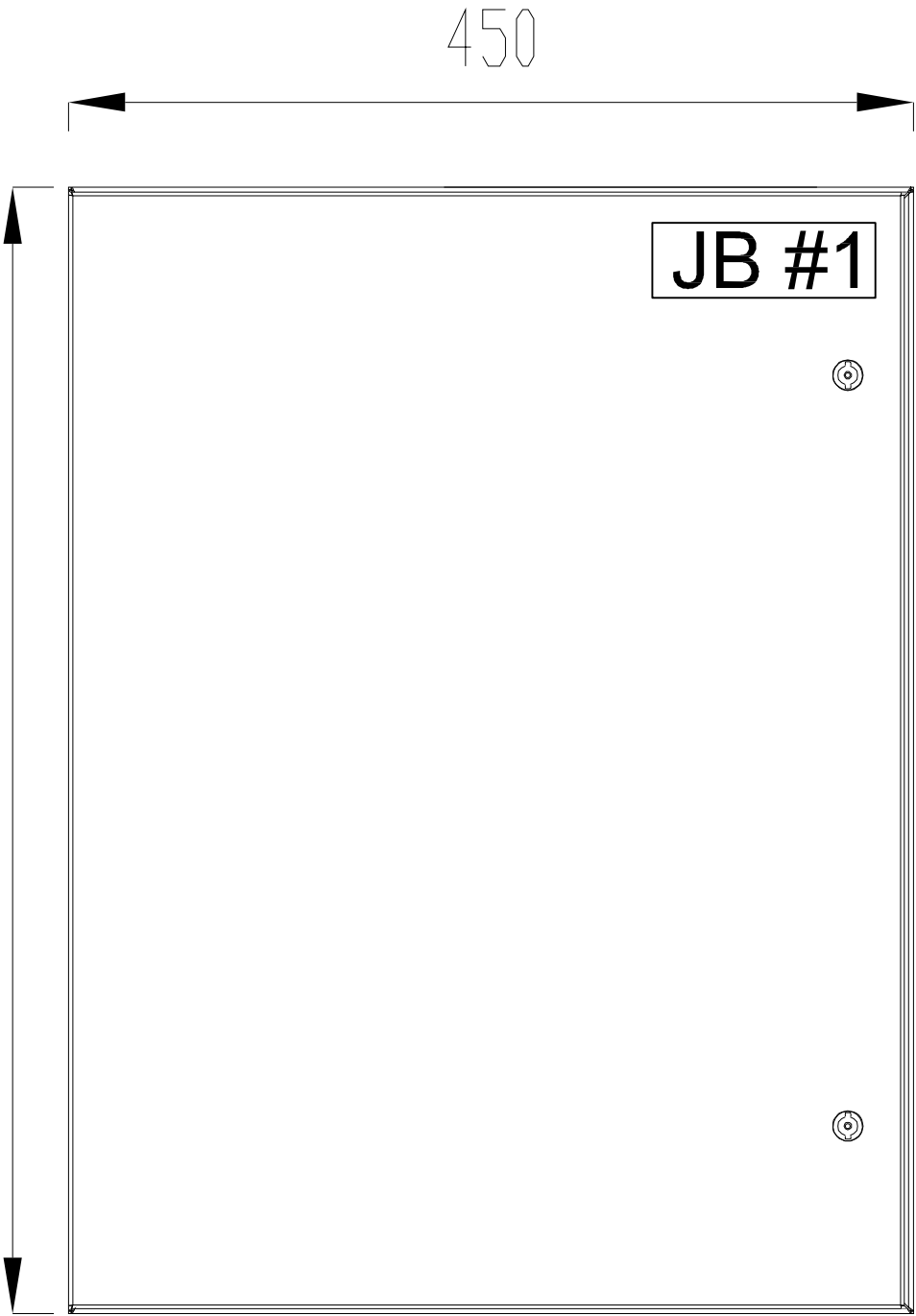
BOTONERA

TIJERA CORTE DE FILM

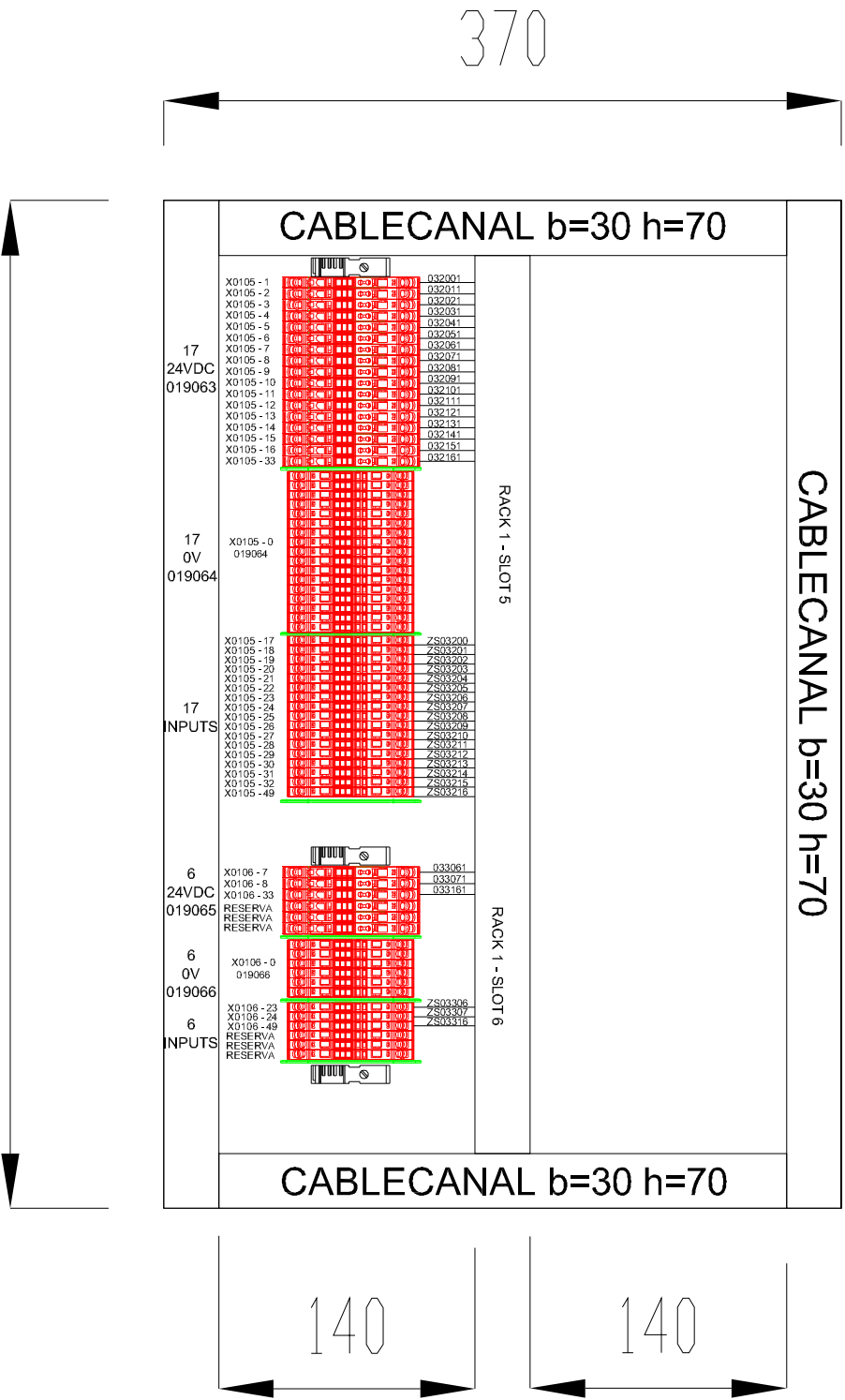


MULTIPAR	ID MULTIPAR	Nº HILO	Nº CABLE	DESCRIPCION
10x1 mm ²	CC80 COMANDO - BOTONERA	1	019067	24VDC
		2	PX03420	PULSADOR - SUBIR TIJERA
		3	PX03421	PULSADOR - BAJAR TIJERA
		4	PX03422	PULSADOR - ABRIR TIJERA
		5	PX03423	PULSADOR - CERRAR TIJERA
		6	PX03424	PULSADOR - CORTAR FILM
		7	-	RESERVA
		8	-	RESERVA
		9	-	RESERVA
		10	-	RESERVA

TABLERO DE DISTRIBUCION I/O
JB#1



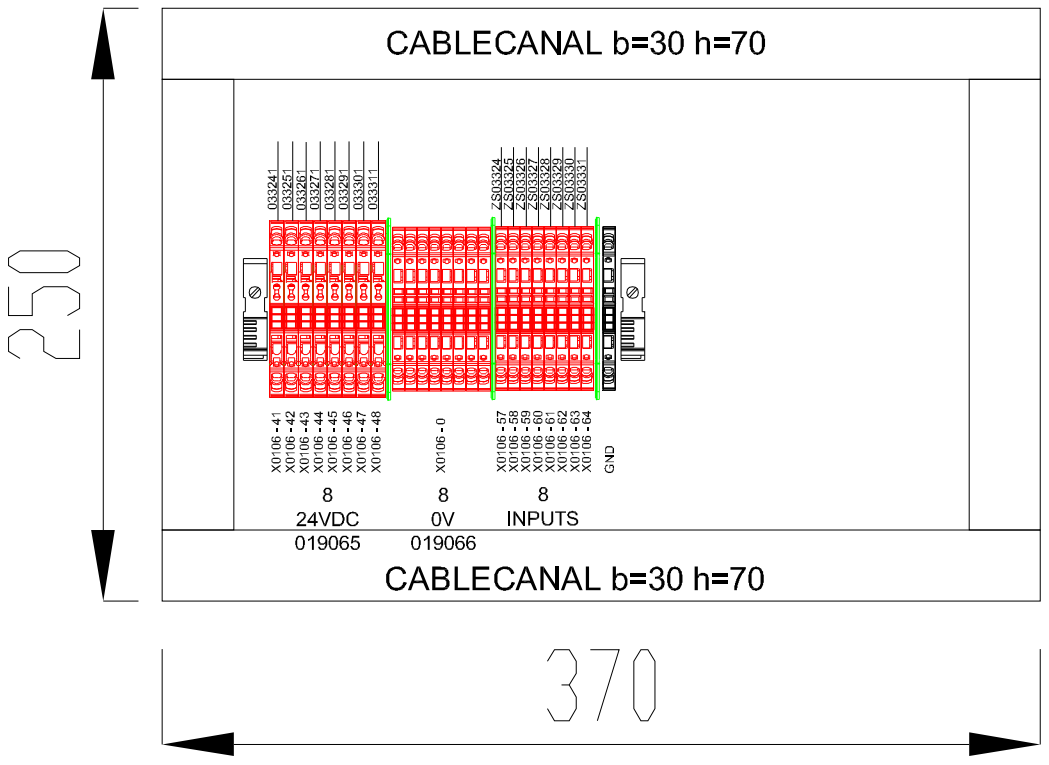
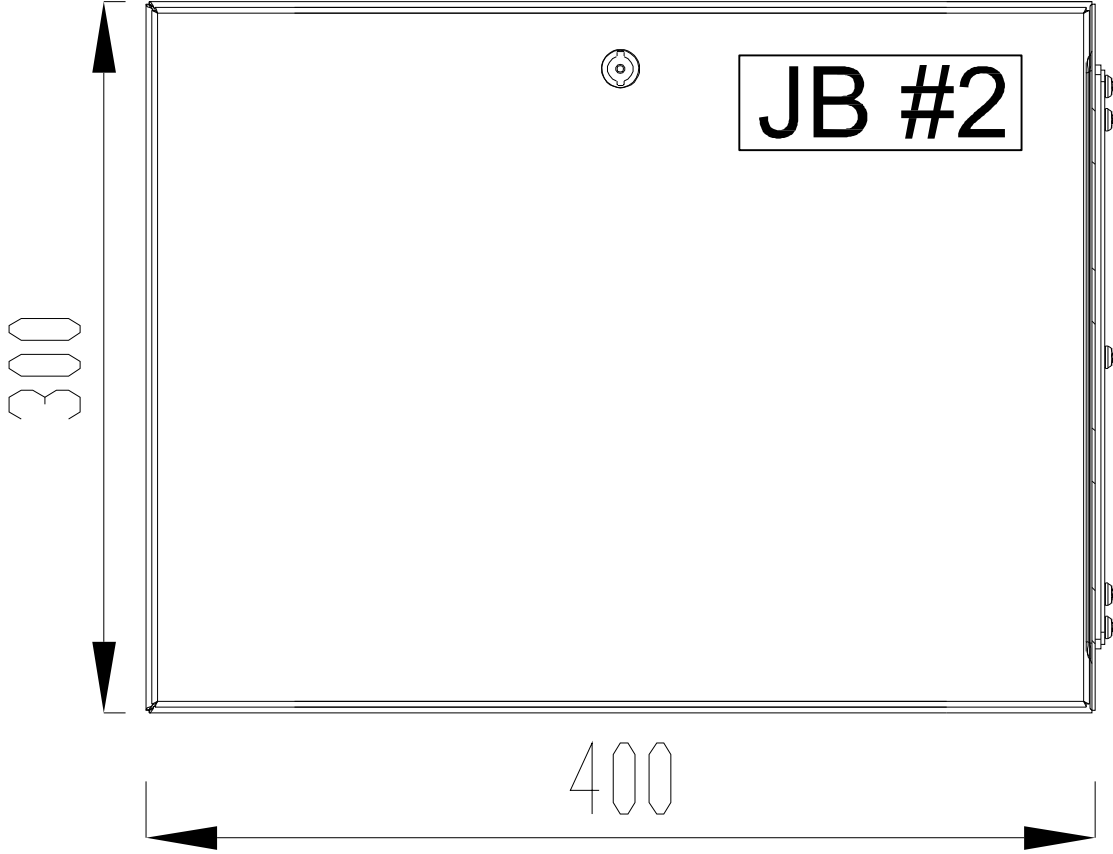
Tablero 450 x 600 x 210
Ristal



Tablero	Multipar	N° Hilo	N° Cable	Tags Bornera	Componente	Descripcion
JB#1	CC8	2	019063		24VDC - Rack 1 / Slot 5	Alimentacion
		3	019064		0VDC - Rack 1 / Slot 5	
		4	ZS03200	X0105 - 17	ZS03200	Sensor Optico
		5	ZS03201	X0105 - 18	ZS03201	Sensor Optico
		6	ZS03202	X0105 - 19	ZS03202	Sensor Optico
		7	ZS03203	X0105 - 20	ZS03203	Sensor Optico
		8	ZS03204	X0105 - 21	ZS03204	Sensor Optico
		9	ZS03205	X0105 - 22	ZS03205	Sensor Optico
		10	ZS03206	X0105 - 23	ZS03206	Sensor Optico
		11	ZS03207	X0105 - 24	ZS03207	Sensor Optico
		12	ZS03208	X0105 - 25	ZS03208	Sensor Optico
	CC9	2	ZS03209	X0105 - 26	ZS03209	Sensor Optico
		3	ZS03210	X0105 - 27	ZS03210	Sensor Optico
		4	ZS03211	X0105 - 28	ZS03211	Sensor Optico
		5	ZS03212	X0105 - 29	ZS03212	Sensor Optico
		6	ZS03213	X0105 - 30	ZS03213	Sensor Optico
		7	ZS03214	X0105 - 31	ZS03214	Sensor Optico
		8	ZS03215	X0105 - 32	ZS03215	Sensor Optico
		9	ZS03216	X0105 - 39	ZS03216	Sensor Optico
		10	-	NO SE CABLEA	RESERVA	
		11	-	NO SE CABLEA	RESERVA	
		12	-	NO SE CABLEA	RESERVA	
	CC10	2	019065		24VDC - Rack 1 / Slot 6	Alimentacion
		3	019066		0VDC - Rack 1 / Slot 6	
		4	ZS03306	X0106 - 23	ZS03306	Sensor Inductivo
		5	ZS03307	X0106 - 24	ZS03307	Sensor Inductivo
		6	SX03316	X0106 - 49	SX03316	Sensor Optico
		7	-	NO SE CABLEA	RESERVA	
		8	-	NO SE CABLEA	RESERVA	
		9	-	NO SE CABLEA	RESERVA	
		10	-	NO SE CABLEA	RESERVA	
		11	-	NO SE CABLEA	RESERVA	
		12	-	NO SE CABLEA	RESERVA	

TABLERO DE DISTRIBUCION I/O

JB#2



Tablero	Multipar	N° Hilo	N° Cable	Tags Bornera	Componente	Descripcion
JB#2	CC11	1	019065			24VDC - Rack 1 / Slot 6
		2	019066			0VDC - Rack 1 / Slot 6
		3	ZS03324	X0106 - 57	ZS03324	Sensor Cilindro
		4	ZS03325	X0106 - 58	ZS03325	Sensor Cilindro
		5	ZS03327	X0106 - 60	ZS03327	Sensor Optico
		6	ZS03328	X0106 - 61	ZS03328	Sensor Optico
		7	ZS03329	X0106 - 62	ZS03329	Sensor Optico
		8	ZS03330	X0106 - 63	ZS03330	Sensor Optico
		9	ZS03331	X0106 - 64	ZS03331	Sensor Cilindro
		10	ZS03326	X0106 - 59	ZS03326	Sensor Optico
		11	-	NO SE CABLEA	RESERVA	
		12	GND			

Tablero 300 x 400 x 210

Forli 0464

5

4

3

2

1

5

4

3

2

1

PROJECT

SIDERCROM EMBALAJE

DRAWING NO.

100

SHEET

131

REV. CLIENTE

REV. INT. P&F

DESIGN BY

EA

CHECKED BY

EA

DATE

29/14/215

SCALE

DIVISION

PLANTA SIDERCROM

LOCATION/ACTIVITY

Av. Belgrano 2272 (CHALUPVD)
San Torcuato Buenos Aires
Argentina

TABLERO DE DISTRIBUCION I/O
JB#3

450

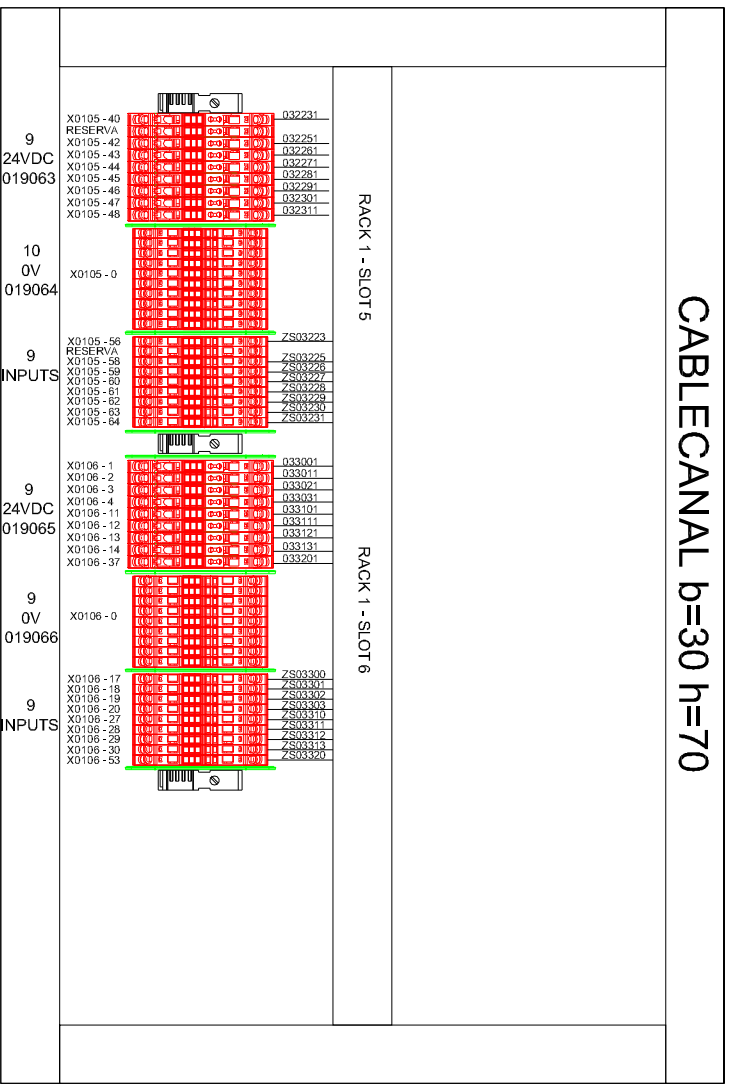
370

JB #3

600

210

Tablero 450 x 600 x 210
Ristal



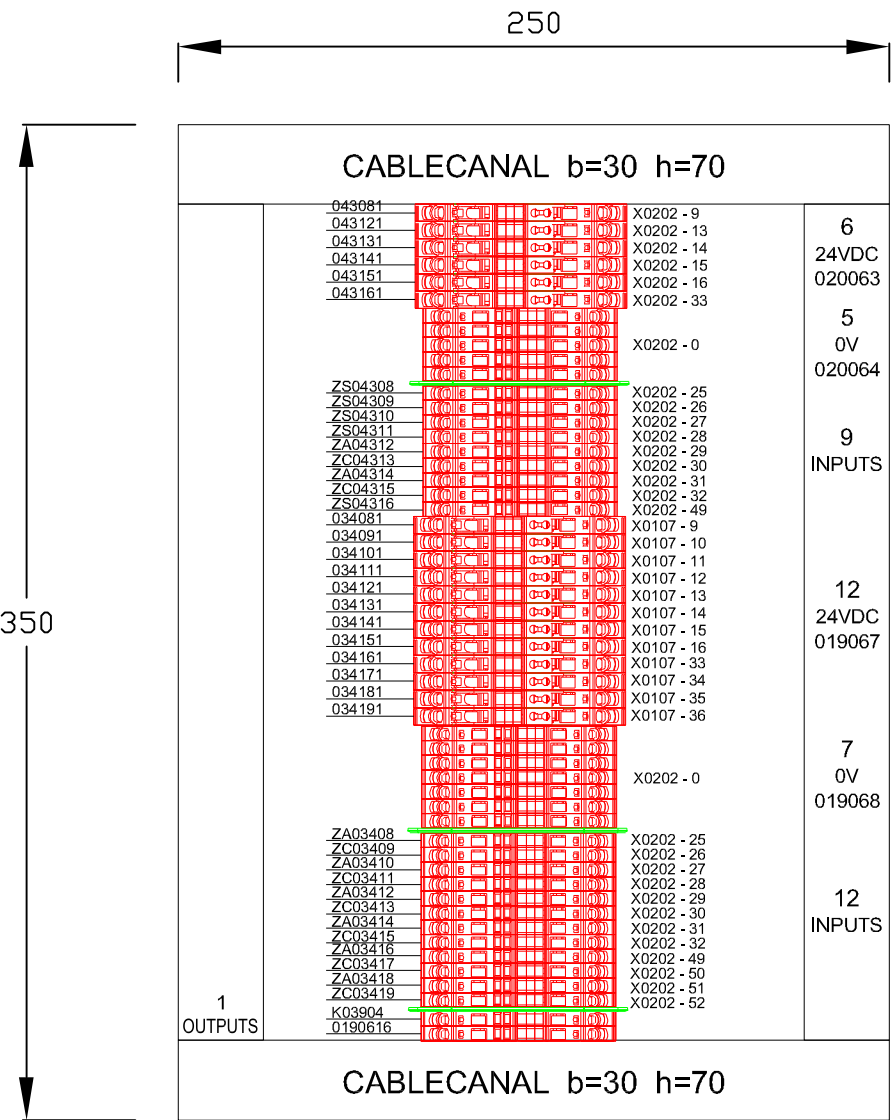
Tablero	Multipar	N° Hilo	N° Cable	Tags	Bornera	Componente	Descripcion
JB#3	CC12	1	019063				24VDC - Rack 1 / Slot 5
		2	019064				0VDC - Rack 1 / Slot 5
		3	ZS03223	X0105 - 56	ZS03223		Sensor Optico
		4	-				
		5	ZS03225	X0105 - 58	ZS03225		Sensor Optico
		6	ZS03226	X0105 - 59	ZS03226		Sensor Optico
		7	ZS03227	X0105 - 60	ZS03227		Sensor Optico
		8	ZS03228	X0105 - 61	ZS03228		Sensor Optico
		9	ZS03229	X0105 - 62	ZS03229		Sensor Optico
		10	ZS03230	X0105 - 63	ZS03230		Sensor Optico
		11	ZS03231	X0105 - 64	ZS03231		Sensor Optico
	CC13	1	ZS03313	X0106 - 30	ZS03313		Sensor Inductivo
		2	019065				24VDC - Rack 1 / Slot 6
		3	019066				0VDC - Rack 1 / Slot 6
		4	ZS03300	X0106 - 17	ZS03300		Sensor Optico
		5	ZS03301	X0106 - 18	ZS03301		Sensor Optico
		6	ZS03302	X0106 - 19	ZS03302		Sensor Optico
		7	ZS03303	X0106 - 20	ZS03303		Sensor Optico
		8	ZS03310	X0106 - 11	ZS03310		Sensor Inductivo
		9	ZS03311	X0106 - 12	ZS03311		Sensor Inductivo
		10	ZS03312	X0106 - 13	ZS03312		Sensor Inductivo
		11	ZS03320	X0106 - 37	ZS03320		Sensor Inductivo

140

140

TABLERO DE DISTRIBUCION I/O

JB#4

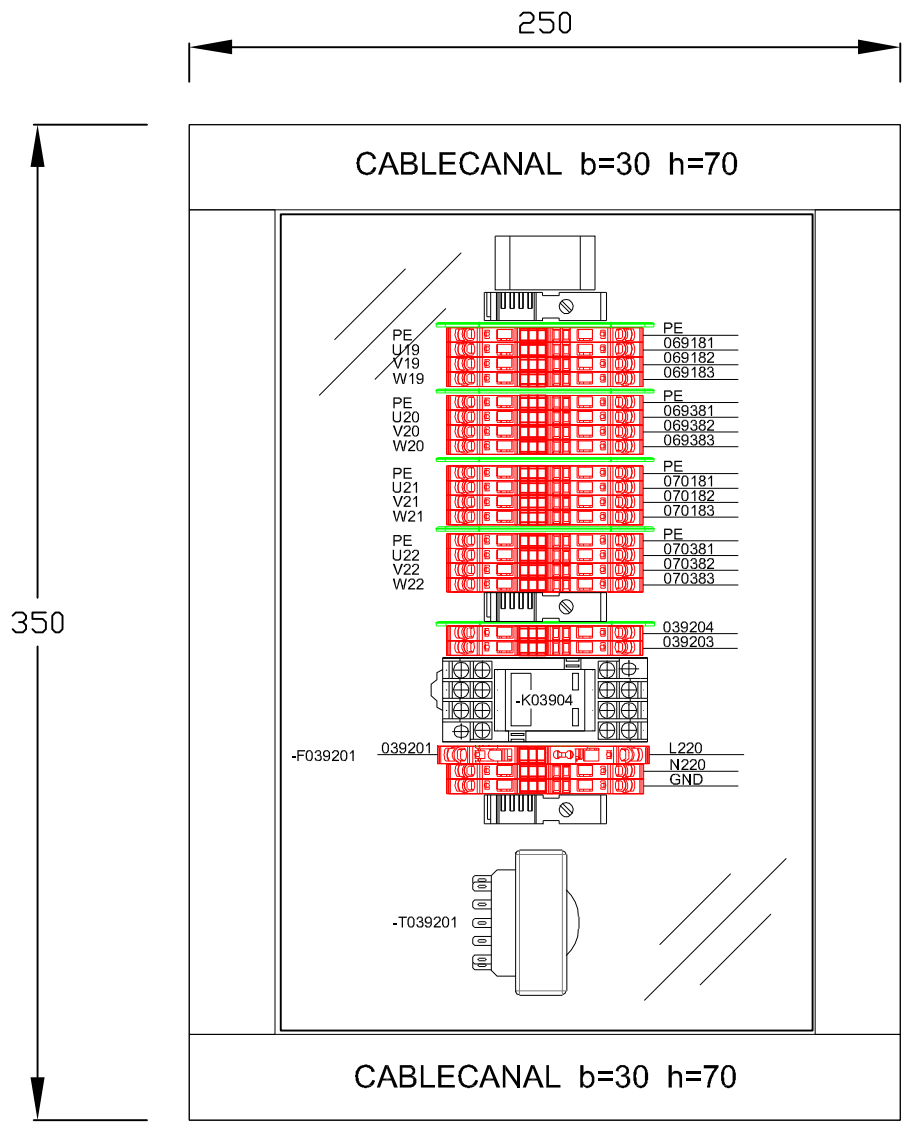


Tablero 300x400x160
Forli 0410

Tablero	Multipar	N° Hilo	Tags Bornera PF	Componente	Bornera Inputs	Descripcion
JB#4	CC14	1	X0202 - 33	ZS04316	X0202 - 49	Sensor Inductivo
		2	N° cable 020063			24VDC - Rack 2 / Slot 2
		3	N° cable 020064			0VDC - Rack 2 / Slot 2
		4	X0202 - 9	ZS04308	X0202 - 25	Sensor Inductivo
		5	X0202 - 10	ZS04309	X0202 - 26	Sensor Inductivo
		6	X0202 - 11	ZS04310	X0202 - 27	Sensor Inductivo
		7	X0202 - 12	ZS04311	X0202 - 28	Sensor
		8	X0202 - 13	ZS04312	X0202 - 29	Sensor Magnetico
		9	X0202 - 14	ZC04313	X0202 - 30	Sensor Magnetico
		10	X0202 - 15	ZS04314	X0202 - 31	Sensor Magnetico
		11	X0202 - 16	ZC04315	X0202 - 32	Sensor Magnetico
		12	Reserva			
	CC15	1		K03904		Rele Corte de Film
		2		0190616		0VDC
		3	Reserva			
		4	N° cable 019067			24VDC - Rack 1 / Slot 7
		5	N° cable 019068			0VDC - Rack 1 / Slot 7
		6	X0107 - 9	ZA03408	X0107 - 25	Limit Switch
		7	X0107 - 10	ZC03409	X0107 - 26	Sensor Cilindro
		8	X0107 - 11	ZA03410	X0107 - 27	Limit Switch
		9	X0107 - 12	ZC03411	X0107 - 28	Sensor Cilindro
		10	X0107 - 13	ZA03412	X0107 - 29	Sensor Inductivo
		11	X0107 - 14	ZC03413	X0107 - 30	Sensor Cilindro
	CC16	1	X0107 - 15	ZA03414	X0107 - 31	Sensor Inductivo
		2	X0107 - 16	ZC03415	X0107 - 32	Sensor Cilindro
		3	X0107 - 33	ZA03416	X0107 - 49	Sensor Inductivo
		4	X0107 - 34	ZC03417	X0107 - 50	Sensor Cilindro
		5	X0107 - 35	ZA03418	X0107 - 51	Sensor Inductivo
		6	X0107 - 36	ZC03419	X0107 - 52	Sensor Cilindro
		7	Reserva			
		8	Reserva			
		9	Reserva			
		10	Reserva			
		11	Reserva			

TABLERO DE DISTRIBUCION I/O

JB#5



Tablero 300x400x160

Forli 0410

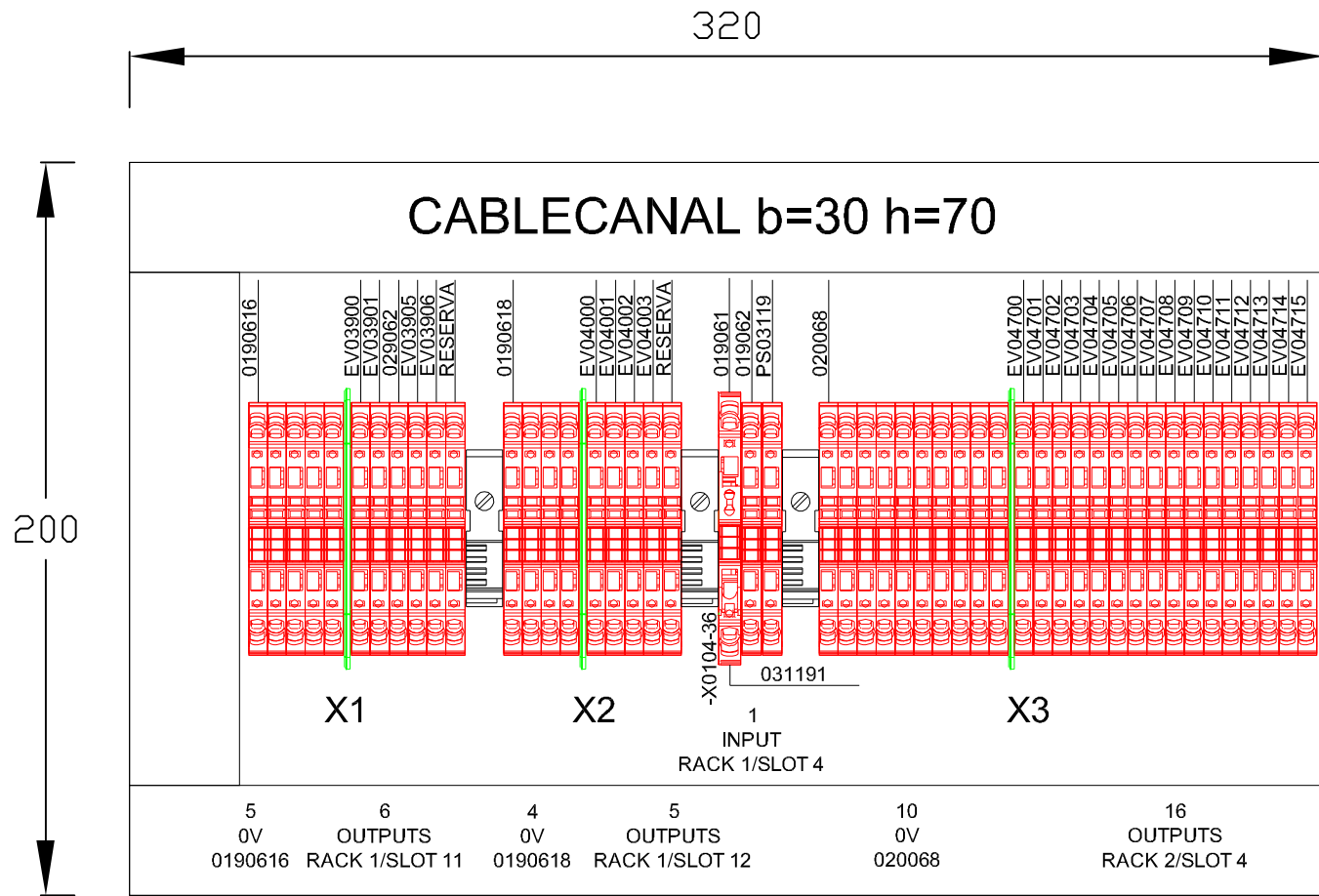
Tablero	TPR 3x2,5	N° Hilo	Componente
JB#5	M015171	PE	-M015171
		069181	
		069182	
	M015172	PE	-M015172
		069381	
		069382	
	M015173	PE	-M015173
		070181	
		070182	
	M015174	PE	-M015174
		070381	
		070382	

Tablero	TPR 4X2,5	COLOR	N° Hilo
JB#5	CC68	Marron	L220
		Celeste	N220
		Negro	-
		V/A	-

Tablero	4X0.35	COLOR	N° Hilo
JB#5	CC67	Blanco	0190616
		Verde	K03904
		Marron	-
		Amarillo	-

TABLERO DE NEUMATICA

JB#6

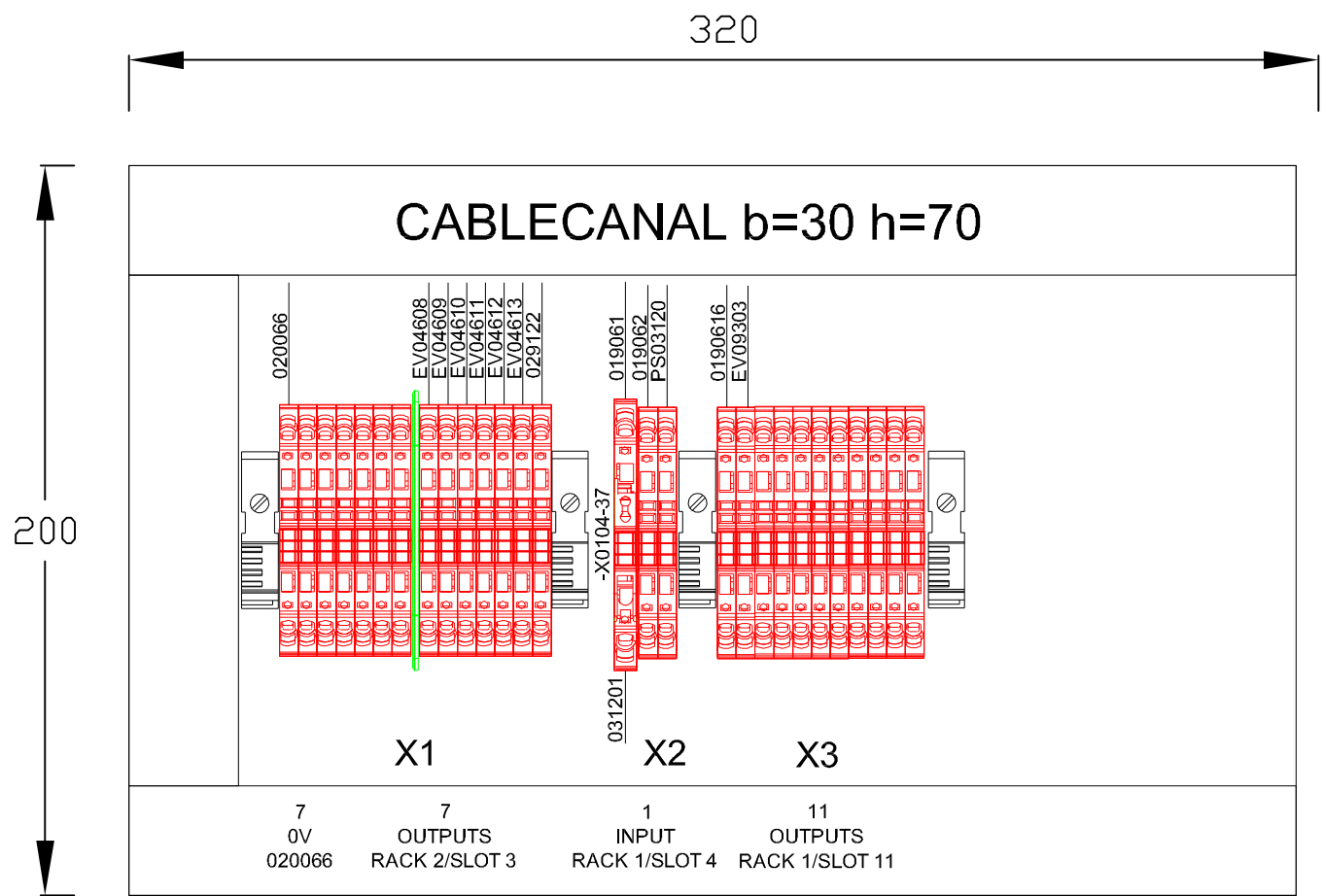


Tablero 300x400x160
Forli 0410

Tablero	Multipar	N° Hilo	N° Cable	Tags Bornera	Descripcion
JB#6	CC17	1	0190616	X1	0VDC – Rack 1 / Slot 11
		2	EV03900		Electrovalvula
		3	EV03901		Electrovalvula
		4	029062		Valvula Seguridad
		5	EV03905		Electrovalvula
		6	EV03906		Electrovalvula
		7	RESERVA		–
		8	0190618	X2	0VDC – Rack 1 / Slot 12
		9	EV04000		Electrovalvula
		10	EV04001		Electrovalvula
		11	EV04002		Electrovalvula
			Electrovalvula		
	CC18	1	EV04003		–
		2	RESERVA		
		3	019061	X4	24VDC – Rack 1 / Slot 4
		4	019062		0VDC – Rack 1 / Slot 4
		5	PS03119		Presostato
		6	020068		0VDC – Rack 2 / Slot 4
		7	EV04700		Electrovalvula
		8	EV04701		Electrovalvula
		9	EV04702		Electrovalvula
		10	EV04703		Electrovalvula
		11	EV04704		Electrovalvula
	CC19	1	EV04705	X3	Electrovalvula
		2	EV04706		Electrovalvula
		3	EV04707		Electrovalvula
		4	EV04708		Electrovalvula
		5	EV04709		Electrovalvula
		6	EV04710		Electrovalvula
		7	EV04711		Electrovalvula
8		EV04712	Electrovalvula		
9		EV04713	Electrovalvula		
10		EV04714	Electrovalvula		
11		EV04715	Electrovalvula		

TABLERO DE NEUMATICA

JB#7



Tablero 300x400x160

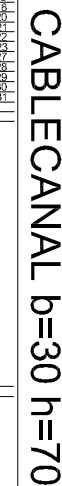
Forli 0410

Tablero	Multipar	N° Hilo	N° Cable	Tags Bornera Señal	Descripcion
JB#7	CC20	1	020066	X1	0VDC – Rack 2 / Slot 3
		2	EV04608		Electrovalvula
		3	EV04609		Electrovalvula
		4	EV04610		Electrovalvula
		5	EV04611		Electrovalvula
		6	EV04612		Electrovalvula
		7	EV04613		Electrovalvula
		8	029122		Valvula Seguridad
		9	019061	X2	24VDC – Rack 1 / Slot 4
	CC79	10	019062		0VDC – Rack 1 / Slot 4
		11	PS03120		Presostato
		1	0190616	X3	0VDC – Rack 1 / Slot 11
		2	EV03903		Electrovalvula
		3	–		Reserva
		4	–		Reserva
		5	–		Reserva
		6	–		Reserva
		7	–		Reserva
		8	–		Reserva
		9	–		Reserva
		10	–		Reserva
		11	–		Reserva

370



140



*En bornera portafusible
Fusibles de 0.25A

CABEZAL SUNCHADOR TITAN K120-19M

PINOUT FICHA			
XSB	TENSION	N° DE CABLE	DESCRIPCION
1	380V	016121	Motor Enrollador Fleje de Plastico M16172
2		016122	
3		016123	
4	380V	016124	Motor Tensionador Fleje M16173
5		016125	
6		016126	
7	380V	016127	Motor Arbol de Levas M16174
8		016128	
9		016129	
10	24VDC	10	Freno Motor Arbol de Levas
11	0V	11	
12	24VAC	12	Reostato
13	0V	13	
14	Señal	SX04227	Serie Termistores - Puerta
15		ZS04210	Sensor Inductivo - ZS04210 (B81)
16		ZS04213	Sensor Inductivo - ZS04213 (B84)
17		ZS04211	Sensor Inductivo - ZS04211 (B82)
18		ZS04212	Sensor Inductivo - ZS04212 (B83)
19			
20			
21			
22			
23	24VDC	042101	Alimentación Sensores
24		020062	

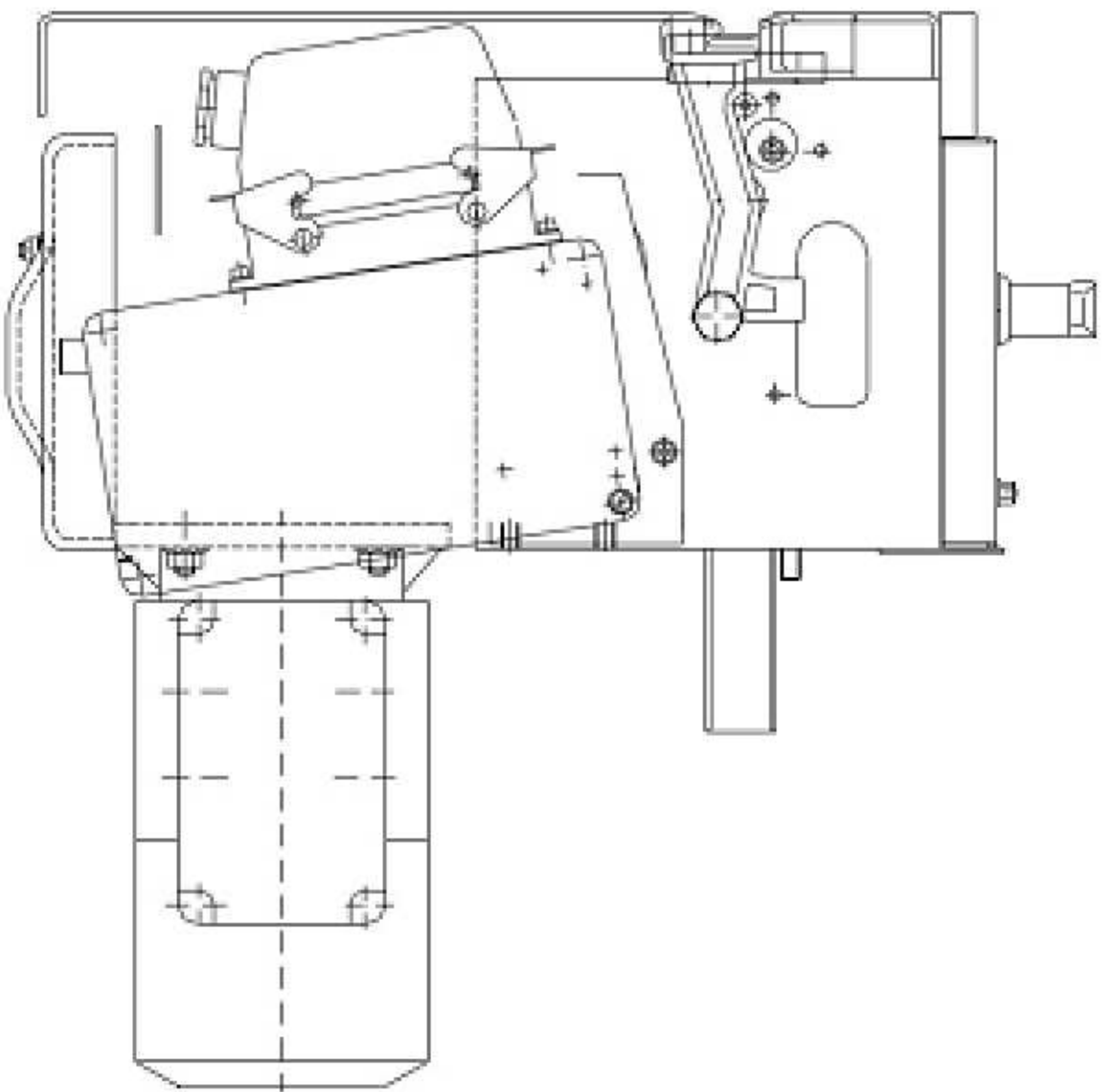
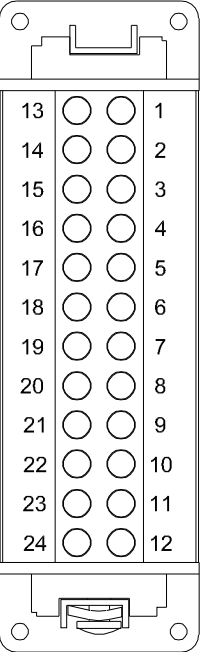
M016172
4x2,5mm²

M016173
4x2,5mm²

M016174
4x2,5mm²

CC74
JB#8 - CABEZAL
10x1mm²

CC76
JB#8 - CABEZAL
3x1,5mm²

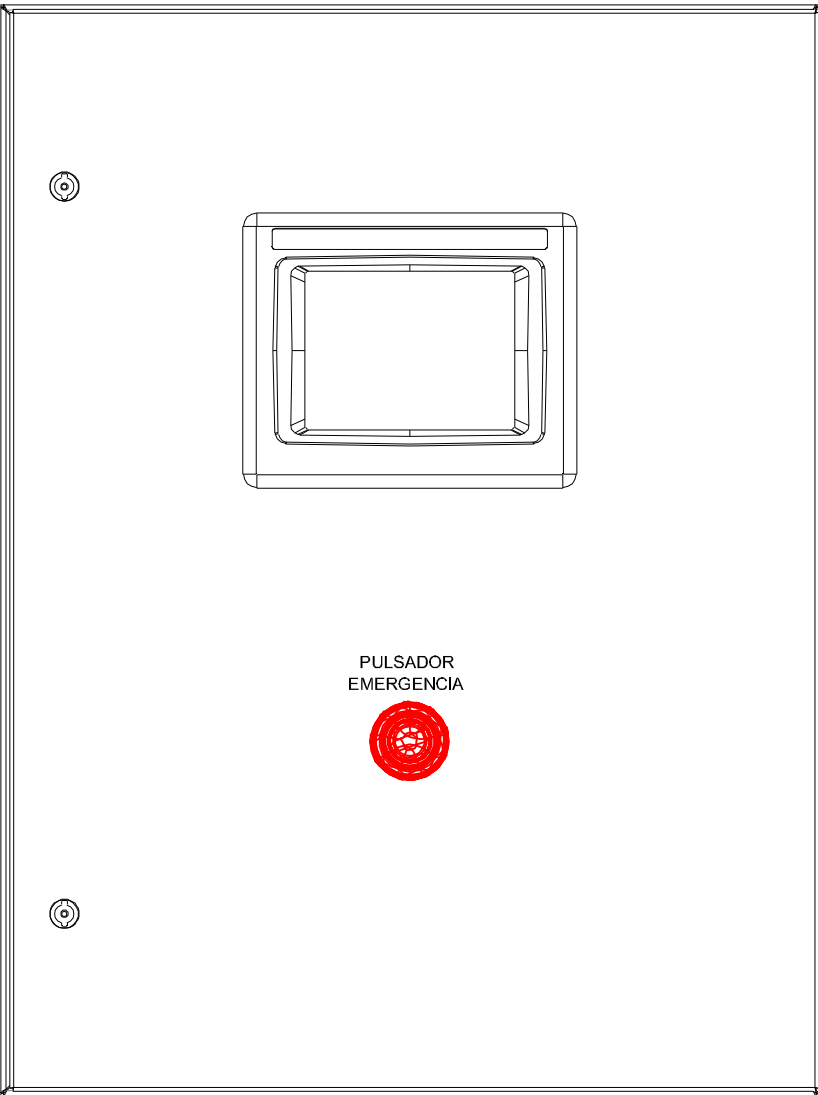
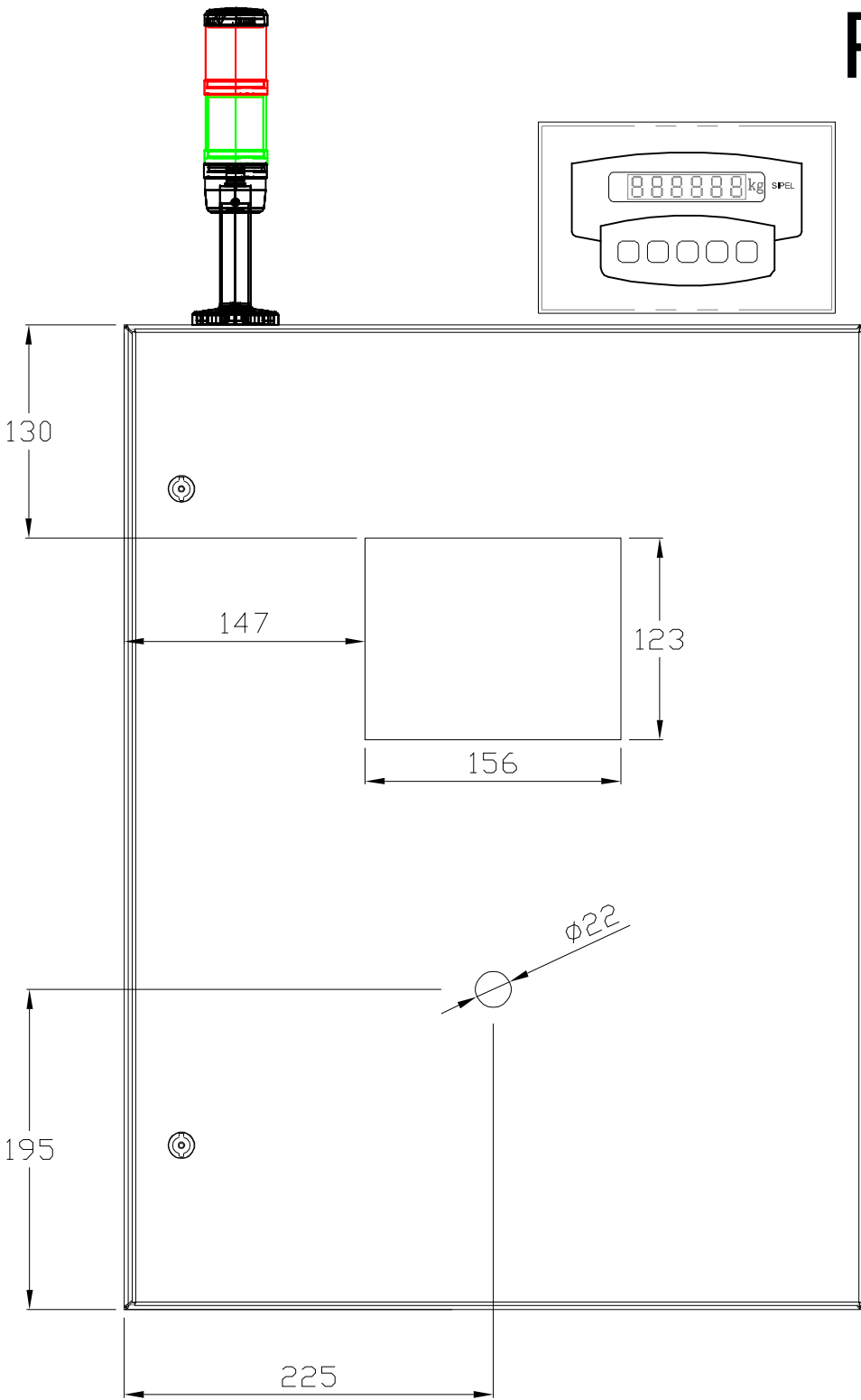


PINOUT MULTIPAR			
ID MULTIPAR	N° HILO	N° CABLE	DESCRIPCION
CC74 JB#8-CABEZAL	1	ZS04210	Sensor Inductivo - ZS04210 (B81)
	2	ZS04213	Sensor Inductivo - ZS04213 (B84)
	3	ZS04211	Sensor Inductivo - ZS04211 (B82)
	4	ZS04212	Sensor Inductivo - ZS04212 (B83)
	5	042101	24VDC - Alimentación Sensores
	6	020062	0V - Alimentación Sensores
	7	10	Freno Motor Arbol de Levas
	8	11	
	9	SX04227	Serie Termistores - Puerta
	10	GND	Puesta a Tierra

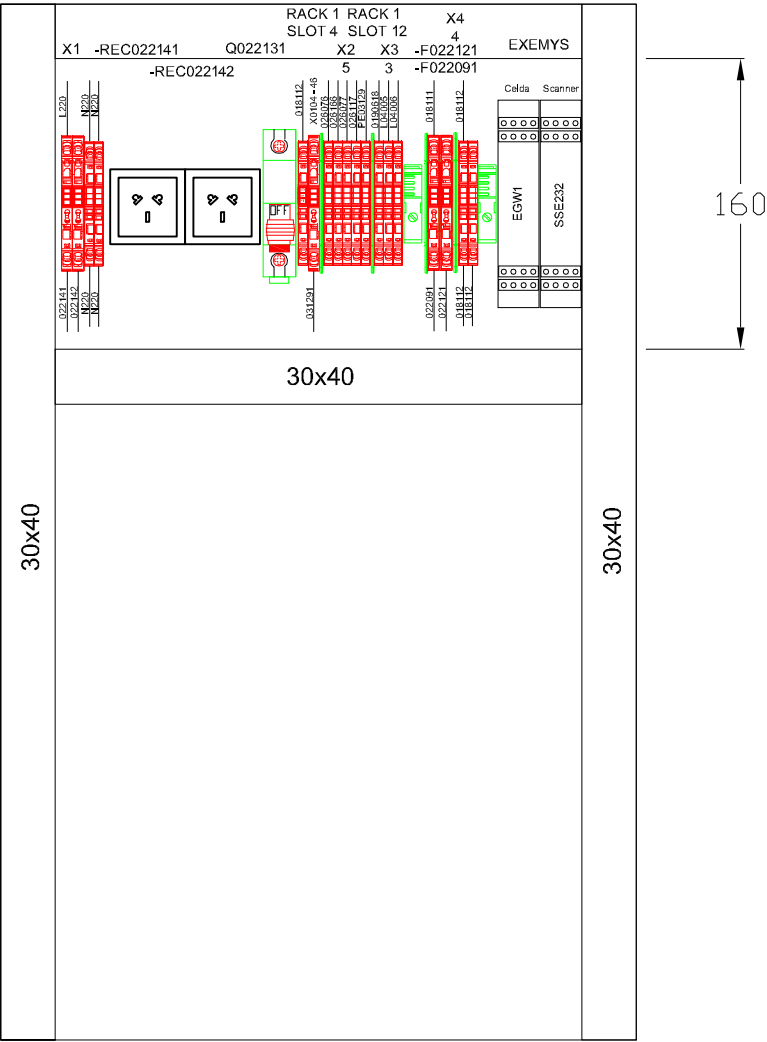
PINOUT MULTIPAR			
ID MULTIPAR	HILO	N° CABLE	DESCRIPCION
CC76 JB#8-CABEZAL	MARRON	12	Calentador de fleje
	CELESTE	13	
	V/A	-	-

PanelView #1 - Estación 1

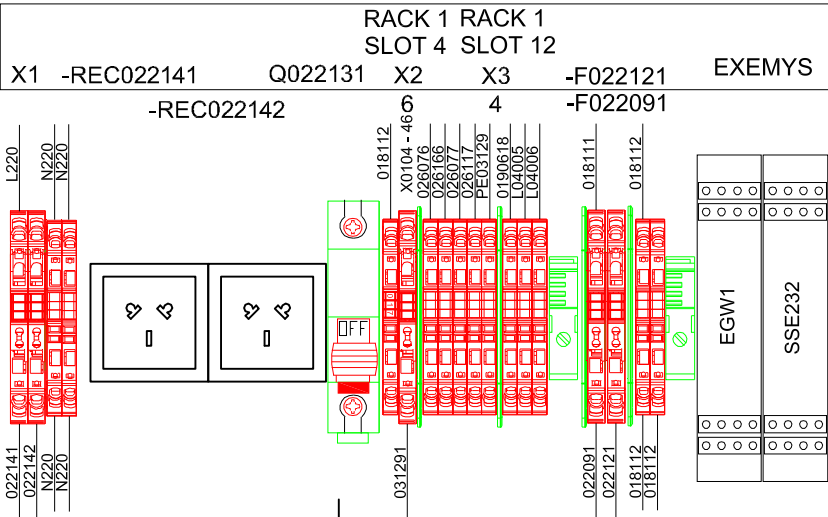
Tablero 450 x 600 x 210
Ristal



DETALLE
A



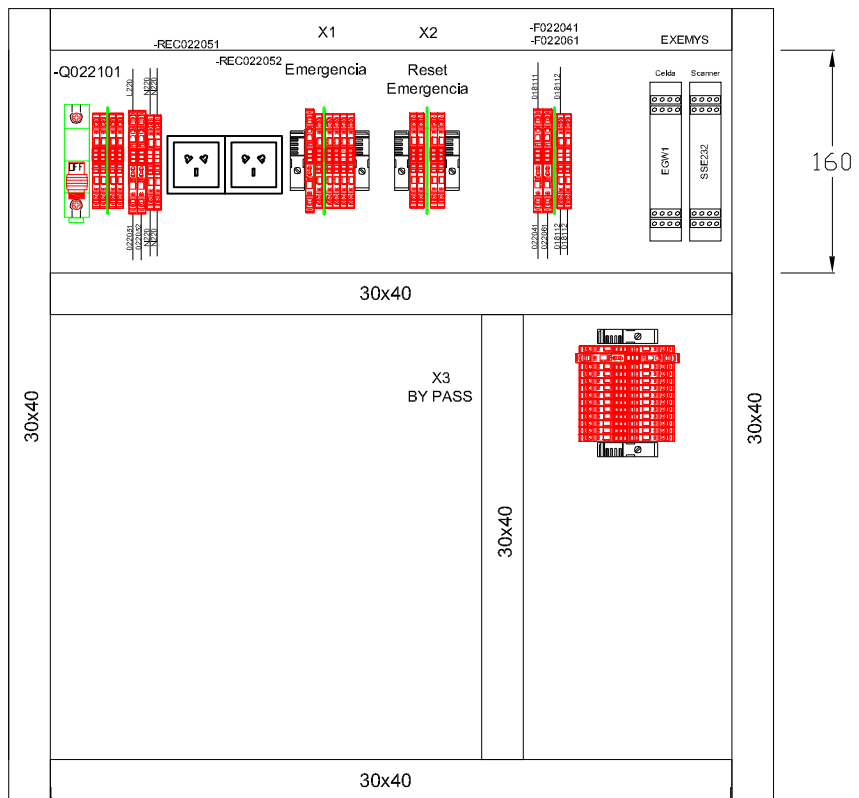
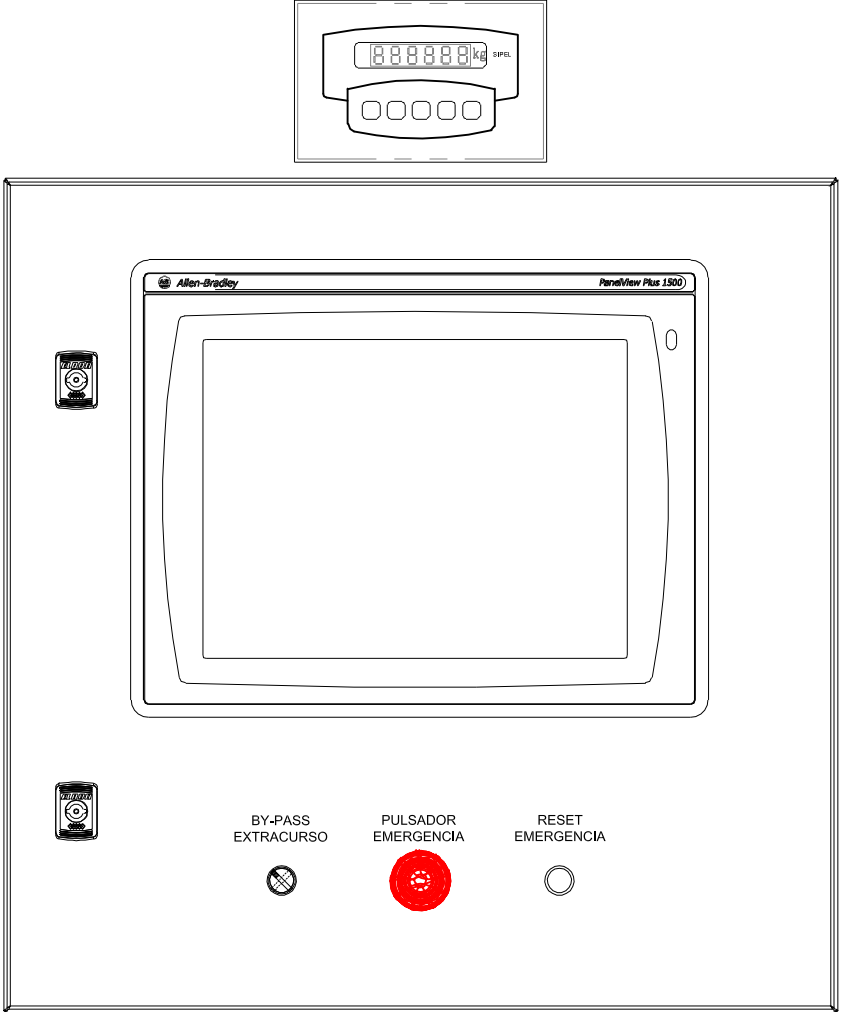
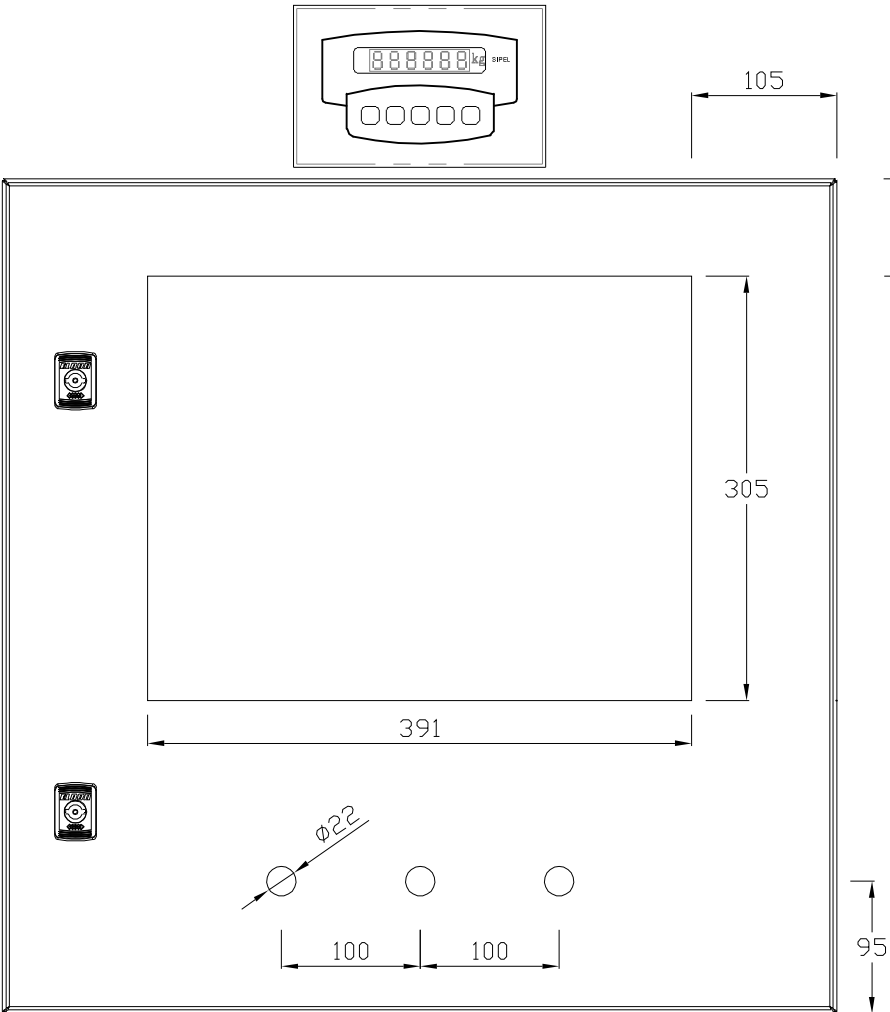
DETALLE
A



Tablero	Multipar	N° Hilo	N° Cable	Tags Bornera	Componente	Descripción
PV#1	CC34	2	019061	X0104 - 46	24VDC - Rack 1 / Slot 4	
		3	026077	X2		IN - Pulsador de Emerg. PV1
		4	026117			IN - Pulsador de Emerg. PV1
		5	026076			OUT - Pulsador de Emerg. PV1
		6	026116			OUT - Pulsador de Emerg. PV1
		7	PE03129			Pulsador de Emergencia PV1
		8	0190618	X3	PE03129	
		9	L04005		OV - Rack 1 / Slot 12	
		10	L04006		L04005	Baliza Verde
		11	018111		L04006	Baliza Roja
		12	018112		24VDC - PV #1	
		CC50	MARRON	L220		OV - PV #1
	CELESTE		N220		220VAC - LECTOR	
	NEGRO		-		OVAC - LECTOR	
A/V	-			-		

PROYECTO	SIDERCROM EMBALAJE	REVISOR	108	SHEET	108
DRAWING NO.	108	REVISOR	108	SHEET	108
REV. CLIENTE	REV. INT. P&F	CREADO POR	SAFETY	TOPOGRAFICO	TABLERO PANEL VIEW
Av. Belgrano 2272 (BOLIVIA)	Don Torcuato Buenos Aires	ARGENTINA	ARGENTINA	PLANTA SIDERCROM	LOCATION/ACTIVITY

PanelView #2 - Estación 4

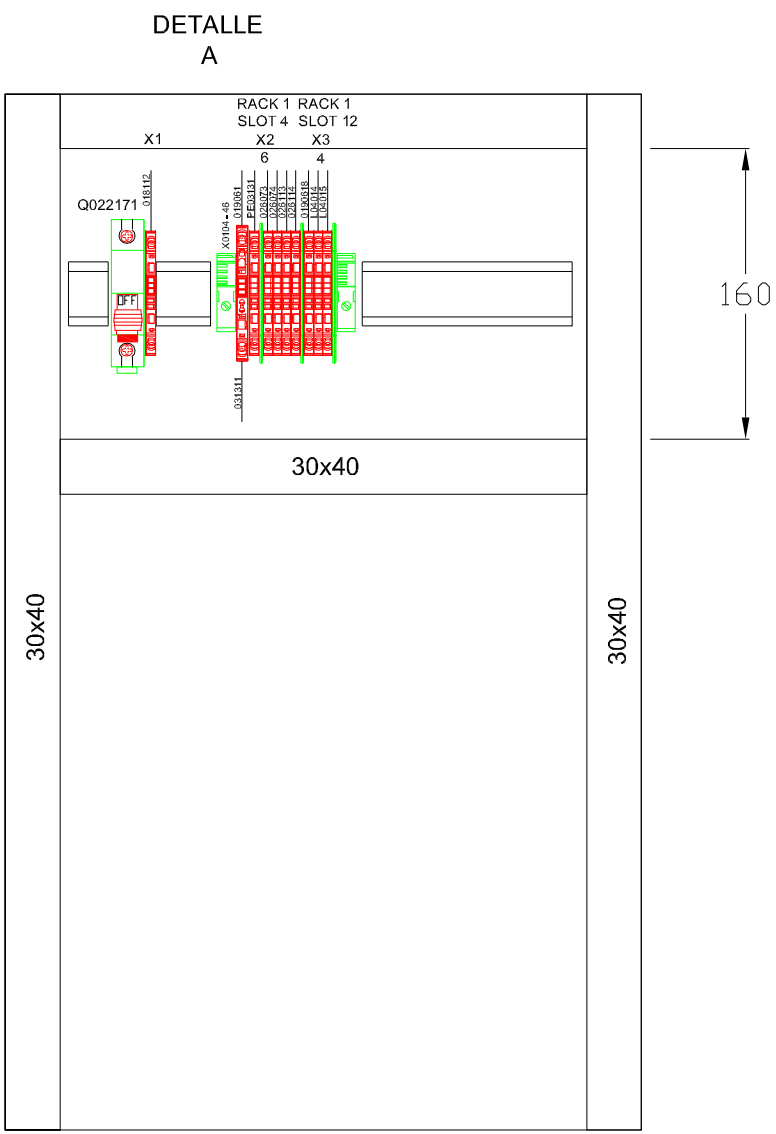
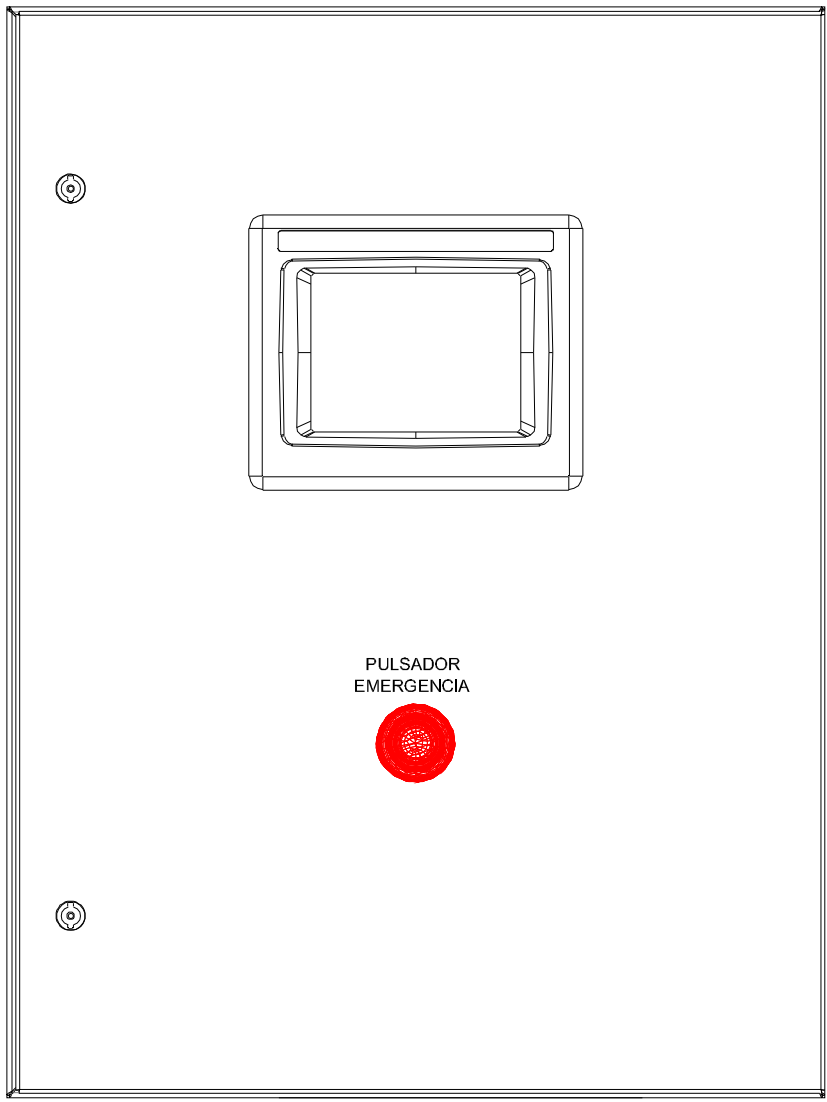
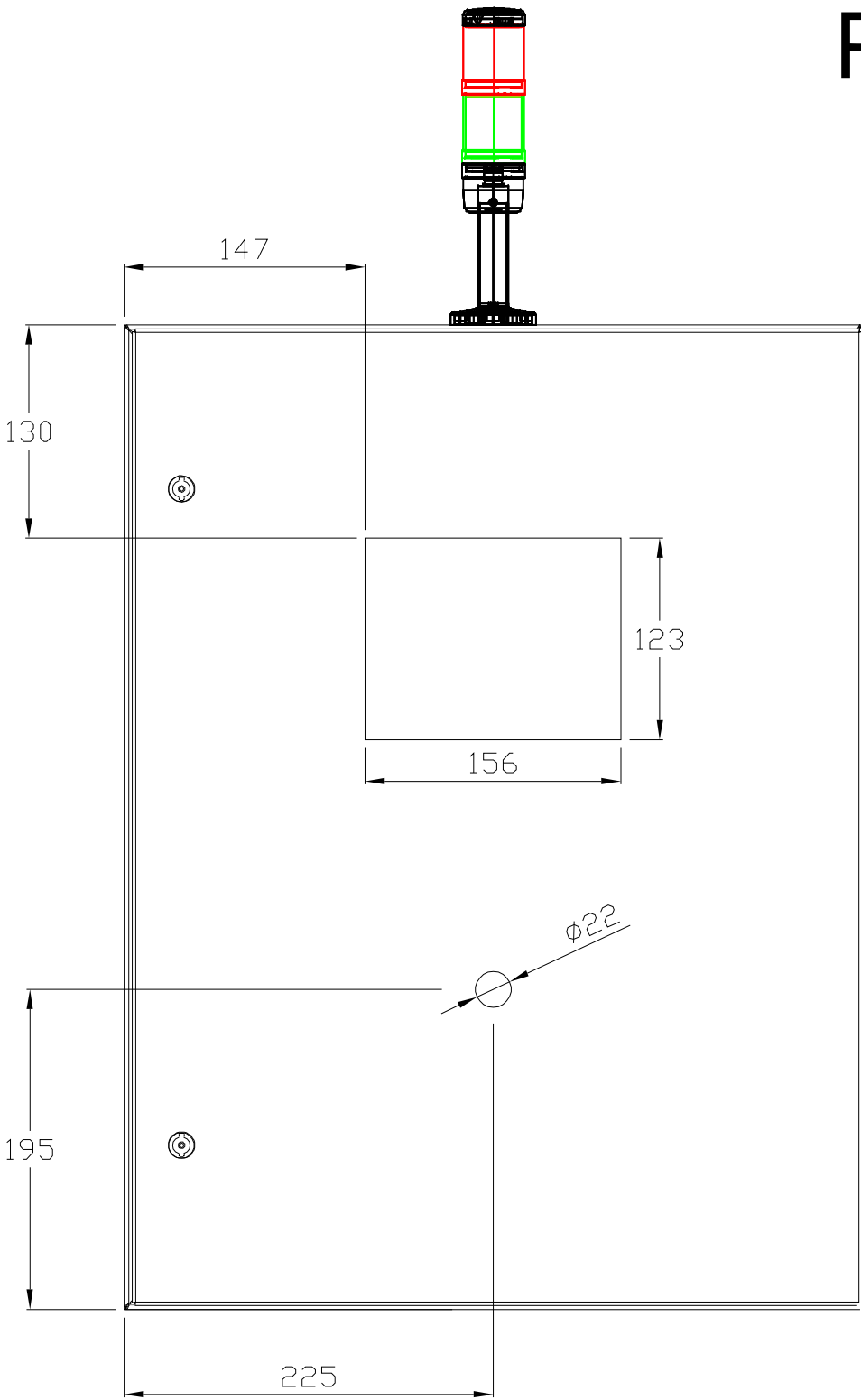


Tablero 600 x 600 x 210
Eldon MAS0606021R5

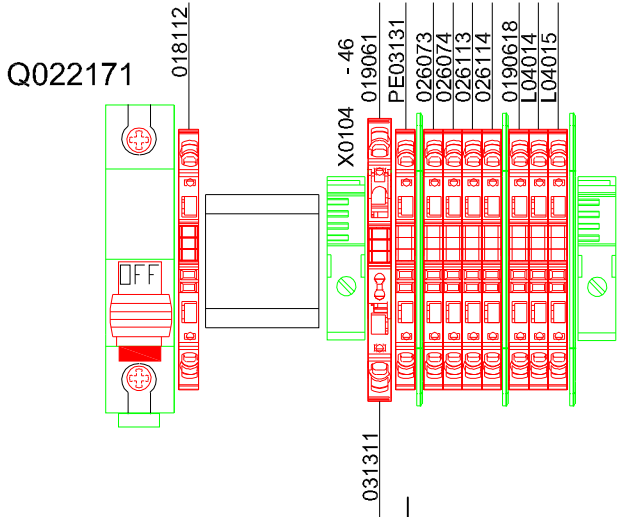
Tablero	Multipar	N° Hilo	N° Cable	Tags Bornera	Componente	Descripción
PV#2	CC35	2	019061	X0104 - 46	24VDC - Rack 1 / Slot 4	
		3	026072			IN - Pulsador de Emerg. PV2
		4	026112			IN - Pulsador de Emerg. PV2
		5	026071	X1	PE03129	OUT - Pulsador de Emerg. PV2
		6	026111			OUT - Pulsador de Emerg. PV2
		7	PE03130			Pulsador de Emergencia PV2
		8	-		RESERVA	
		9	057111			Pulsador Reset de Emerg.
		10	026161	X2	PB02616	
		11	L03907			Luz Pulsador Reset Emerg.
		12	0190616			
PV#2	CC51	MARRON	24VDC		24VDC - EXEMYS	
		CELESTE	0V		0V - EXEMYS	
		NEGRO	-		-	
		A/V	-		-	
		MARRON	L220		220VAC - LECTOR	
		CELESTE	N220		0VAC - LECTOR	
		NEGRO	-		-	
		A/V	-		-	
PV#2	CC52	MARRON	24VDC		24VDC - EXEMYS	
		CELESTE	0V		0V - EXEMYS	
		NEGRO	-		-	
		A/V	-		-	
		MARRON	L220		220VAC - LECTOR	
		CELESTE	N220		0VAC - LECTOR	
		NEGRO	-		-	
		A/V	-		-	
PV#2	CC77	2	020061			0V
		3	LR04217			BYPASS-Selectora 2 Pos.
		4	060071			
		5	060072			BYPASS
		6	071072			Movimiento Transversal Sunchadora
		7	071074			
		8	060071			
		9	060072			BYPASS
		10	071272			Elevador Puente Sunchadora
		11	071274			
		12	-		RESERVA	

PanelView #3 - Estación 6

Tablero 450 x 600 x 210
Ristal



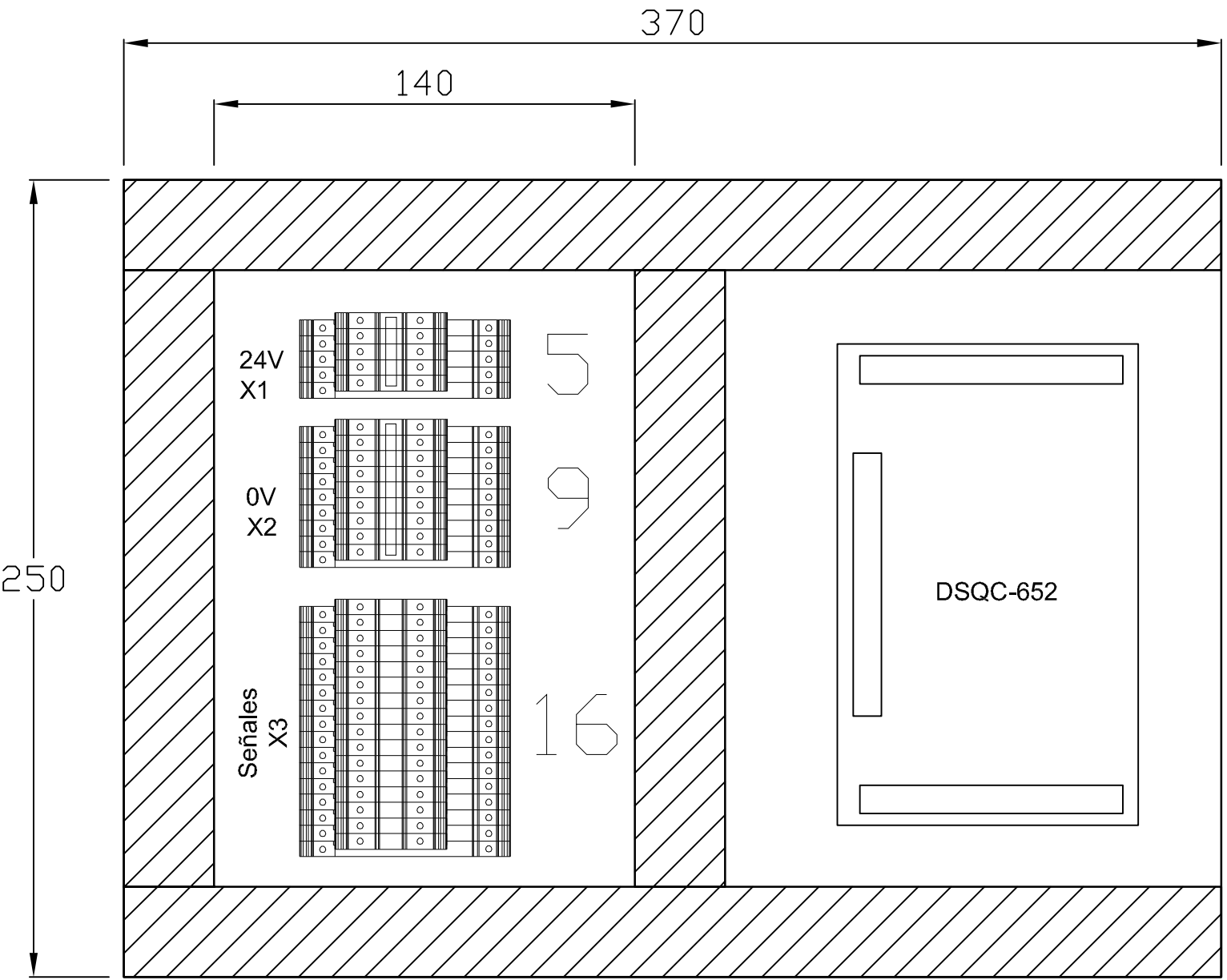
DETALLE
A



Tablero	Multipar	N° Hilo	N° Cable	Tags Bornera	Componente	Descripción
PV#3	CC36	2	019061	X0104 - 46	24VDC - Rack 1 / Slot 4	IN - Pulsador de Emerg. PV3
		3	026074	X2		IN - Pulsador de Emerg. PV3
		4	026114			OUT - Pulsador de Emerg. PV3
		5	026073			OUT - Pulsador de Emerg. PV3
		6	026113			OUT - Pulsador de Emerg. PV3
		7	PE03131	X3	PE03131	Pulsador de Emergencia PV3
		8	0190618		0V - Rack 1 / Slot 12	
		9	L04014		L04014	Baliza Verde
		10	L04015		L04015	Baliza Roja
		11	018111		24VDC - PV#3	
		12	018112		0V - PV#3	

TABLERO ROBOT I/O

JB ROBOT #1

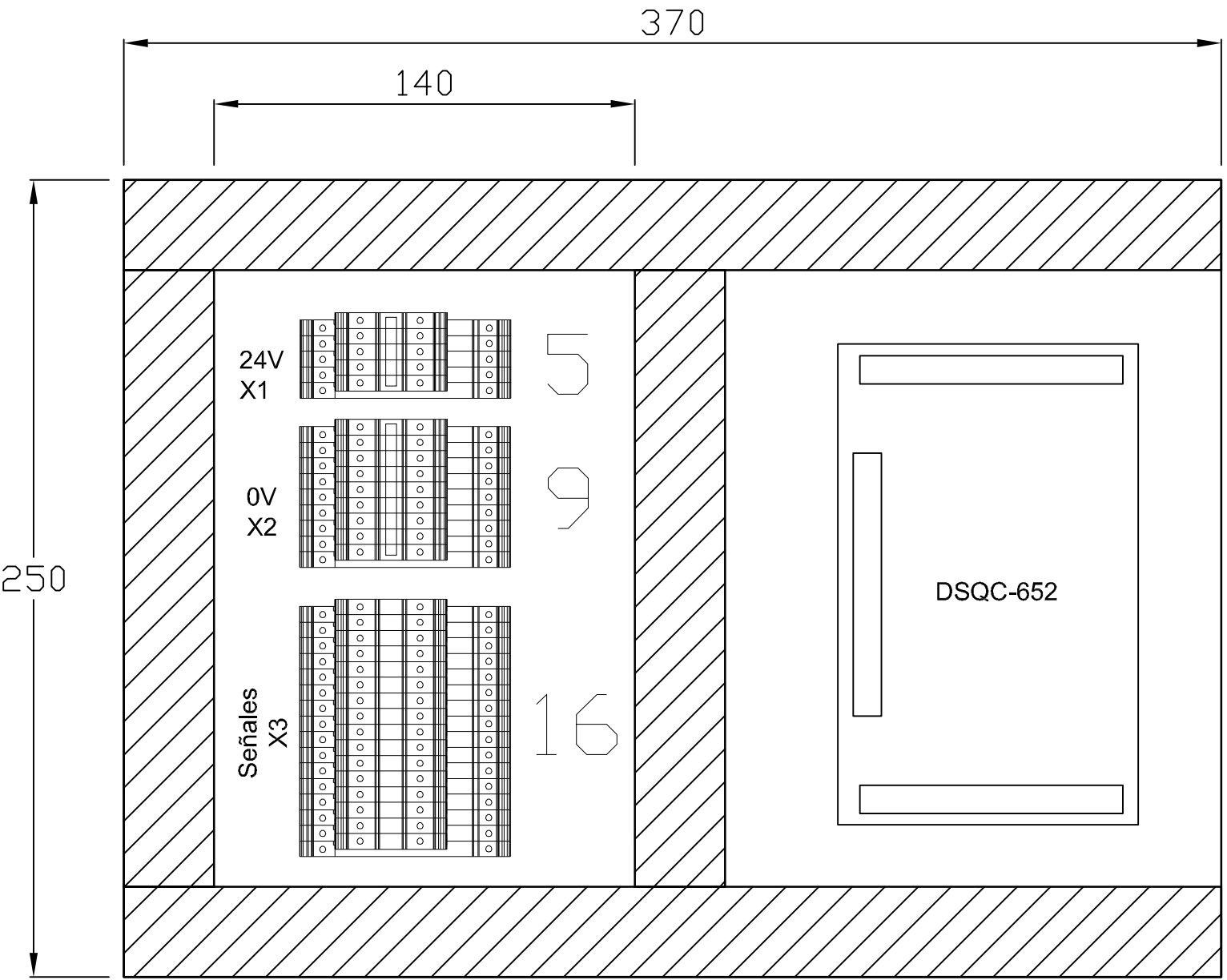


Tablero 300x400x160
Forli 0464
Cablecanal 30x70

IRC - X1	BORNERA	N° CABLE		IRC - X3	BORNERA	N° CABLE
1	X3 PISO ABAJO	ZA04900		NC	—	—
2		ZC04901		0V	X2	018112
3		ZA04902		8	X3 PISO ARRIBA	MC04925
4		ZC04903		7		MA04924
5		ZA04904		6		MC04923
6		ZC04905		5		MA04922
7		ZA04906		4		MC04921
8		ZC04907		3		MA04920
0V	X2	018112		2		MC04919
24V	X1	018111		1		MA04918
IRC - X2	BORNERA	N° CABLE		IRC - X4	BORNERA	N° CABLE
9	X3 PISO ABAJO	SA04909		NC	—	—
10		SA04910		0V	—	—
11		SA04911		16	—	04933
12		SA04912		15	—	04932
13		SX04913		14	—	04931
14		SX04914		13	—	04930
15		SX04915		12	X3 PISO ARRIBA	MC04929
16		SX04916		11		MC04928
0V		—		10		MC04927
24V		—		9		MC04926

TABLERO ROBOT I/O

JB ROBOT #2

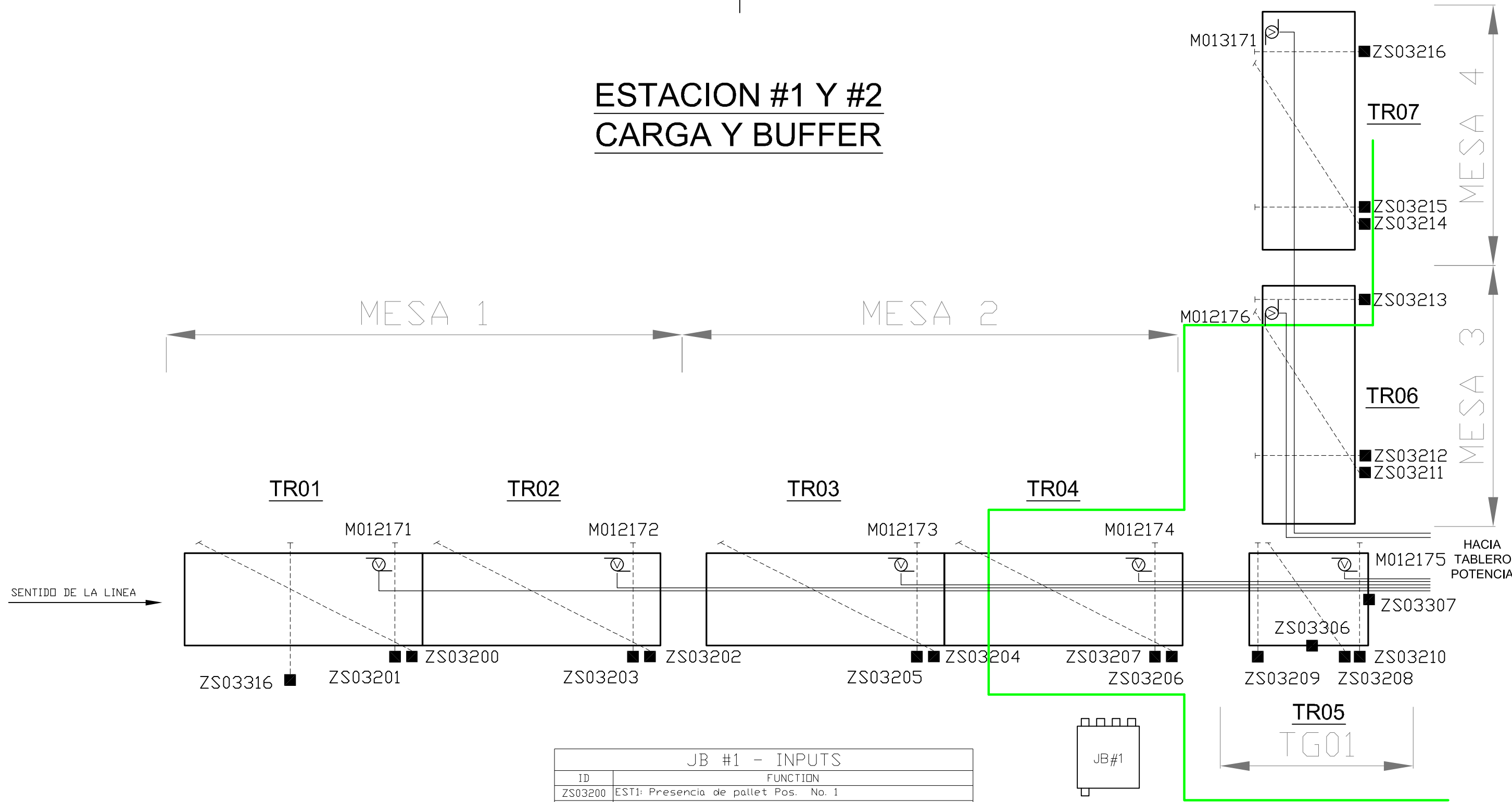


IRC - X1	BORNERA	N° CABLE		IRC - X3	BORNERA	N° CABLE
1	X3	ZA05000		NC	-	-
2	PISO ABAJO	ZA05001		0V	X2	018112
3	-	05002		8	-	05025
4	-	05003		7	-	05024
5	-	05004		6	-	05023
6	-	05005		5	-	05022
7	-	05006		4	-	05021
8	-	05007		3	-	05020
0V	X2	018112		2	X3	MC05019
24V	X1	018111		1	PISO ARRIBA	MA05018

IRC - X2	BORNERA	N° CABLE		IRC - X4	BORNERA	N° CABLE
9		SA05009		NC	-	-
10		SA05010		0V	-	-
11		SA05011		16	-	05033
12		SA05012		15	-	05032
13	X3	SA05013		14	-	05031
14	PISO ABAJO	SA05014		13	-	05030
15		SA05015		12		MA05029
16		SA05016		11	X3	MA05028
0V		-		10	PISO ARRIBA	MA05027
24V		-		9		MA05026

Tablero 300x400x160
Forli 0464
Cablecanal 30x70

ESTACION #1 Y #2
CARGA Y BUFFER



LEGEND

- LLAVE DE CORTE

MOTOR (CONSTANT SPEED)

MOTOR (ADJUSTABLE SPEED)

MOTOR (VARIABLE SPEED)

MOTOR BRAKE

SOLENOID

AUTO OFF P.B. STATION

CYLINDER SENSOR

AIR STOP

FIXED STOP

LIGHT

START/STOP P.B. STATION
- LIMIT SWITCH

■ SENSOR PRESENCIA

■---■ SENSOR OPTICO CON ESPEJO

JB# JUNCTION BOX & NUMBER

— CABLE CONNECTION & HOME RUN TO PANEL
MCx = MOTOR CABLE & NUMBER (380V)
CCx = CONTROL CABLE & NUMBER (24VDC)

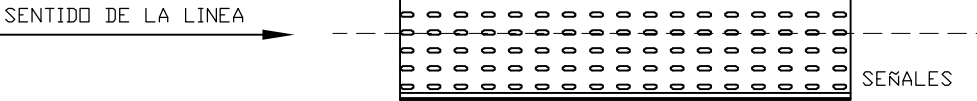
--- CONDUIT AND WIRE

— CABLE

▢ CONTROL PANEL/MIMIC STATION

▢ OPERATOR STATION

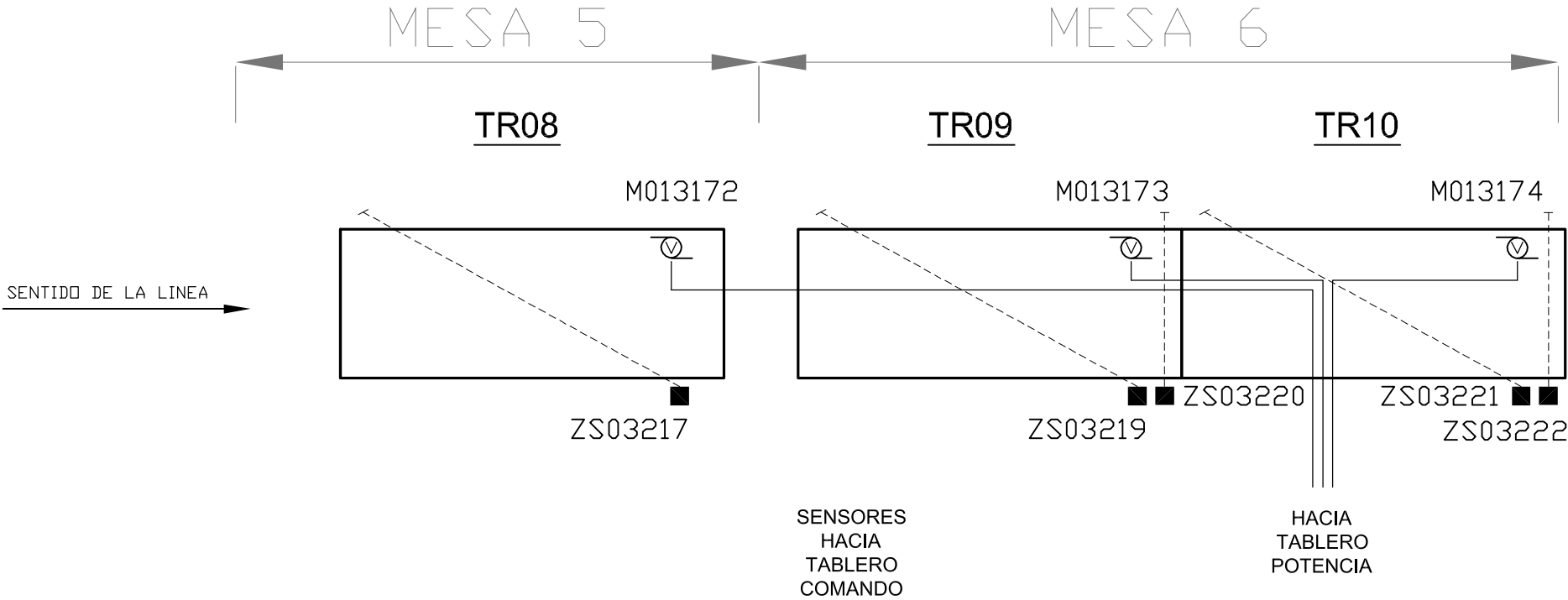
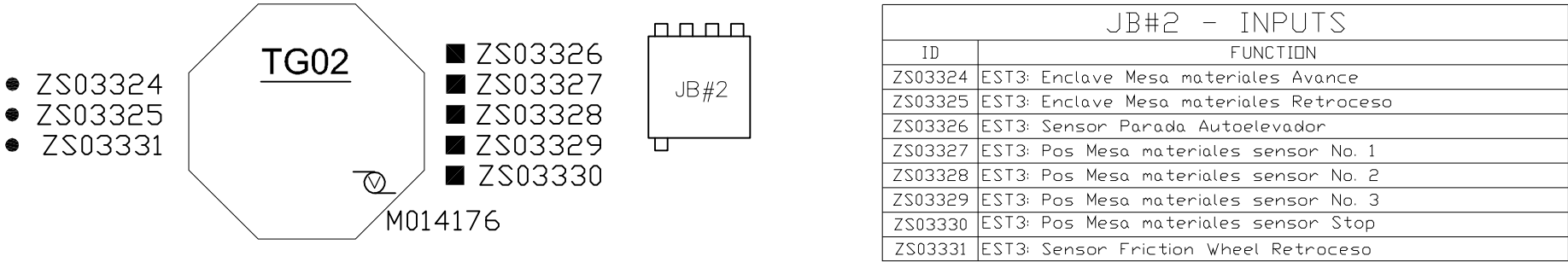
JB #1 - INPUTS	
ID	FUNCTION
ZS03200	EST1: Presencia de pallet Pos. No. 1
ZS03201	EST1: Stop de pallet Pos. No. 1
ZS03202	EST1: Presencia de pallet Pos. No. 2
ZS03203	EST1: Stop de pallet Pos. No. 2
ZS03204	EST1: Presencia de pallet Pos. No. 3
ZS03205	EST1: Stop de pallet Pos. No. 3
ZS03206	EST1: Presencia de pallet Pos. No. 4
ZS03207	EST1: Stop de pallet Pos. No. 4
ZS03208	EST2: Presencia de pallet Pos. No. 5 (TG)
ZS03209	EST2: Pos. Stop Trasero Pos. No. 5 (TG)
ZS03210	EST2: Pos. Stop Delantero Pos. No. 5 (TG)
ZS03211	EST2: Presencia de pallet Pos. No. 6
ZS03212	EST2: Stop Trasero pallet Pos. No. 6
ZS03213	EST2: Stop Delantero pallet Pos. No. 6
ZS03214	EST2: Presencia de pallet Pos. No. 7
ZS03215	EST2: Stop Trasero pallet Pos. No. 7
ZS03216	EST2: Stop Delantero pallet Pos. No. 7
ZS03316	EST1: Sensor sobre altura de entrada pallet
ZS03306	EST2: TG en Pos. de Línea
ZS03307	EST2: TG en Pos. de Descarte



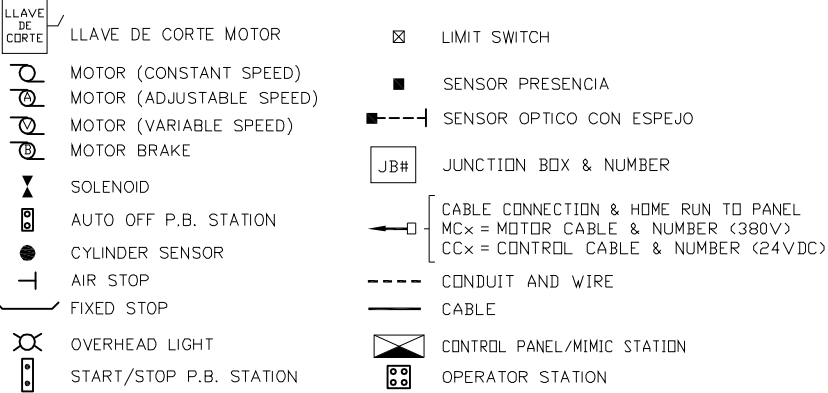
5	5	PROJECT	SIDERCROM EMBALAJE
4	4	DRAWING NO.	114
3	3	REVISOR BY	REVISOR BY
2	2	DESIGNER BY	DESIGNER BY
1	1	CHECKED BY	CHECKED BY
REV. CLIENTE	REV. INT. P&F	Av. Belgrano 2272 (BOLIVAR)	ELECTRICAL INSTALLATION
P&F	Don Torcuato Buenos Aires	Argentina	EST #1 Y #2
JOB	DATE	13/10/2015	DIVISION
			PLANTA SIDEROCROM
			LOCATION/ACTIVITY

ESTACION #3 Y #4

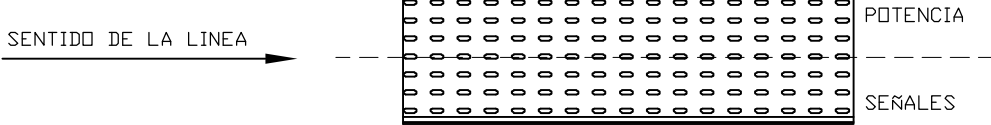
ESTRECHADO - BUFFER - MESA GIRATORIA DE MATERIALES



LEGEND

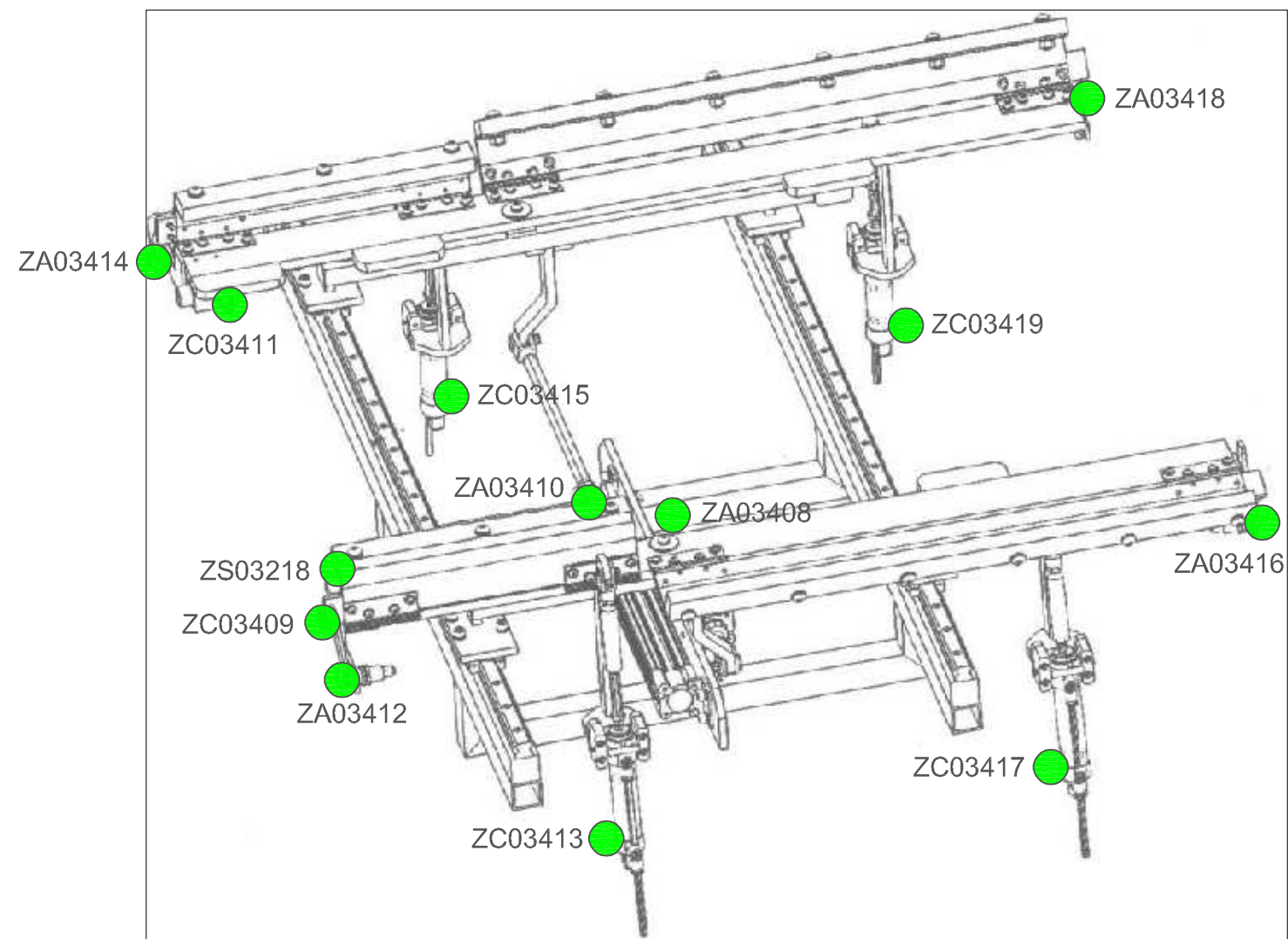


TBP COMANDO - INPUTS	
ID	FUNCTION
ZS03217	EST3: Presencia de pallet Pos. No. 8
ZS3219	EST4: Presencia de pallet Pos. No. 9
ZS3220	EST4: Stop de pallet Pos. No. 9
ZS3221	EST4: Presencia de pallet Pos. No. 10
ZS3222	EST4: Stop de pallet Pos. No. 10



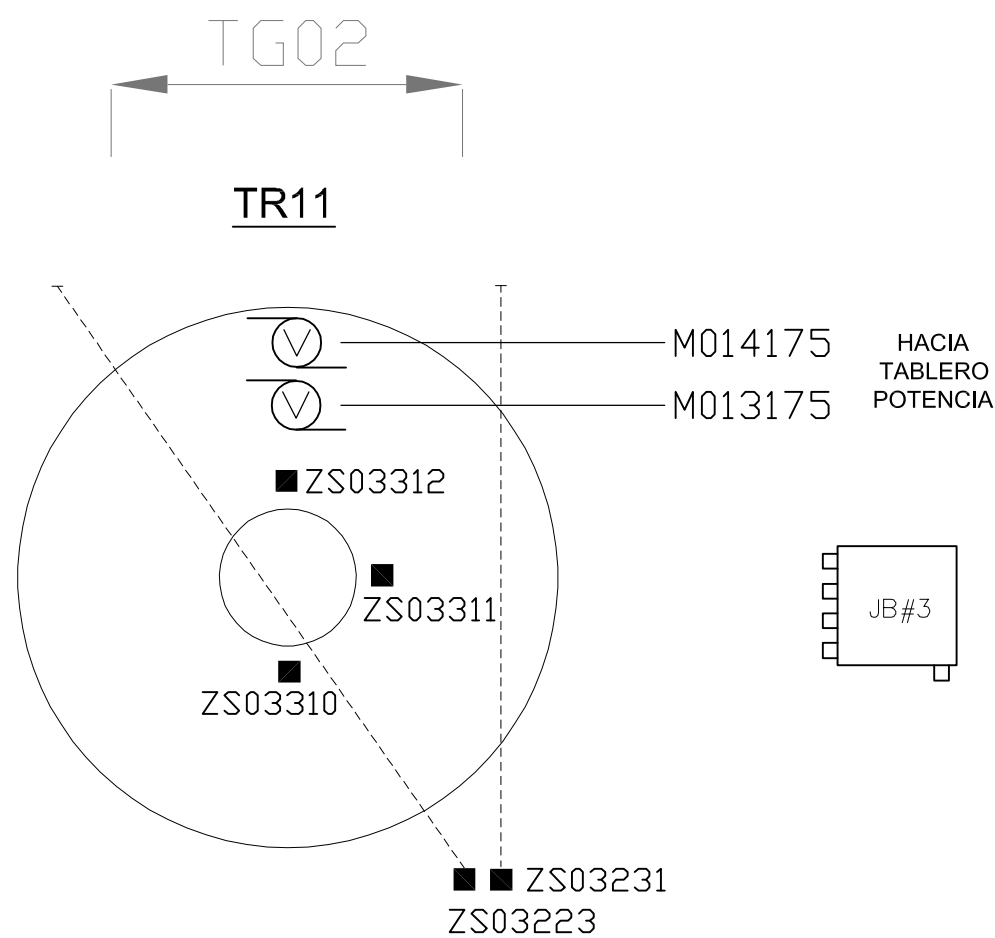
5	5	PROJECT	SIDERCROM EMBALAJE
4	4	DRAWING NO.	115
3	3	DESIGN BY	DESIGN BY
2	2	REV. CLIENTE	REV. INT. P&F
1	1	ORDER BY	ORDER BY
		Av. Belgrano 2272 CHESLUPVD	Av. Belgrano 2272 CHESLUPVD
		Bon Torcuato Buenos Aires	Bon Torcuato Buenos Aires
		DATE	DATE
		29/14/215	29/14/215
		DIVISION	DIVISION
		PLANTA SIDERCRON	PLANTA SIDERCRON
		LOCATION/ACTIVITY	LOCATION/ACTIVITY

ESTACION #3
RETENEDOR DE ESQUINEROS



Sensores	Descripcion
ZS03218	EST3: Stop de pallet Pos. No. 8
ZS03408	EST3: Carro Sujetador Esqueros Izquierdo Abierto
ZC03409	EST3: Carro Sujetador Esqueros Izquierdo Cerrado
ZA03410	EST3: Carro Sujetador Esqueros Derecho Abierto
ZC03411	EST3: Carro Sujetador Esqueros Derecho Cerrado
ZA03412	EST3: Sujetador Esqueros #1 Abierto
ZC03413	EST3: Sujetador Esqueros #1 Cerrado
ZA03414	EST3: Sujetador Esqueros #2 Abierto
ZC03415	EST3: Sujetador Esqueros #2 Cerrado
ZA03416	EST3: Sujetador Esqueros #3 Abierto
ZC03417	EST3: Sujetador Esqueros #3 Cerrado
ZA03418	EST3: Sujetador Esqueros #4 Abierto
ZC03419	EST3: Sujetador Esqueros #4 Cerrado

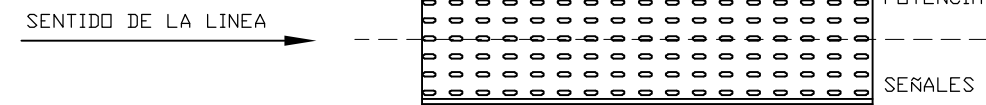
ESTACION #5
TG SUNCHADORA



LEGEND

	LLAVE DE CORTE MOTOR		LIMIT SWITCH
	MOTOR (CONSTANT SPEED)		SENSOR PRESENCIA
	MOTOR (ADJUSTABLE SPEED)		SENSOR OPTICO CON ESPEJO
	MOTOR (VARIABLE SPEED)		JUNCTION BOX & NUMBER
	MOTOR BRAKE		CABLE CONNECTION & HOME RUN TO PANEL MCx = MOTOR CABLE & NUMBER (380V) CCx = CONTROL CABLE & NUMBER (24VDC)
	SOLENOID		CONDUIT AND WIRE
	AUTO OFF P.B. STATION		CABLE
	CYLINDER SENSOR		CONTROL PANEL/MIMIC STATION
	AIR STOP		OPERATOR STATION
	FIXED STOP		
	OVERHEAD LIGHT		
	START/STOP P.B. STATION		

JB #3 - INPUTS	
ID	FUNCTION
ZS03223	EST5: Presencia de pallet Pos. No. 11 (TG)
ZS03231	EST5: Stop pallet corto No. 11 (TG)
ZS03310	EST5: Transporte giratorio sensor N°1
ZS03311	EST5: Transporte giratorio sensor N°2
ZS03312	EST5: Transporte giratorio sensor N°3
ZS03313	EST5: Transporte giratorio sensor parada



5	5	PROJECT	SIDERCROM EMBALAJE	
4	4	DRAWING NO.	117	SHEET SHEETS
3	3			117 1131
2	2	DESIGN BY: DESIGNED BY: TITLE		
1	1	REV. CLIENTE	REV. INT. P&F	ELECTRICAL INSTALLATION
			Av. Belgrano 2272 CHESUPVD Buenos Aires	TG SUNCHADORA
		CHECKED BY: D.F.	SAFETY BY:	
		DATE: 29/14/2015	SCALE:	DIVISION: PLANTA SUNCHADORA
				LOCATION/ACTIVITY:

ESTACION #5 Y #6
ZUNCHACHADO - BUFFER

MESA 7 MESA 8 MESA 9

SENTIDO DE LA LINEA →

HACIA TABLERO POTENCIA

TR12 TR13 TR14 TR15 TR16

M013176 M014171 M014172 M014173 M014174

ZS03225 ZS03226 ZS03227 ZS03228 ZS03229 ZS03230 ZS03300 ZS03301 ZS03302 ZS03303

JB#3

LEGEND

- LLAVE DE CORTE MOTOR
- MOTOR (CONSTANT SPEED)
- MOTOR (ADJUSTABLE SPEED)
- MOTOR (VARIABLE SPEED)
- MOTOR BRAKE
- SOLENOID
- AUTO OFF P.B. STATION
- CYLINDER SENSOR
- AIR STOP
- FIXED STOP
- OVERHEAD LIGHT
- START/STOP P.B. STATION
- LIMIT SWITCH
- SENSOR PRESENCIA
- SENSOR OPTICO CON ESPEJO
- JUNCTION BOX & NUMBER
- CABLE CONNECTION & HOME RUN TO PANEL
MCx = MOTOR CABLE & NUMBER (380V)
CCx = CONTROL CABLE & NUMBER (24VDC)
- CONDUIT AND WIRE
- CABLE
- CONTROL PANEL/MIMIC STATION
- OPERATOR STATION

JB #3 - INPUTS

ID	FUNCTION
ZS03225	EST5: Presencia de pallet Pos. No. 12
ZS03226	EST5: Stop de pallet Pos. No. 12
ZS03227	EST6: Presencia de pallet Pos. No. 13
ZS03228	EST6: Stop de pallet Pos. No. 13
ZS03229	EST6: Presencia de pallet Pos. No. 14
ZS03230	EST6: Stop de pallet Pos. No. 14
ZS03300	EST6: Presencia de pallet Pos. No. 15
ZS03301	EST6: Stop de pallet Pos. No. 15
ZS03302	EST6: Presencia de pallet Pos. No. 16
ZS03303	EST6: Stop de pallet Pos. No. 16

SENTIDO DE LA LINEA →

POTENCIA

SEÑALES

PROYECT: SIDEROCROM EMBALAJE

DRAWING NO: 118

DESIGNER BY: J.L.H. CHECKED BY: D.F. DATE: 20/14/2015

REV. CLIENTE REV. INT. P&F

Av. Belgrano 2272 (ENGLUVD) San Torcuato Buenos Aires Argentina.

ELECTRICAL INSTALLATION EST #5 Y #6

DIVISION: PLANTA SIDEROCROM LOCATION/ACTIVITY:

ESTACION #5 Y #6
ZUNCHACHADO - BUFFER

MESA 7 MESA 8 MESA 9

SENTIDO DE LA LINEA →

HACIA TABLERO POTENCIA

TR12 TR13 TR14 TR15 TR16

M013176 M014171 M014172 M014173 M014174

ZS03225 ZS03226 ZS03227 ZS03228 ZS03229 ZS03230 ZS03300 ZS03301 ZS03302 ZS03303

JB#3

JB #3 - INPUTS	
ID	FUNCTION
ZS03225	EST5: Presencia de pallet Pos. No. 12
ZS03226	EST5: Stop de pallet Pos. No. 12
ZS03227	EST6: Presencia de pallet Pos. No. 13
ZS03228	EST6: Stop de pallet Pos. No. 13
ZS03229	EST6: Presencia de pallet Pos. No. 14
ZS03230	EST6: Stop de pallet Pos. No. 14
ZS03300	EST6: Presencia de pallet Pos. No. 15
ZS03301	EST6: Stop de pallet Pos. No. 15
ZS03302	EST6: Presencia de pallet Pos. No. 16
ZS03303	EST6: Stop de pallet Pos. No. 16

SENTIDO DE LA LINEA →

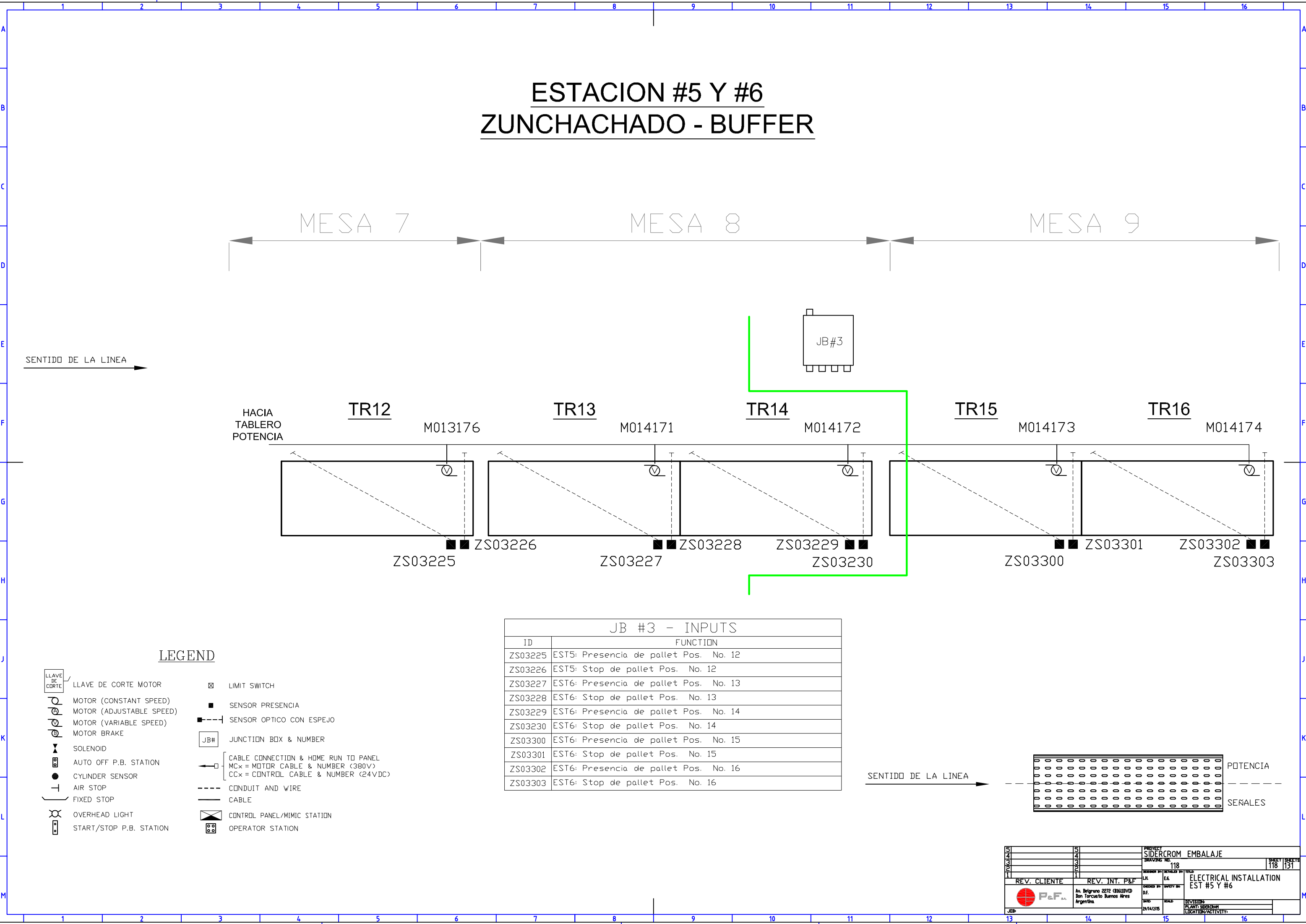
POTENCIA

SEÑALES

LEGEND

- LLAVE DE CORTE MOTOR
- MOTOR (CONSTANT SPEED)
- MOTOR (ADJUSTABLE SPEED)
- MOTOR (VARIABLE SPEED)
- MOTOR BRAKE
- SOLENOID
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- CYLINDER SENSOR
- AIR STOP
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- OVERHEAD LIGHT
- START/STOP P.B. STATION
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- SENSOR PRESENCIA
- SENSOR OPTICO CON ESPEJO
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- CABLE CONNECTION & HOME RUN TO PANEL
MCx = MOTOR CABLE & NUMBER (380V)
CCx = CONTROL CABLE & NUMBER (24VDC)
- CONDUIT AND WIRE
- CABLE
- CONTROL PANEL/MIMIC STATION
- OPERATOR STATION

PROYECTO: SIDERCROM EMBALAJE
DRAWING NO.: 118
SHEET 118 OF 131
REV. CLIENTE
REV. INT. P&F
DESIGNER BY: J.L.H.
CHECKED BY: D.F.
DATE: 20/14/2015
E.E.
SAFETY NO.
DIVISION: PLANTA SIDERCROM
LOCATION/ACTIVITY:



ESTACION #5 Y #6
ZUNCHACHADO - BUFFER

MESA 7 MESA 8 MESA 9

SENTIDO DE LA LINEA →

HACIA TABLERO POTENCIA

TR12 TR13 TR14 TR15 TR16

M013176 M014171 M014172 M014173 M014174

ZS03225 ZS03226 ZS03227 ZS03228 ZS03229 ZS03230 ZS03300 ZS03301 ZS03302 ZS03303

JB#3

LEGEND

- LLAVE DE CORTE MOTOR
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- JUNCTION BOX & NUMBER
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MCx = MOTOR CABLE & NUMBER (380V)
CCx = CONTROL CABLE & NUMBER (24VDC)
- CONDUIT AND WIRE
- CABLE
- CONTROL PANEL/MIMIC STATION
- OPERATOR STATION

JB #3 - INPUTS

ID	FUNCTION
ZS03225	EST5: Presencia de pallet Pos. No. 12
ZS03226	EST5: Stop de pallet Pos. No. 12
ZS03227	EST6: Presencia de pallet Pos. No. 13
ZS03228	EST6: Stop de pallet Pos. No. 13
ZS03229	EST6: Presencia de pallet Pos. No. 14
ZS03230	EST6: Stop de pallet Pos. No. 14
ZS03300	EST6: Presencia de pallet Pos. No. 15
ZS03301	EST6: Stop de pallet Pos. No. 15
ZS03302	EST6: Presencia de pallet Pos. No. 16
ZS03303	EST6: Stop de pallet Pos. No. 16

SENTIDO DE LA LINEA →

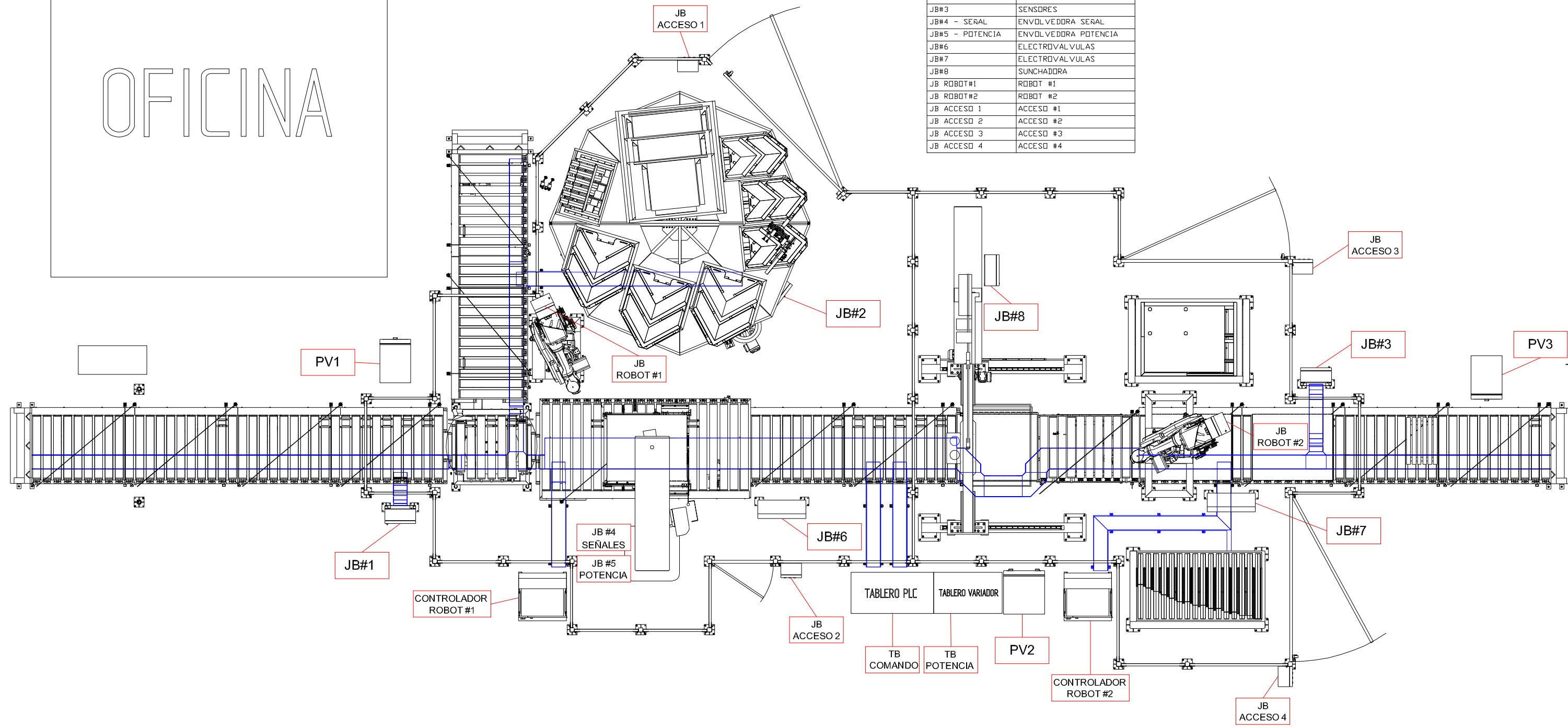
POTENCIA

SEÑALES

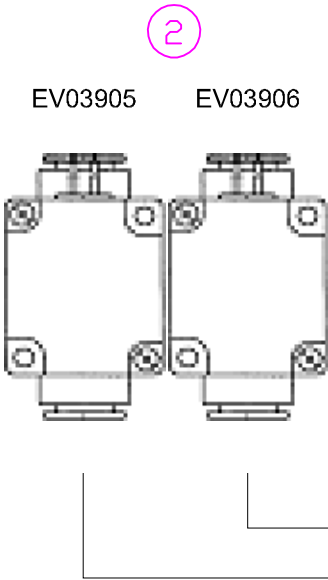
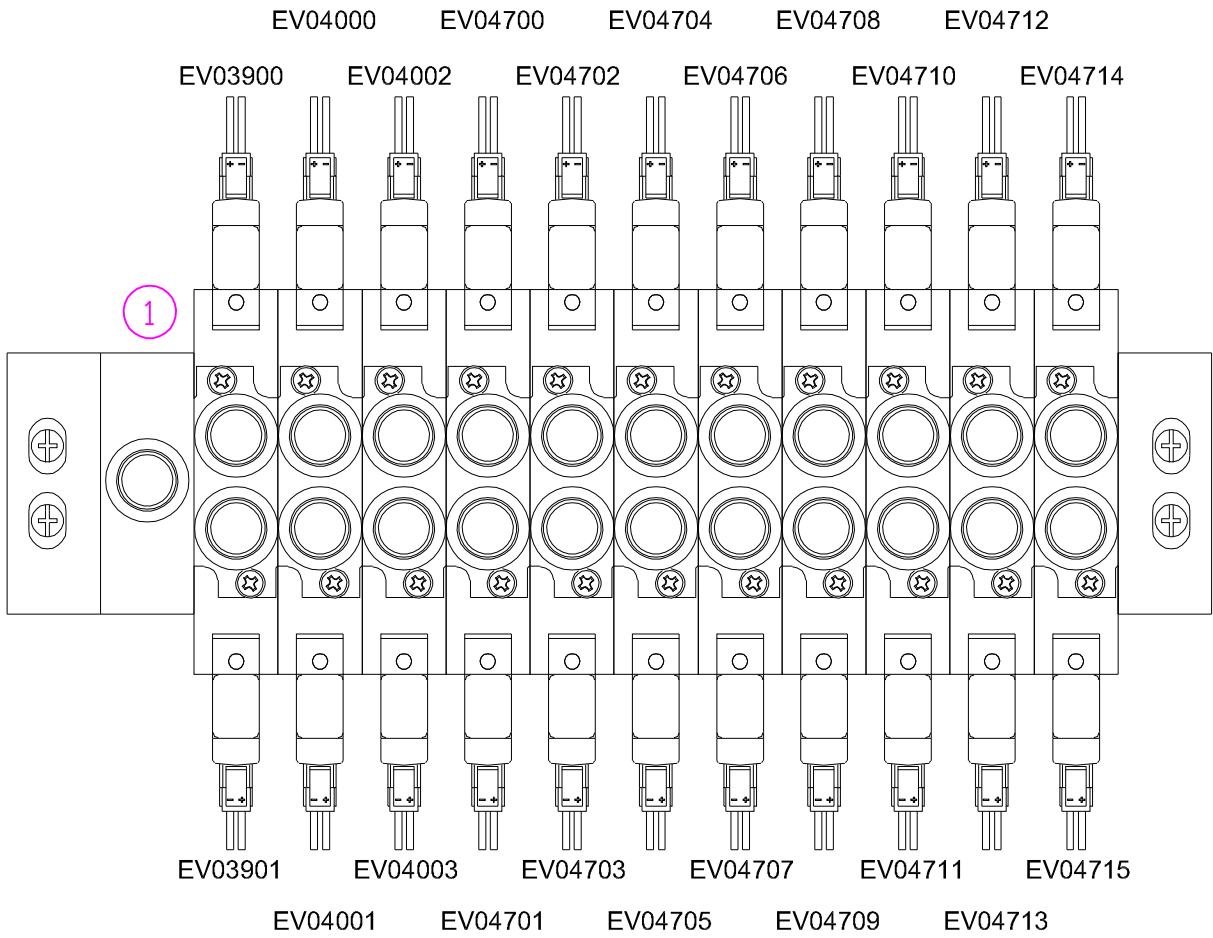
PROYECT: SIDERCROM EMBALAJE
DRAWING NO: 118
SHEET 118 OF 131
REV. CLIENTE REV. INT. P&F
DESIGN BY: J.L. E.A. CHECKED BY: D.F. SAFETY BY:
DATE: 29/11/2015
JOB:
PLANT: SIDERCROM
LOCATION/ACTIVITY:
ELECTRICAL INSTALLATION
EST #5 Y #6

OFICINA

ID	TABLERO
TB COMANDO	COMANDO
TB POTENCIA	POTENCIA
PV1	PANELVIEW #1
PV2	PANELVIEW #2
PV3	PANELVIEW #3
JB#1	SENSORES
JB#2	SENSORES
JB#3	SENSORES
JB#4 - SEÑAL	ENVOLVEDORA SEÑAL
JB#5 - POTENCIA	ENVOLVEDORA POTENCIA
JB#6	ELECTROVALVULAS
JB#7	ELECTROVALVULAS
JB#8	SUNCHADORA
JB ROBOT#1	ROBOT #1
JB ROBOT#2	ROBOT #2
JB ACCESO 1	ACCESO #1
JB ACCESO 2	ACCESO #2
JB ACCESO 3	ACCESO #3
JB ACCESO 4	ACCESO #4



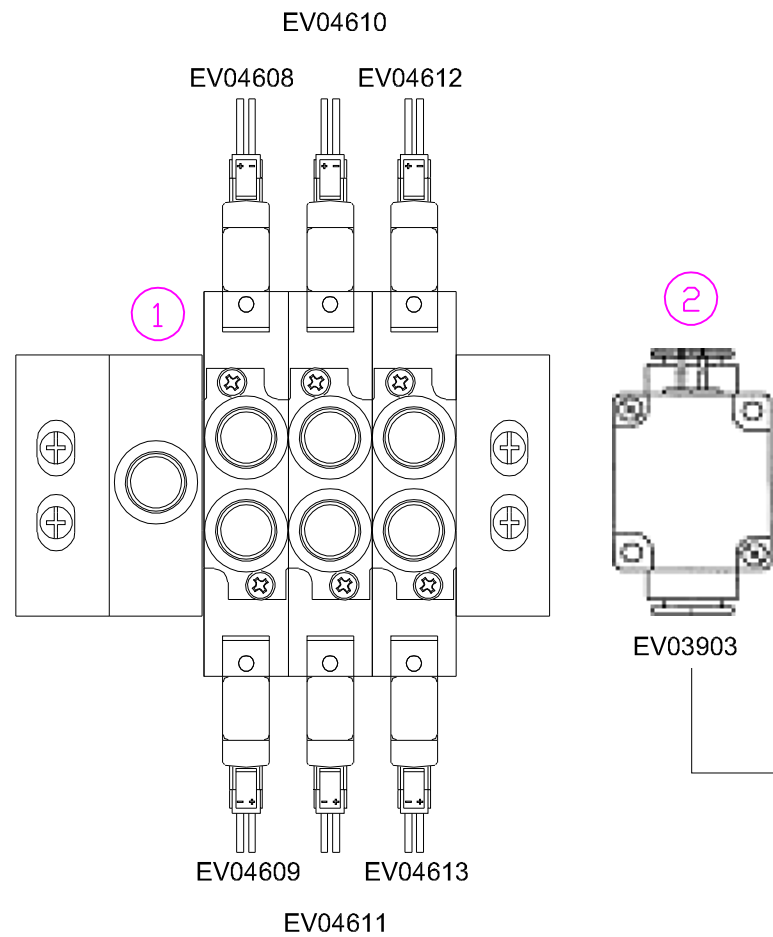
NEUMÁTICA - ESTACION #3



- EST2:TG Giro 90°
- EST3: Mesa Materiales Friction Wheel
- EST3: Enclave Mesa materiales
- EST3: Cabezal corte
- EST3: Cabezal film
- EST3: Carro sujetador esquineros izquierdo
- EST3: Carro sujetador esquineros derecho
- EST3: Presionador Esquinero No. 1
- EST3: Presionador Esquinero No. 2
- EST3: Presionador Esquinero No. 3
- EST3: Presionador Esquinero No. 4

ITEM	DESCRIPCION	N° CATALOGO	FABRICANTE
1	Electrovalvula 5/2	SY7260-5LZ-02	SMC
2	Electrovalvula 2/2	VQ21A1-5G-C6-F	SMC

NEUMÁTICA - ESTACION #5



EST5:Soplador de etiquetas

EST5: Presionador Esquinero No. 1

EST5: Presionador Esquinero No. 2

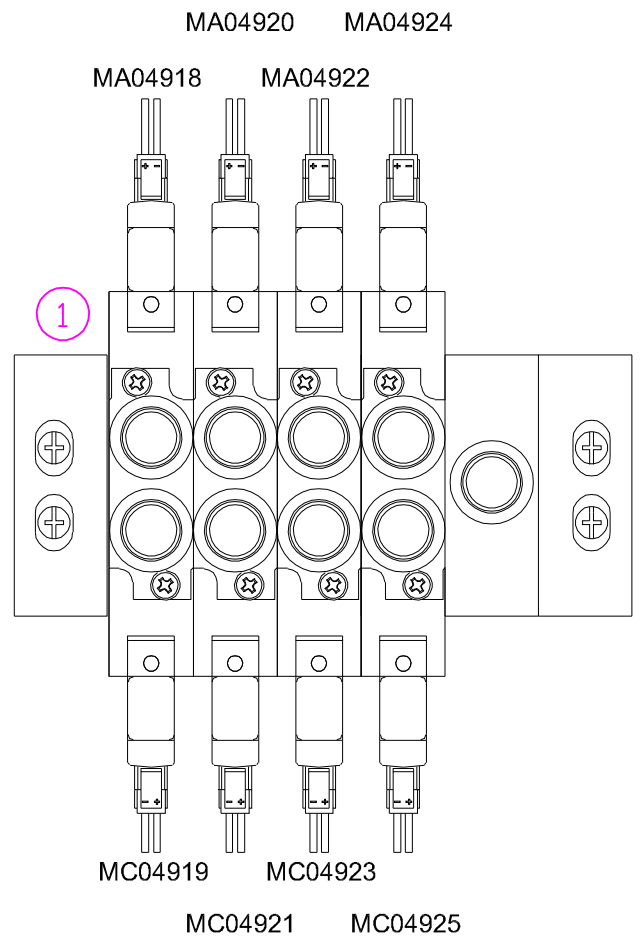
EST5: Friction Wheel Sunchadora

ITEM	DESCRIPCION	N° CATALOGO	FABRICANTE
1	Electrovalvula 5/2	SY7260-5LZ-02	SMC
2	Electrovalvula 2/2	VQ21A1-5G-C6-F	SMC

5	5	PROJECT	SIDERCROM EMBALAJE
4	4	DRAWING NO.	123
3	3	REVISION	123 1131
2	2	REVISION	123 1131
1	1	REVISION	123 1131
REV. CLIENTE	REV. INT. P&F	CHK.	E.G.
Av. Belgrano 2272 (BOLIVAR)	Don Torcuato Buenos Aires	CHKED BY	SAFETY NO.
ARGENTINA		DATE	20/11/2015
JOB		FIELD	
		DIVISION	PLANTA SIDERCROM
		LOCATION/ACTIVITY	

NEUMÁTICA - GRIPPER

ROBOT #1

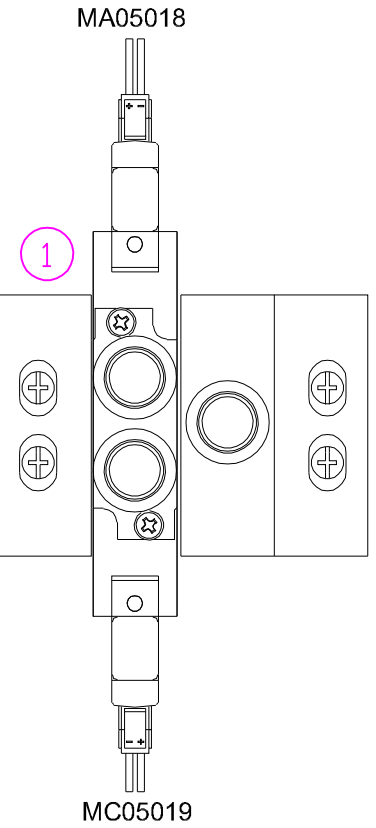


- Cilindro Pivotante N°1
- Cilindro Pivotante N°2
- Presionador Papel
- Extensor Gripper

ITEM	DESCRIPCION	N° CATALOGO	FABRICANTE
1	Electrovalvula 5/2	SY7260-5LZ-02	SMC

5	5	PROJECT	SIDERCROM EMBALAJE
4	4	DRAWING NO.	124
3	3	DESIGNED BY	124
2	2	DETAILS BY	124
1	1	REV. CLIENTE	REV. INT. P&F
		Av. Belgrano 2272 (BELLEROS)	Don Torcuato Buenos Aires
		ARGENTINA	ARGENTINA
		DATE	20/04/2015
		SOLD	
		DIVISION	PLANTA SIDERCROM
		LOCATION/ACTIVITY	

NEUMÁTICA - GRIPPER
ROBOT #2



Cilindro Pivotante N°1

ITEM	DESCRIPCION	N° CATALOGO	FABRICANTE
1	Electrovalvula 5/2	SY7260-5LZ-02	SMC

PROJECT: SIDERCROM EMBALAJE		DRAWING NO. 125		SHEET 1131	
DESIGN BY: J.L.		CHECKED BY: D.F.		DATE: 29/10/2015	
REV. CLIENTE		REV. INT. P&F		DIVISION: PLANTA SIDERCROM	
P&F		Av. Belgrano 2272 C1121JVD Buenos Aires, Argentina		LOCATION/ACTIVITY:	

BILL OF MATERIAL

PROJECT SIDERCROM EMBALAJE	
DRAWING NO. 130	
SHEET SHEETS 130 131	
DESIGNED BY LH	DETAILS BY EA
CHECKED BY D.F.	SAFETY BY D.F.
DATE 29/14/2015	SCALE DIVISION PLANTA SIDERCROM LOCATION/ACTIVITY
REV. CLIENTE	
REV. INT. P&F	
Av. Belgrano 2272 C1810JVD Don Torcuato Buenos Aires Argentina	
P&F S.A.	

ITEM	DESCRIPCION 1	DESCRIPCION 2	FABRICANTE	CATÁLOGO	CANTIDAD	STOCK RECOMENDADO
1	MODULE I/O ROBOT		ABB	DSQC 652	2	-
2	GUARDAMOTOR	MPCB, GM 1.6a 2.5 A	Allen Bradley	140M-C2E-B25	7	1
3	GUARDAMOTOR	MPCB, GM 2.5 - 4 A	Allen Bradley	140M-C2E-B40	16	3
4	GUARDAMOTOR	MPCB, GM 6,3a 10 A	Allen Bradley	140M-C2E-C10	5	1
5	INTERRUPTOR 100A		Allen Bradley	194E-A100-1756	1	-
6	CONTACTOR	MCS-C CONTACTOR, IEC, 9A, 24V DC	Allen Bradley	100-C09EJ10	33	5
7	SWITCH PUERTA	GUARD LOCKING SWITCH - 440G-MT, 24VDC	Allen Bradley	440G-MT47044	4	1
8	RELE 24 VDC		Allen Bradley	700HC24Z24	7	1
9	LED VERDE		Allen Bradley	800F-N3G	4	-
10	LED ROJO		Allen Bradley	800F-N3R	4	-
11	LED AZUL		Allen Bradley	800F-N3W	1	-
12	BALIZA ROJA		Allen Bradley	855E-24TL4	7	-
13	BALIZA AMBAR		Allen Bradley	855E-24TL5	1	-
14	BALIZA VERDE		Allen Bradley	855E-24TL3	3	-
15	PULSADOR CIEGO	800F PUSH BUTTON - METAL, FLUSH, BLACK	Allen Bradley	800FM-F2	7	-
16	PULSADOR LUMINOSO	800F MOMENTARY PUSH BUTTON - METAL, ILLUMINATED, FLUSH, YELLOW	Allen Bradley	800FM-LF5	4	-
17	PULSADOR LUMINOSO	800F MOMENTARY PUSH BUTTON - METAL, ILLUMINATED, FLUSH, CLEAR	Allen Bradley	800FM-LF7	4	-
18	PARADA EMERGENCIA	800F ILLUMINATED MUSHROOM OPERATORS	Allen Bradley	800FM-LMT44	7	-
19	MODULO ENTRADA		Allen Bradley	1756-IB32	7	1
20	MODULO SALIDA		Allen Bradley	1756-OW16I	6	1
21	FUENTE ALIMENTACION 24VDC, 10A	POWER SUPPLY 24-28V DC, 240 W, 120/240V AC INPUT VOLTAGE	Allen Bradley	1606-XLE240EP	4	1
22	MODULO ETHERNET	ETHERNET 10-100M INTERFACE MODULE	Allen Bradley	1756-EN2T	2	1
23	MODULO ETHERNET	ETHERNET DUAL PORT 10-100M INTERFACE MODULE	Allen Bradley	1756-EN2TR	2	1
24	FUENTE PLC	19-32 VDC FUENTE DE ALIMENTACION (10 Amp @ 5V)	Allen Bradley	1756-PB72	2	1
25	CHASSIS 13 SLOTS	13 SLOT CONTROLLOGIX CHASSIS	Allen Bradley	1756-A13	2	-
26	CONTROLADOR PLC	CONTROLADOR CONTROLLOGIX5572 CON 4 MB DE MEMORIA	Allen Bradley	1756-L72	1	1
27	PANEL VIEW PLUS 1500	PANELVIEW PLUS 6 TERMINAL, MODELO 1500, PANTALLA TACTIL, COLOR	Allen Bradley	2711F-T15C4A8	1	-
28	PANEL VIEW PLUS 6	PANELVIEW PLUS 6 TERMINAL, MODELO 600, PANTALLA TACTIL, COLOR	Allen Bradley	2711F-T6C20D8	2	-
29	SWITCH ETHERNET	STRATIX 6000SWITCH, DE NIVEL DE ENTRADA GESTIONADO, 8 PUERTOS	Allen Bradley	1783-EMS08T	1	-
30	RELE DE SEGURIDAD	MONITOREO DE RELE DE SEGURIDAD -,ENTRADAS Y SALIDAS, AUXILIARES 1NC	Allen Bradley	440R-N23132	3	1
31	POWER FLEX 525 - 3HP	DRIVE, CON ETHERNET INCORPORADO / IP y SEGURIDAD, 480 VCA, 3 FASES, 3 HP, 2.2 Kw	Allen Bradley	25B-D6PON114	5	1
32	POWER FLEX 525 - 1HP	DRIVE, CON ETHERNET INCORPORADO / IP y SEGURIDAD, 480 VCA, 3 FASES, 1.0 HP, 0,75 Kw	Allen Bradley	25B-D2P3N114	16	3
33	POWER FLEX 525 - 0,5HP	DRIVE, CON ETHERNET INCORPORADO / IP y SEGURIDAD, 480 VCA, 3 FASES, 0,5 HP, 0,4 kW	Allen Bradley	25B-D1P4N114	4	1
34	ETHERNET/IP TAP3 COPPER PORTS		Allen Bradley	1783-ETAP	2	1
35	ETHERNET/IP COMMUNICATION ADAPTOR - DUAL PORT		Allen Bradley	25-COMM-E2P	25	2
36	RELE 24VDC		Allen Bradley	700-HC24Z24	4	1
37	LLAVE SELECTORA 2 POSICIONES		Allen Bradley	800FM-KM21	1	-
38	BOTONERA DE PLASTICO GRIS, CON 6 AGUJEROS		Allen Bradley	800F-6PM	1	-
39	SENSOR INDUCTIVO		Balluf	M18MI-PSC80B-S04G	31	5
40	SENSOR DIFUSO	RETRORFLECTIVE SENSOR WITH POLARIZING FILTER	Banner	QS18VP6LV	42	8
41	SENSOR DIFUSO	RANGE: 450mm; INPUT: 10-30V DC	Banner	QS18VP6DQ8	1	1
42	SENSOR LASER DIFUSO	RANGE: 300 mm; INPUT: 10-30V DC	Banner	QS18V/P6LDQ8	3	1
43	SENSOR LASER POLAR RETRO	RANGE: 0.1 to 10 m; INPUT: 10-30V DC	Banner	QS18V/P6LLPQ8	2	1
44	SENSOR FOTOELECTRICO		Datalogic	S62-PA-2-C11-PN		
45	SENSOR MAGNETICO	SENSOR, SOLID STATE, DIRECT MOUNT, GROMMET CONNECTION	Festo	SME-8M-DS-24V-K0,3-MBD	6	1
46	TOMACORRIENTE	220VCA GABINETE	Cambre	6904	1	-
47	CONVERSOR SERIAL TCP-IP Y ENTRADAS - SALIDAS DIGITALES		Exemys	SSE232-1C4C-1A	2	1
48	EGW1-MB MODBUS TCP A MODBUS ASCII / RTU CONVERTIDOR		Exemys	EGW1-1C4C-1A-MB	2	1
49	SWITCH ETHERNET	5 PUERTOS	Hirschmann	SPIDER 5TX	1	1
50	TUBO FLUORESCENTE	ILUMINACION GABINETE	Lumenac	Micro 18	4	-
51	RELE 24VDC	RELE DE SEGURIDAD	Omron	G75A-5A1B-DC24	6	1
52	REFRIGERACION	GABINETE	RKL	RKL32800-1M	2	-
53	INTERRUPTOR TERMOMAGNETICO	2X2A	Schneider	24332	11	1
54	INTERRUPTOR TERMOMAGNETICO	2X6A	Schneider	24335	9	1
55	INTERRUPTOR TERMOMAGNETICO	1X2A	Schneider	24396	4	1
56	INTERRUPTOR TERMOMAGNETICO	1X6A	Schneider	24399	3	1
57	INTERRUPTOR TERMOMAGNETICO	3x16A	Schneider	24350	2	1
58	VACUOSTATO		SMC	ZSE3CAF-01-F-L	8	1
59	VALVULA DE CORTE GENERAL		SMC	AV4000-04-5DZ	3	-
60	ELECTROVALVULA		SMC	SY7260-5LZ-02	22	3
61	SENSOR MAGNETICO	SENSOR, SOLID STATE, DIRECT MOUNT, GROMMET CONNECTION	SMC	D-M9B	22	3
62	INTERRUPTOR 100A		Terasaki	S125NJ100P3	1	-
63	BORNE PORTAFUSIBLE		Zoloda	BKF-SI-520-LL024	245	-
64	BORNE DE PASO 1,5		Zoloda	BKF-1.5	362	-
65	BORNE DE PASO 2,5		Zoloda	BKF-2,5	88	-
66	FUSIBLE 5X20MM. 2A				96	10
67	FUSIBLE 5X20MM. 0,25A				149	20
68	TRANSMISOR Y 3 CONTROL REMOTO		Saga	Saga-L4	1	-
69	SWITCH MECANICO		Neumann	FK-1002	4	1
70	SWITCH MECANICO		Neumann	CB-144	4	
71	MICRO SWITCH		Neumann	MPH-3R	6	

5	4	3	2	1	5	4	3	2	1	PROJECT SIDERCROM EMBALAJE	SHEET SHEETS 131 131				
										REVISION BY LJA	REVISION BY E.G.	REVISION BY FIELD	BILL OF MATERIAL		
										REV. CLIENTE	REV. INT. P&F				
										P&F		Av. Belgrano 2272 (BIBI)VD Don Torcuato Buenos Aires Argentina			
										DATE 29/06/2015	SCALE	DIVISION PLANT/SIDERCROM		LOCATION/ACTIVITY	